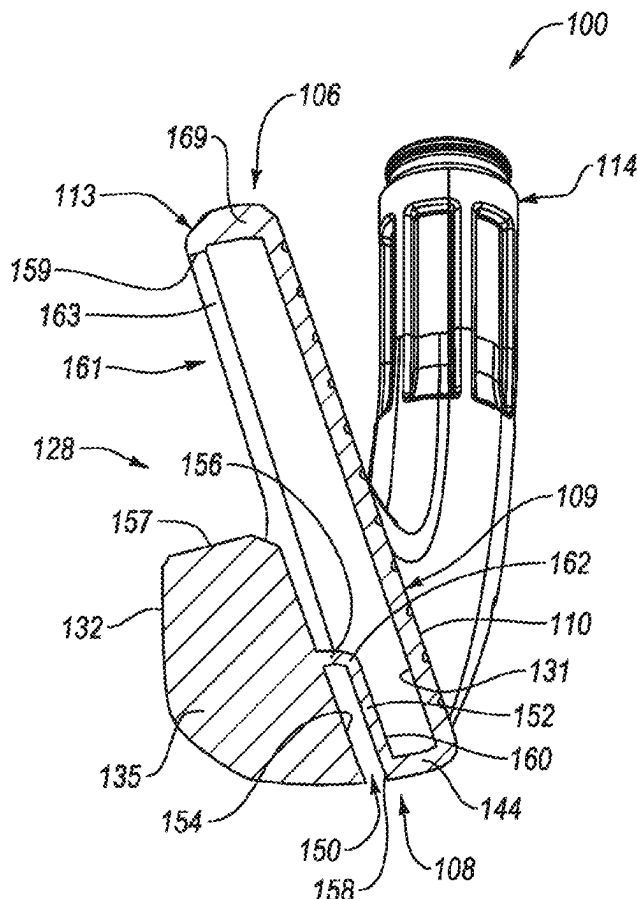


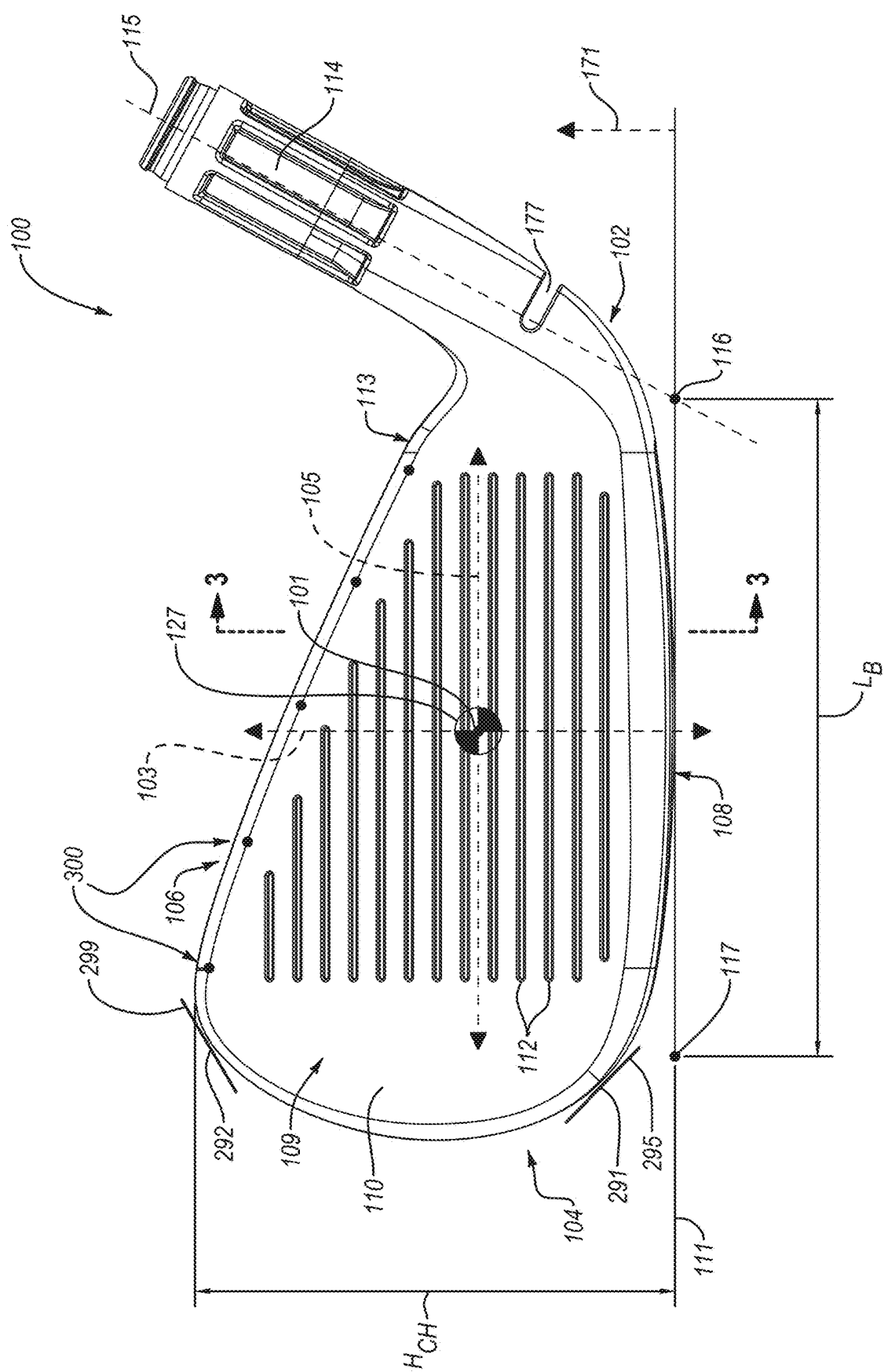


US 20250281803A1

(19) **United States**(12) **Patent Application Publication****Halberg et al.**(10) **Pub. No.: US 2025/0281803 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **GOLF CLUB HEAD**(71) Applicant: **Taylor Made Golf Company, Inc.**,
Carlsbad, CA (US)(72) Inventors: **Connor Halberg**, Vista, CA (US);
Adam Warren, Carlsbad, CA (US);
Allan Saliba, Carlsbad, CA (US); **Scott Taylor**,
Bonita, CA (US); **Brian Hill**, Carlsbad, CA (US);
Conner Sarich, Carlsbad, CA (US); **Mike Walker**,
Vista, CA (US)(21) Appl. No.: **19/040,820**(22) Filed: **Jan. 29, 2025****Related U.S. Application Data**(63) Continuation-in-part of application No. 18/506,843,
filed on Nov. 10, 2023, which is a continuation-in-part
of application No. 17/889,187, filed on Aug. 16, 2022,
now Pat. No. 11,951,365, which is a continuation-in-
part of application No. 17/368,520, filed on Jul. 6,
2021, now Pat. No. 11,458,374, which is a continu-
ation-in-part of application No. 17/330,033, filed on
May 25, 2021, now Pat. No. 11,413,510, which is a
continuation-in-part of application No. 17/132,541,
filed on Dec. 23, 2020, now Pat. No. 11,400,351,which is a continuation-in-part of application No.
16/870,714, filed on May 8, 2020, now Pat. No.
11,406,882.(60) Provisional application No. 62/846,492, filed on May
10, 2019, provisional application No. 62/954,211,
filed on Dec. 27, 2019, provisional application No.
62/954,211, filed on Dec. 27, 2019, provisional ap-
plication No. 63/616,507, filed on Dec. 29, 2023.**Publication Classification**(51) **Int. Cl.**
A63B 53/04 (2015.01)(52) **U.S. Cl.**
CPC **A63B 53/0433** (2020.08); **A63B 53/047**
(2013.01)(57) **ABSTRACT**

An iron-type golf club head that comprises a heel portion, a toe portion, a sole portion, a top portion, a front portion comprising a strike face and a front interior surface opposite the strike face, and a rear portion. The golf club head also comprises a discrete deflection impeding surface spaced apart from the front interior surface such that a gap, perpendicular to the strike face, is defined between the front interior surface and the discrete deflection impeding surface. The gap is between, and inclusive of, 75% and 125% of a minimum face thickness of the front portion.





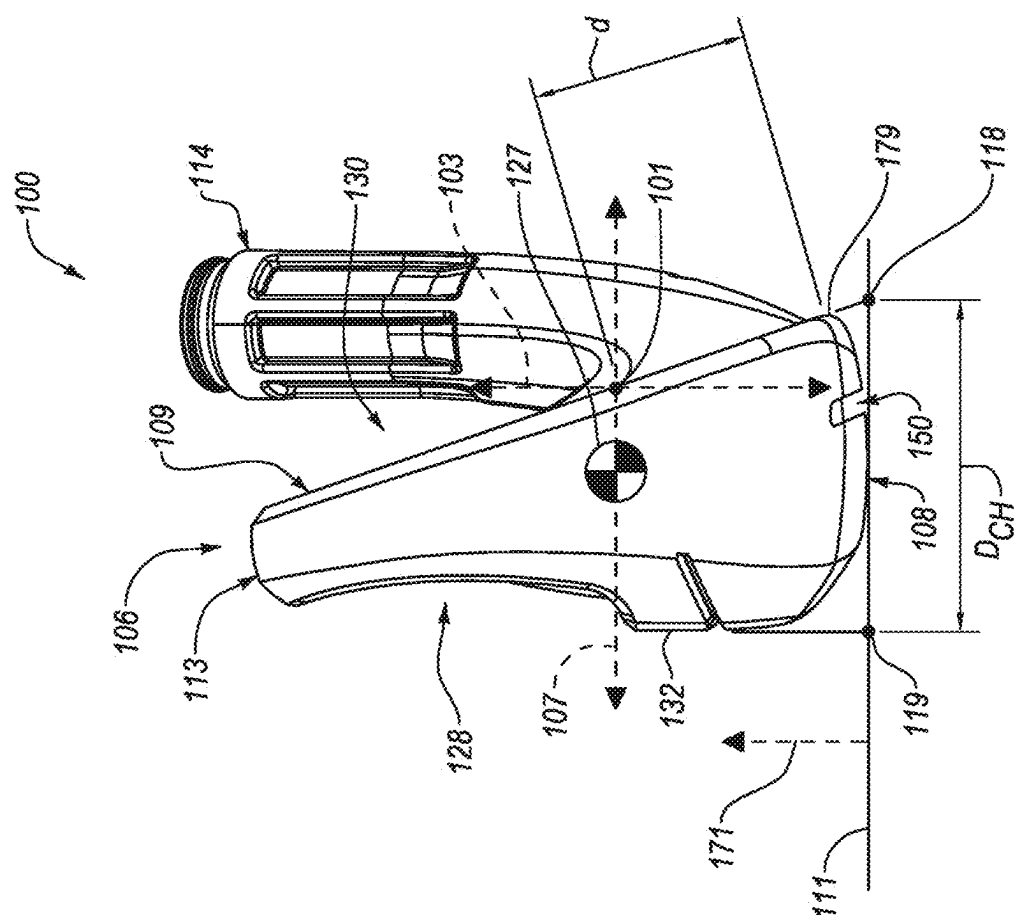


FIG. 2

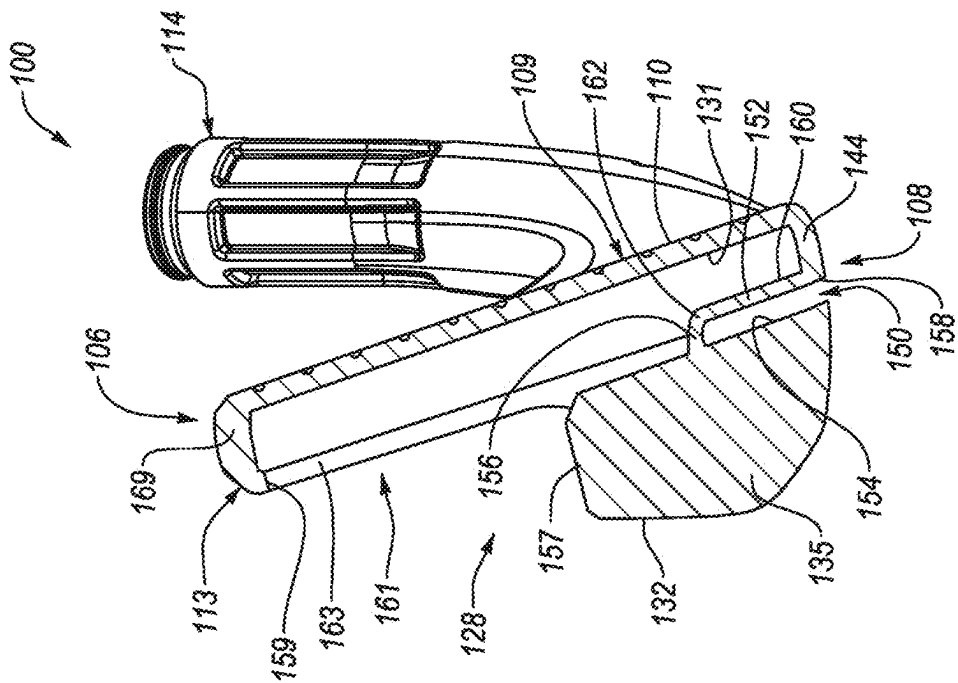
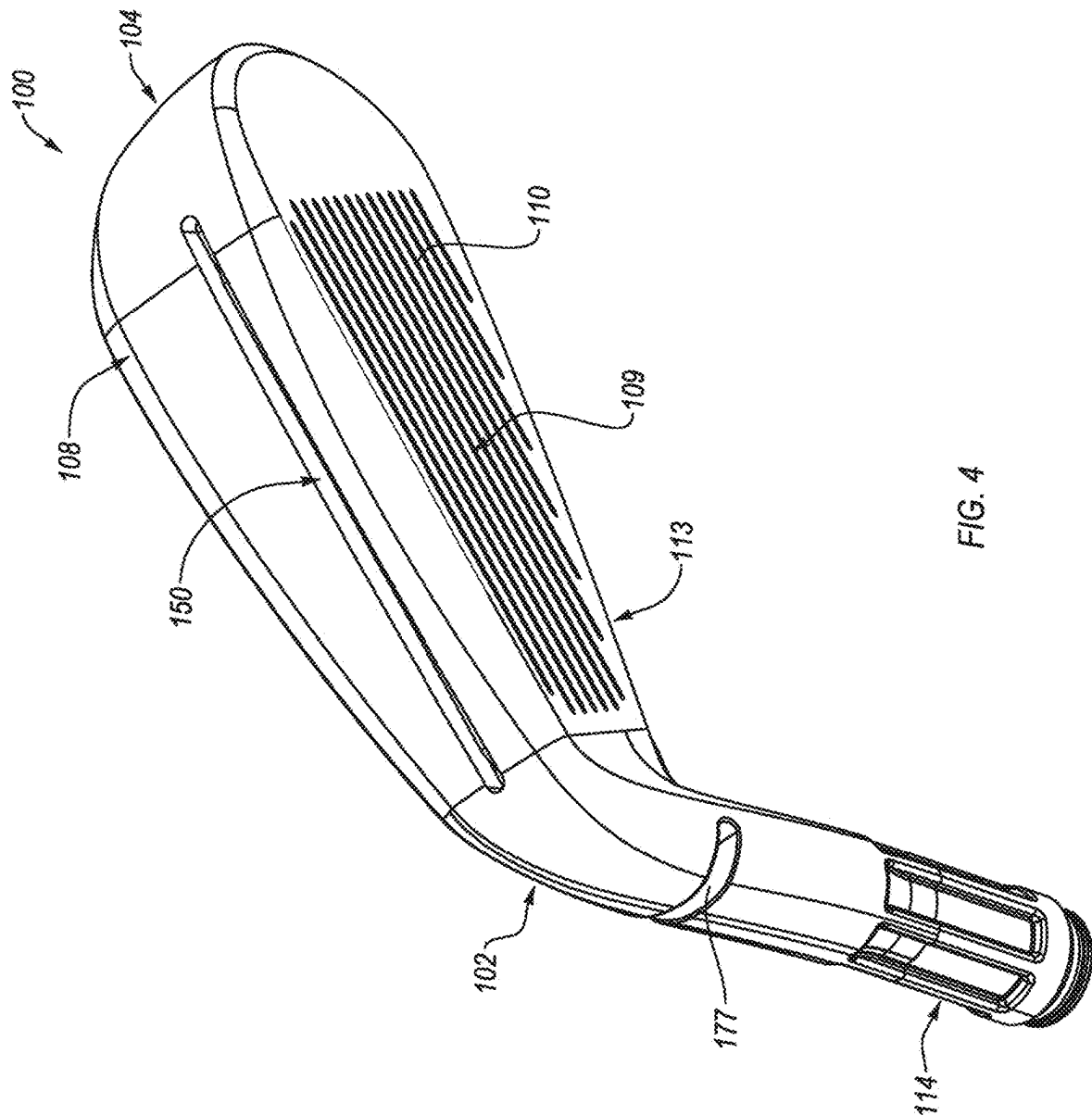


FIG. 3



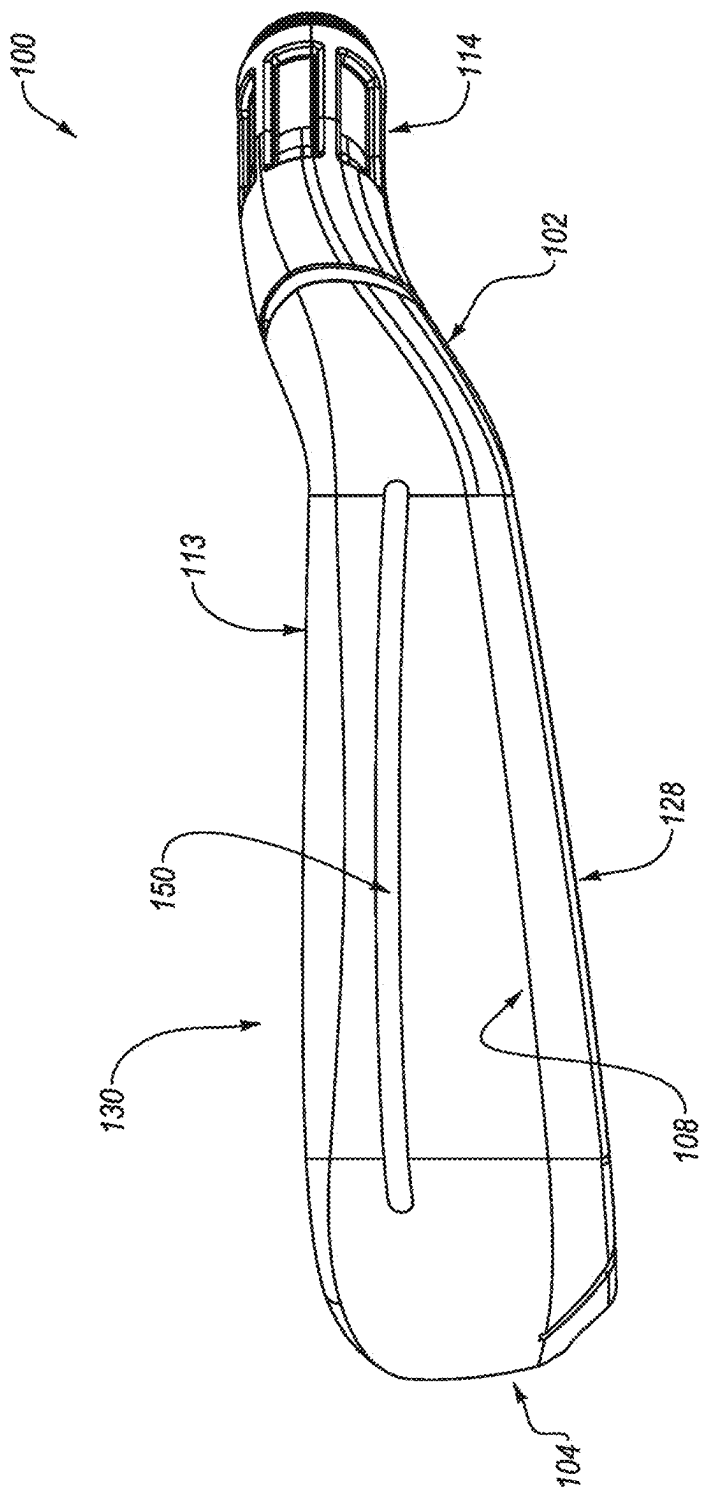


FIG. 5

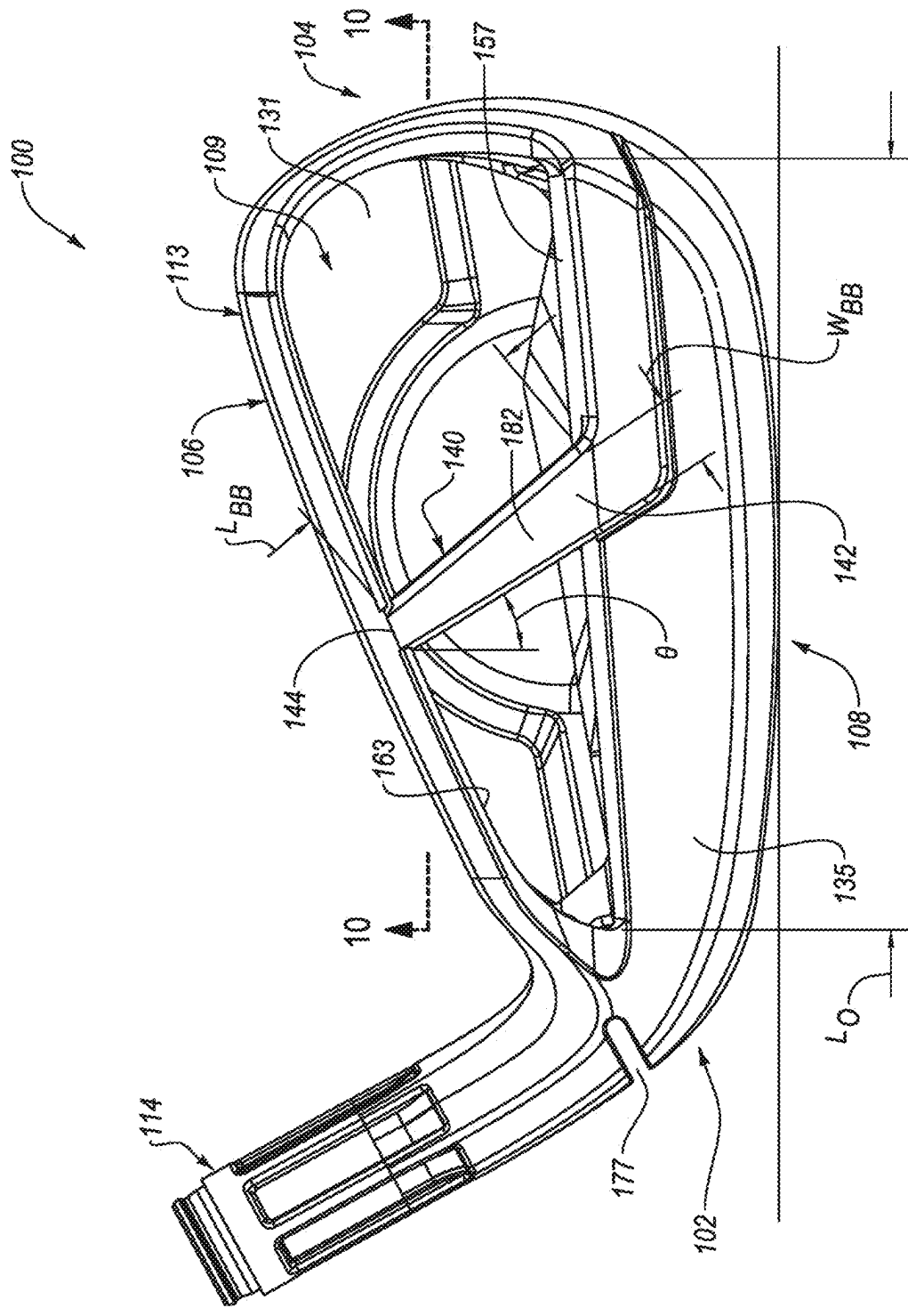


FIG. 6

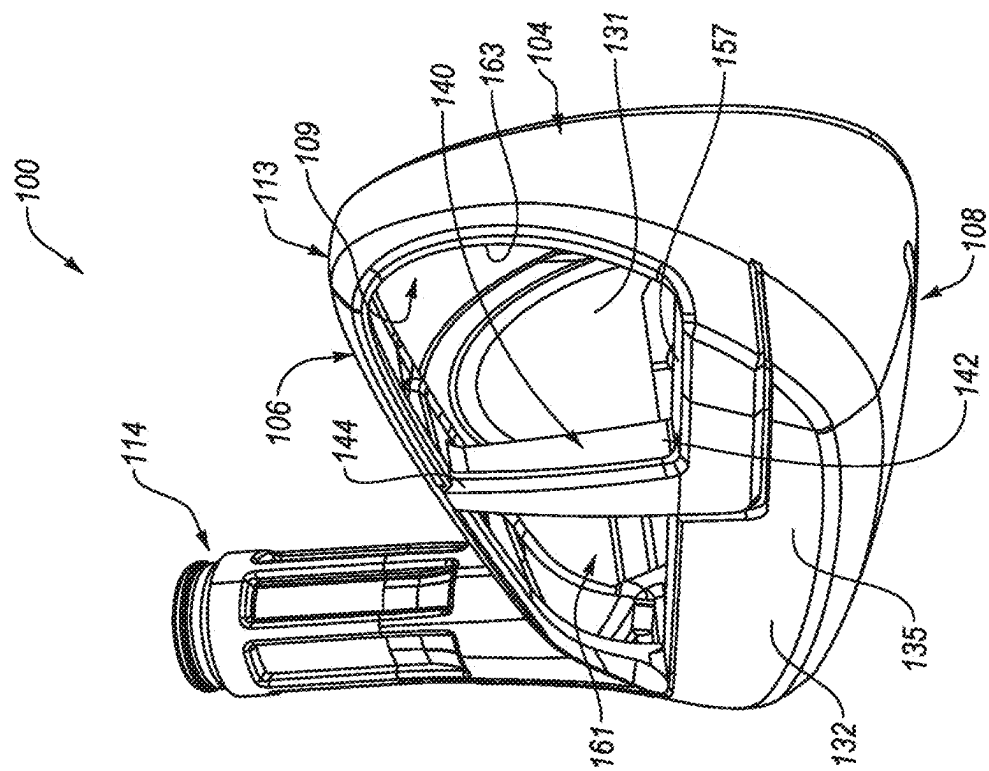


FIG. 7

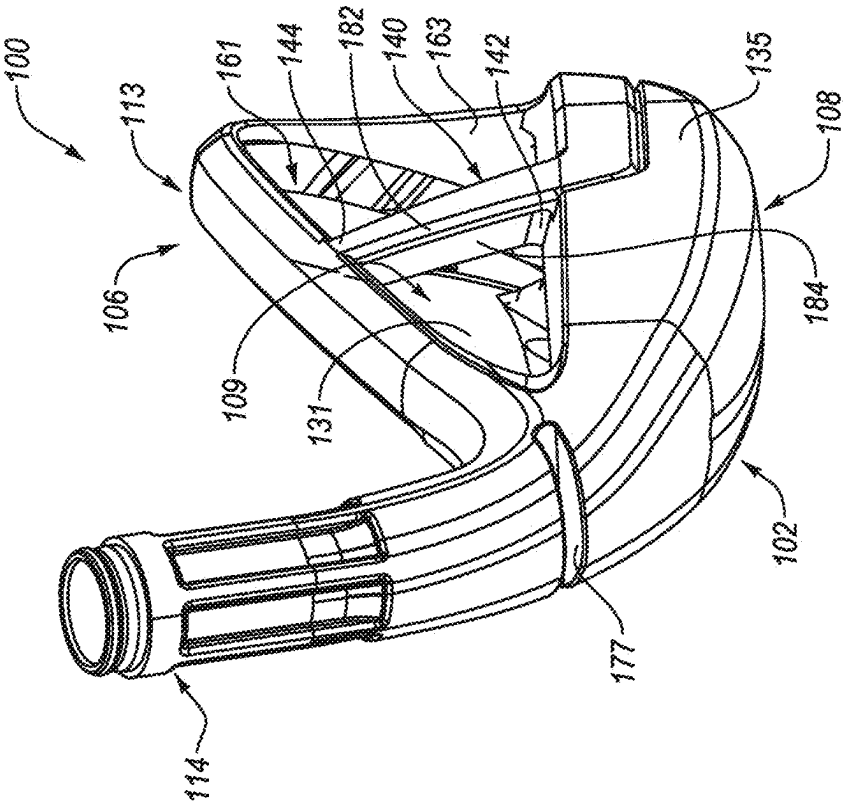


FIG. 8

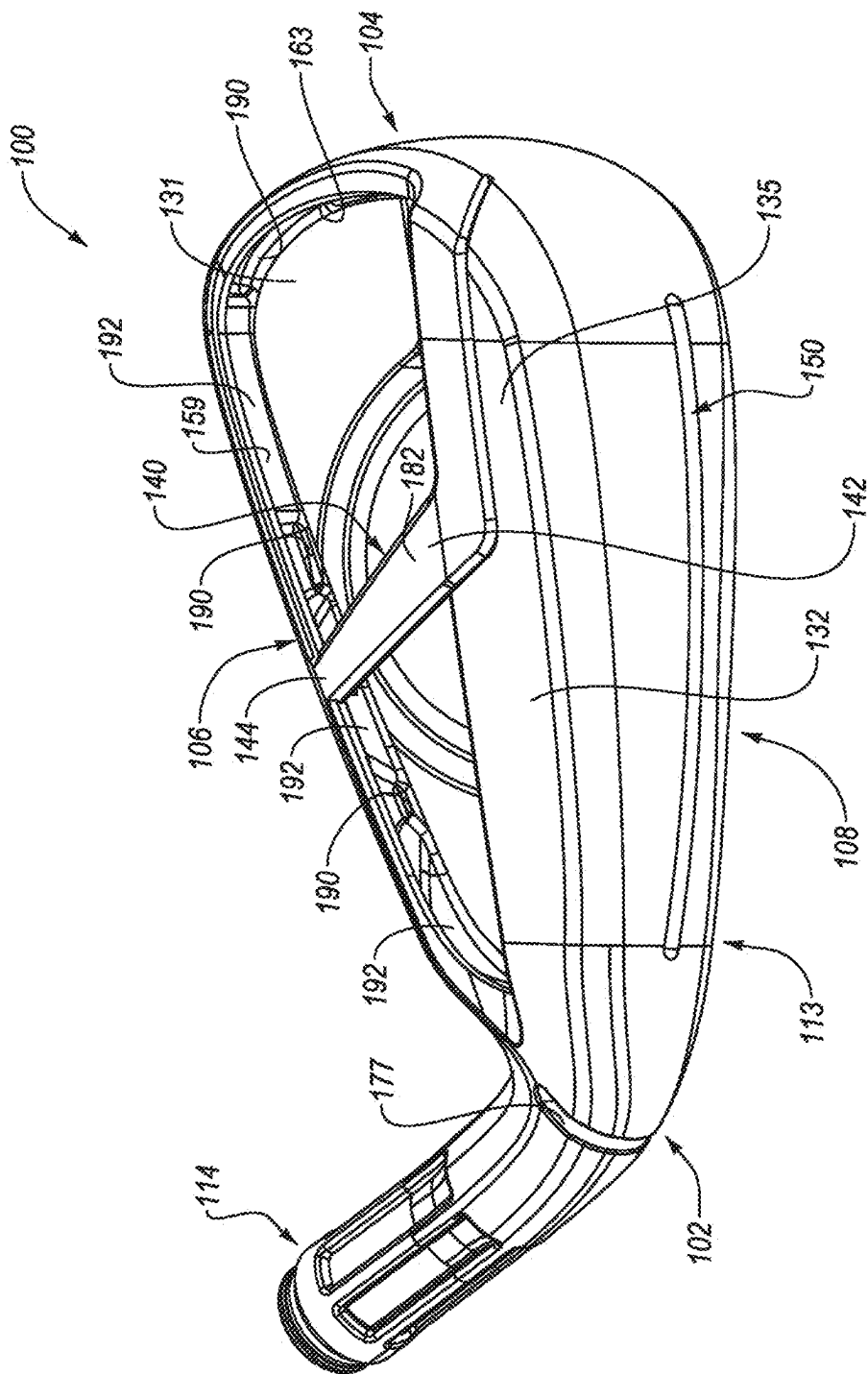


FIG. 9

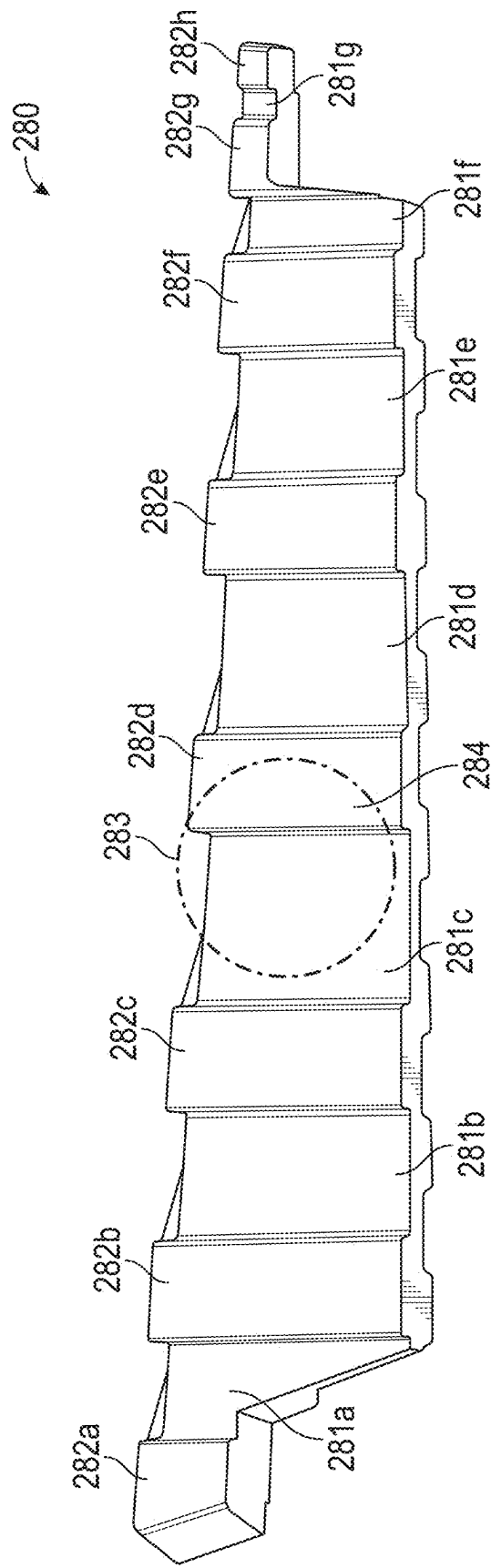


FIG. 10

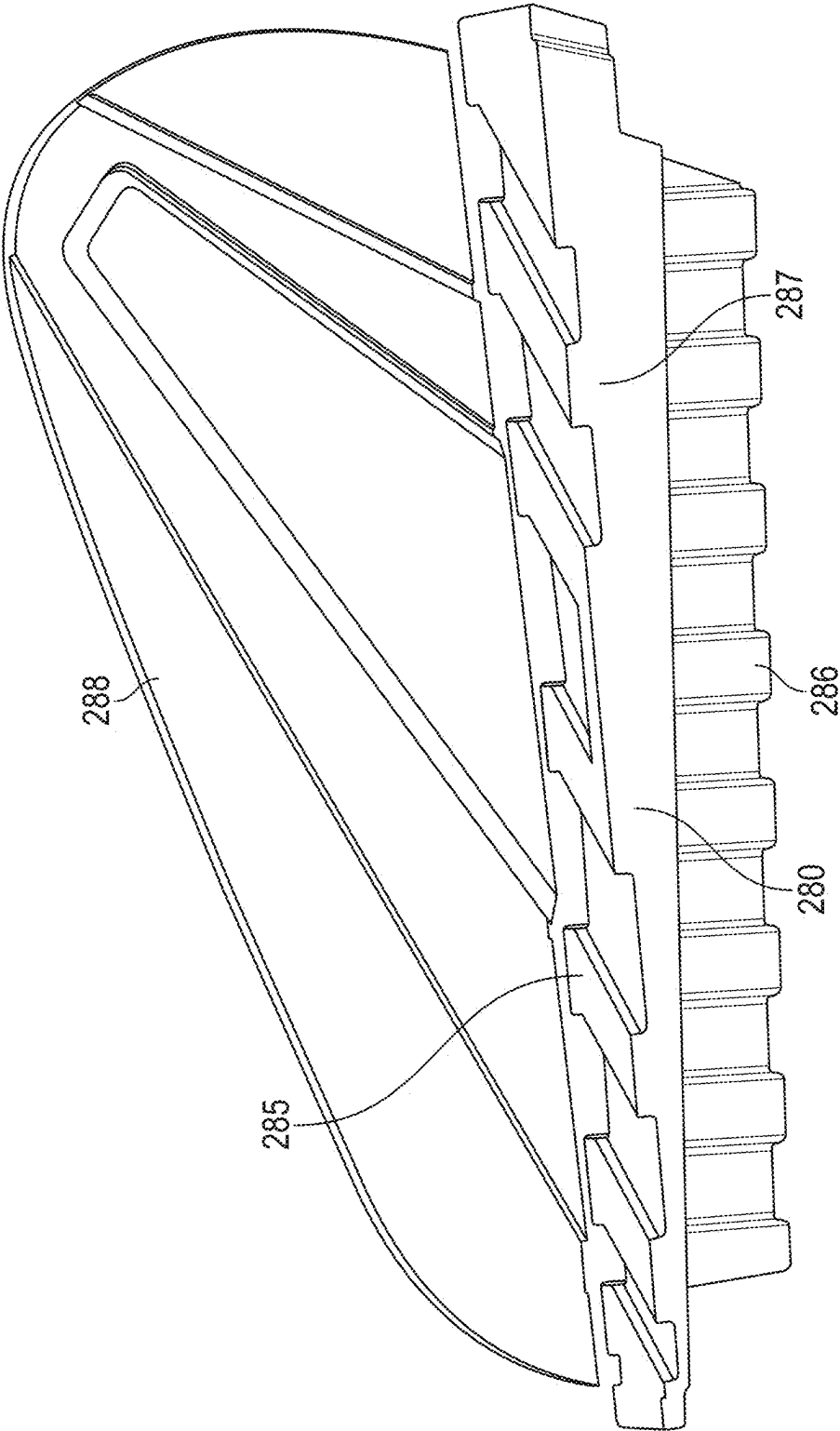


FIG. 11

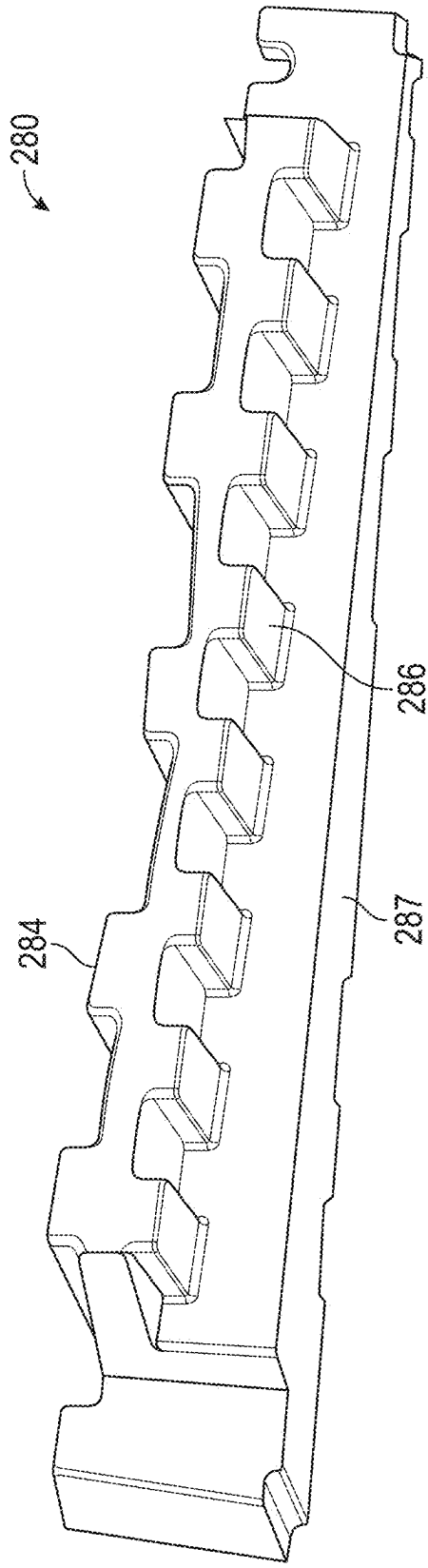


FIG. 12

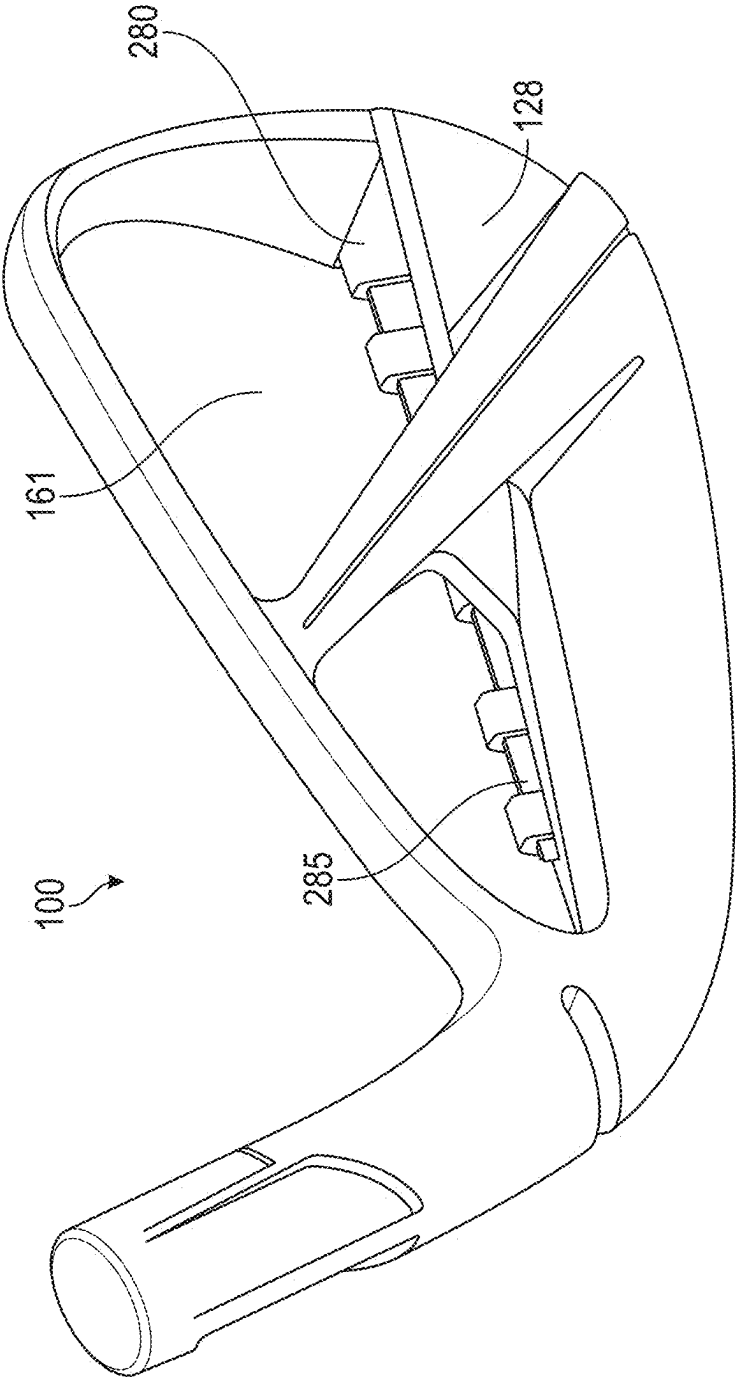


FIG. 13

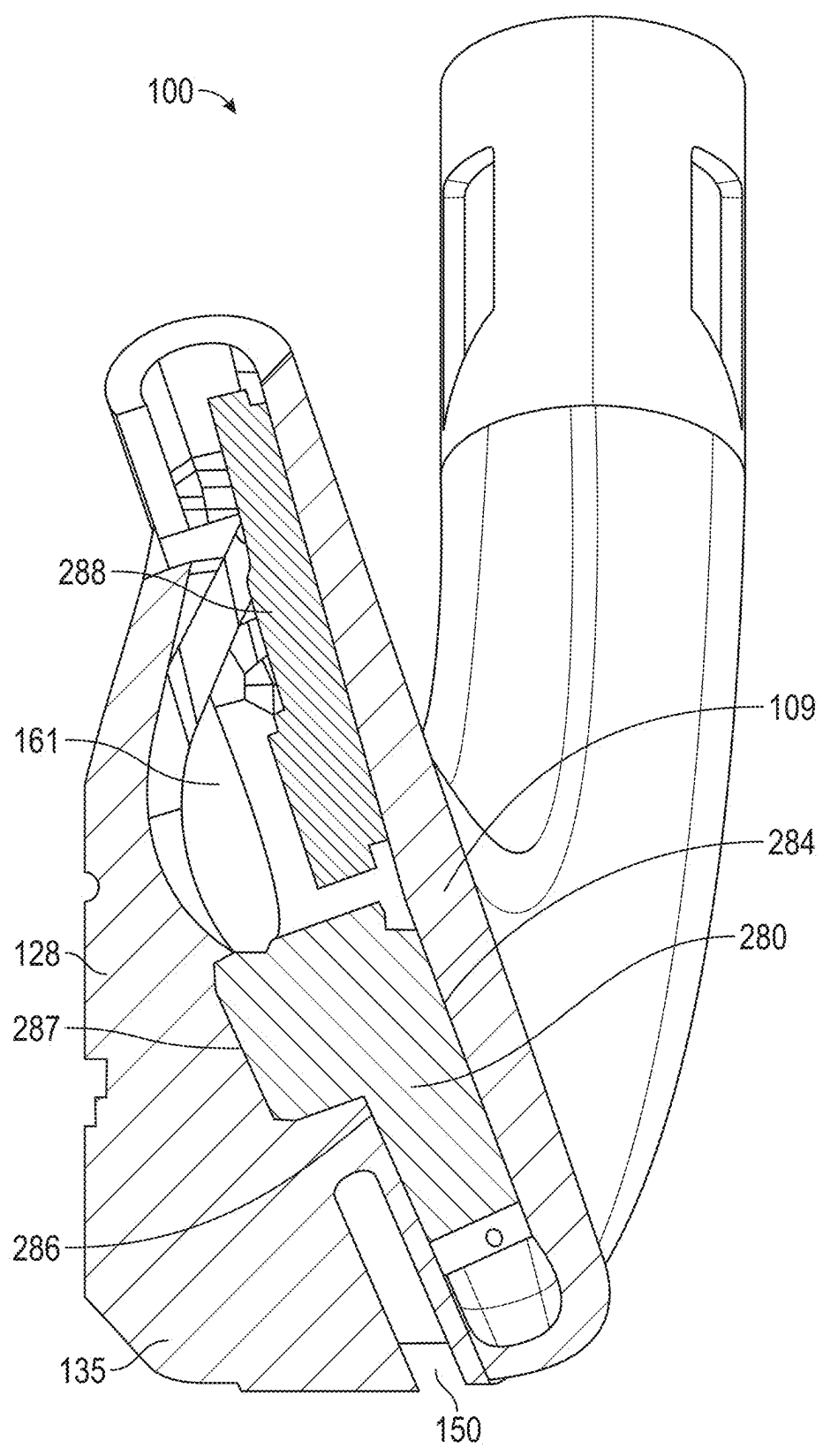


FIG. 14

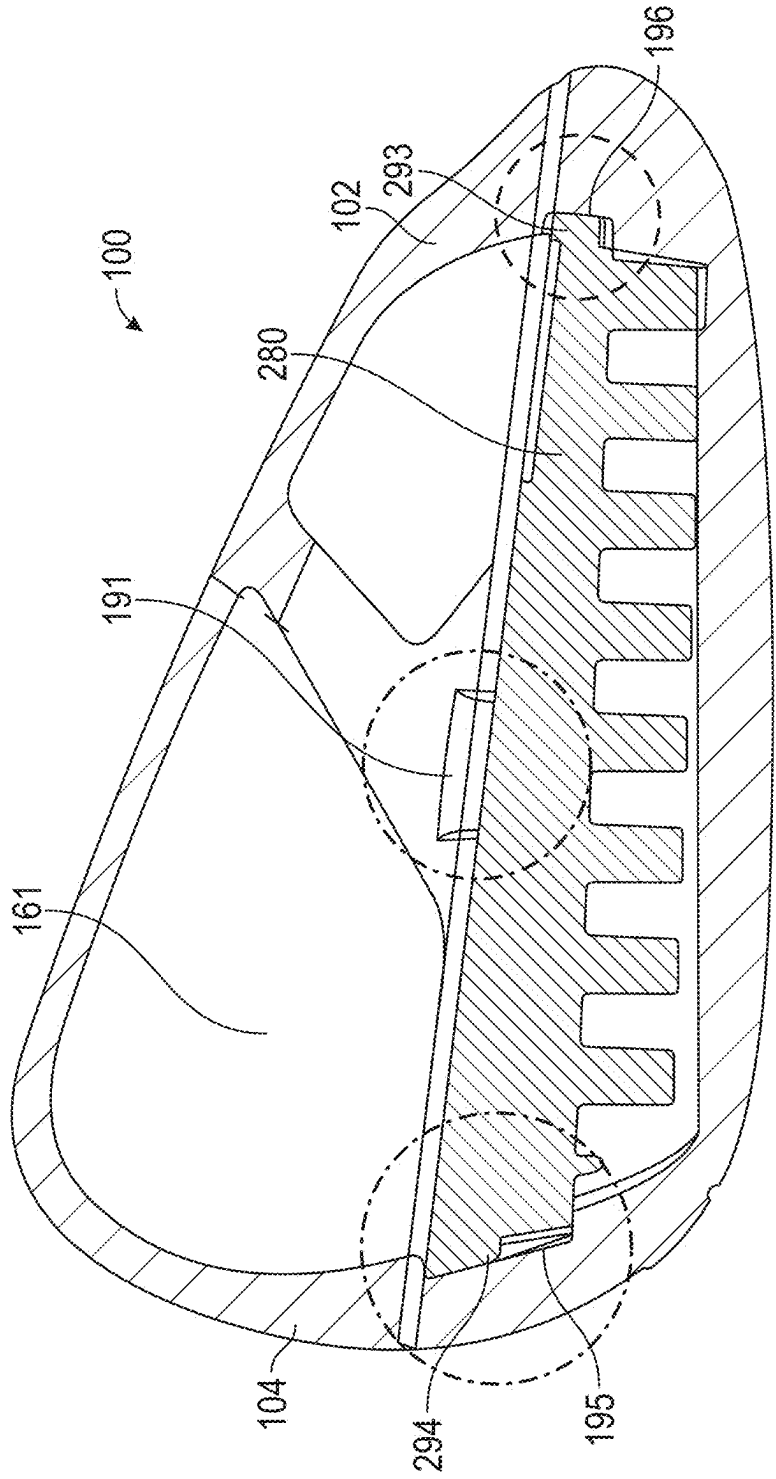


FIG. 15

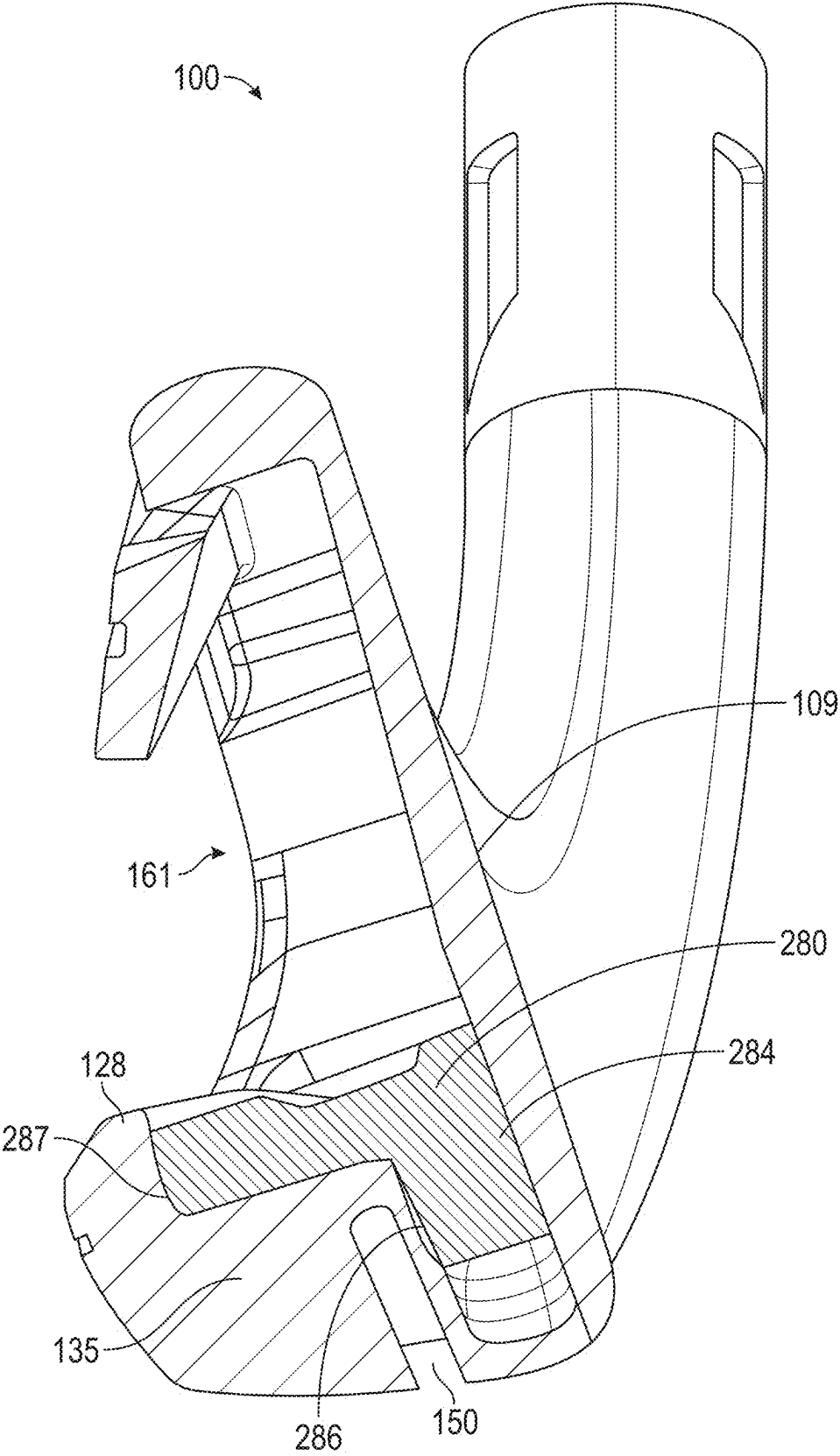


FIG. 16

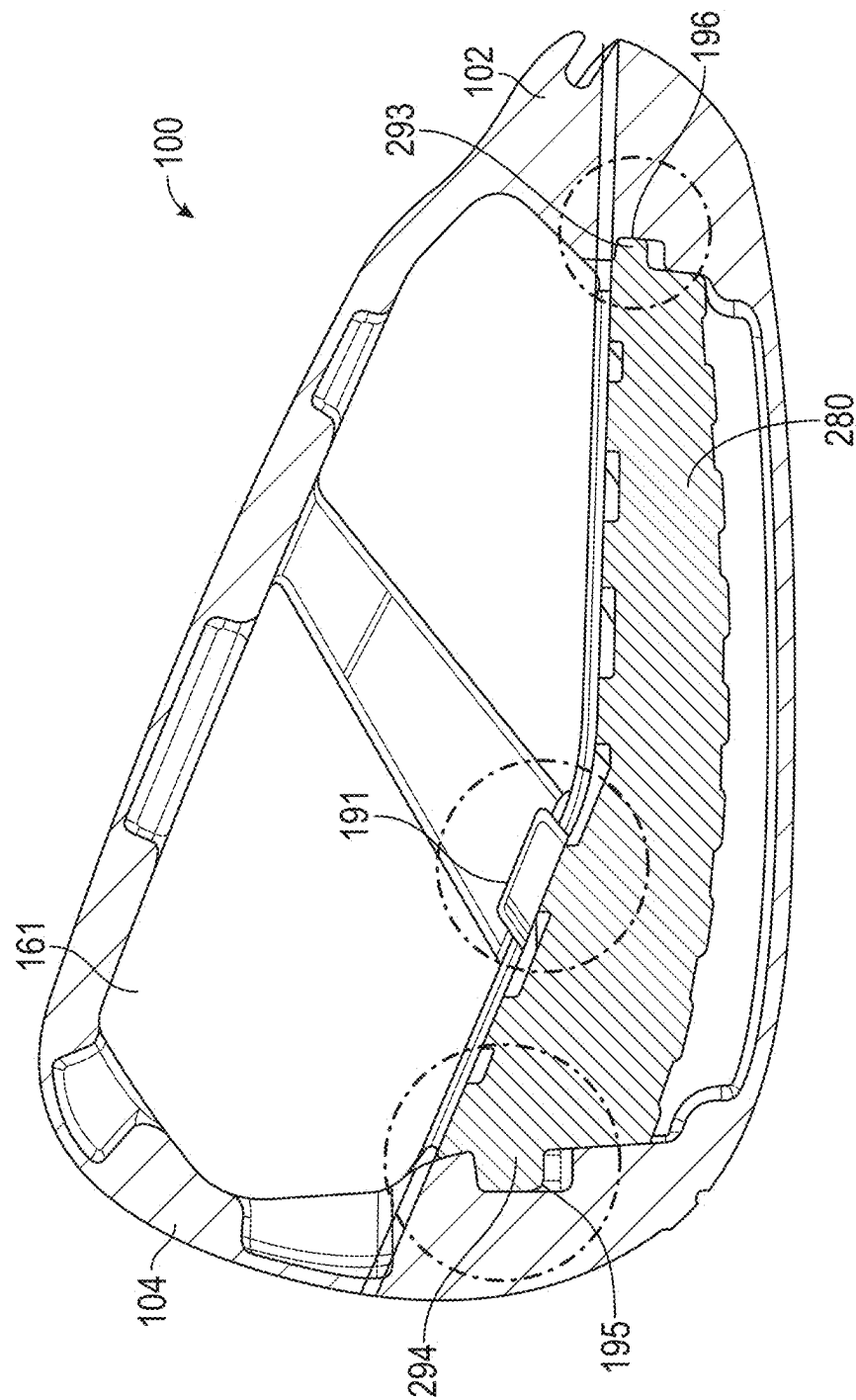


FIG. 17

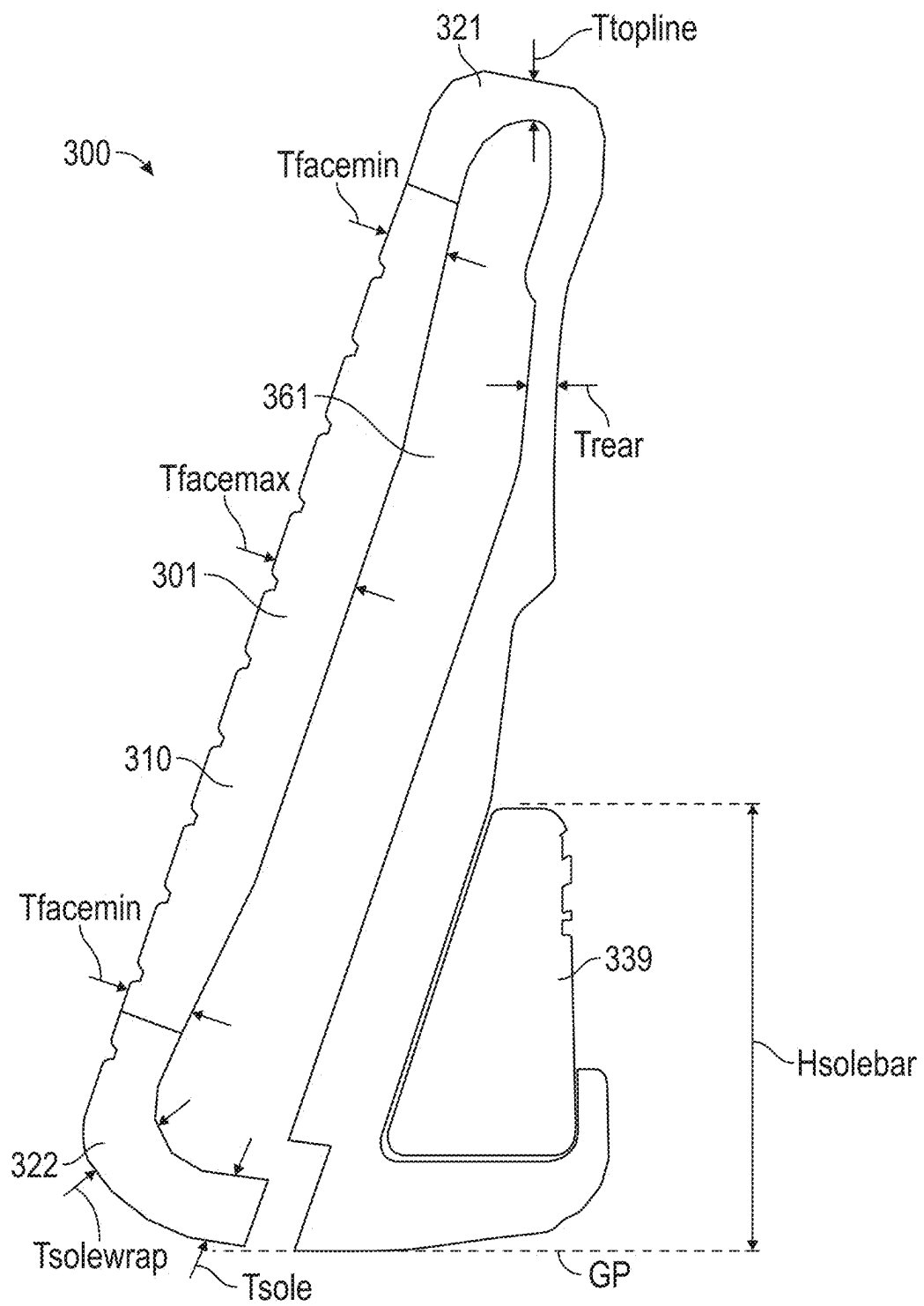


FIG. 18

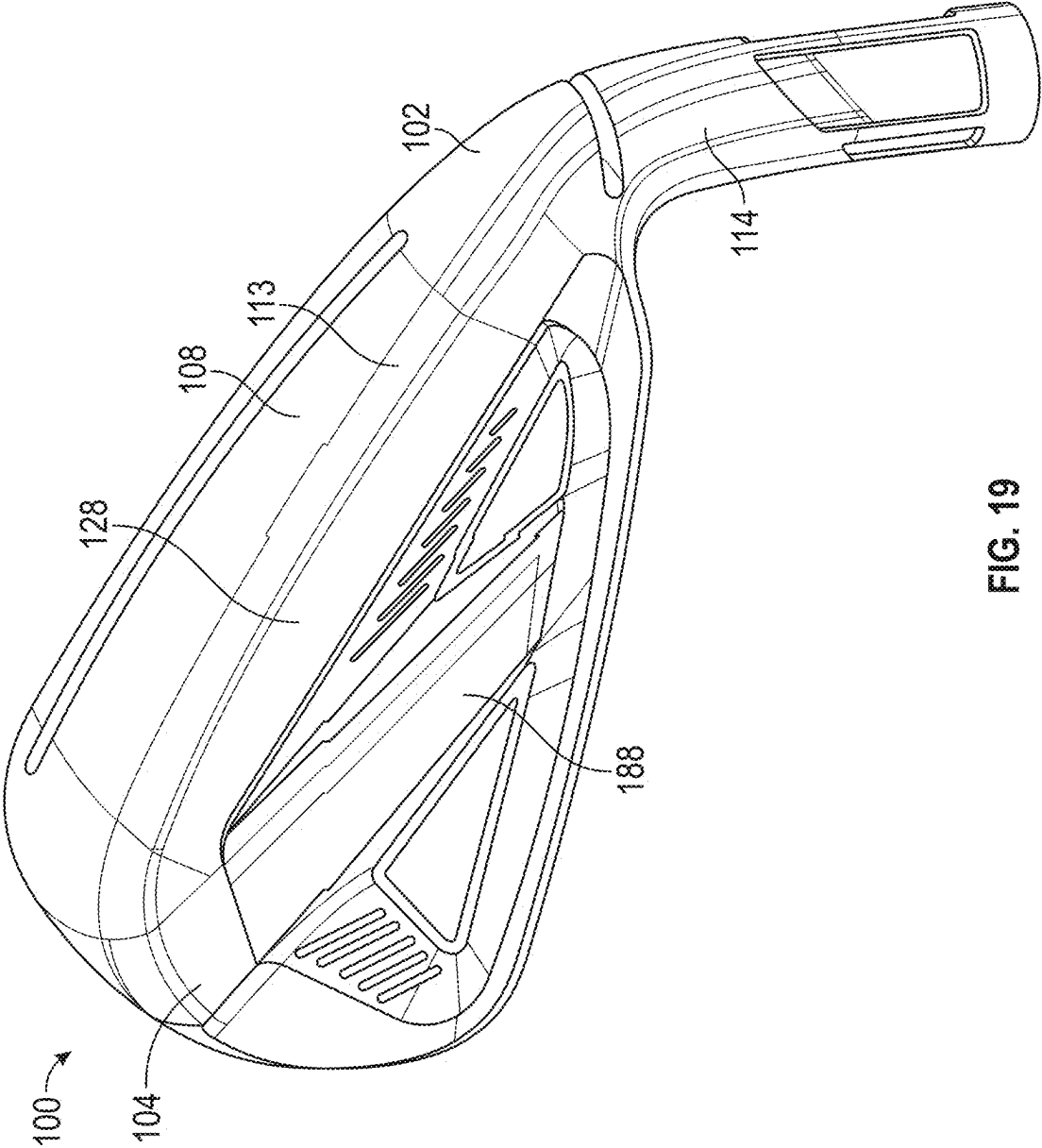
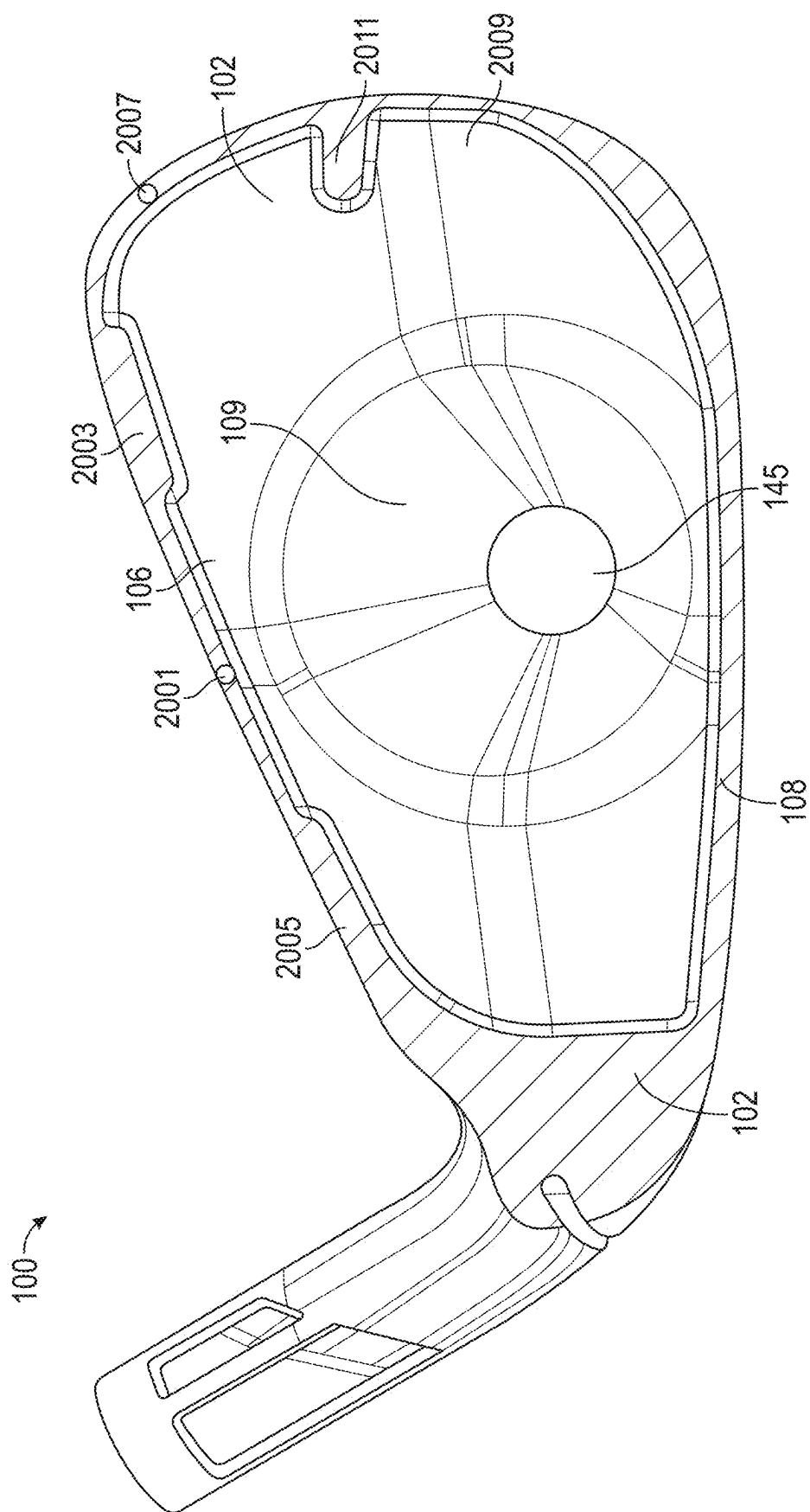


FIG. 19



2000

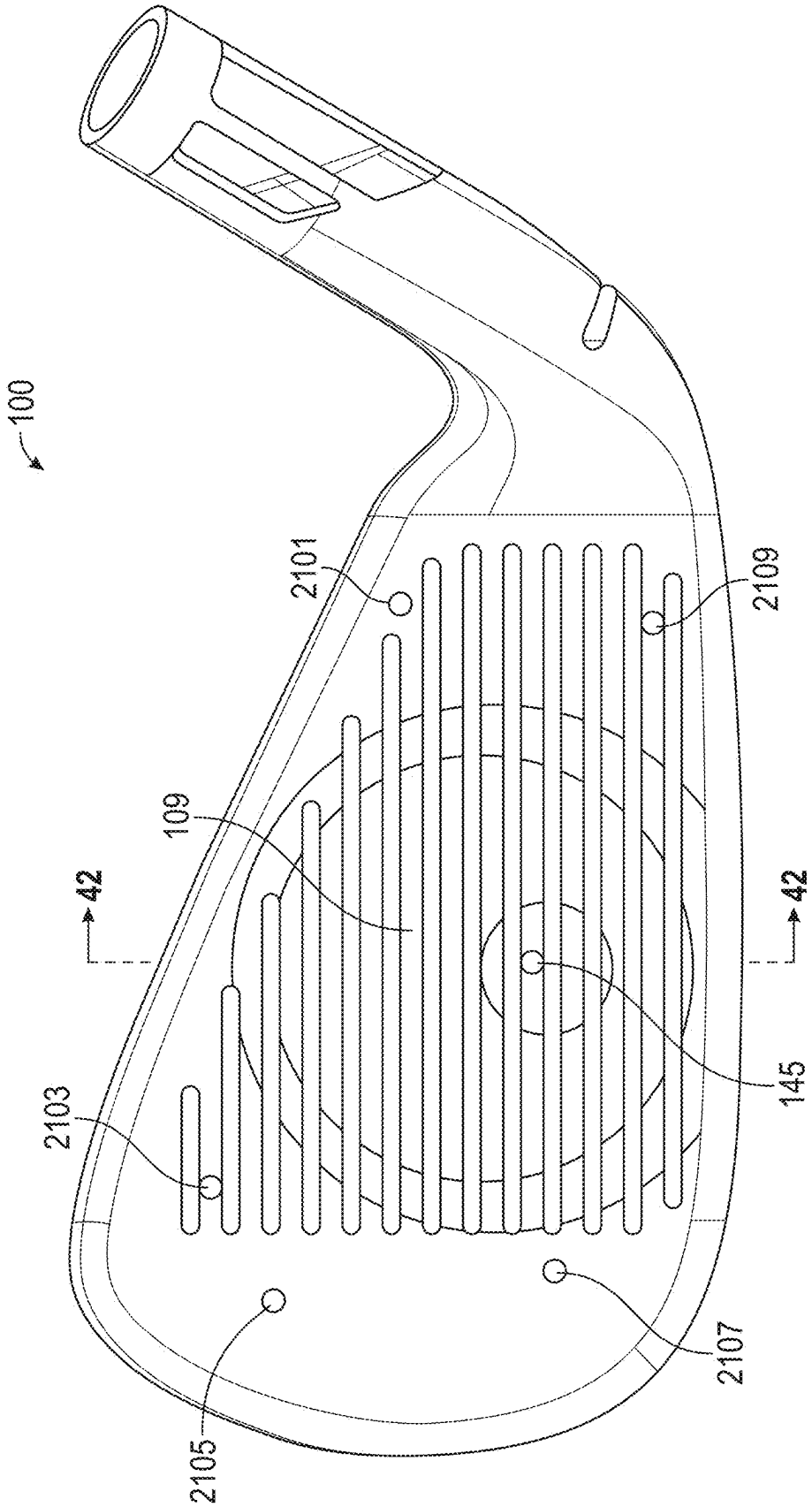
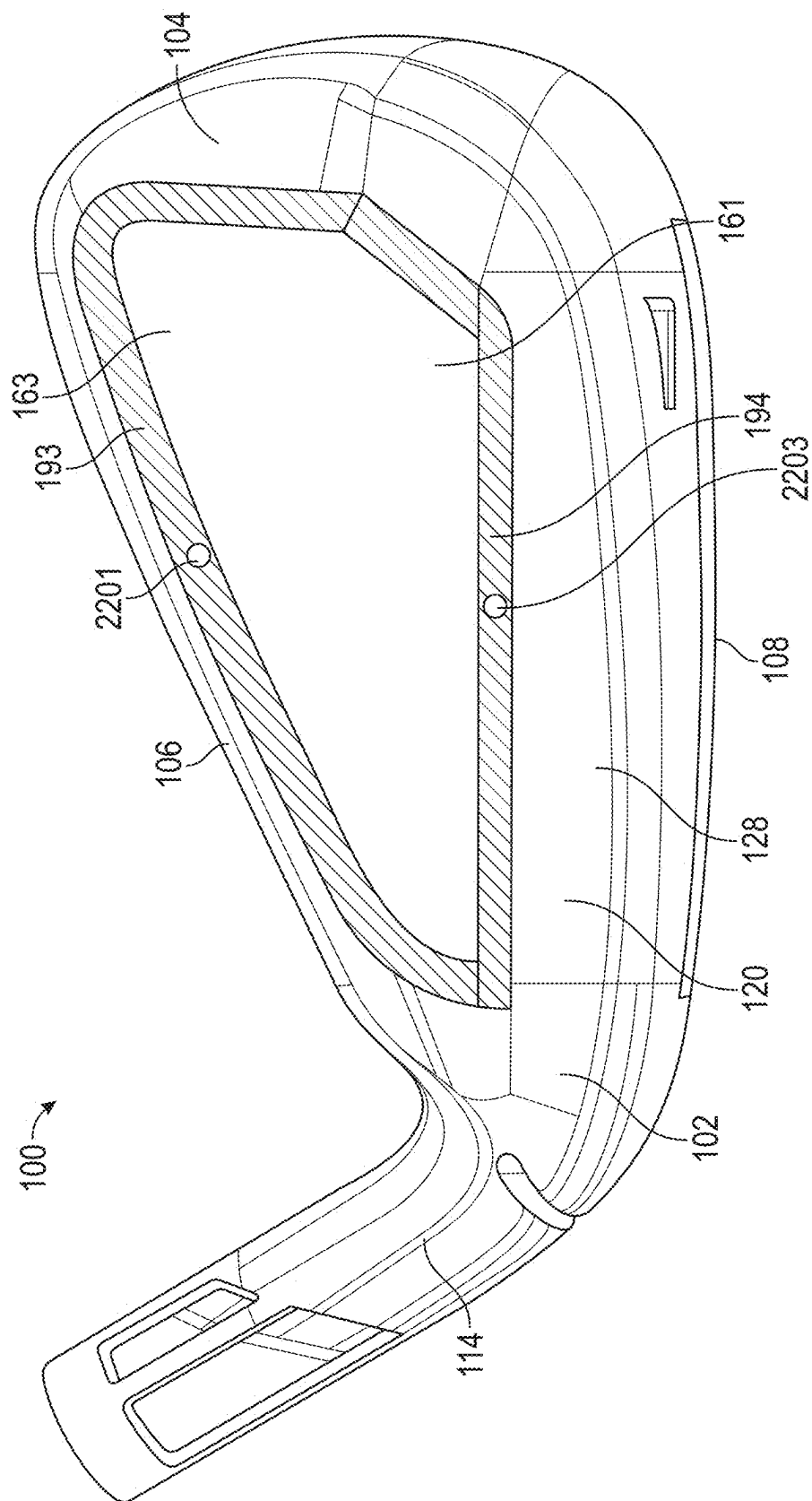


FIG. 21



22
C.
L

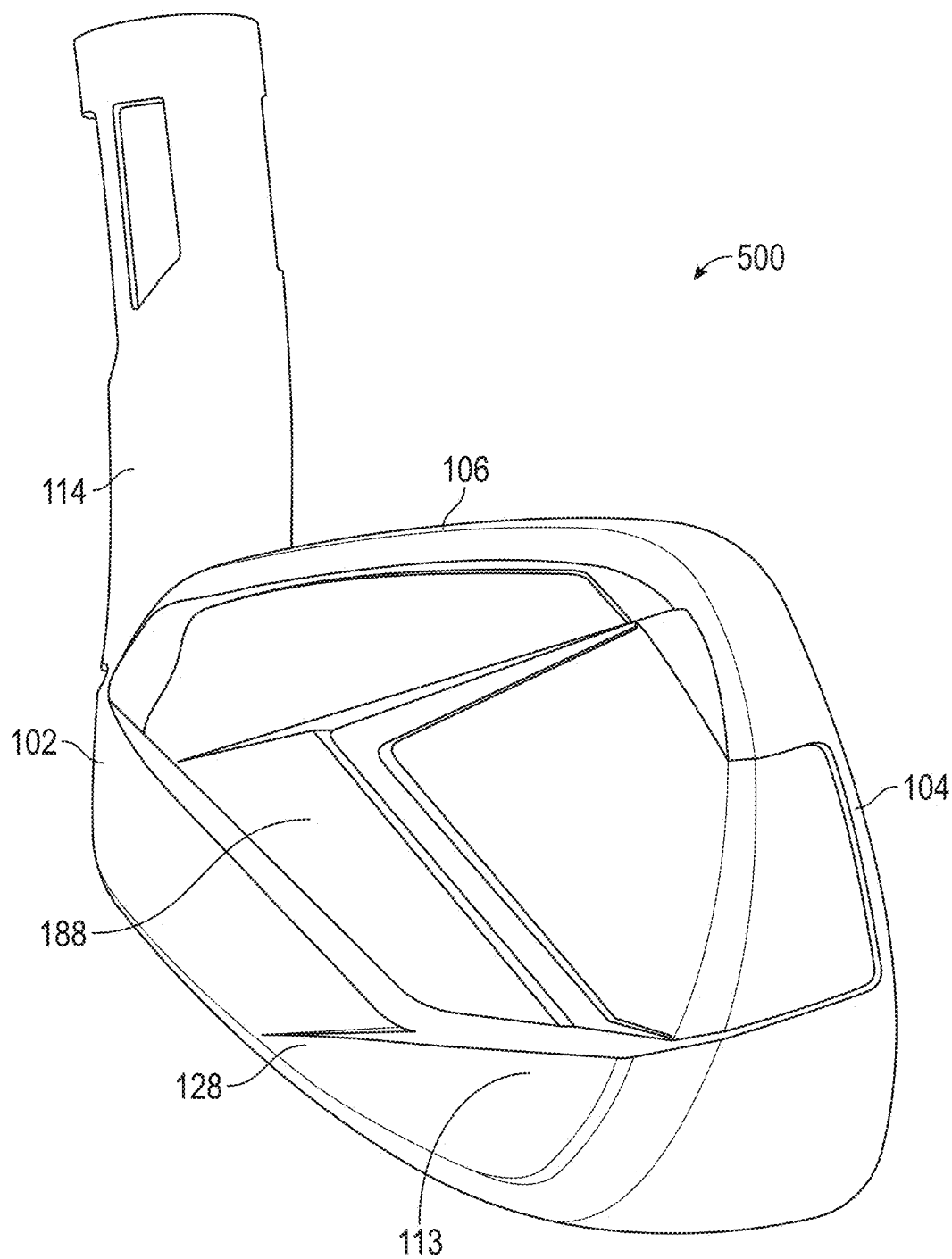


FIG. 23

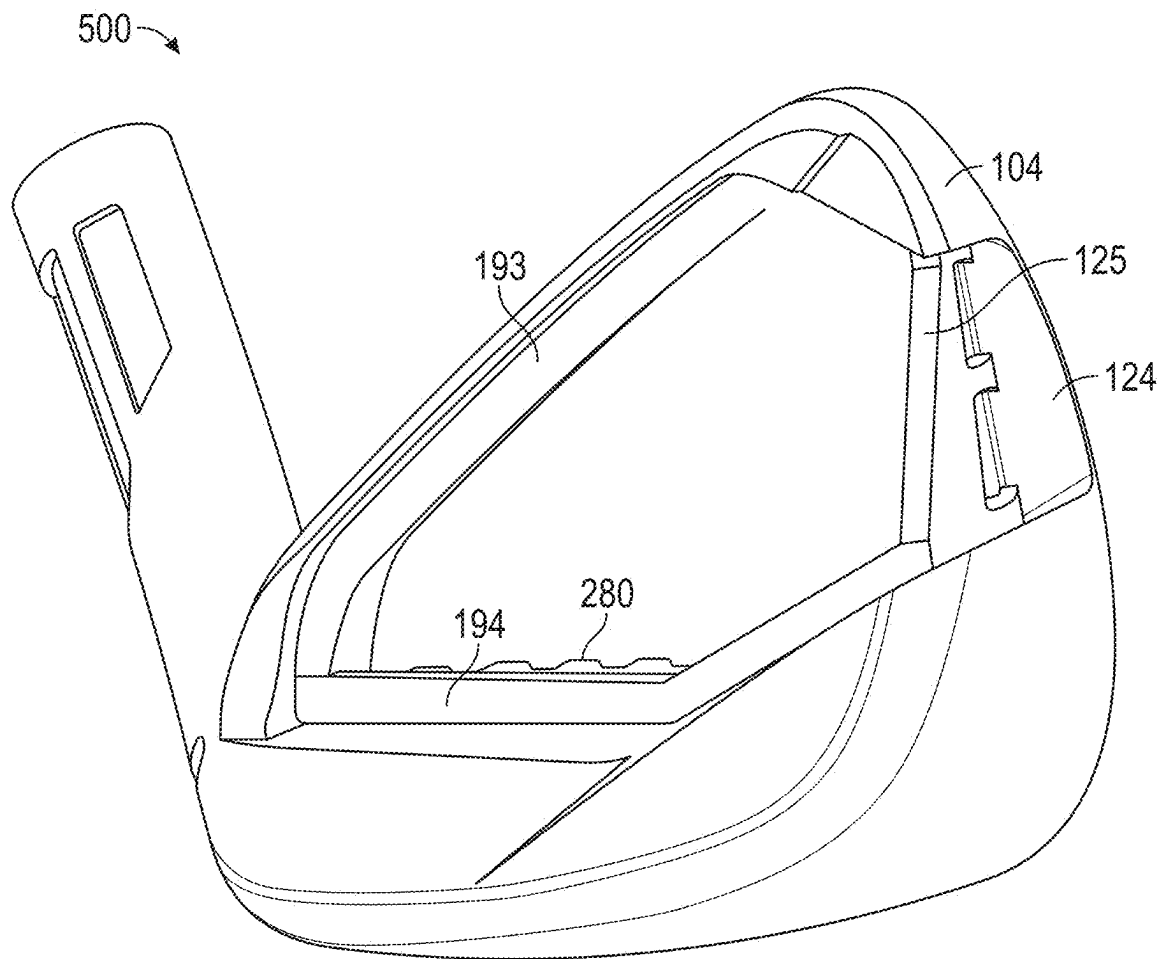


FIG. 24

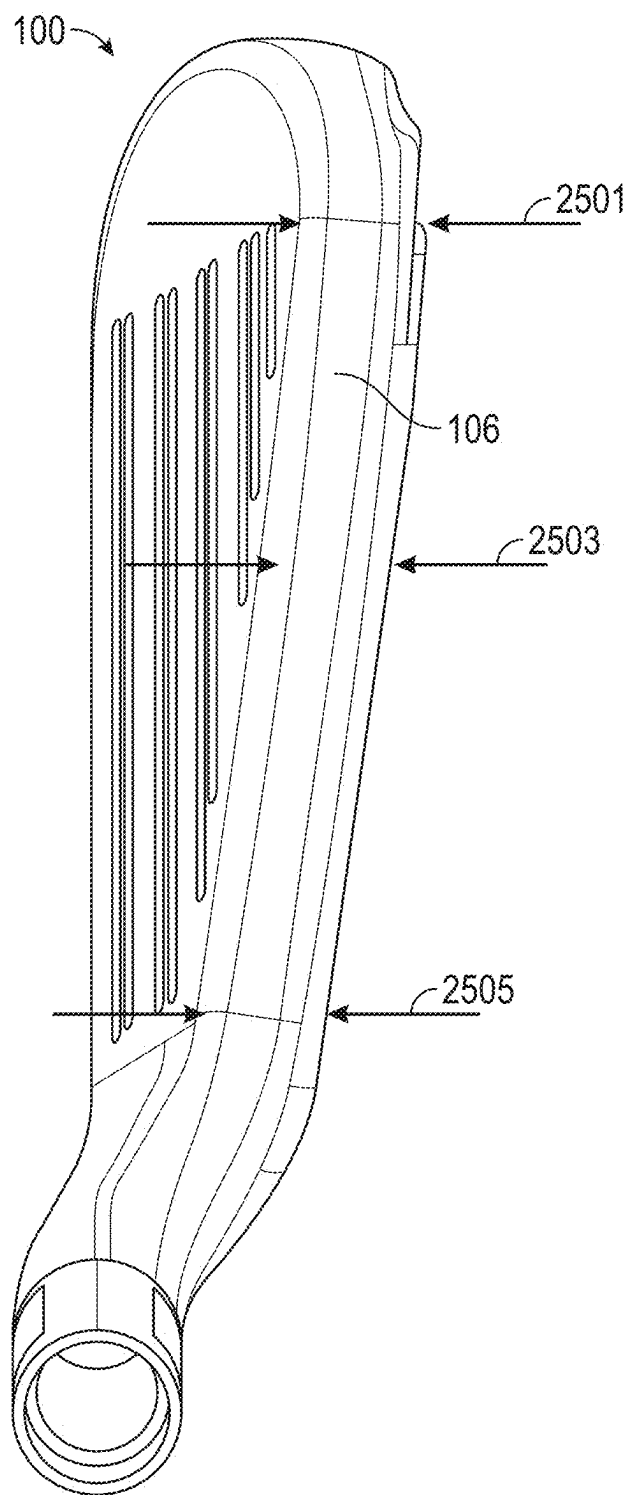


FIG. 25

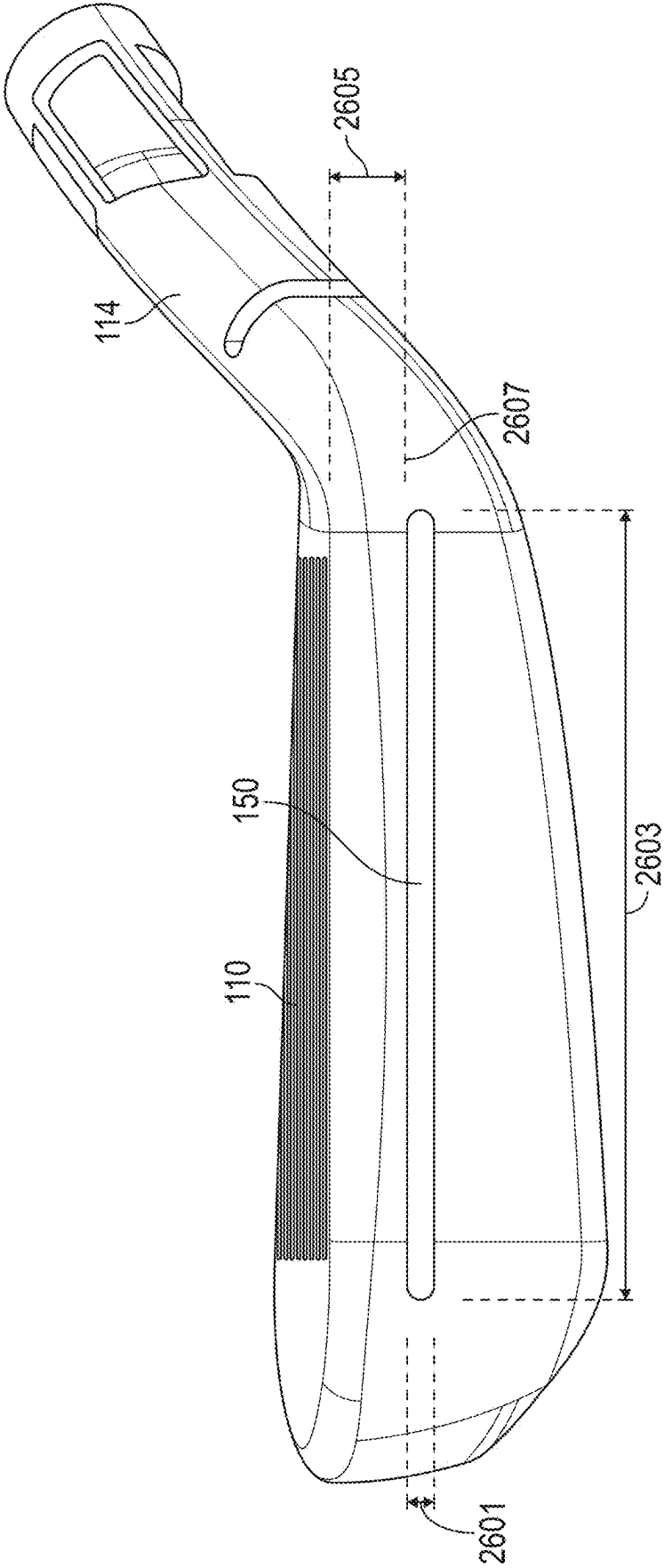


FIG. 26

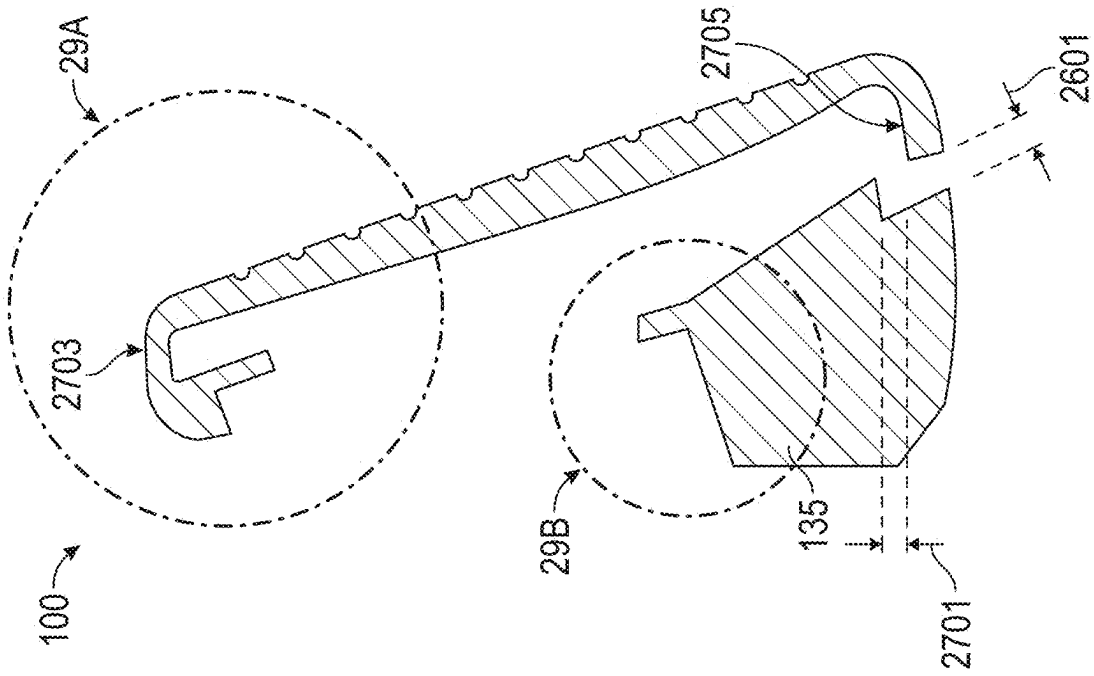


FIG. 27

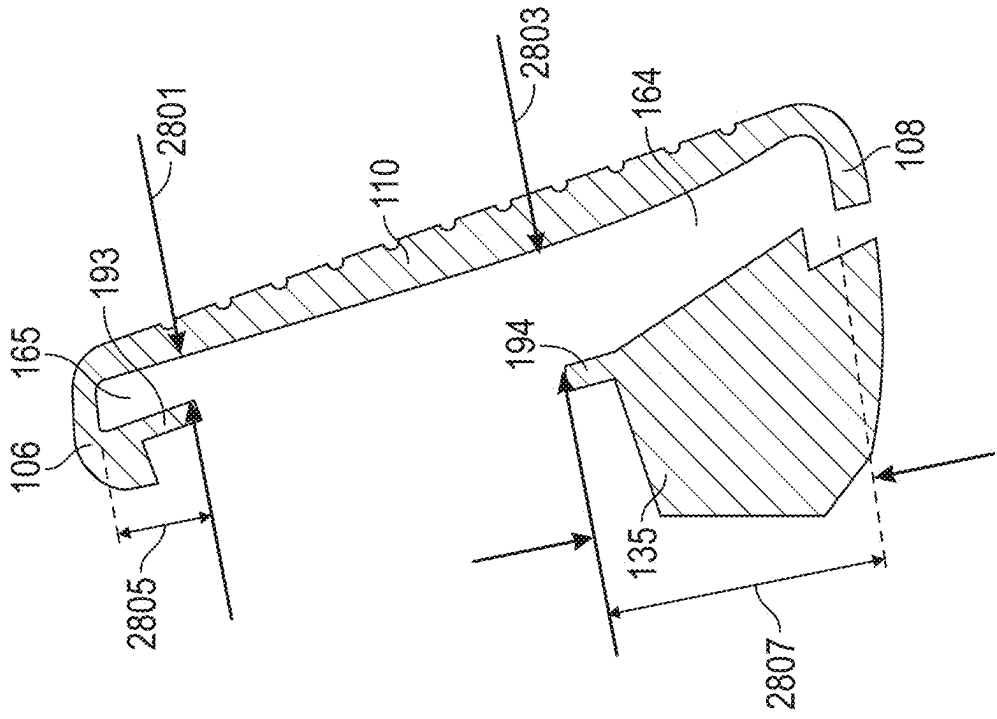


FIG. 28

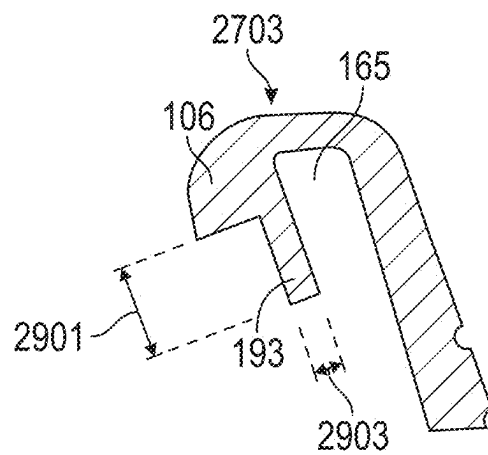


FIG. 29A

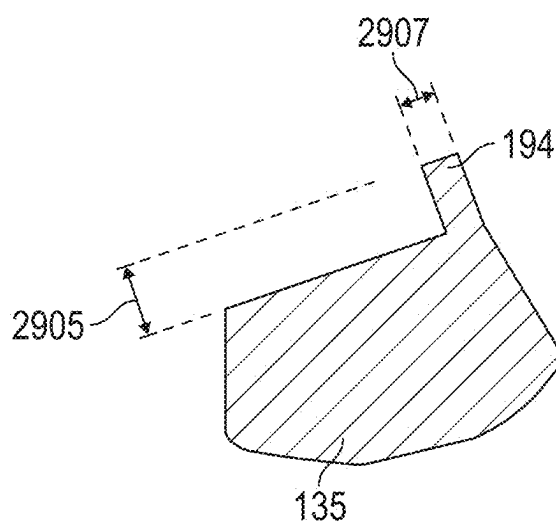


FIG. 29B

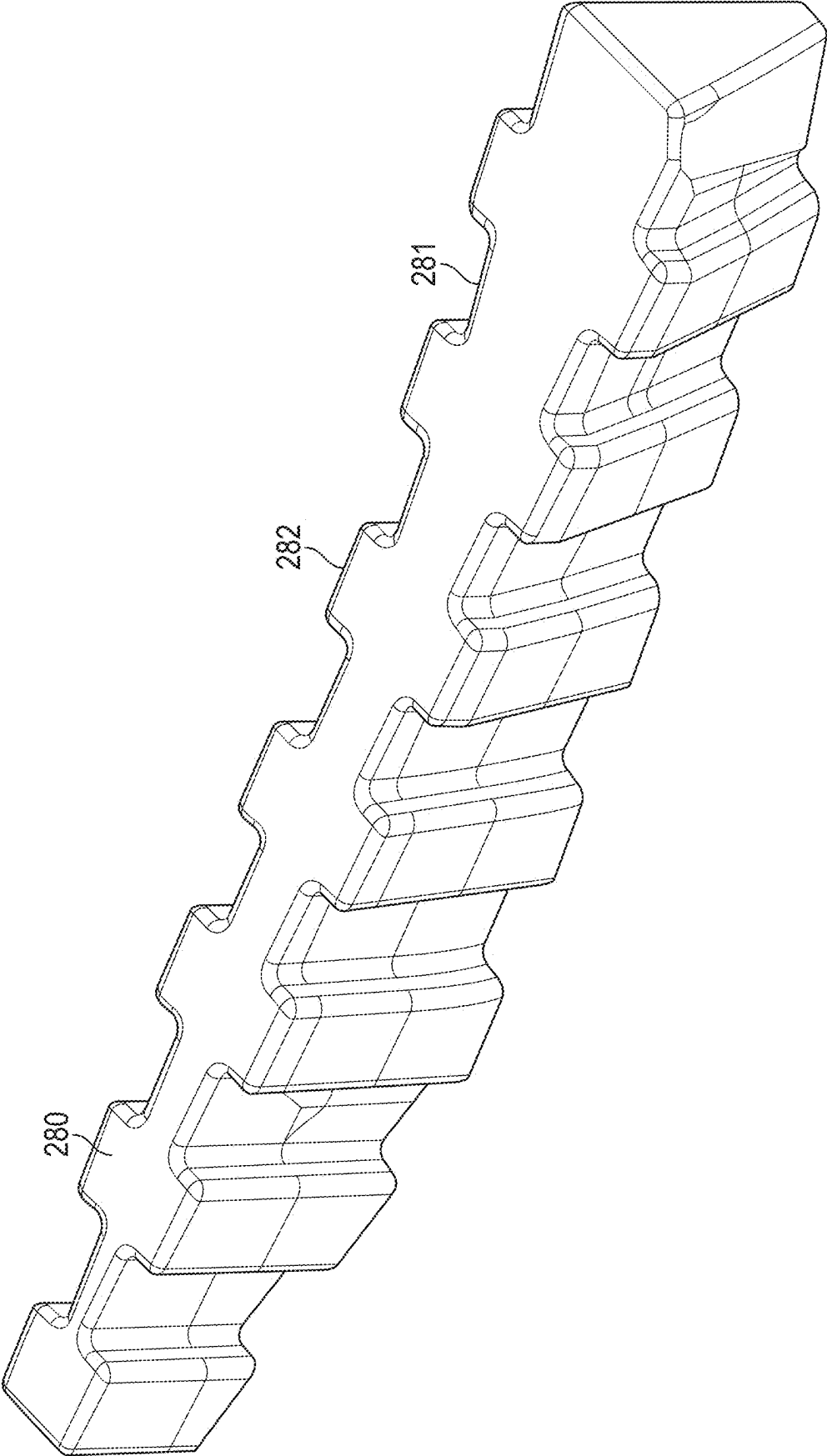


FIG. 30

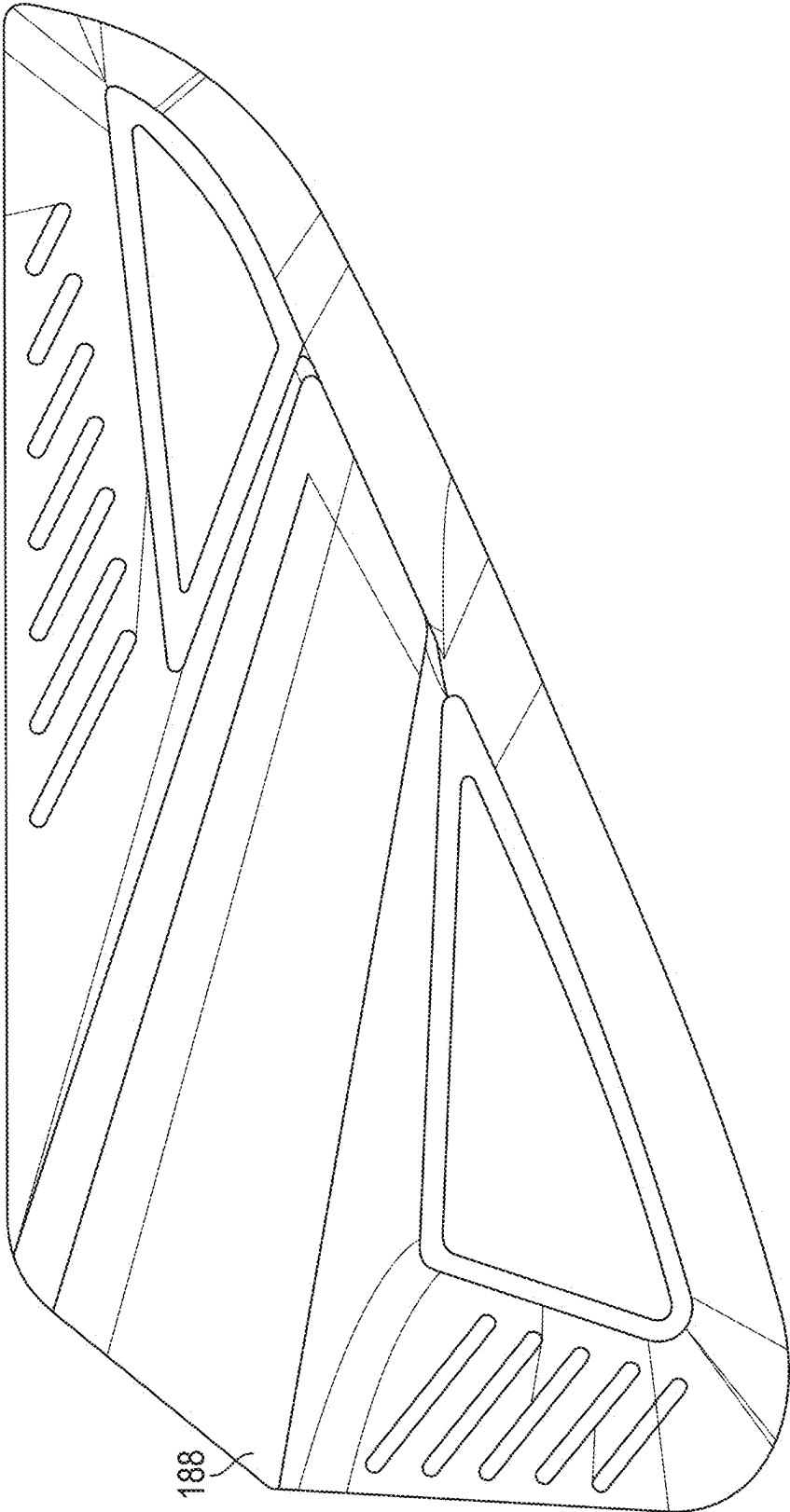


FIG. 31

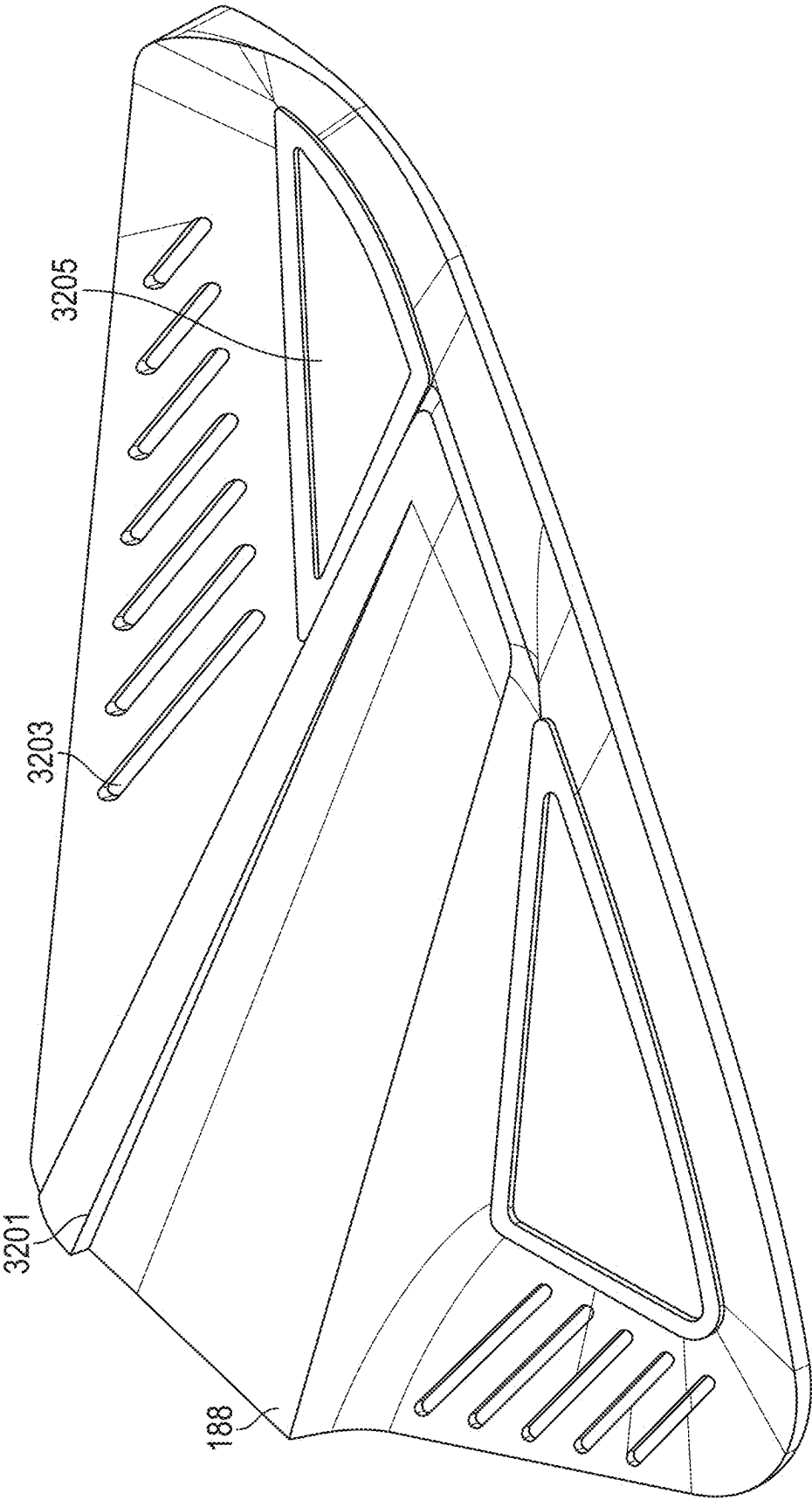


FIG. 32

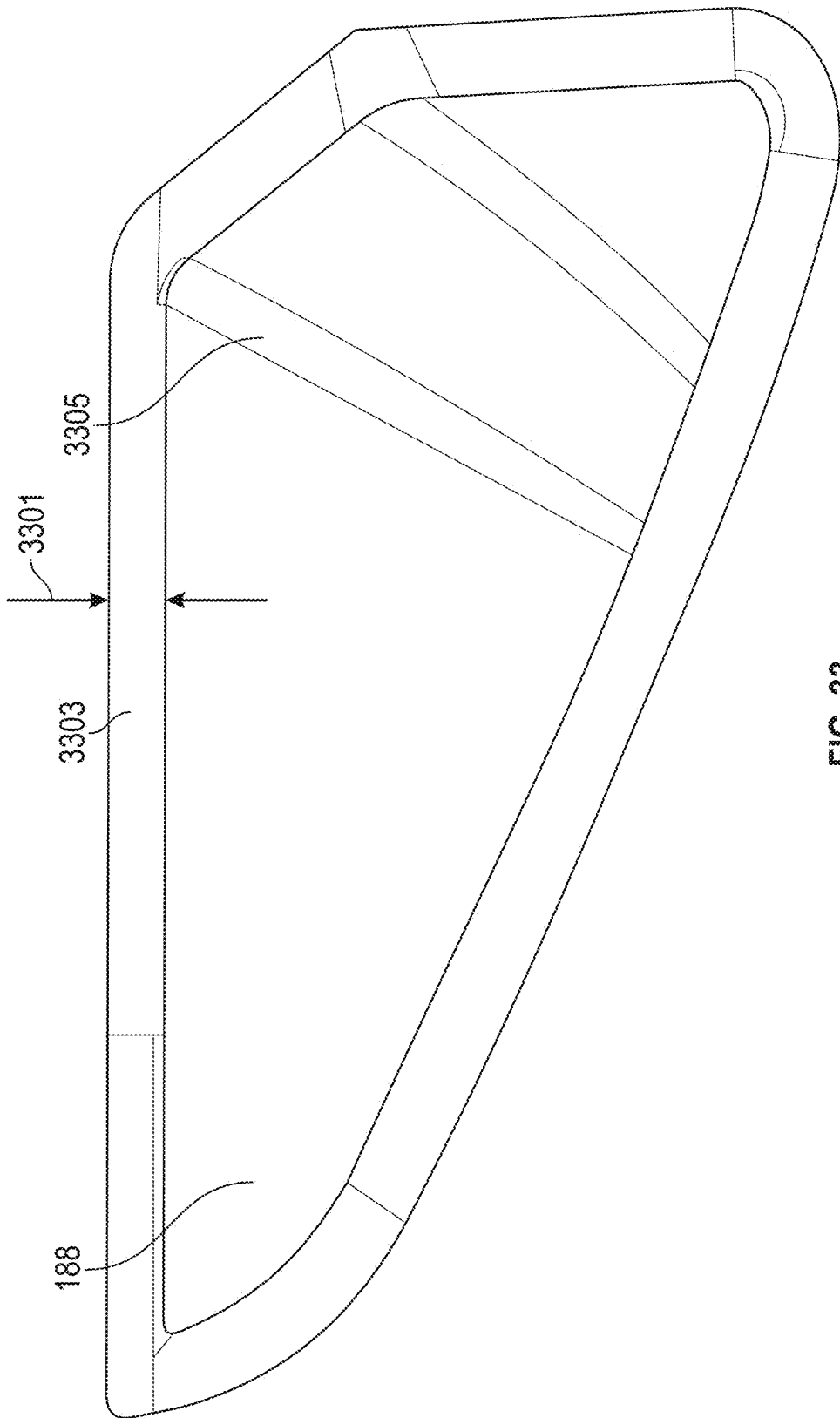


FIG. 33

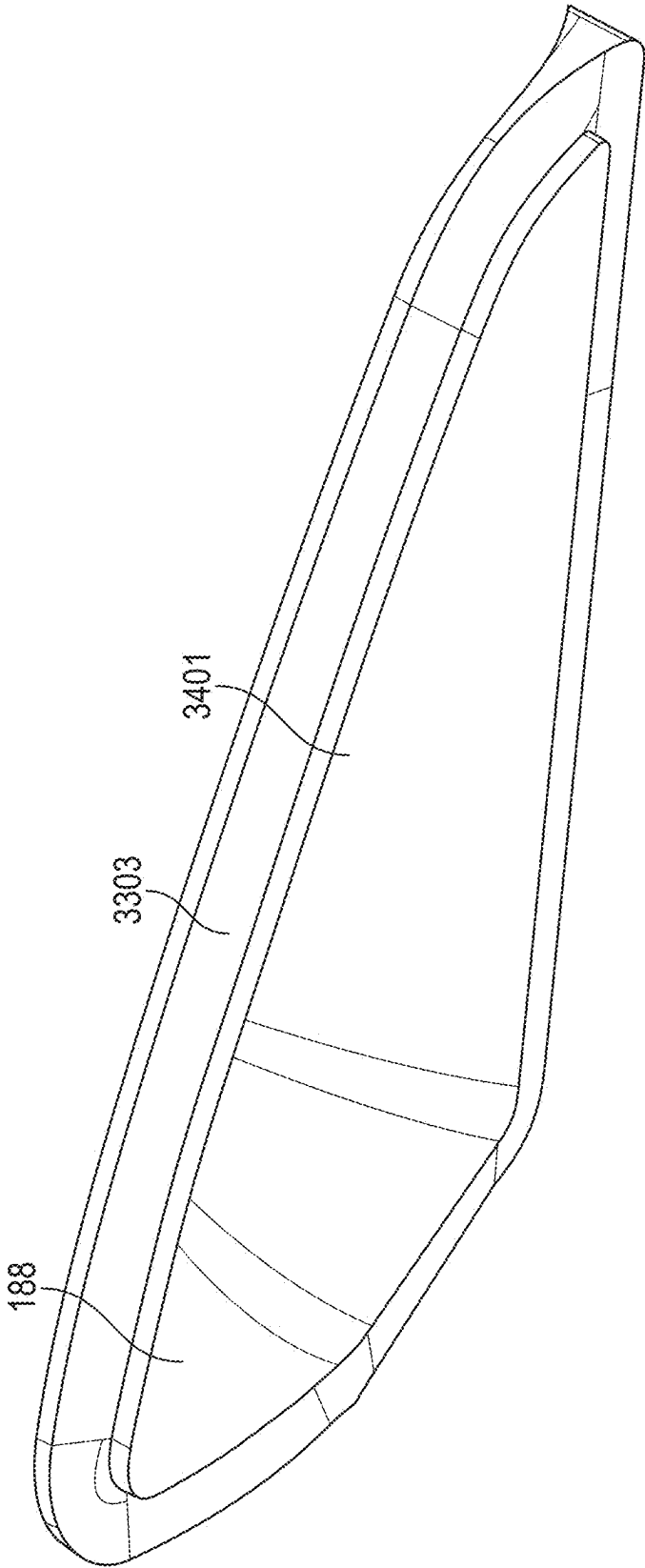


FIG. 34

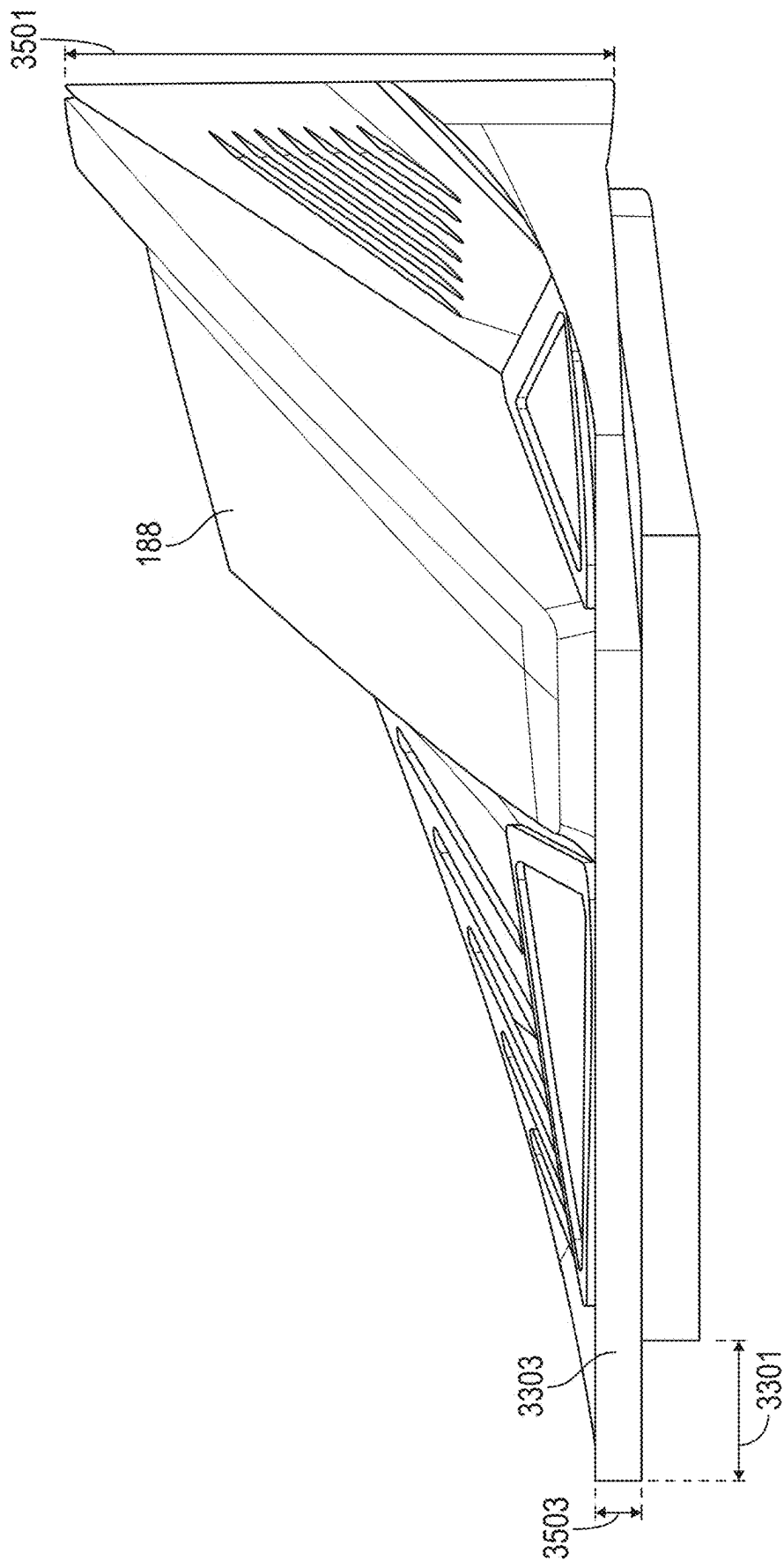


FIG. 35

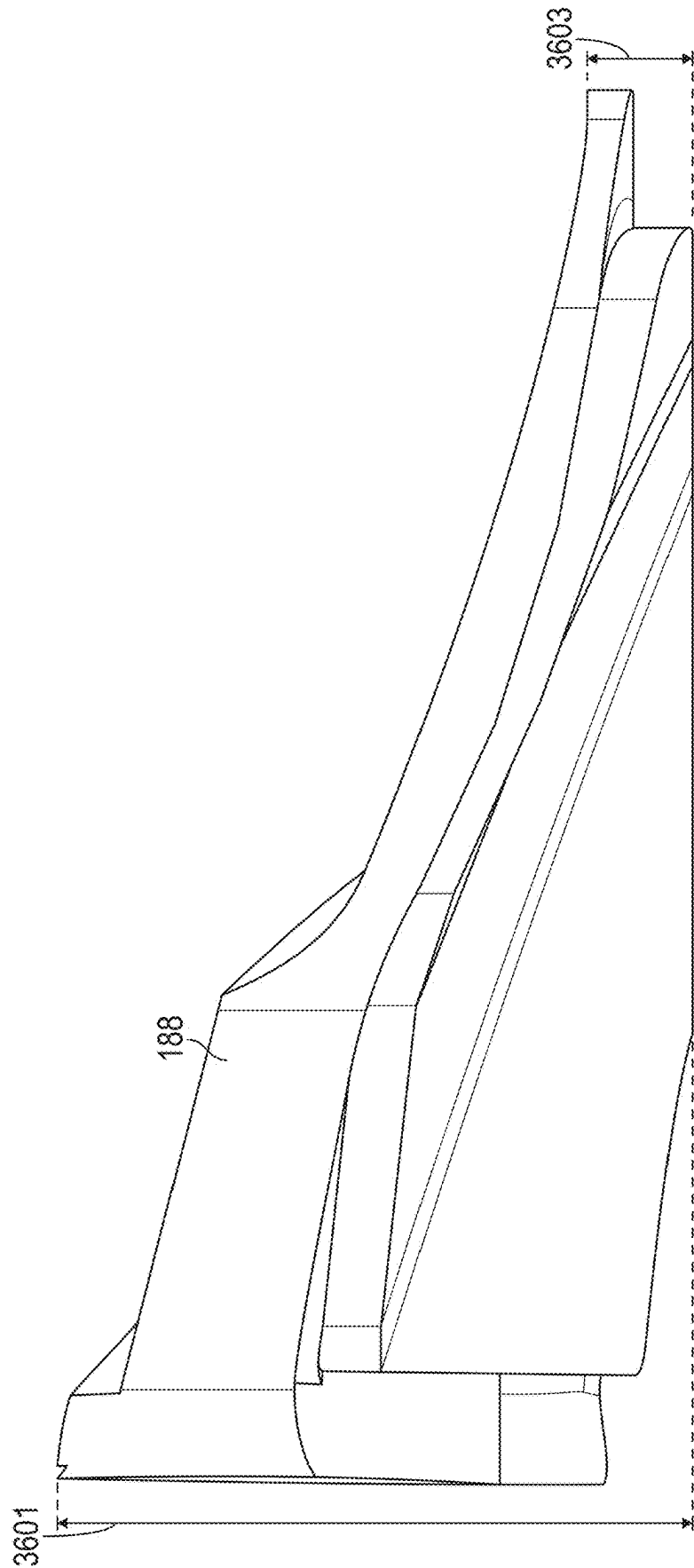


FIG. 36

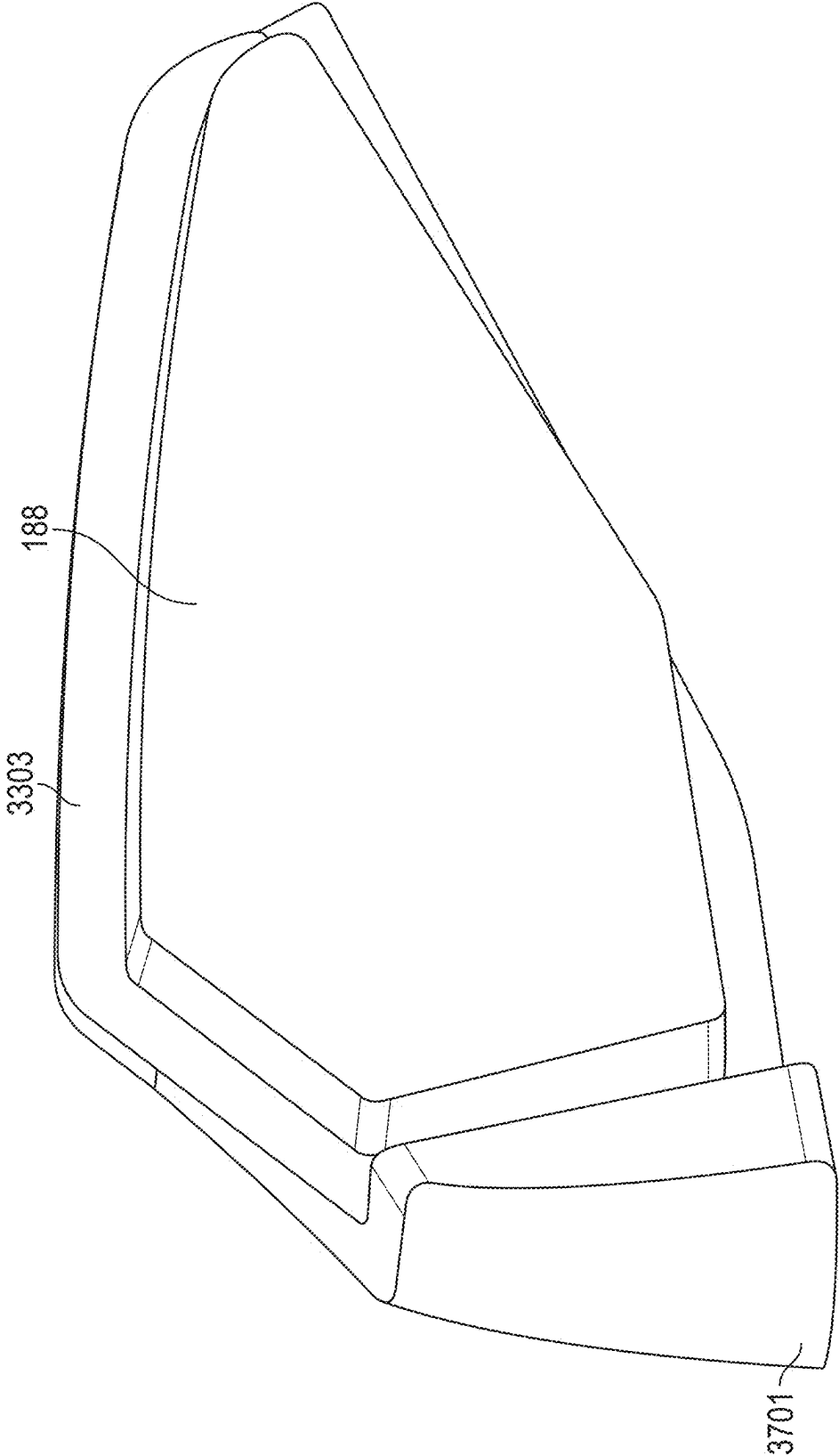


FIG. 37

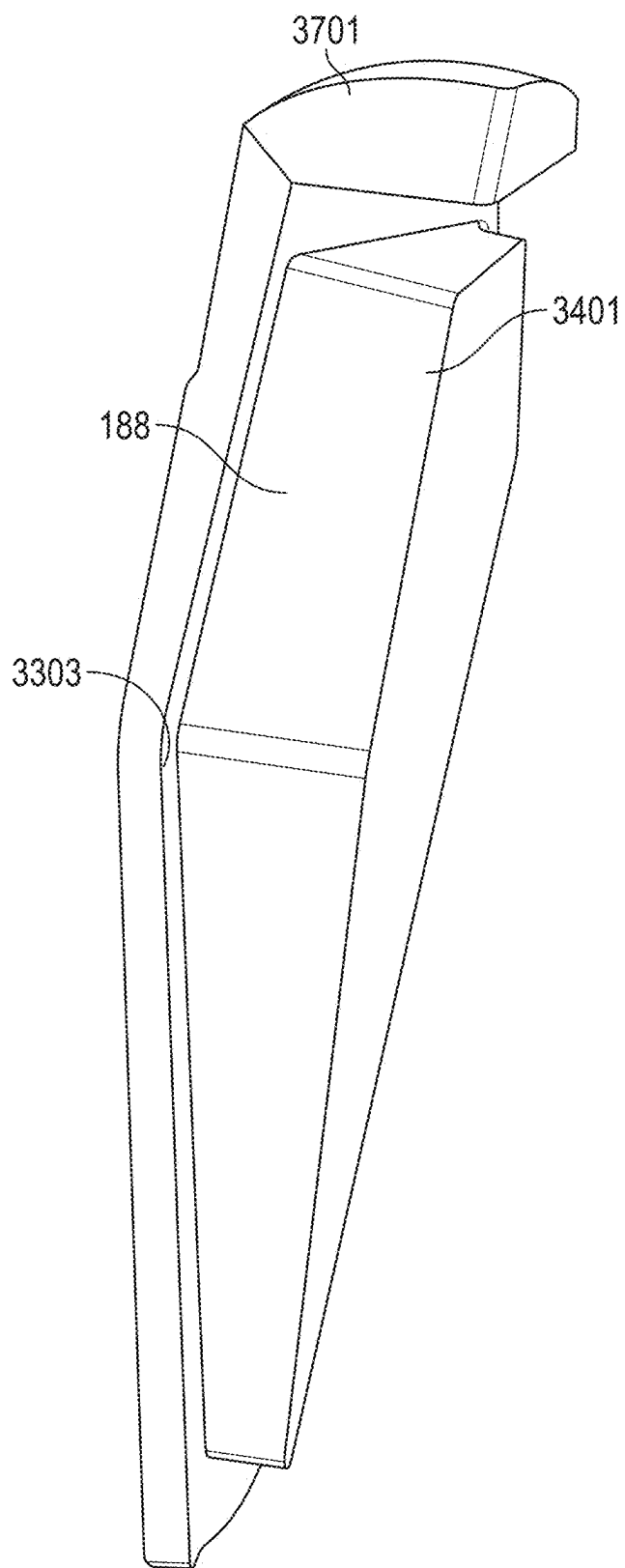


FIG. 38

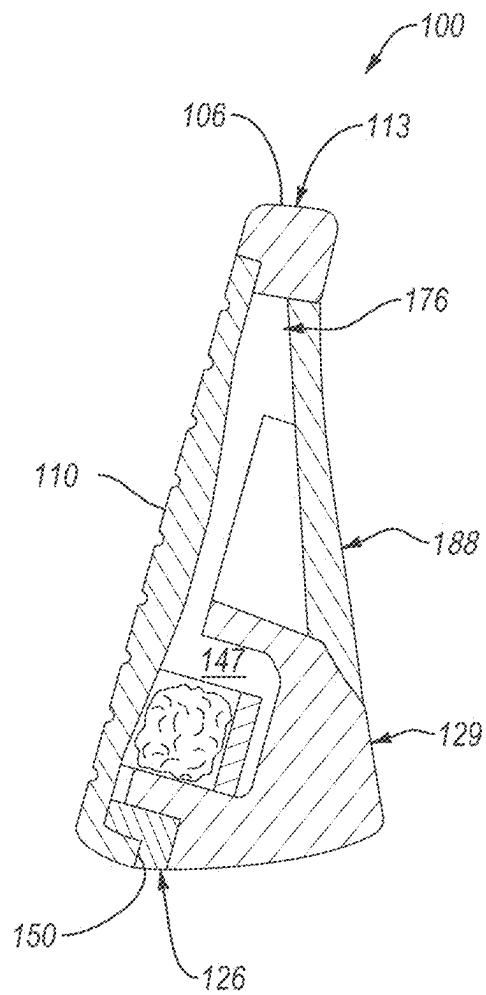


FIG. 39

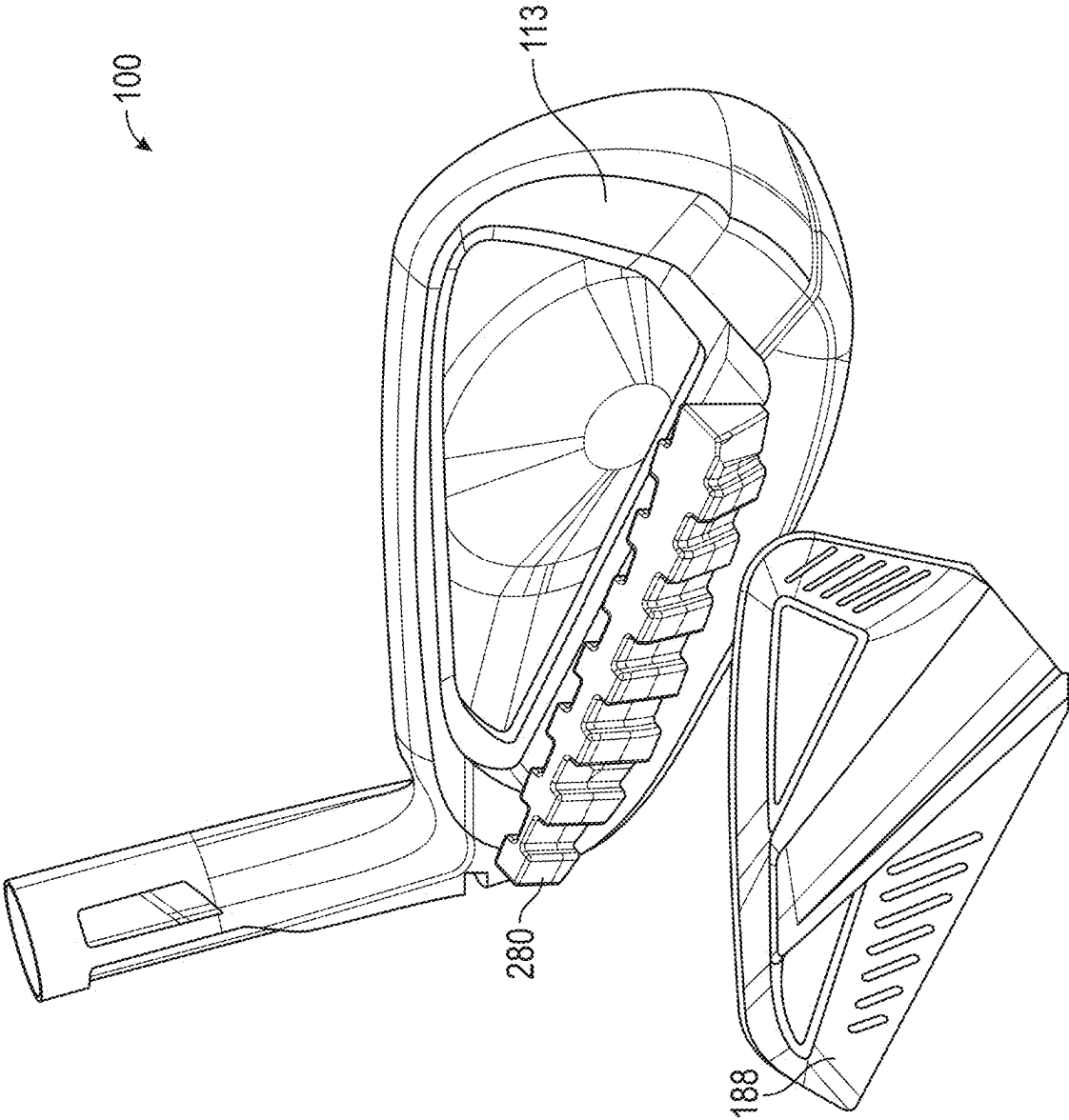


FIG. 40

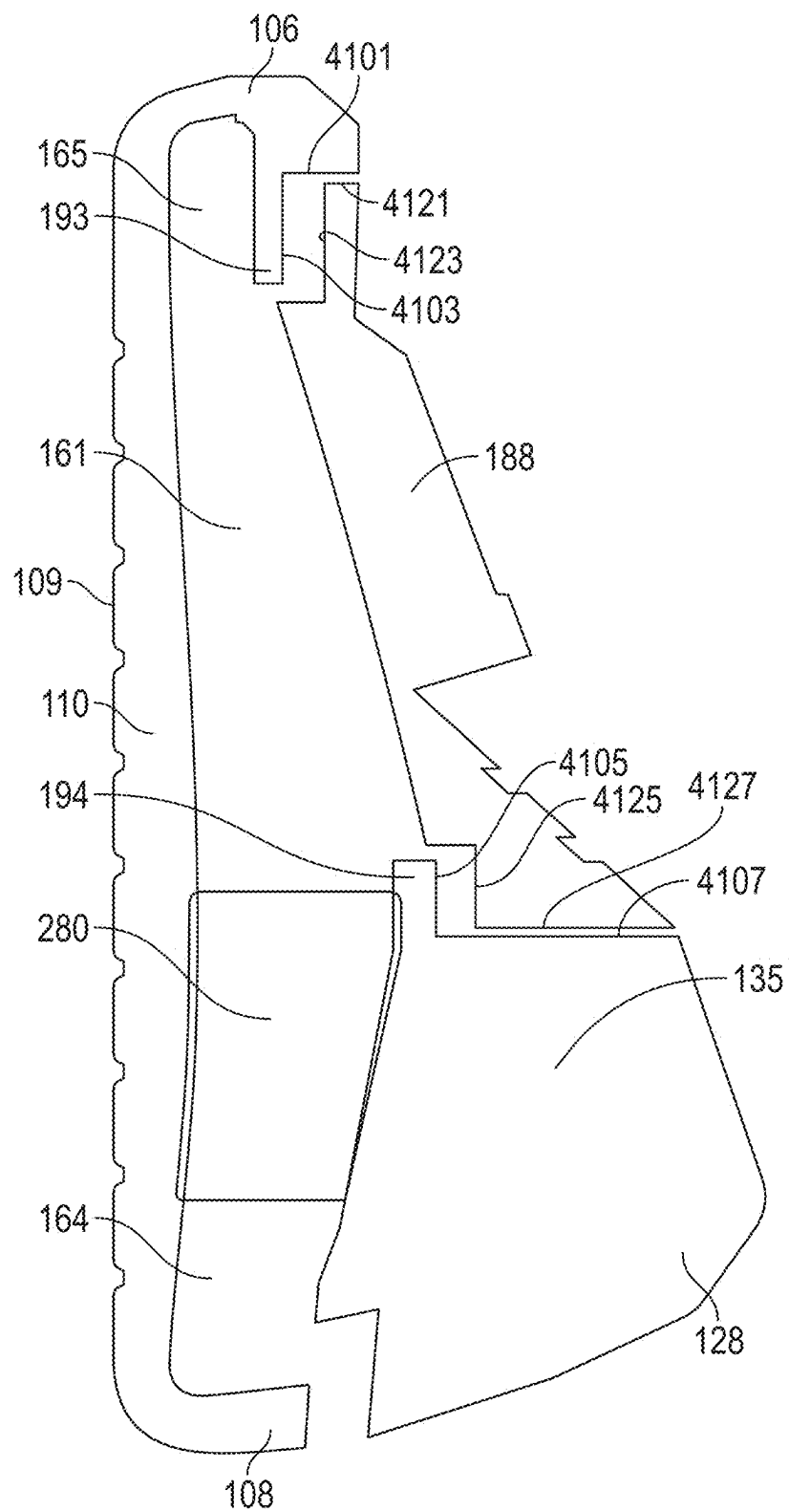


FIG. 41

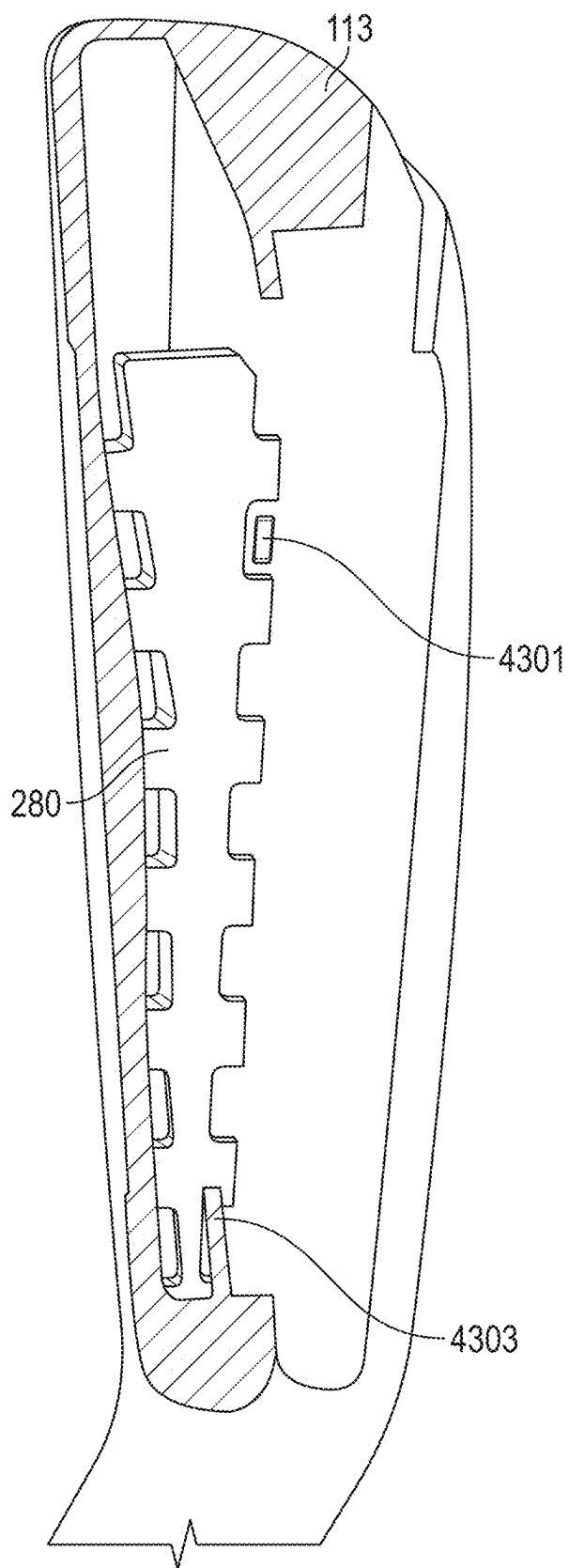


FIG. 43

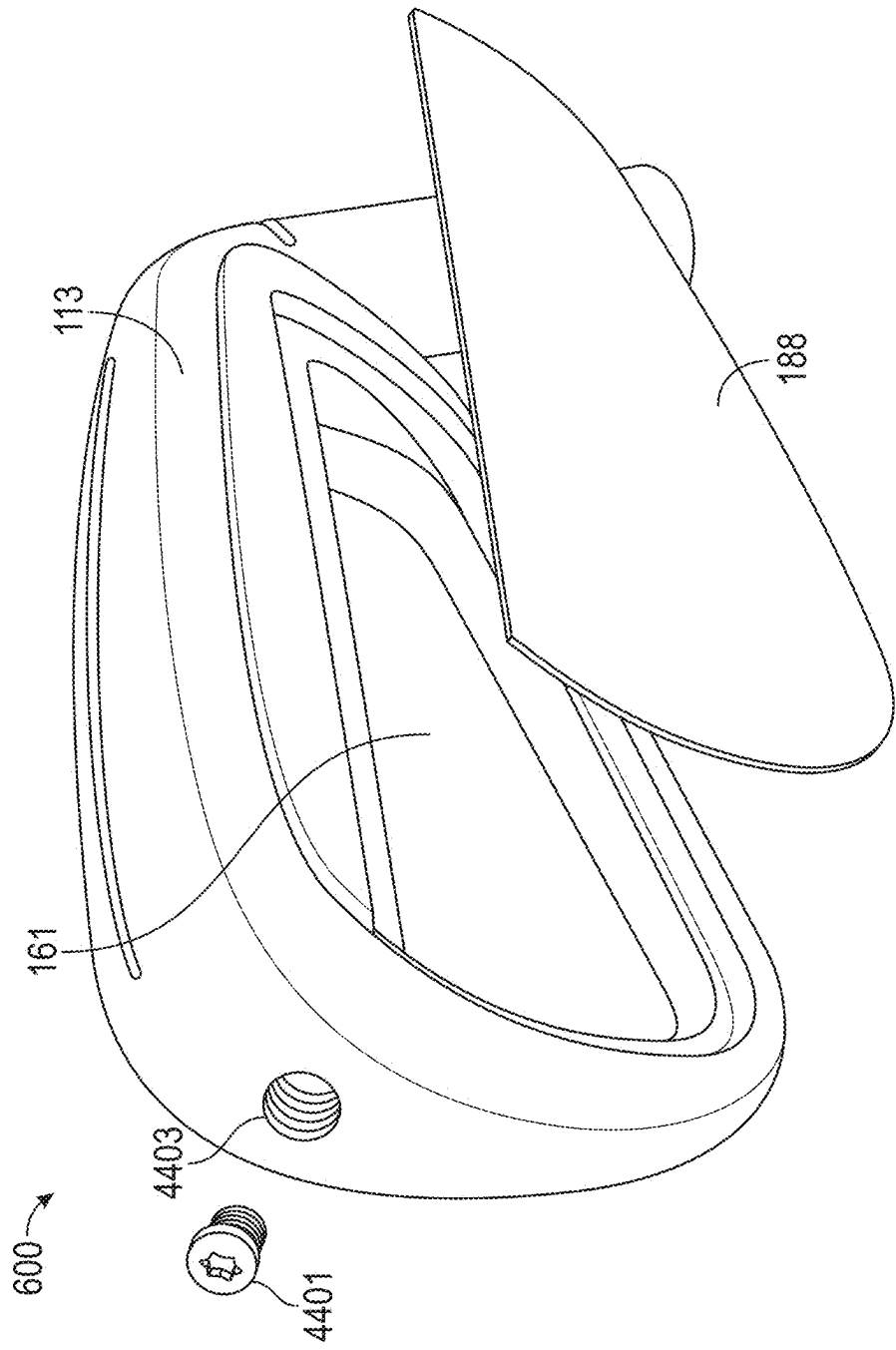


FIG. 44

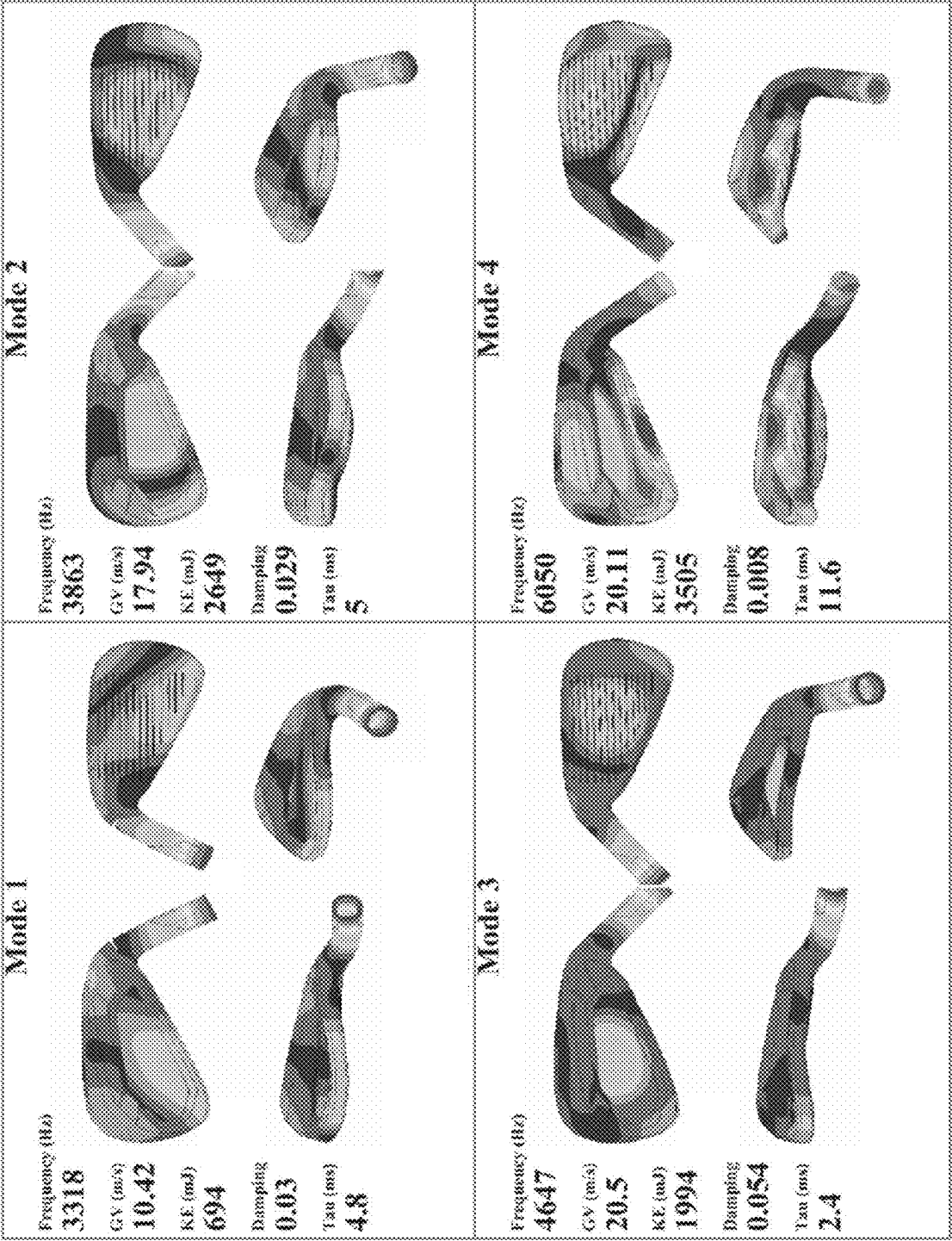


FIG. 45

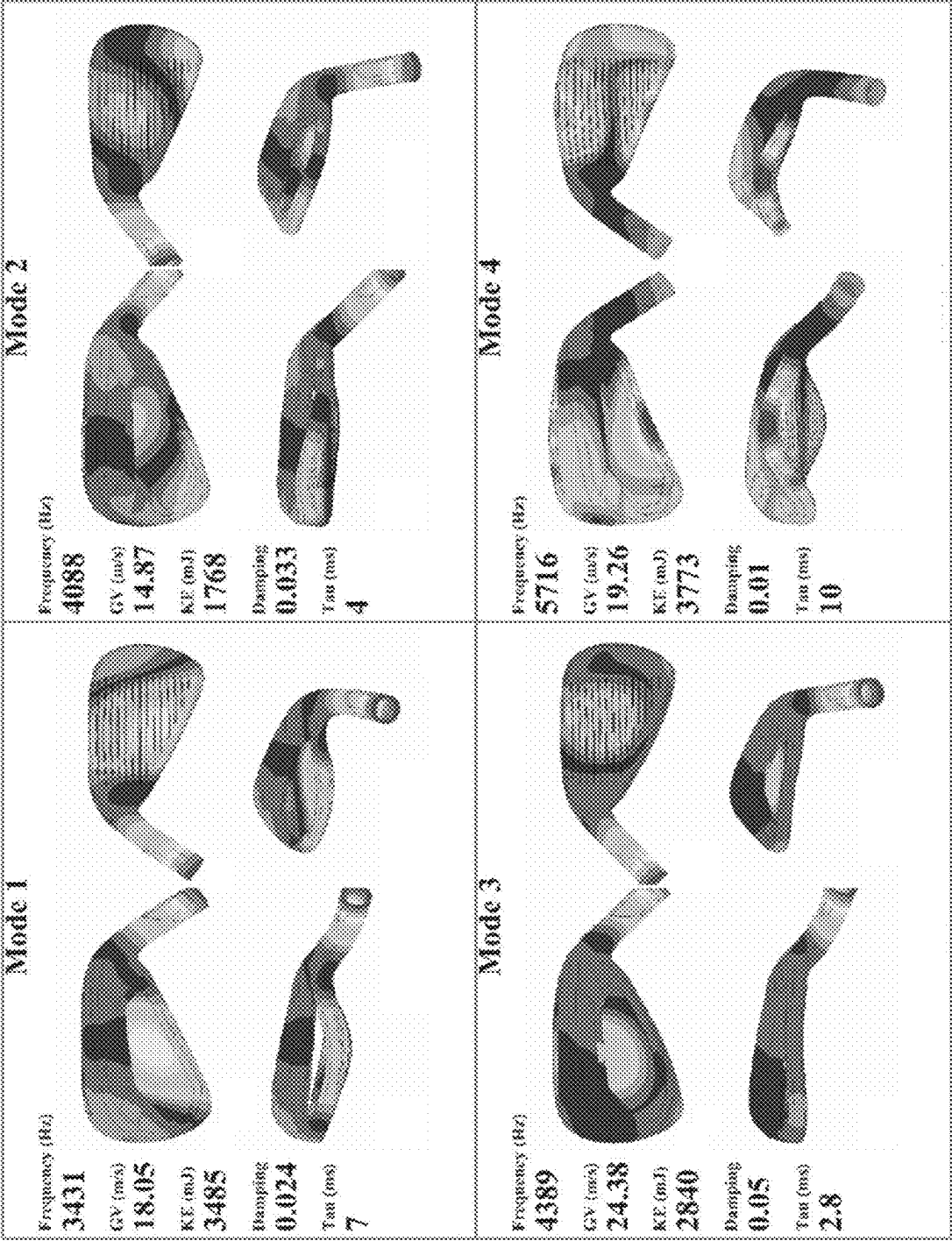


FIG. 46

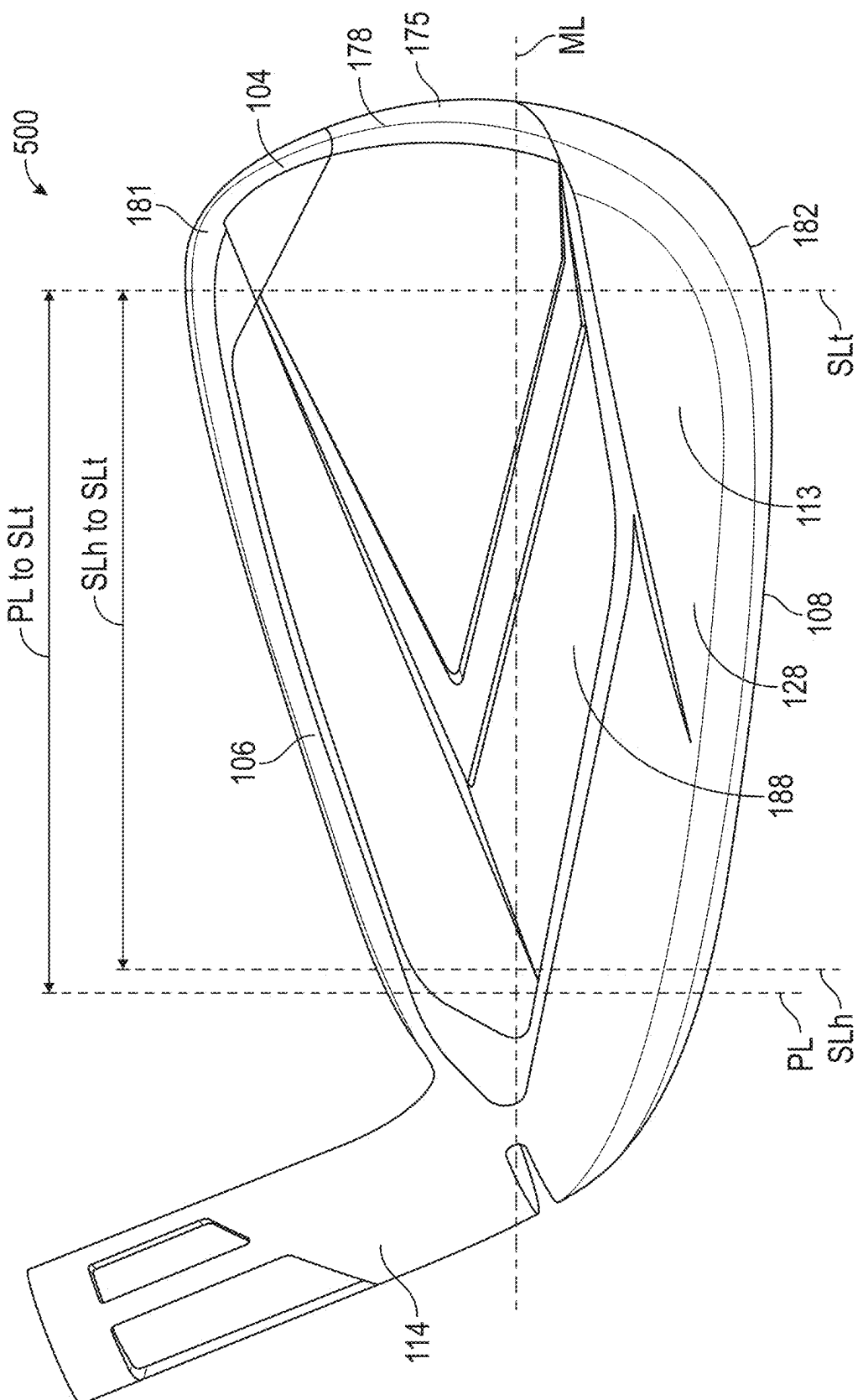


FIG. 47

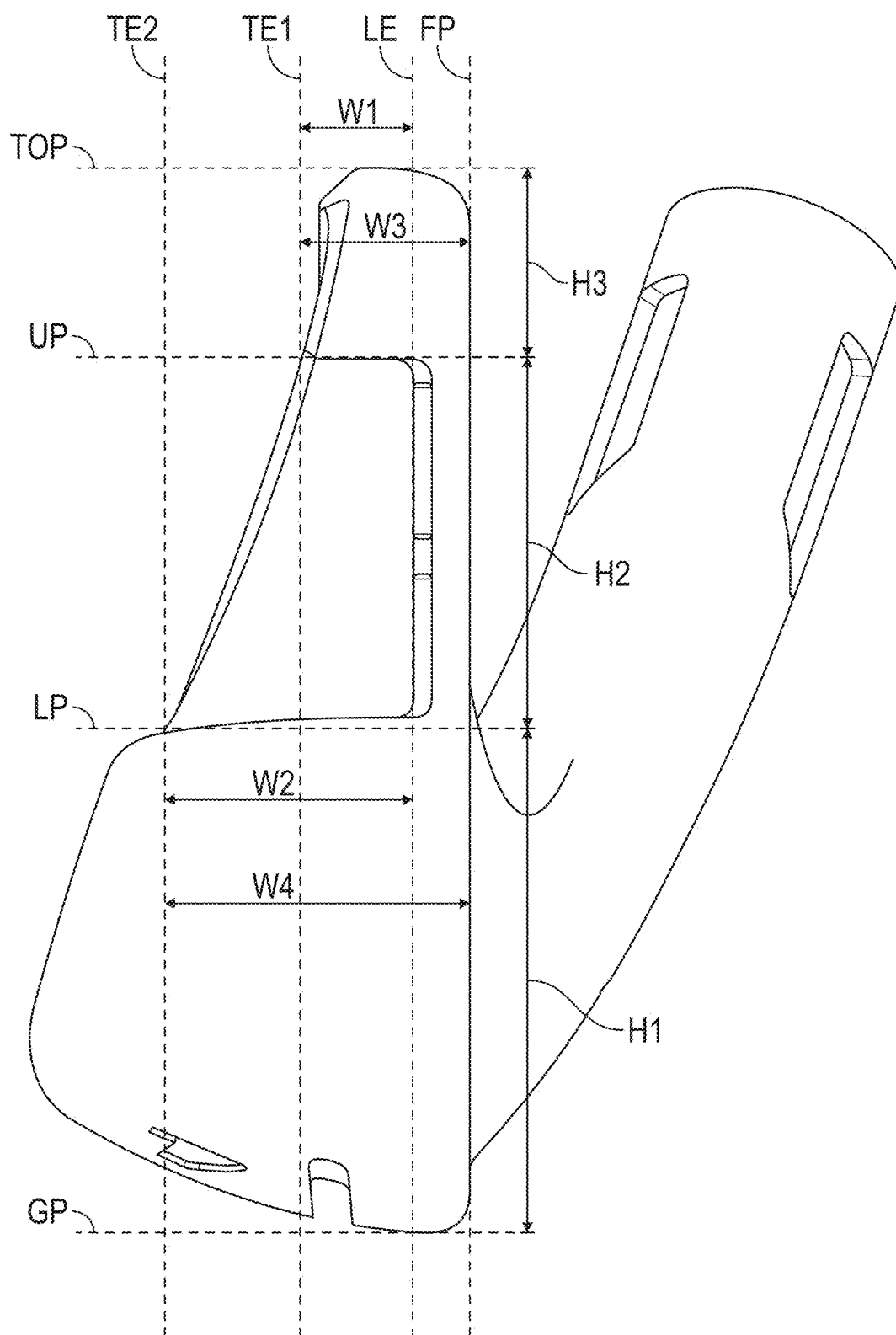


FIG. 48

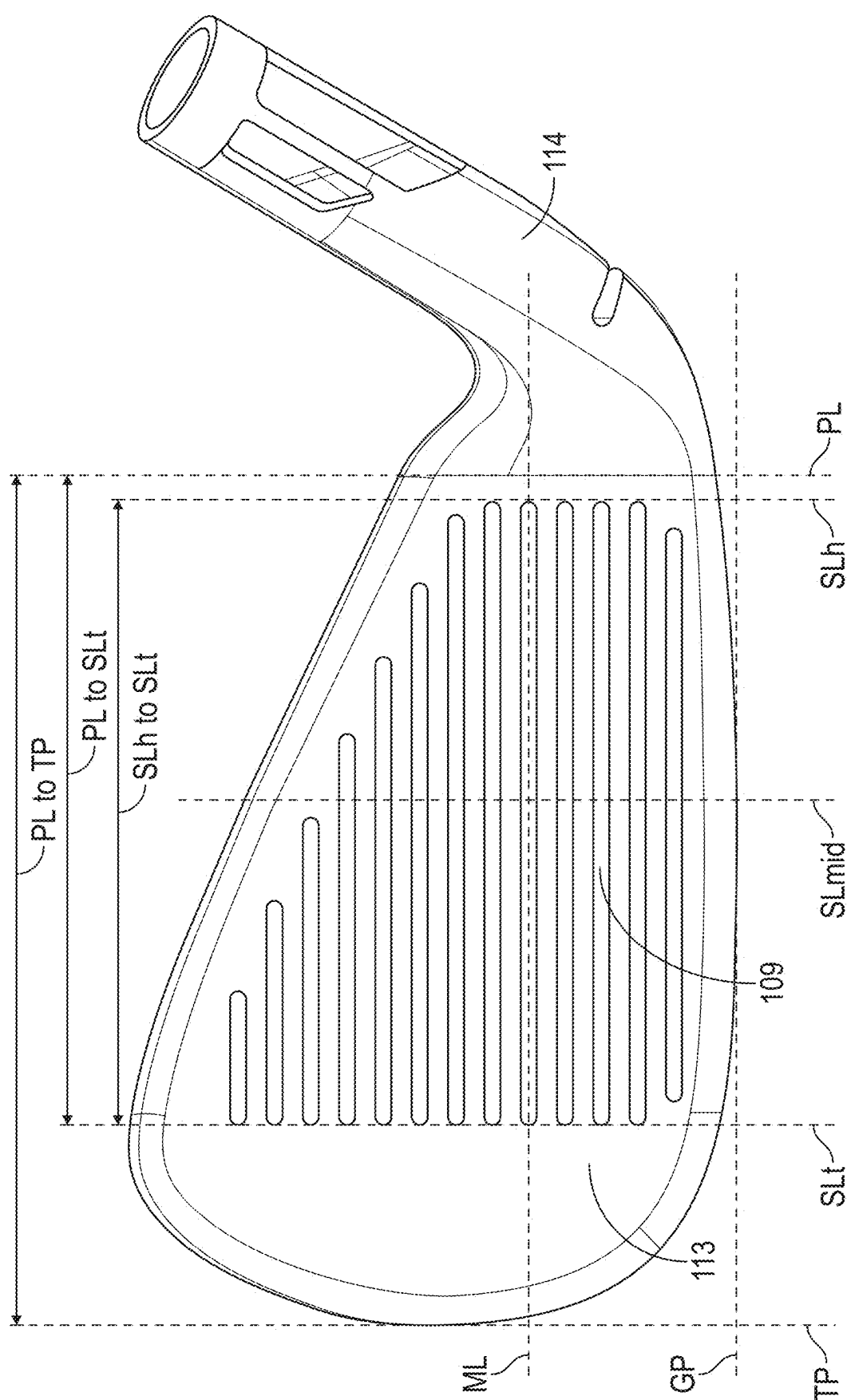


FIG. 49

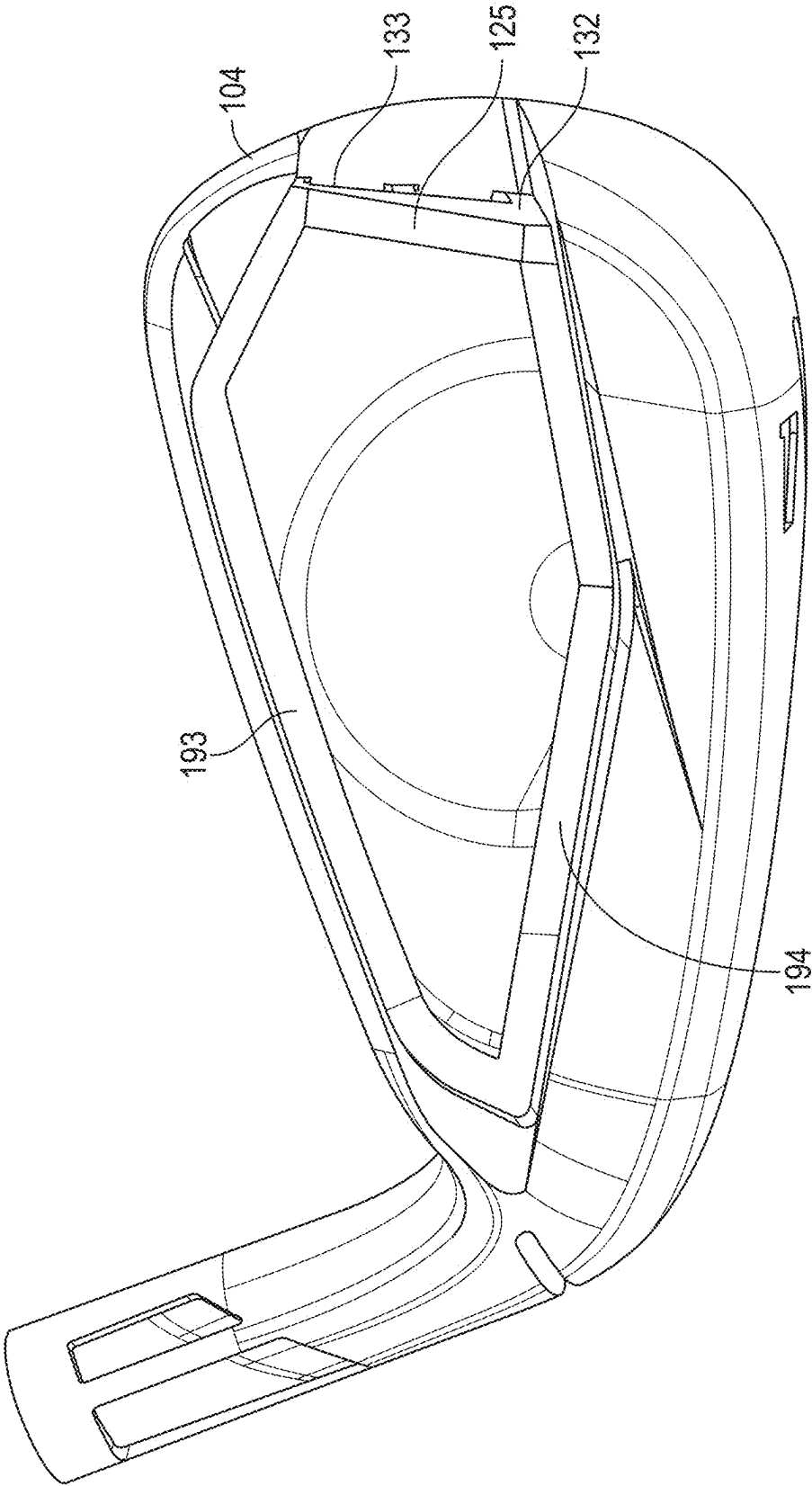


FIG. 50

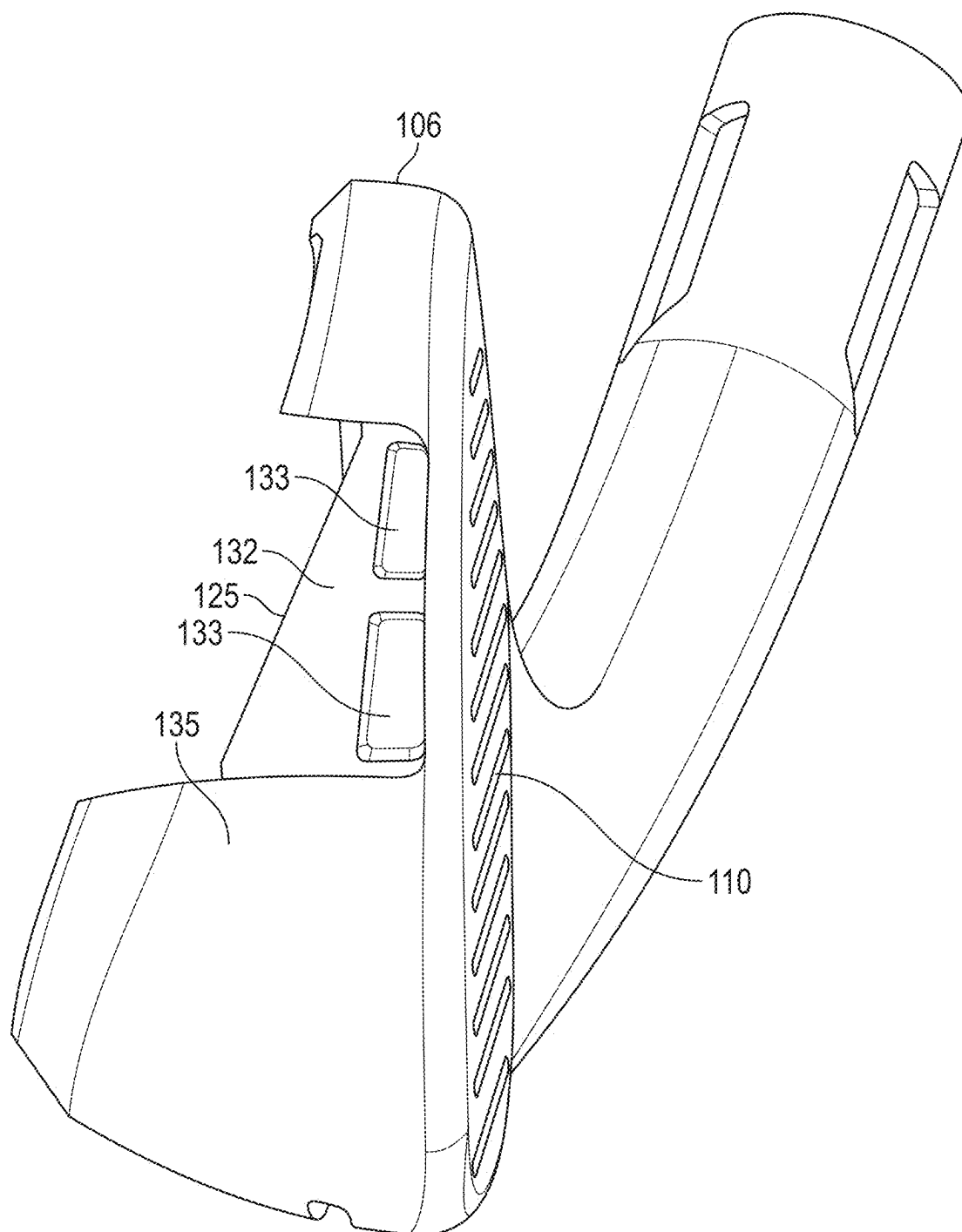


FIG. 51

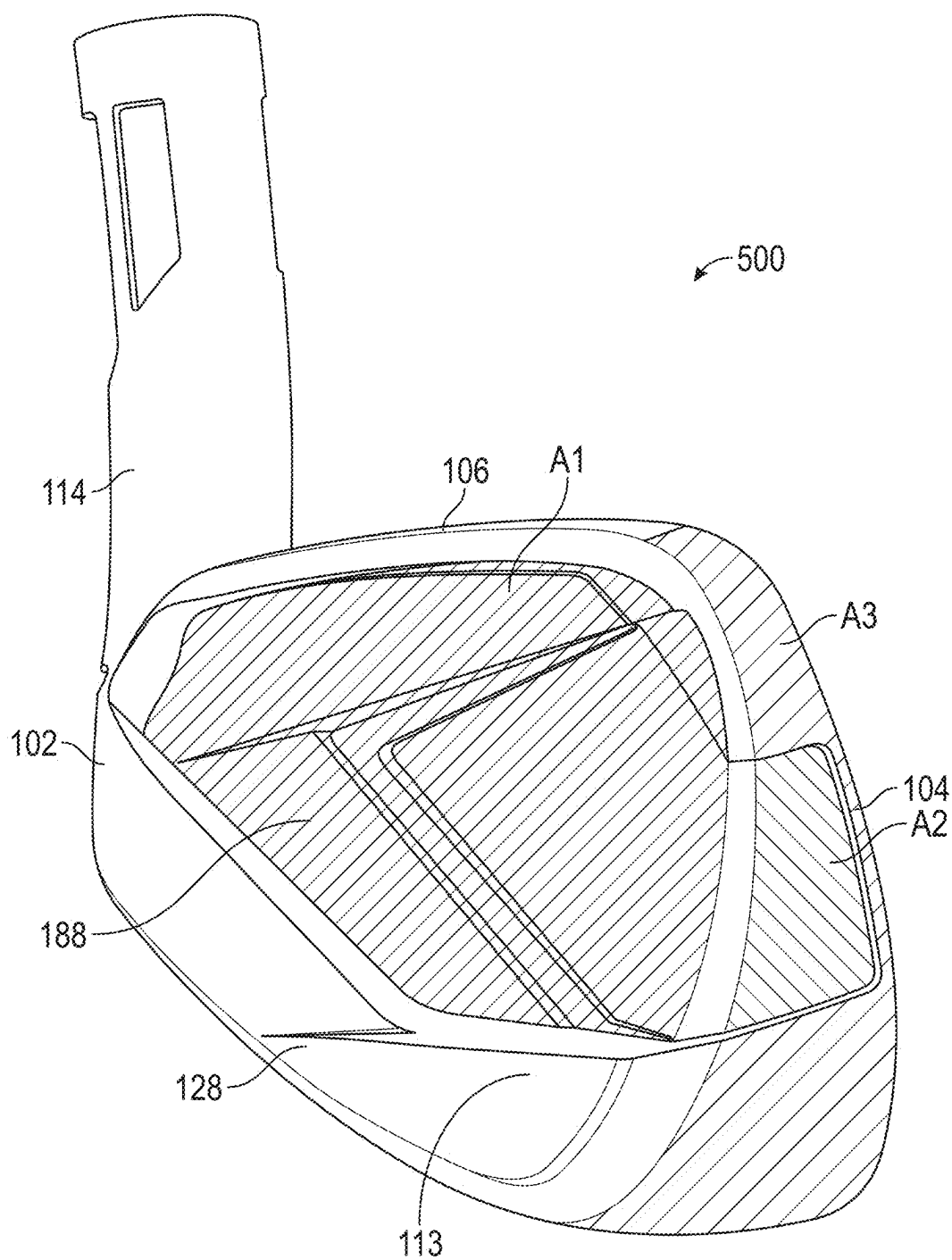


FIG. 52

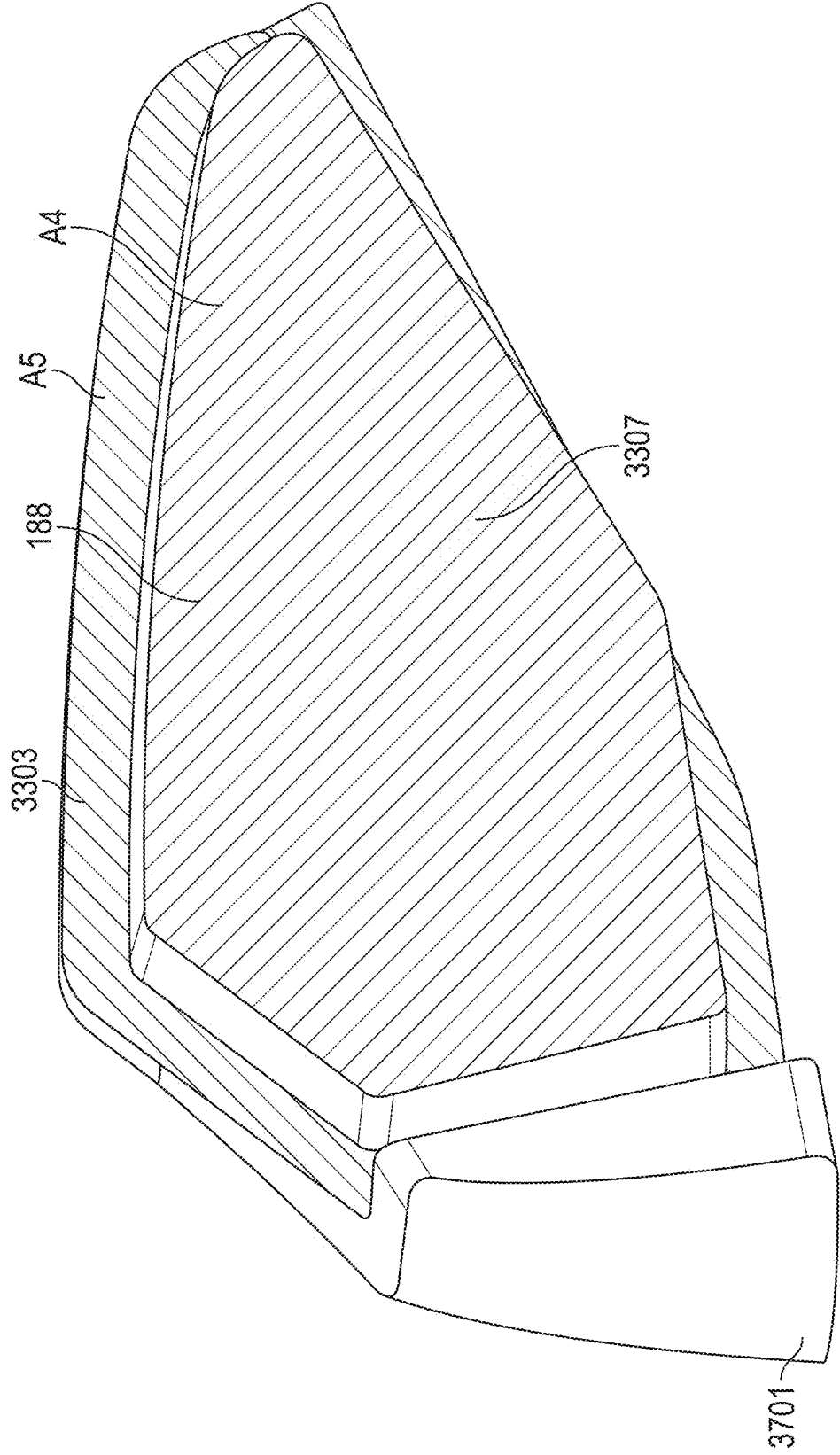


FIG. 53

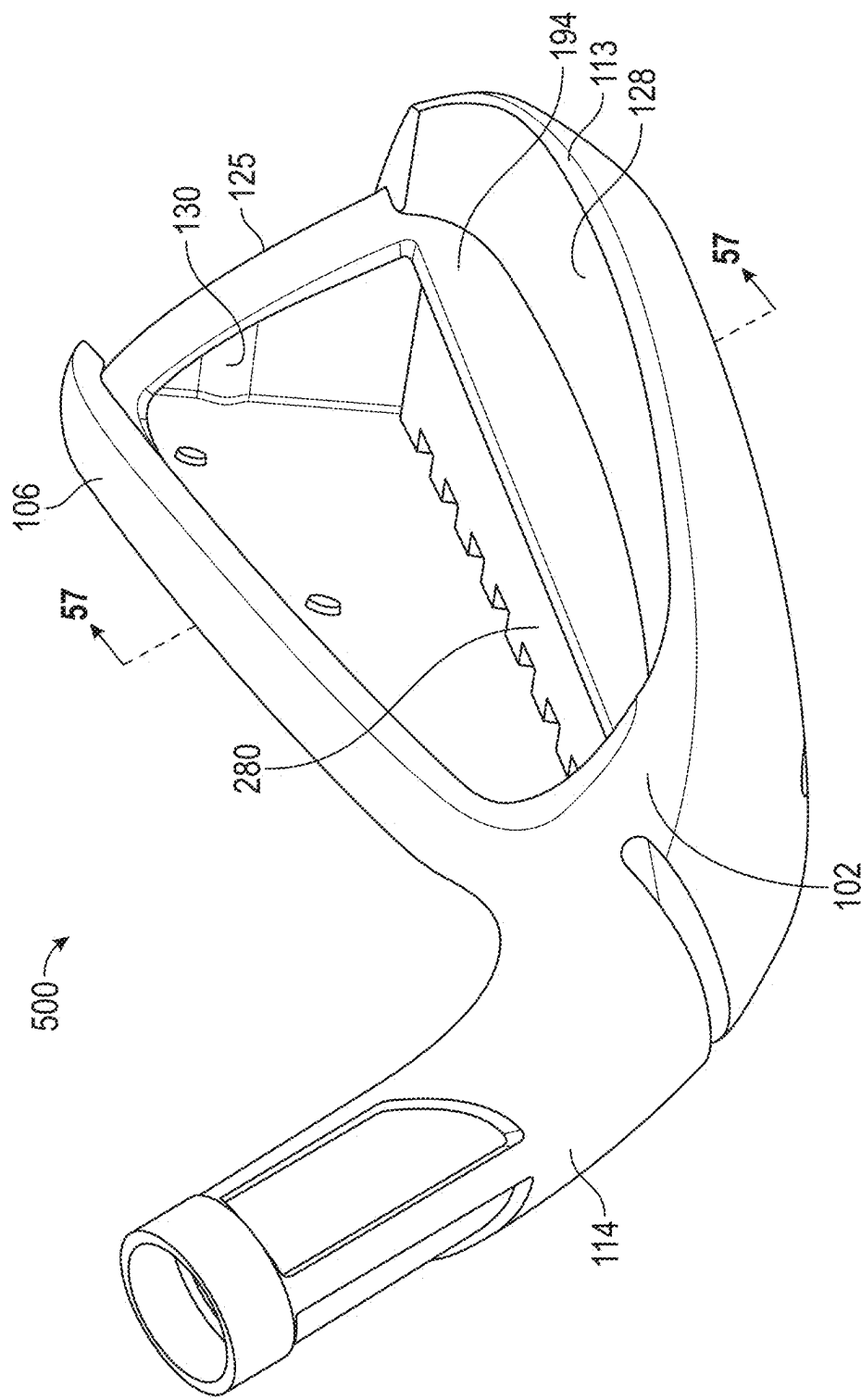


FIG. 54

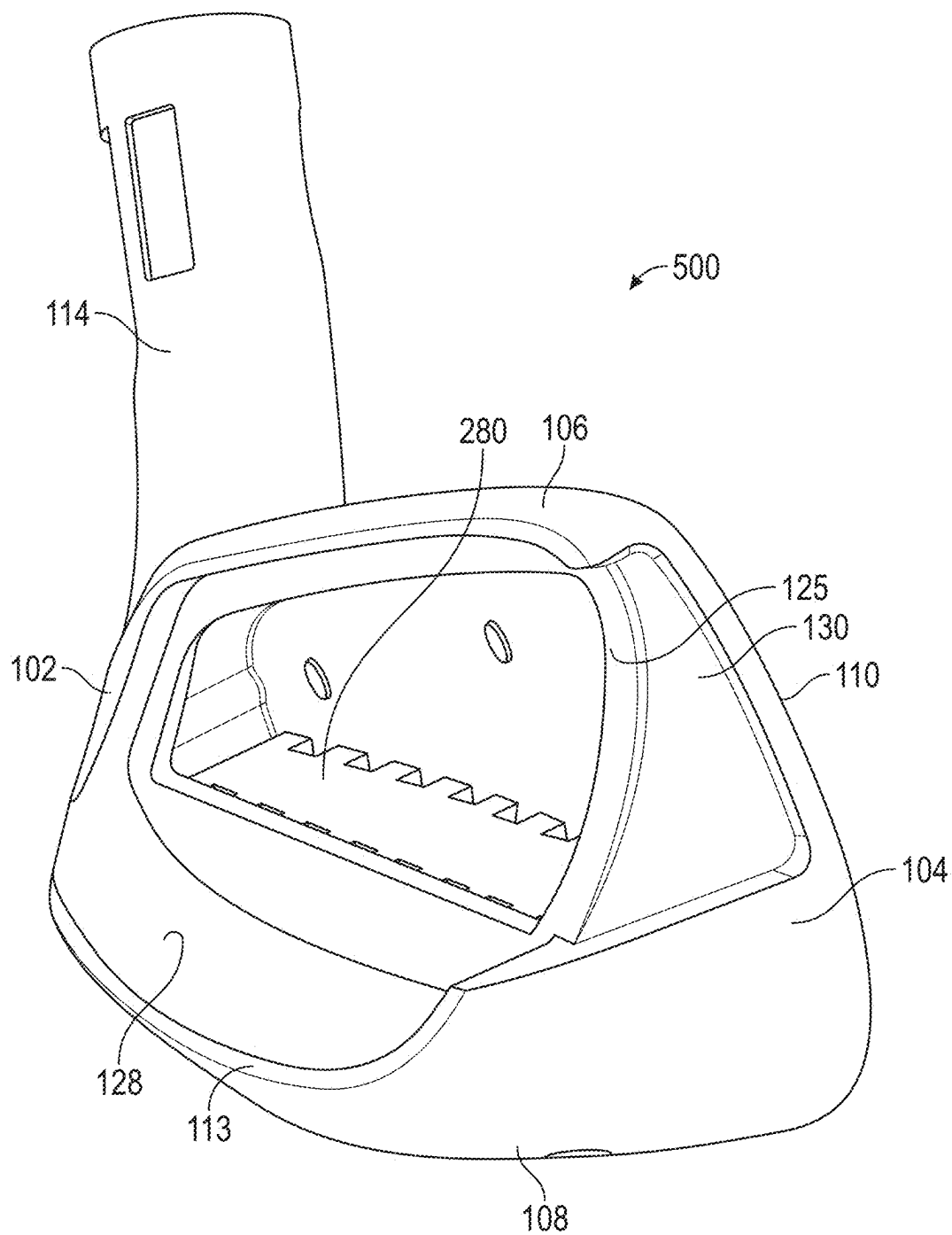


FIG. 55

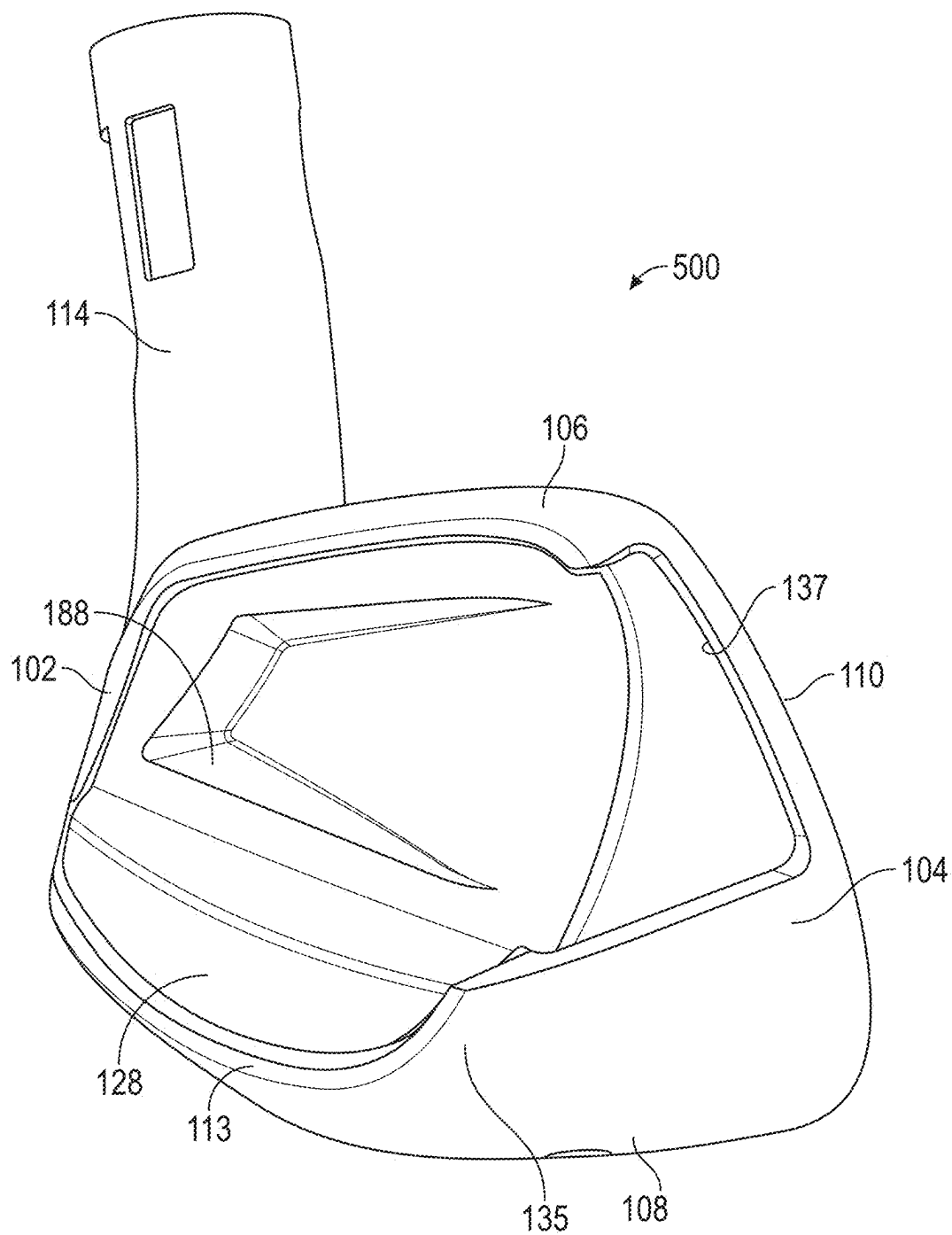
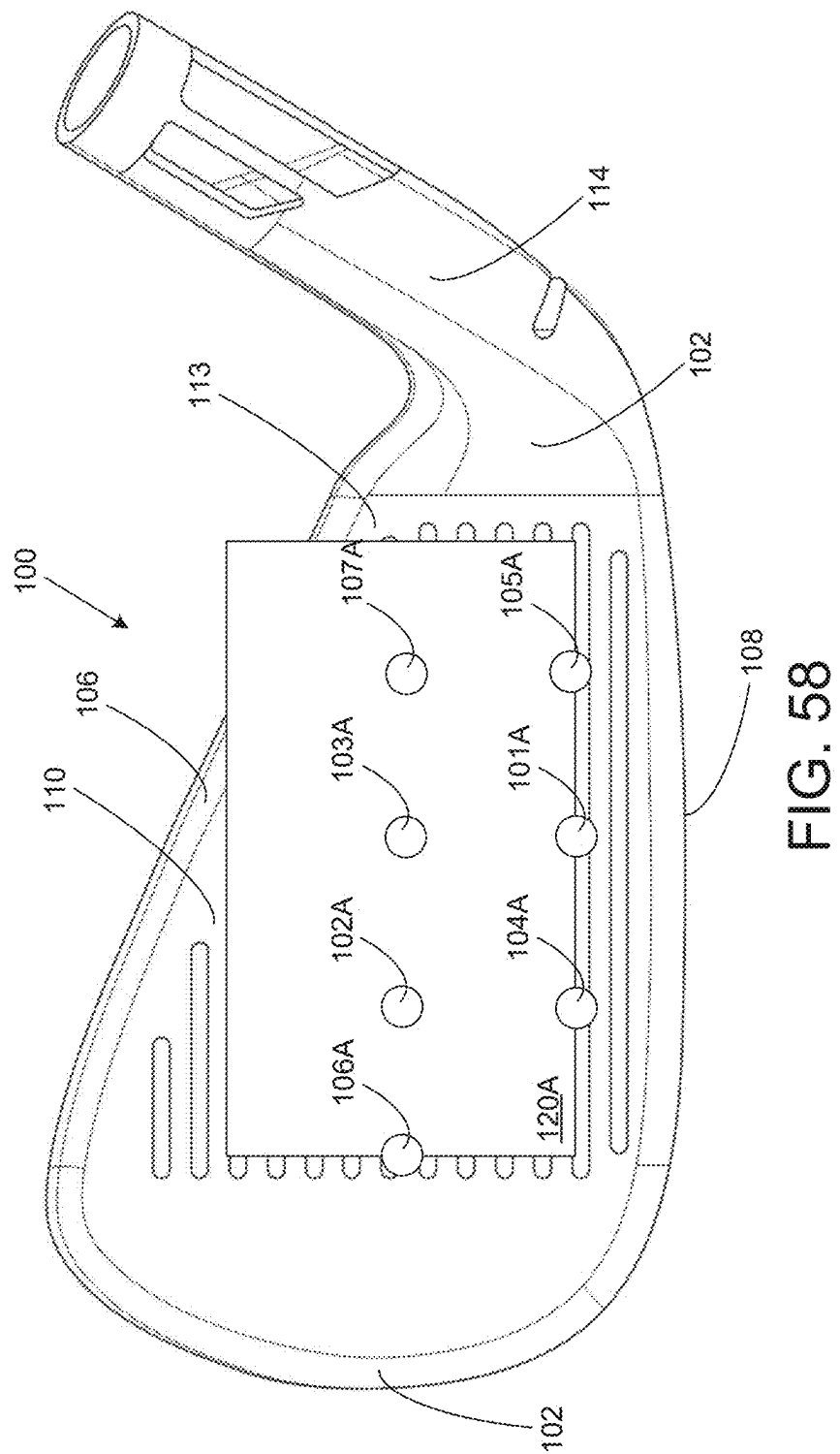


FIG. 56



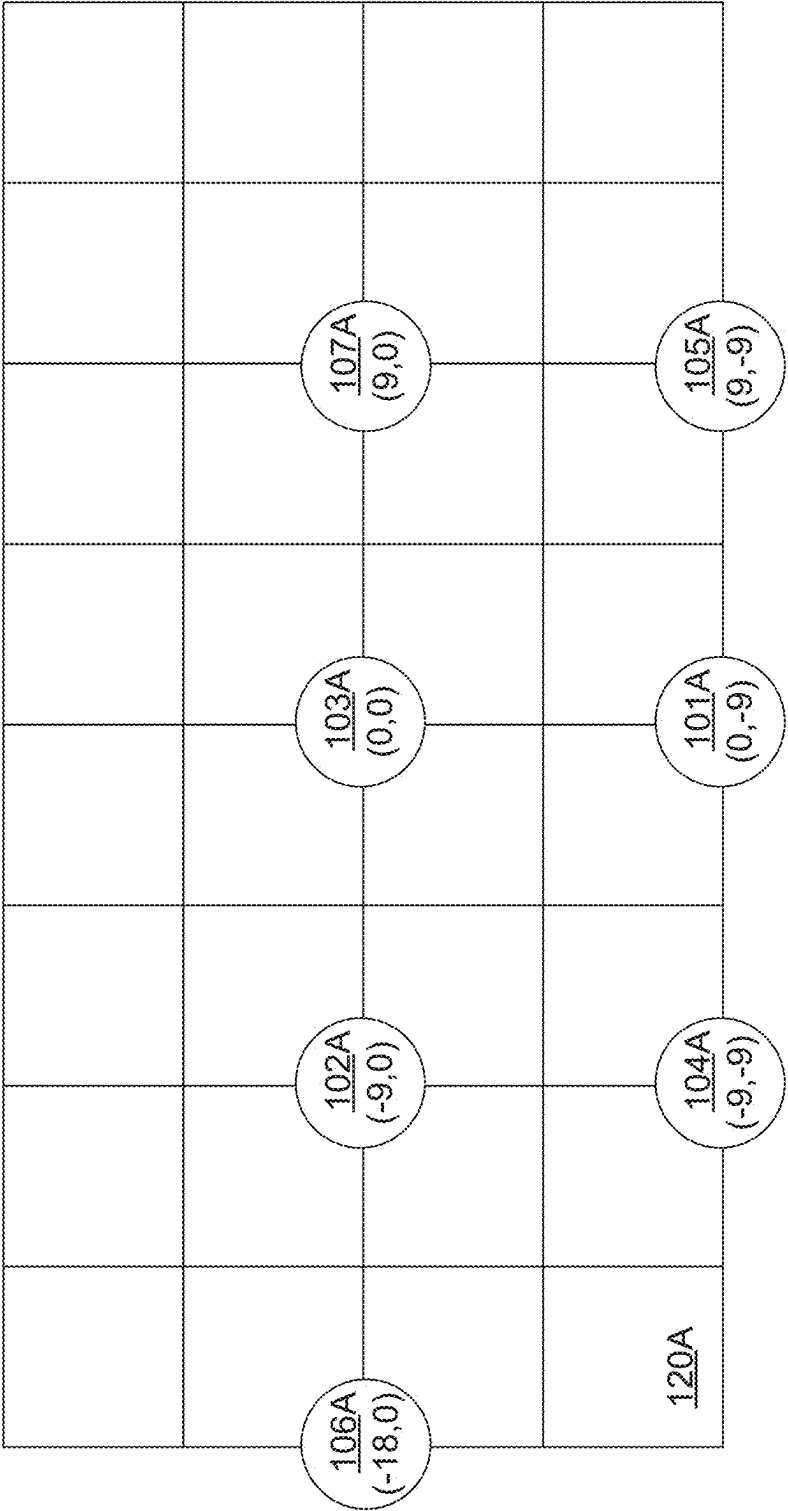


FIG. 59

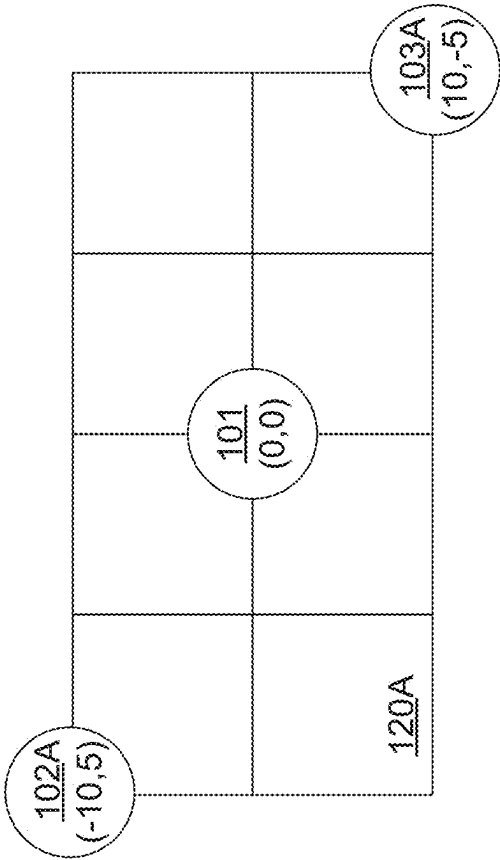


FIG. 60

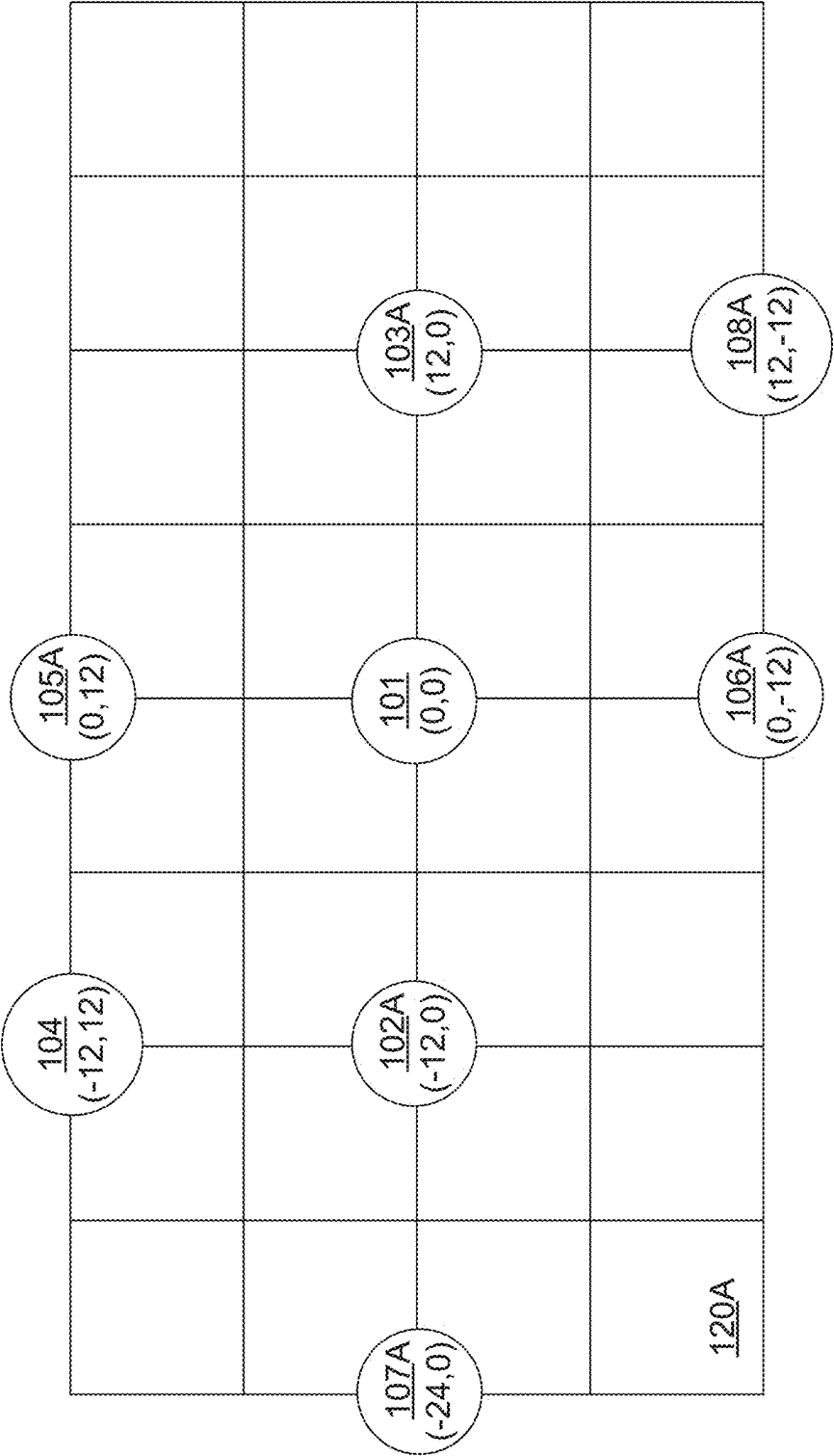


FIG. 61

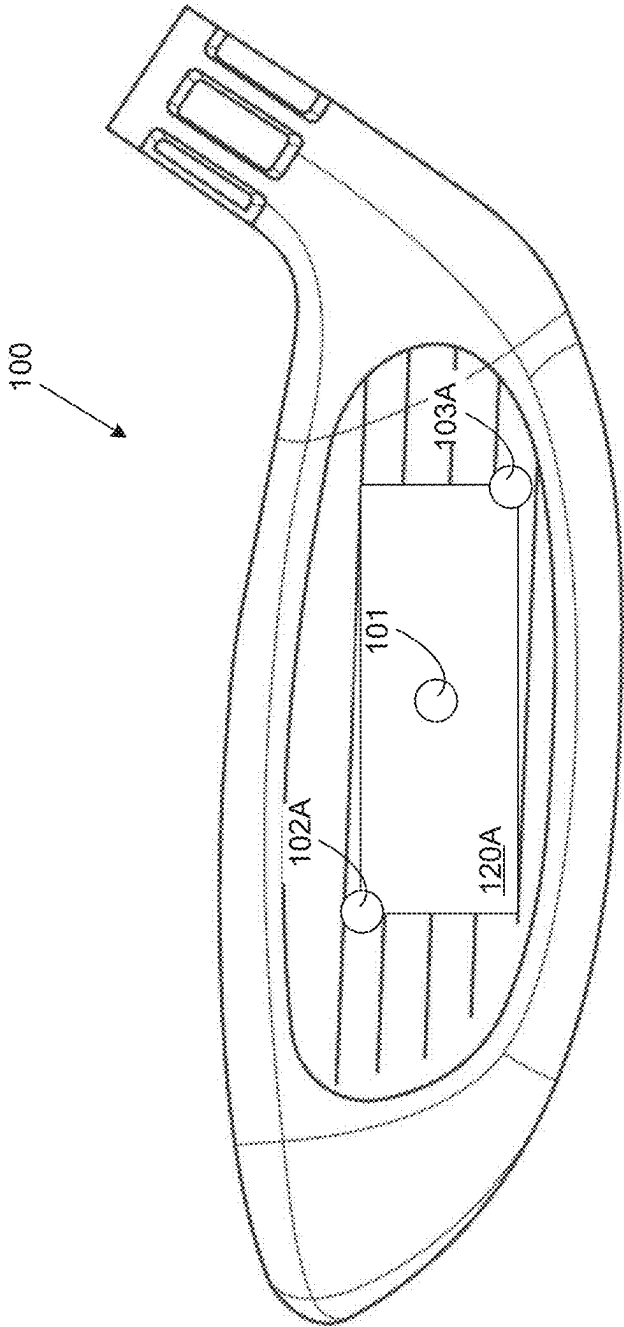


FIG. 62

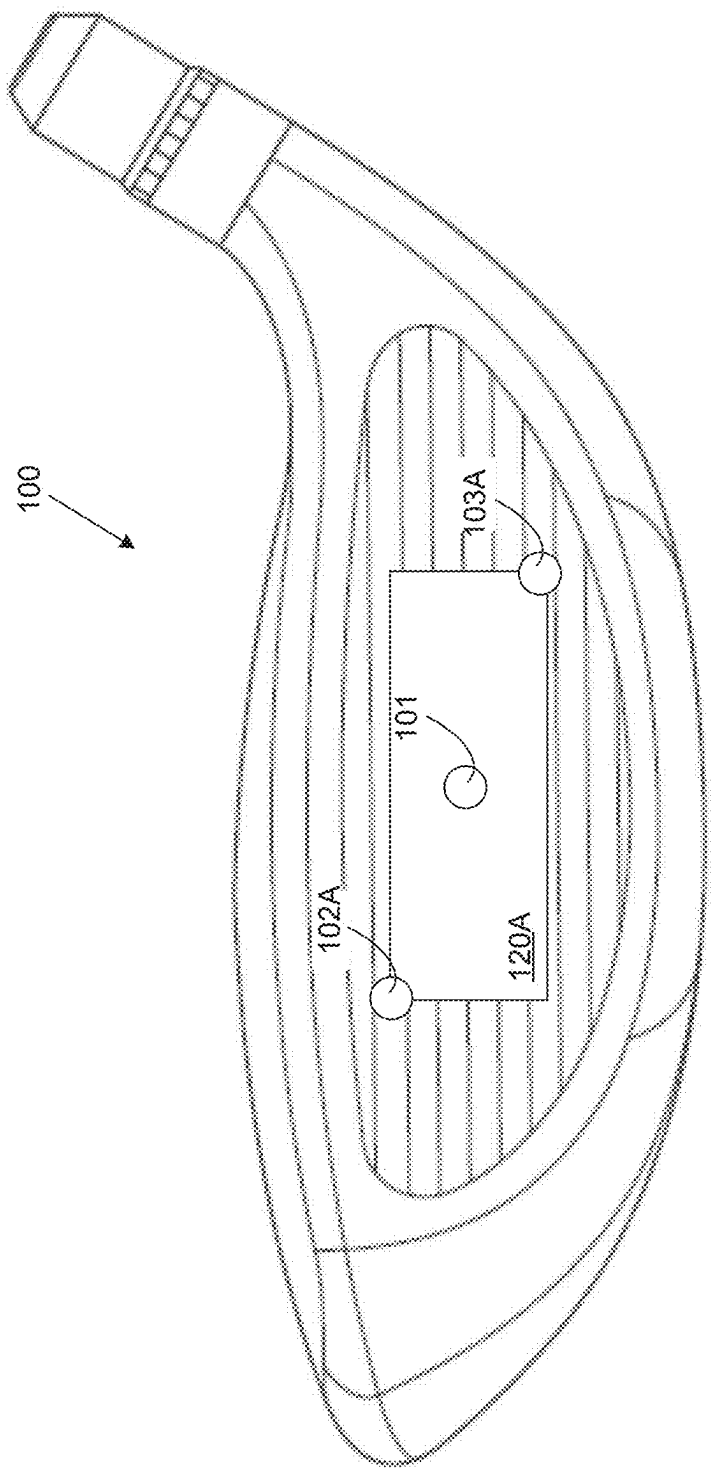


FIG. 63

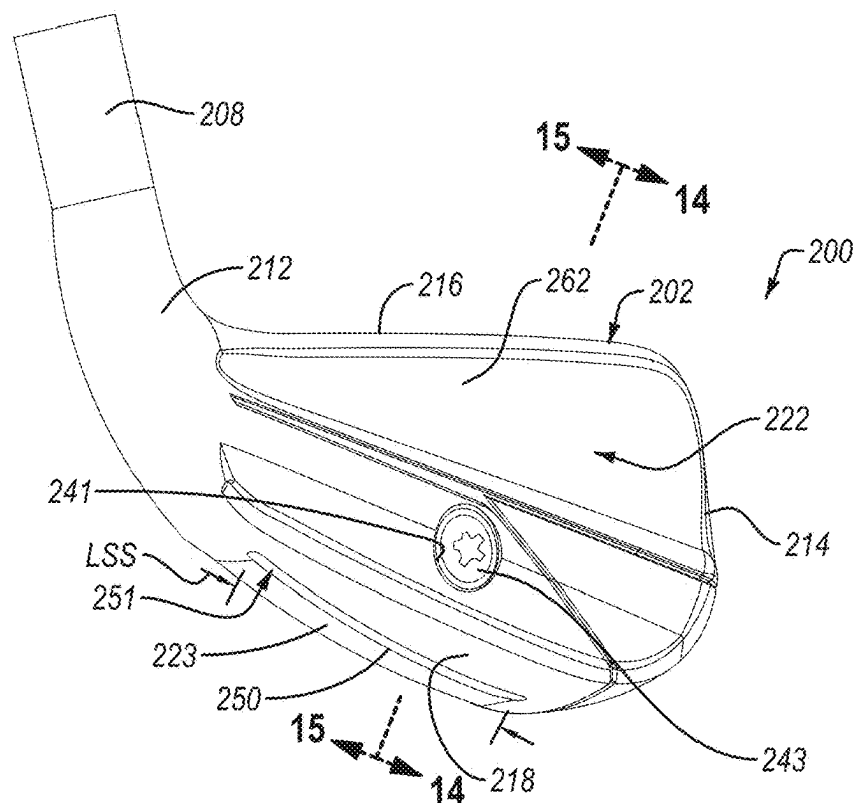


FIG. 64

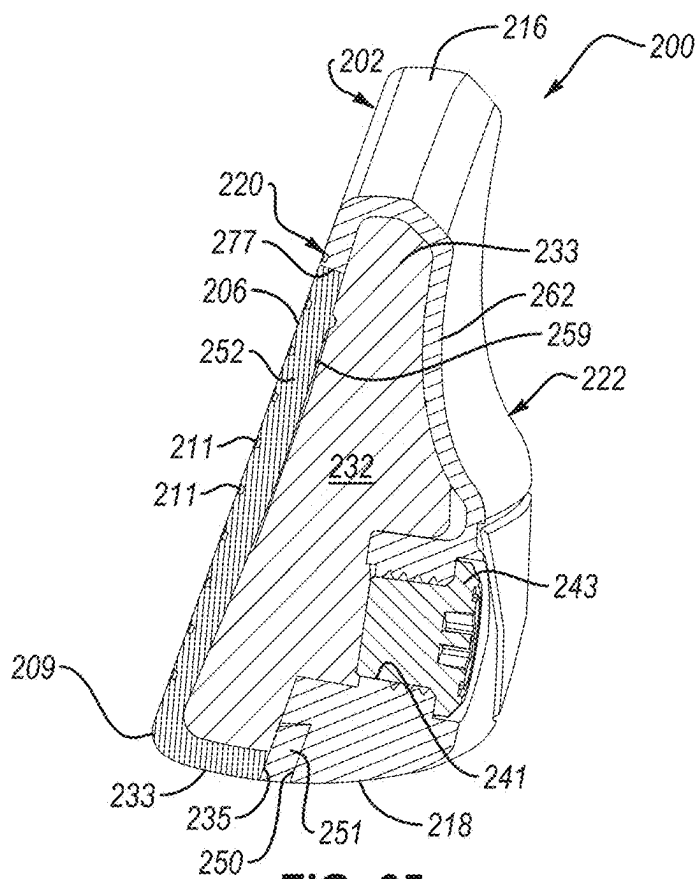


FIG. 65

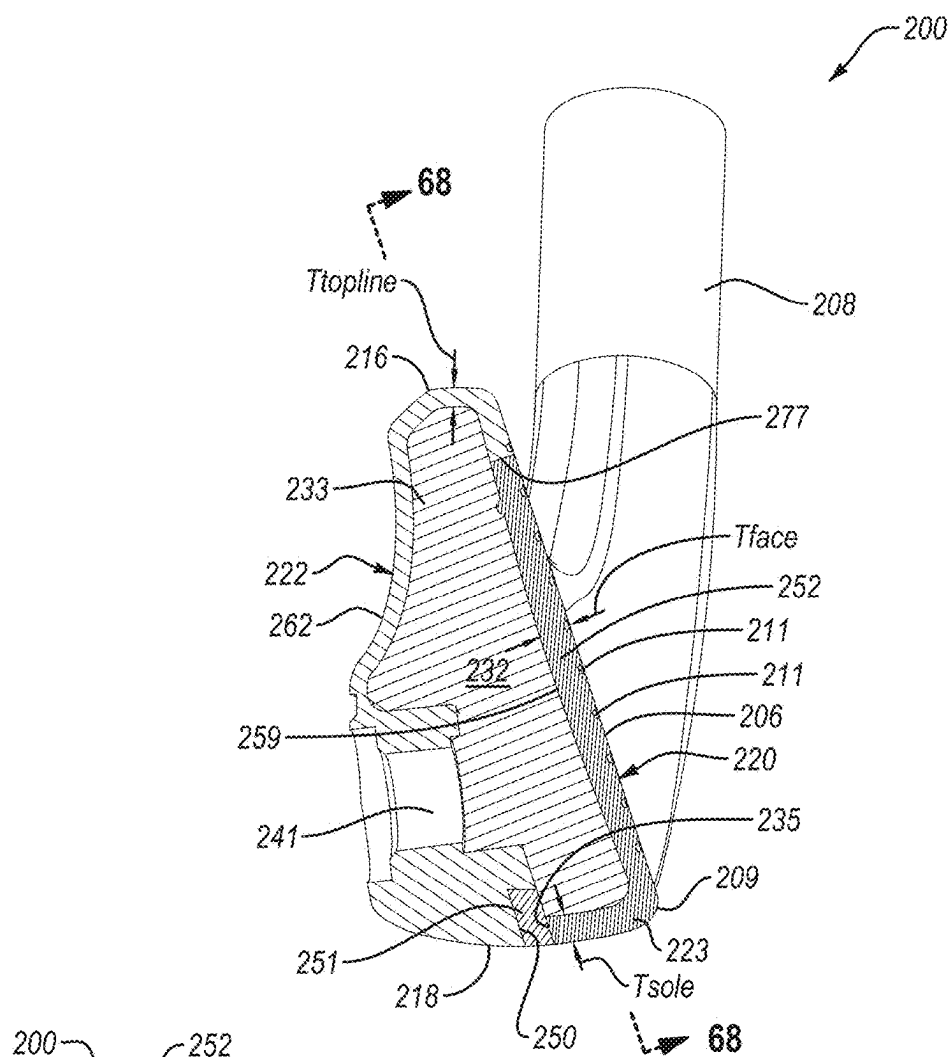


FIG. 66

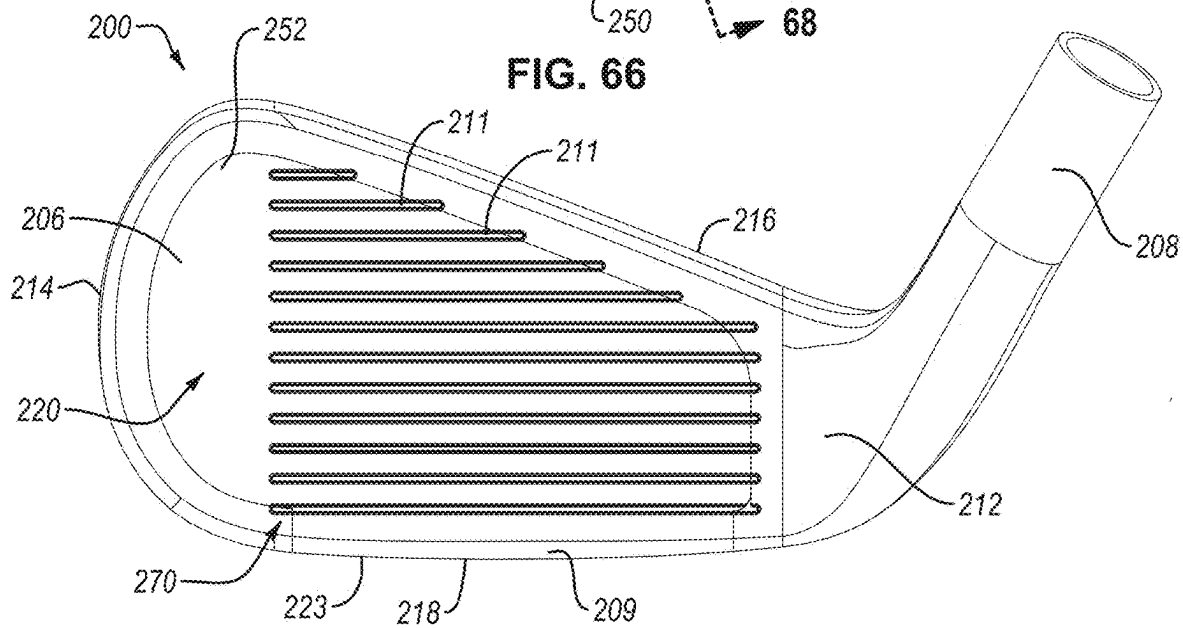
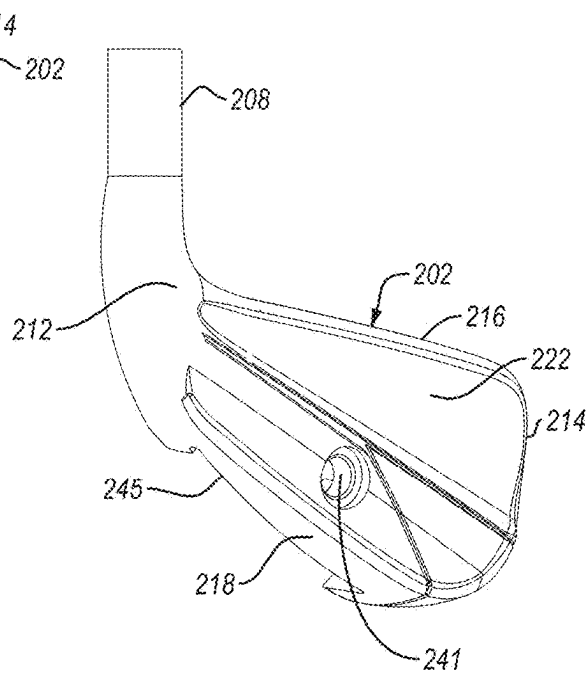
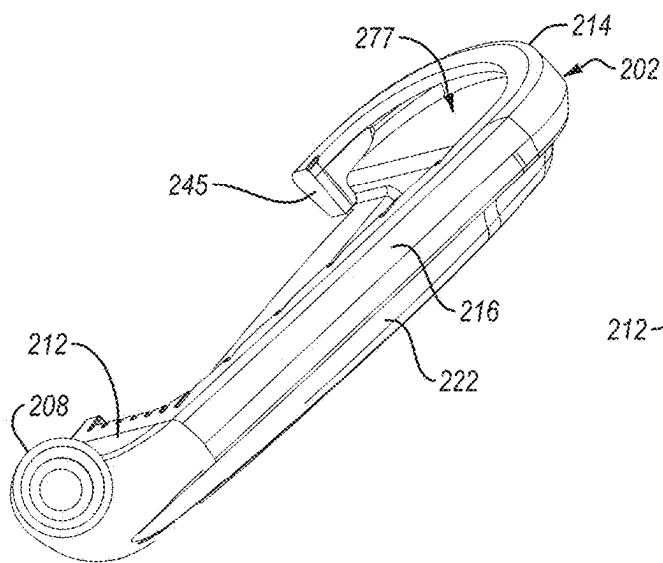
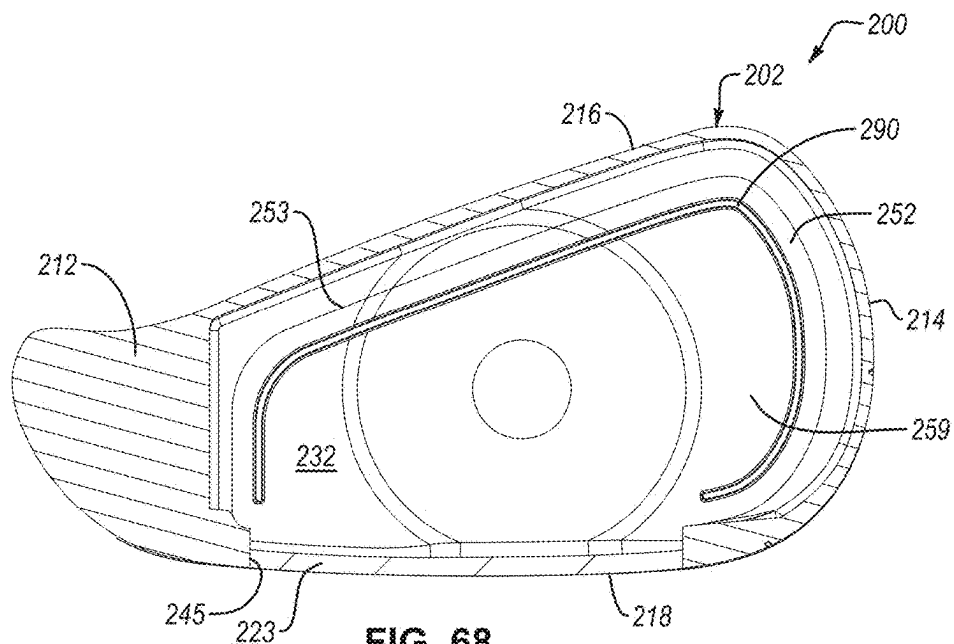
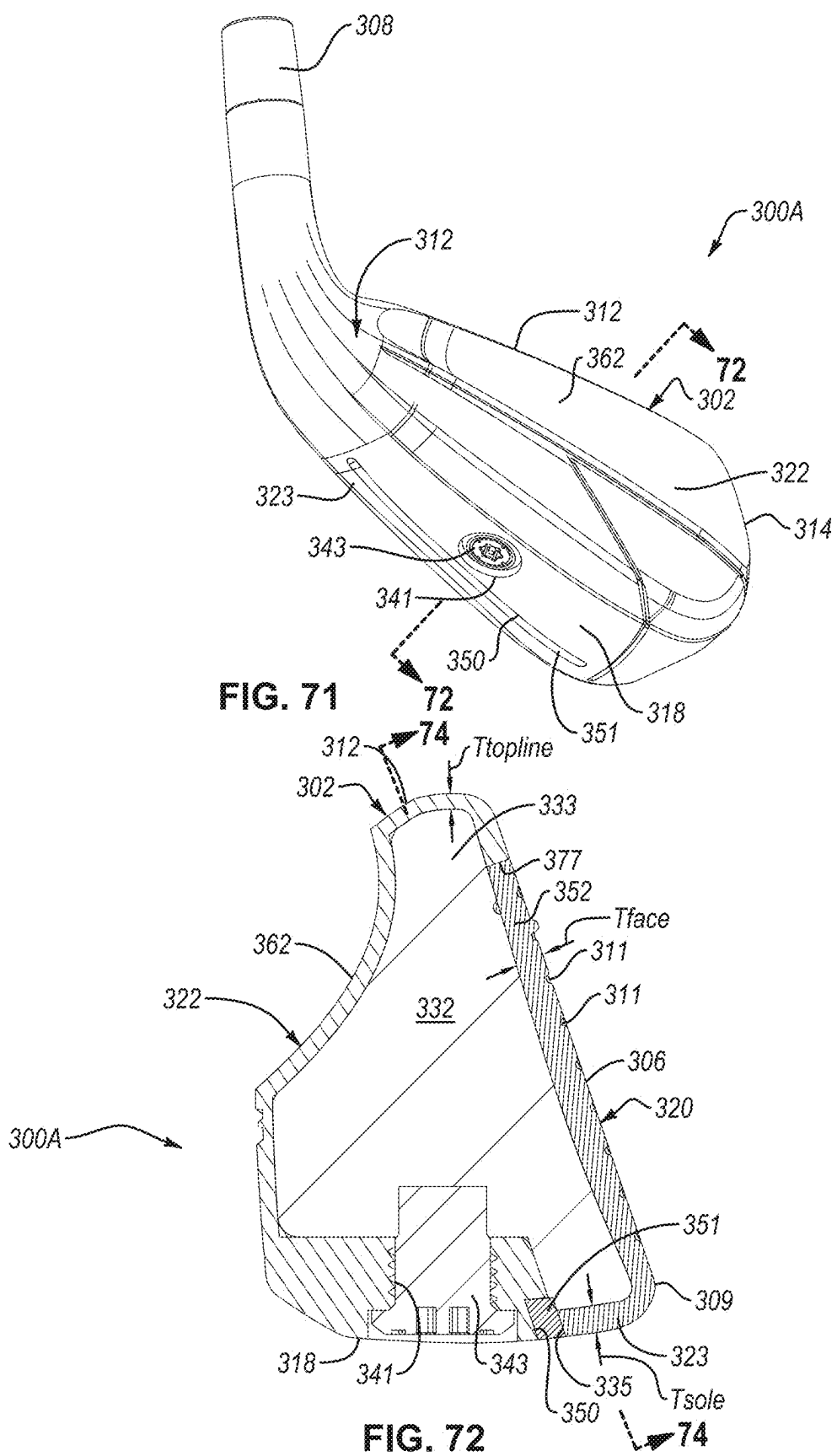


FIG. 67





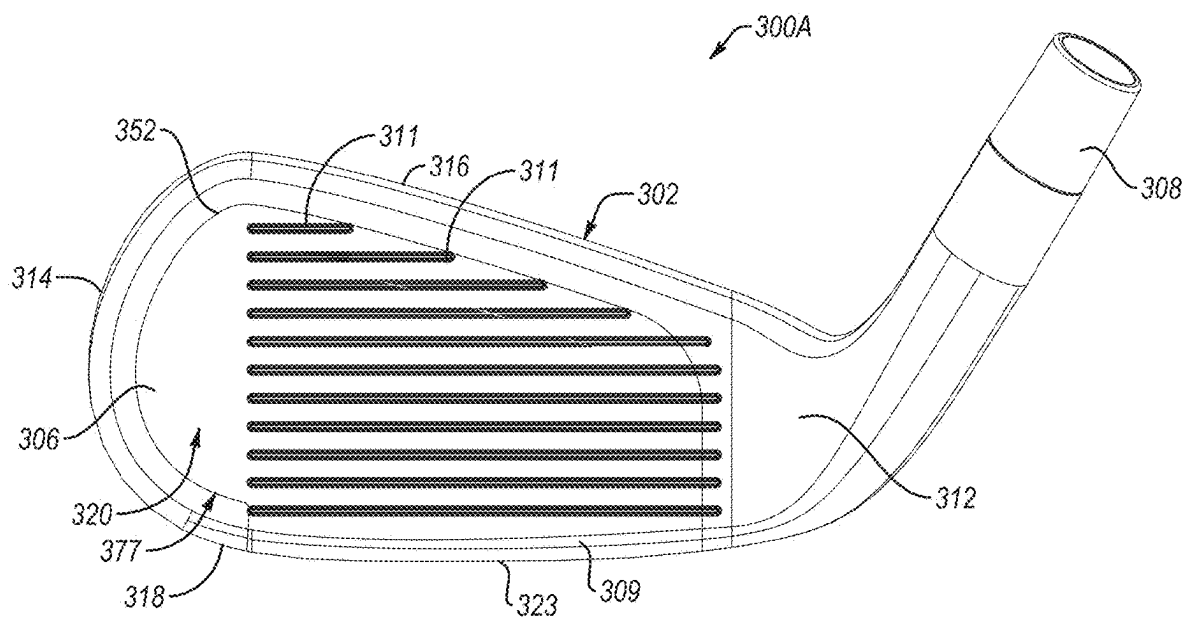


FIG. 73

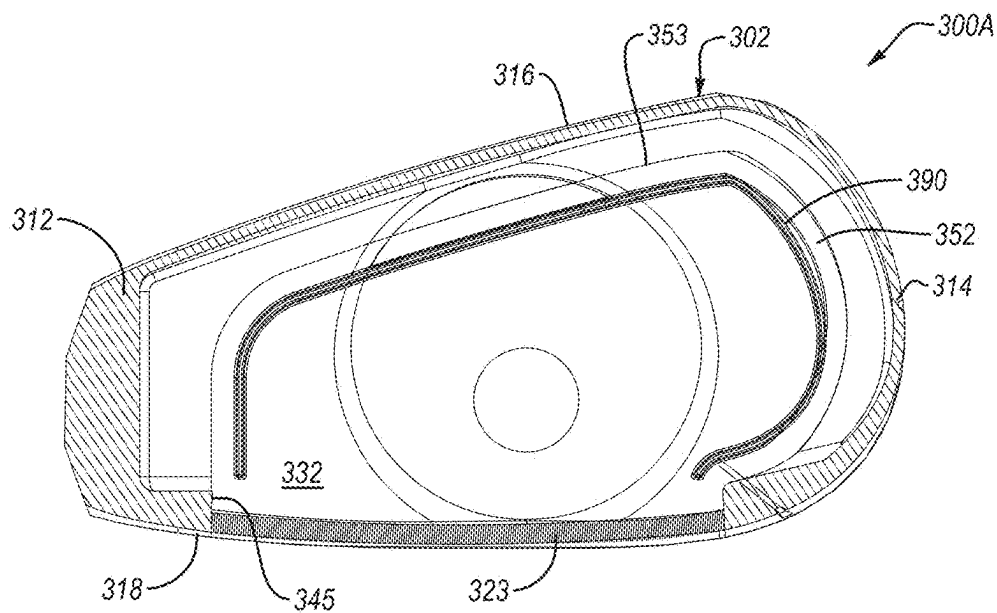


FIG. 74

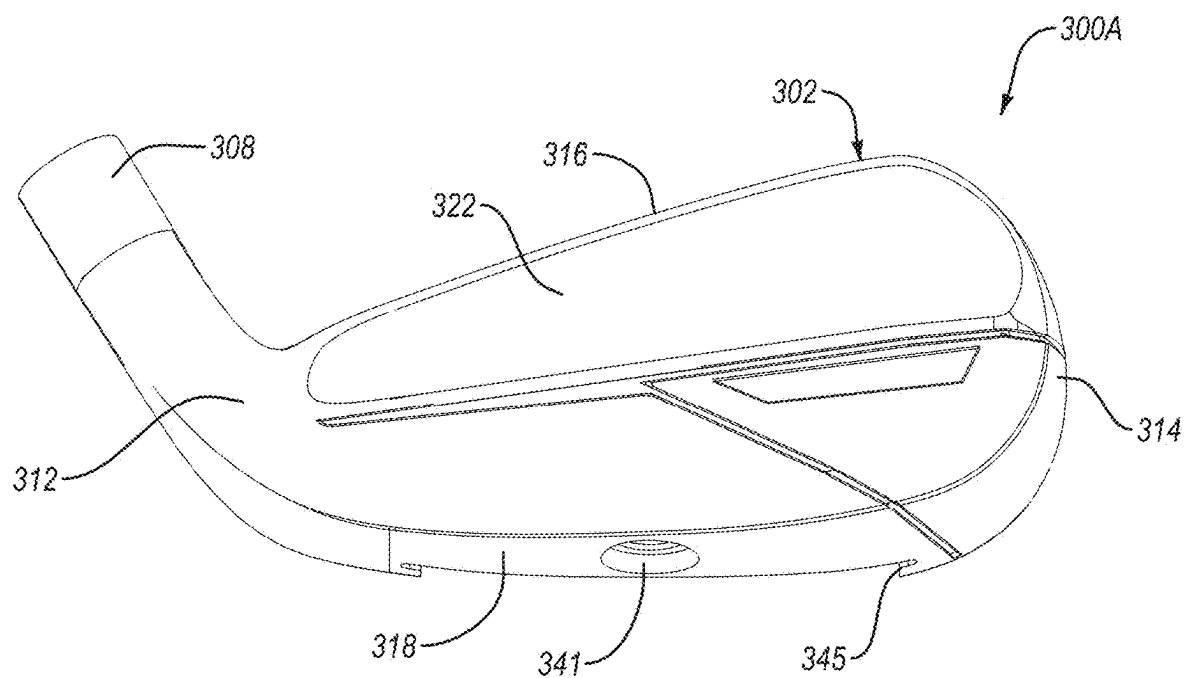
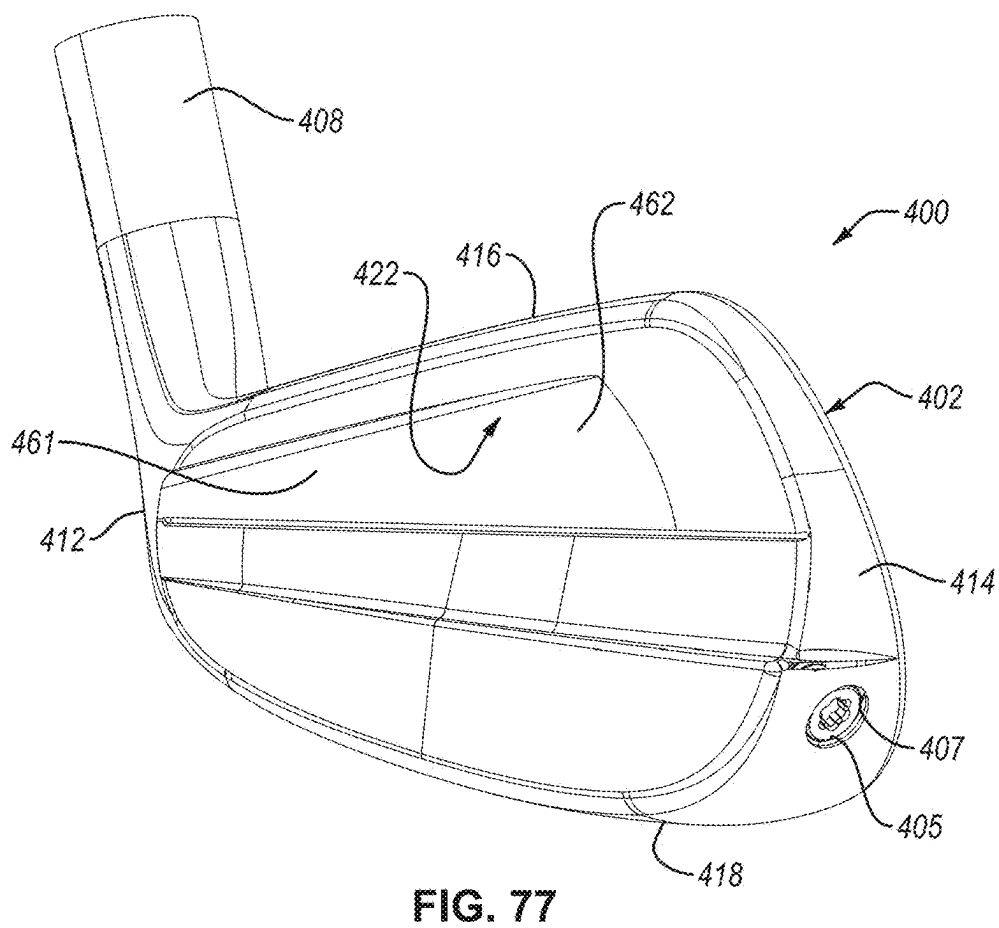
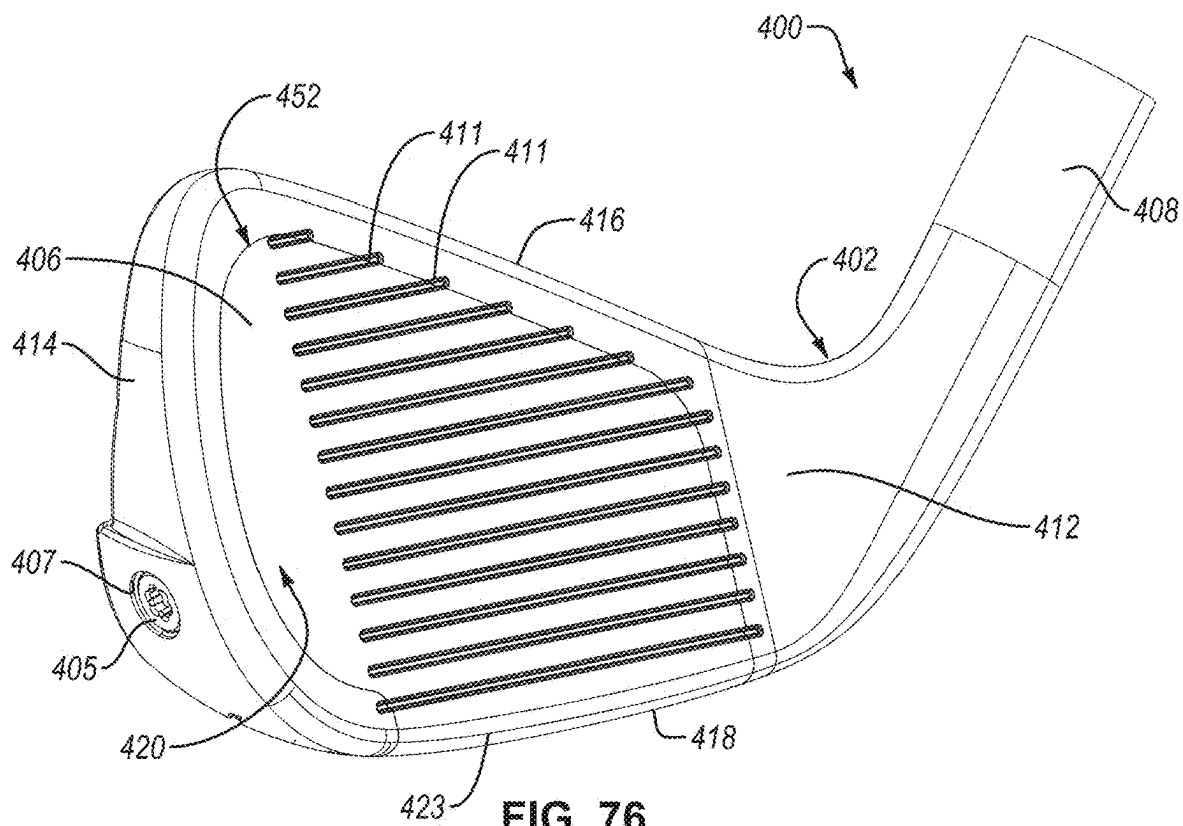


FIG. 75



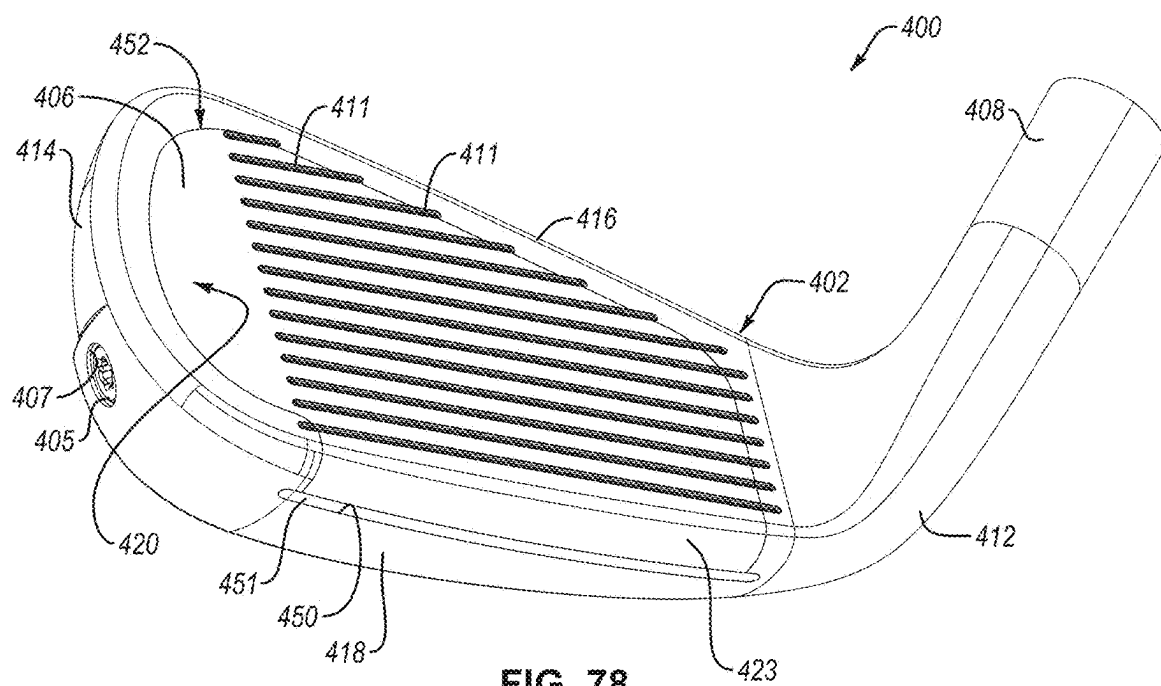


FIG. 78

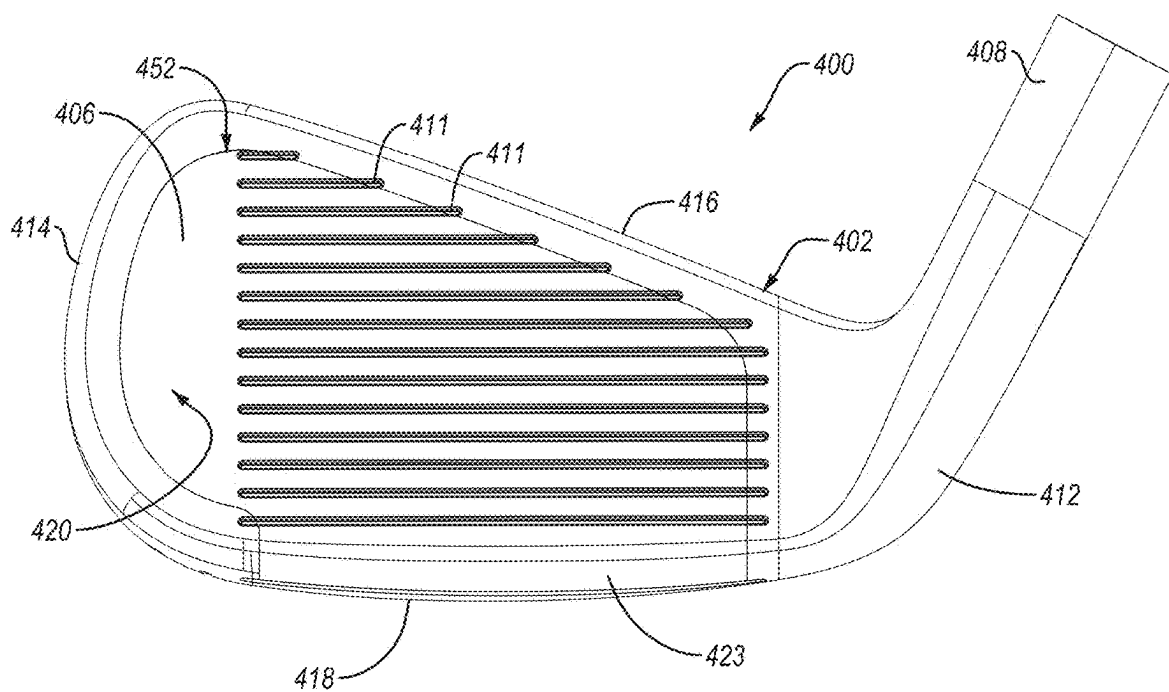


FIG. 79

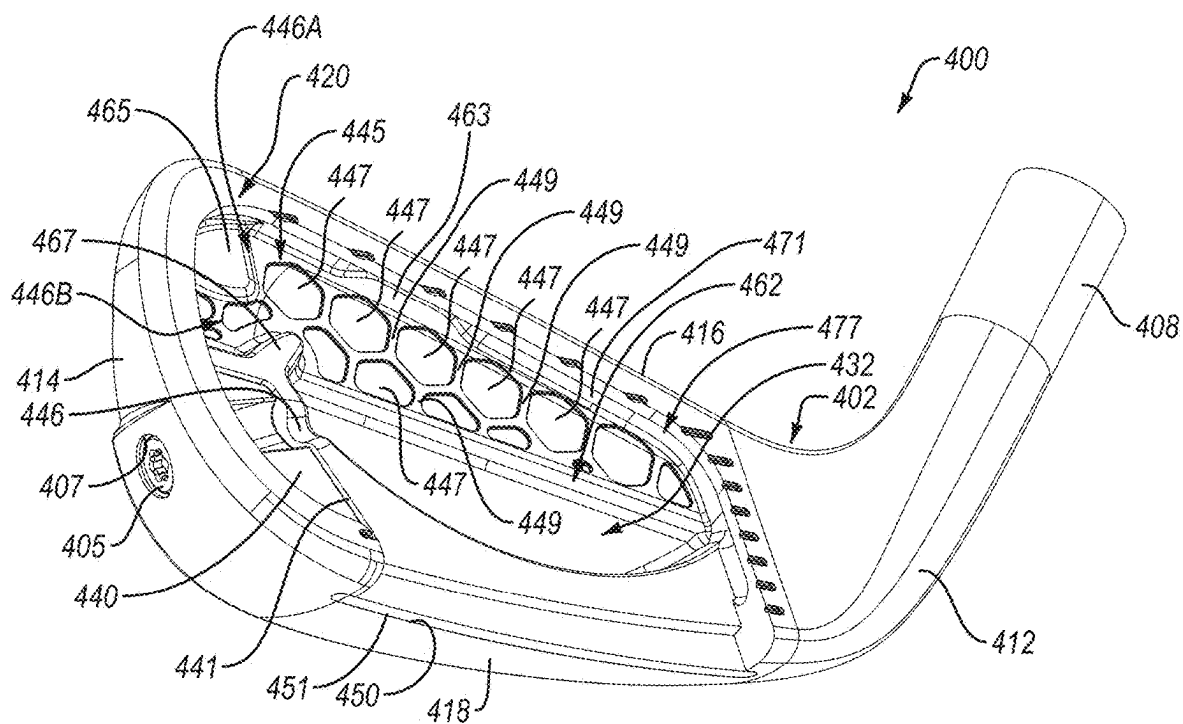


FIG. 80

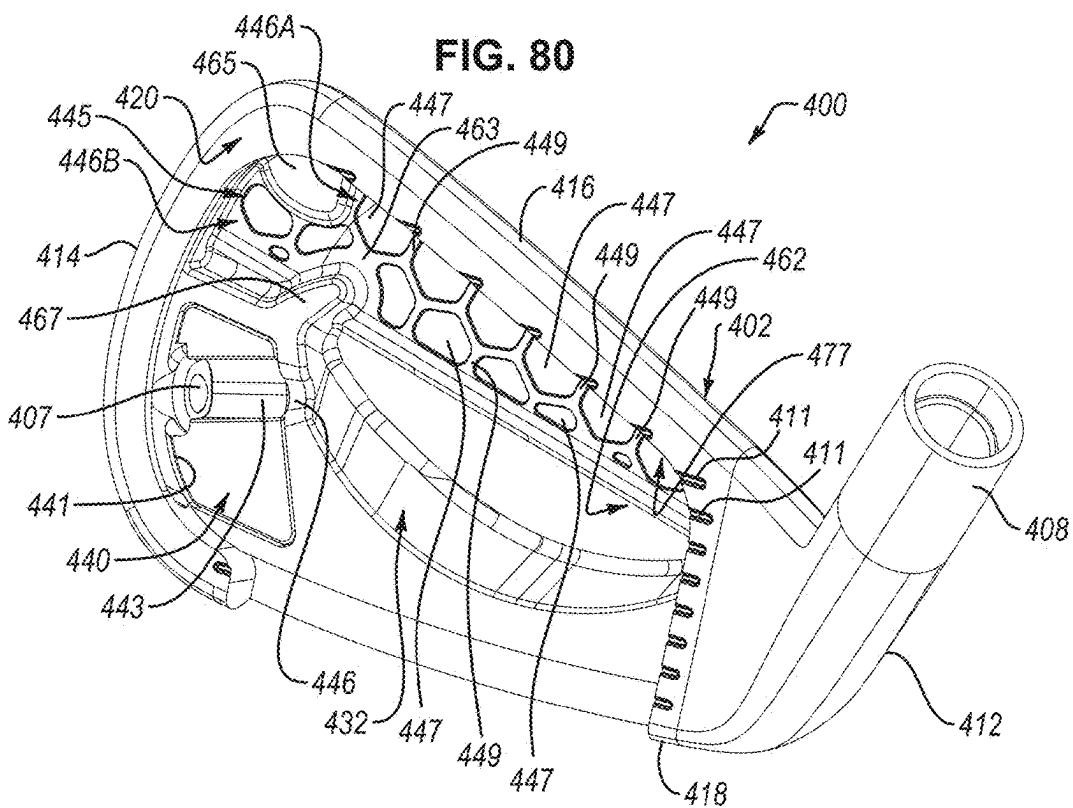


FIG. 81

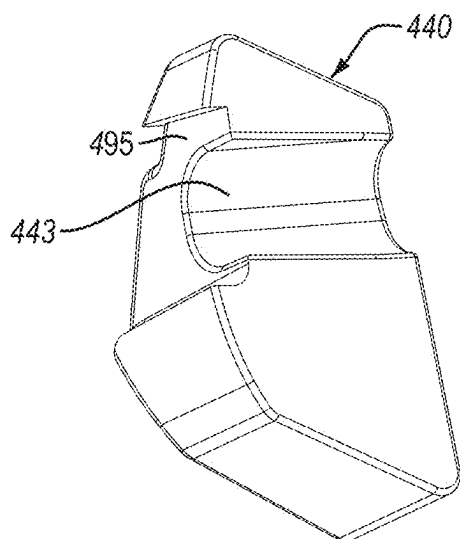


FIG. 82

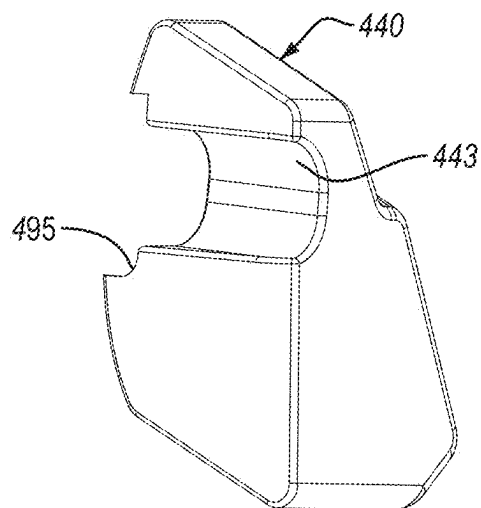


FIG. 83

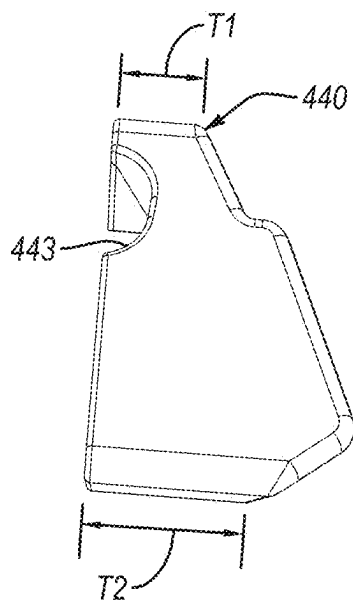


FIG. 84

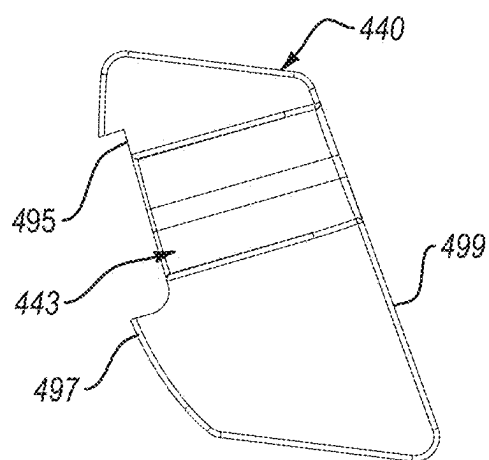


FIG. 85

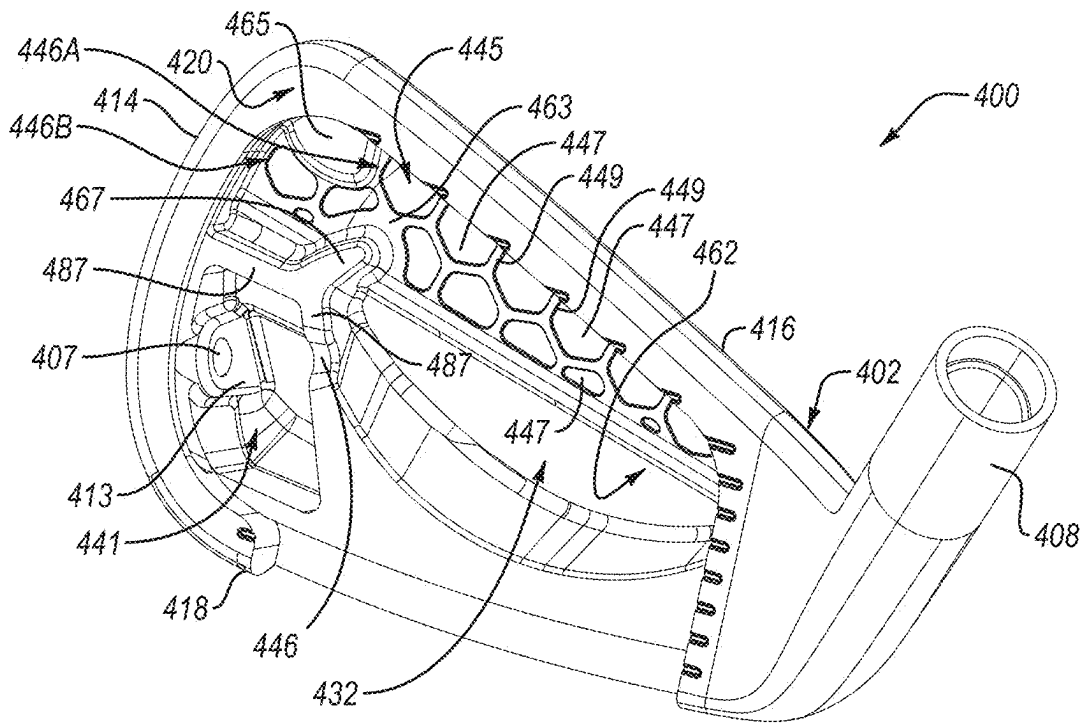


FIG. 86

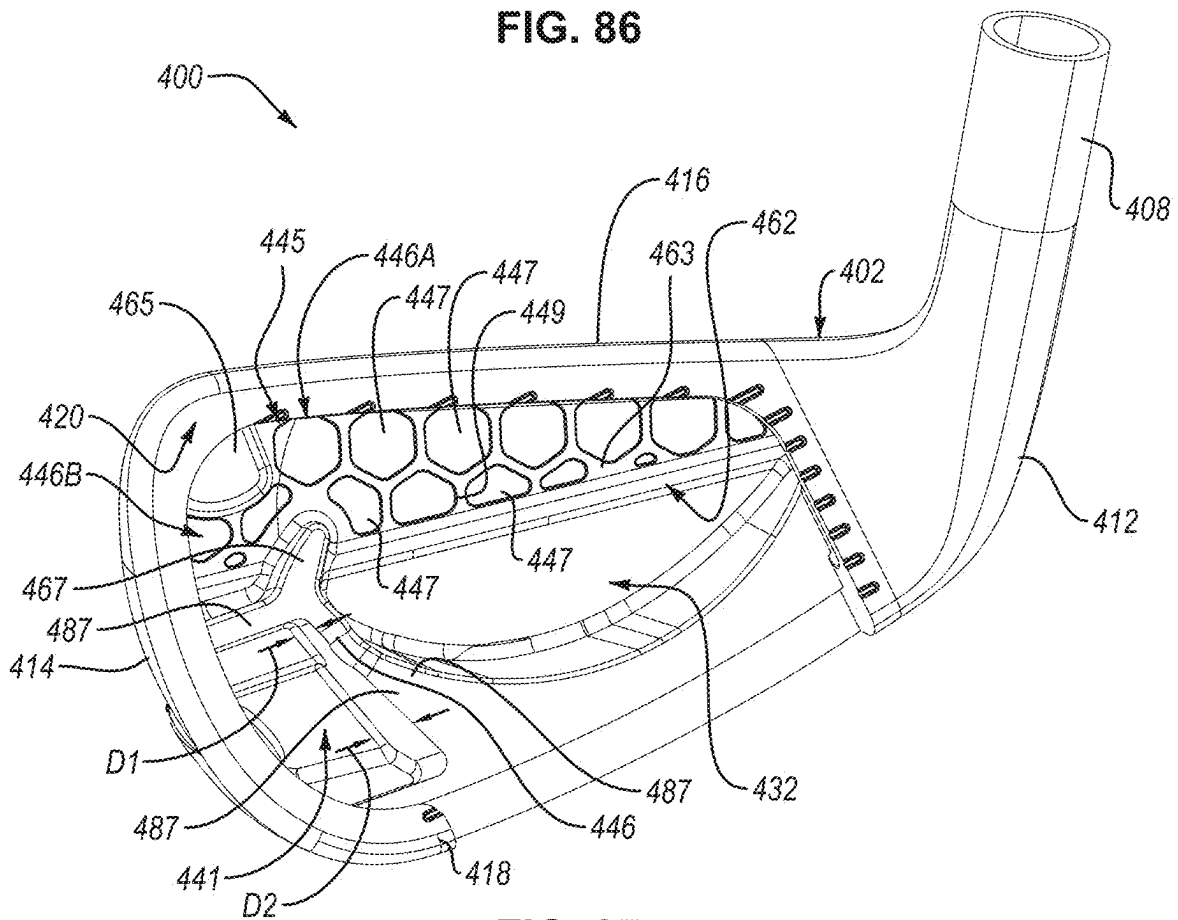


FIG. 87

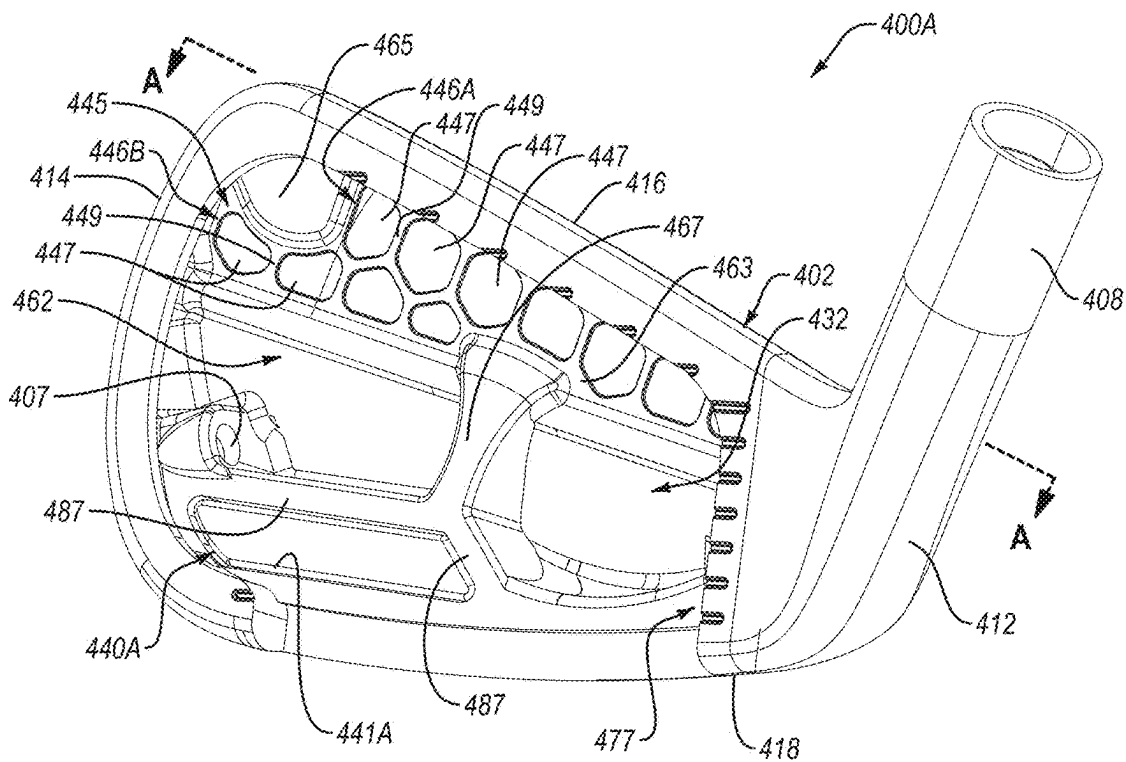


FIG. 88

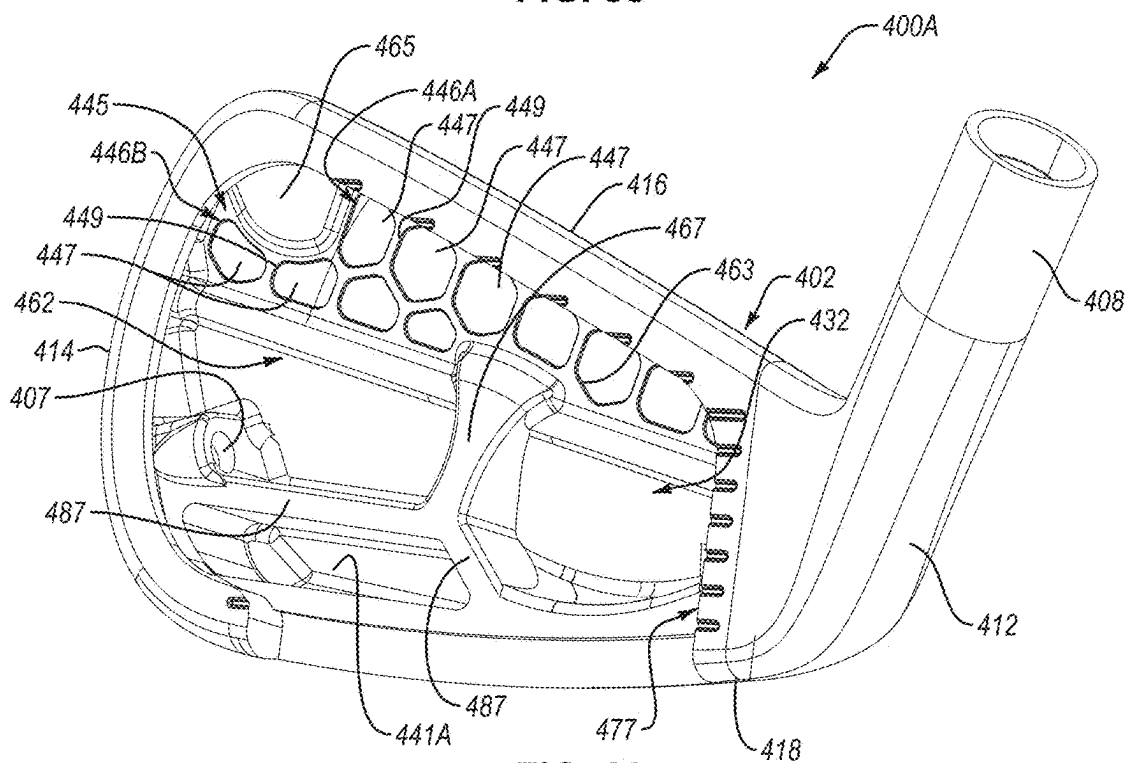
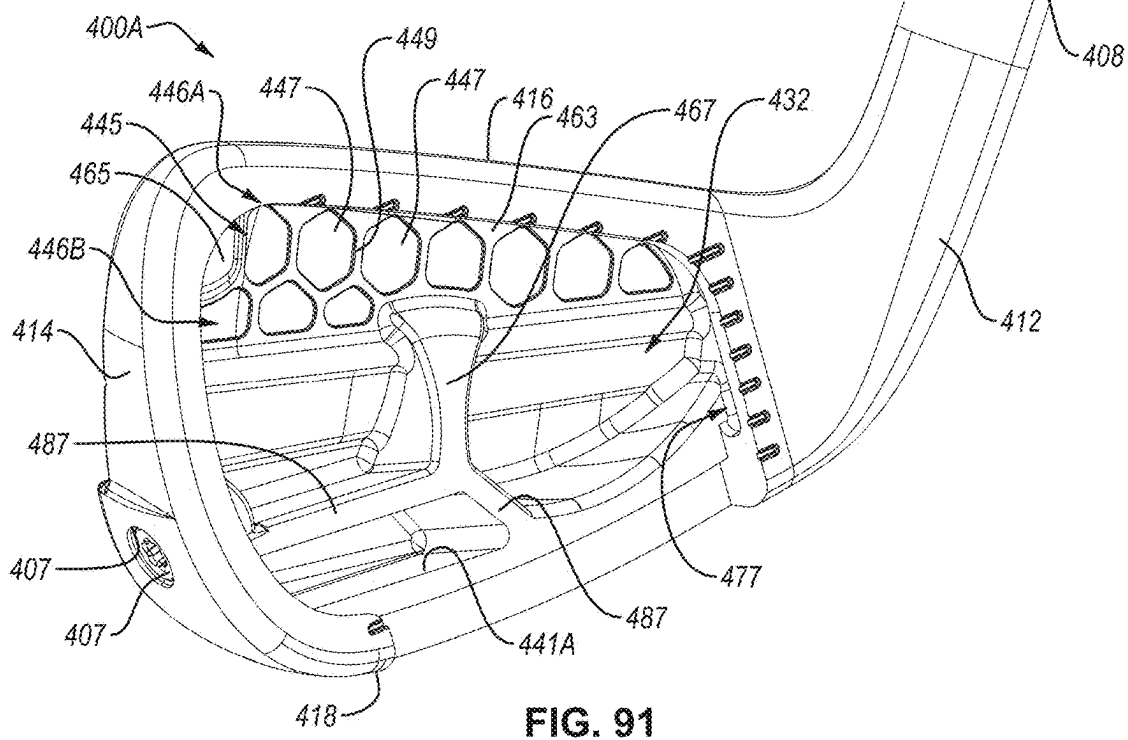
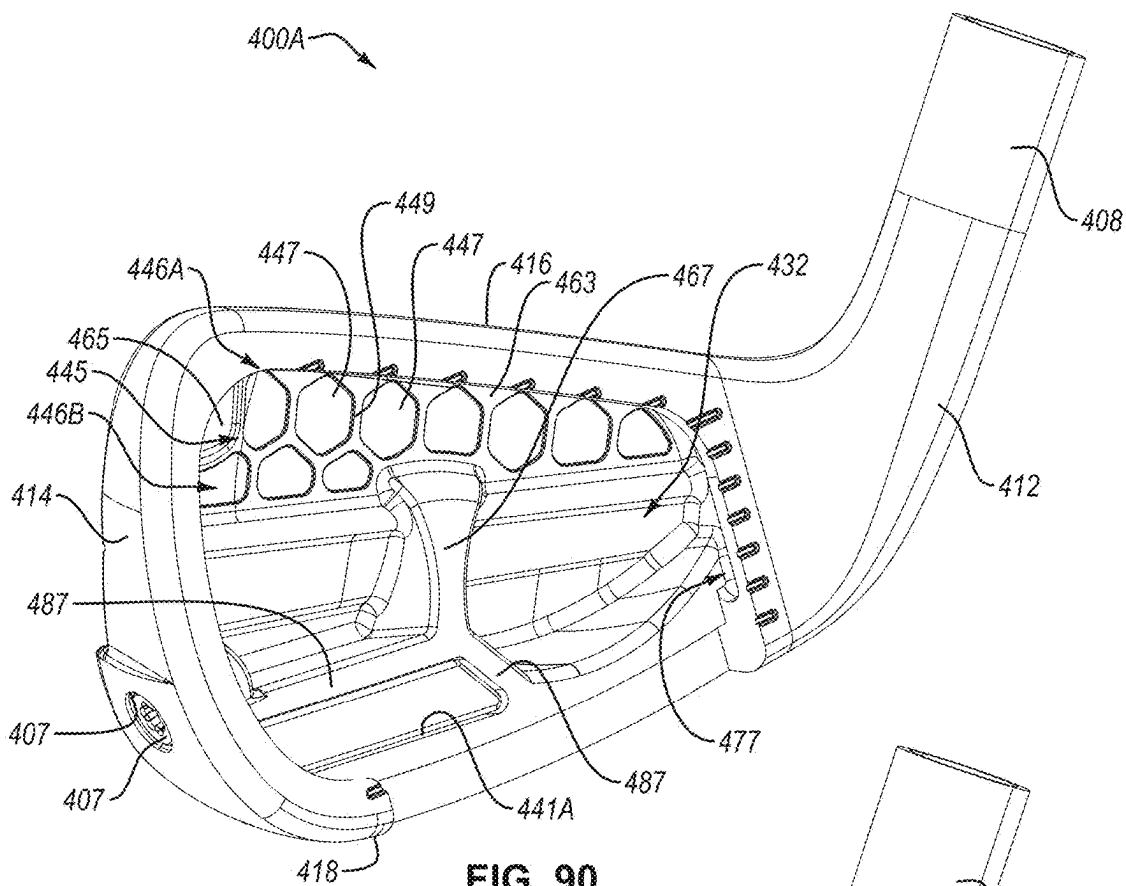


FIG. 89



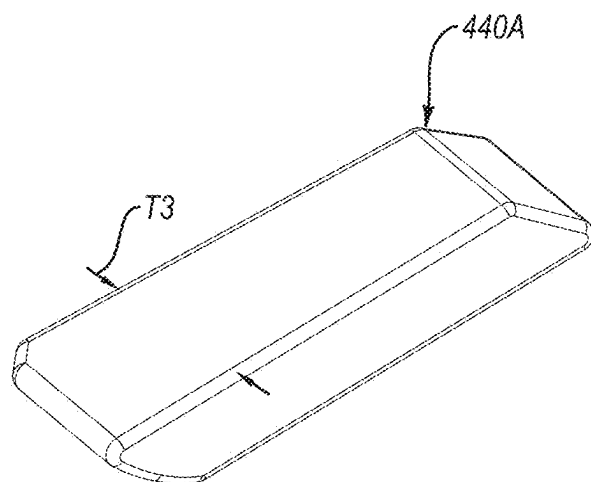
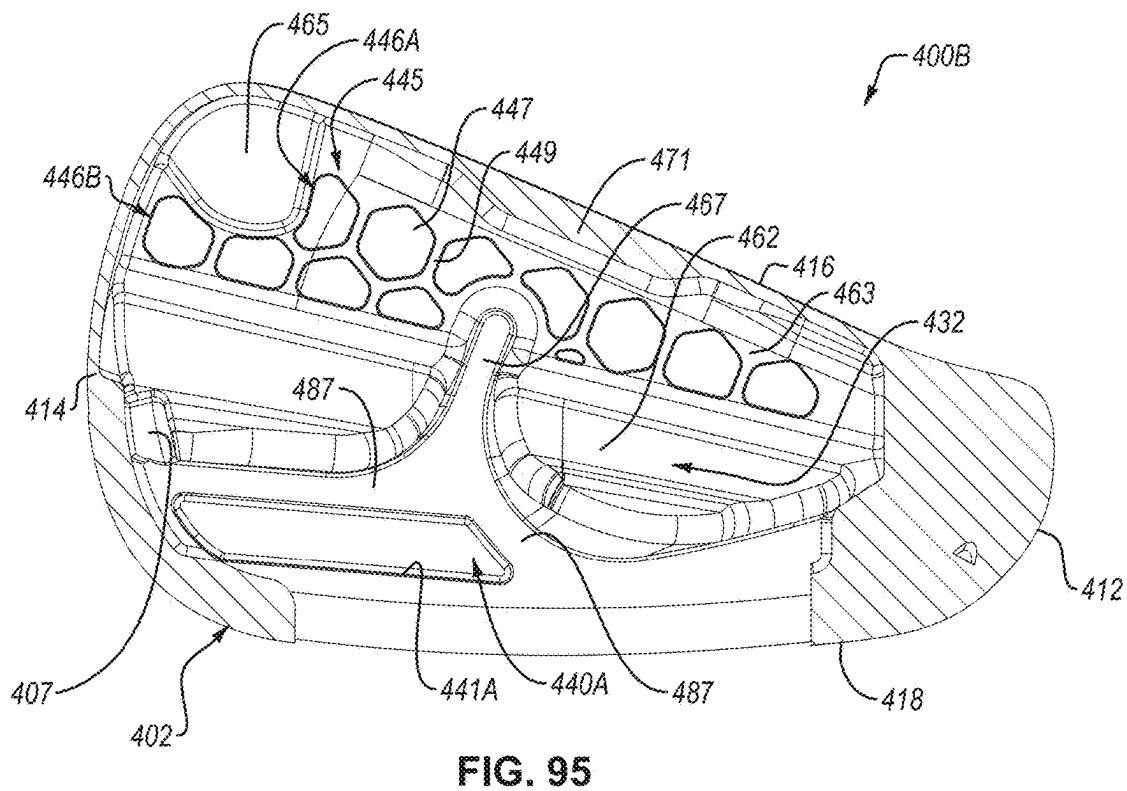
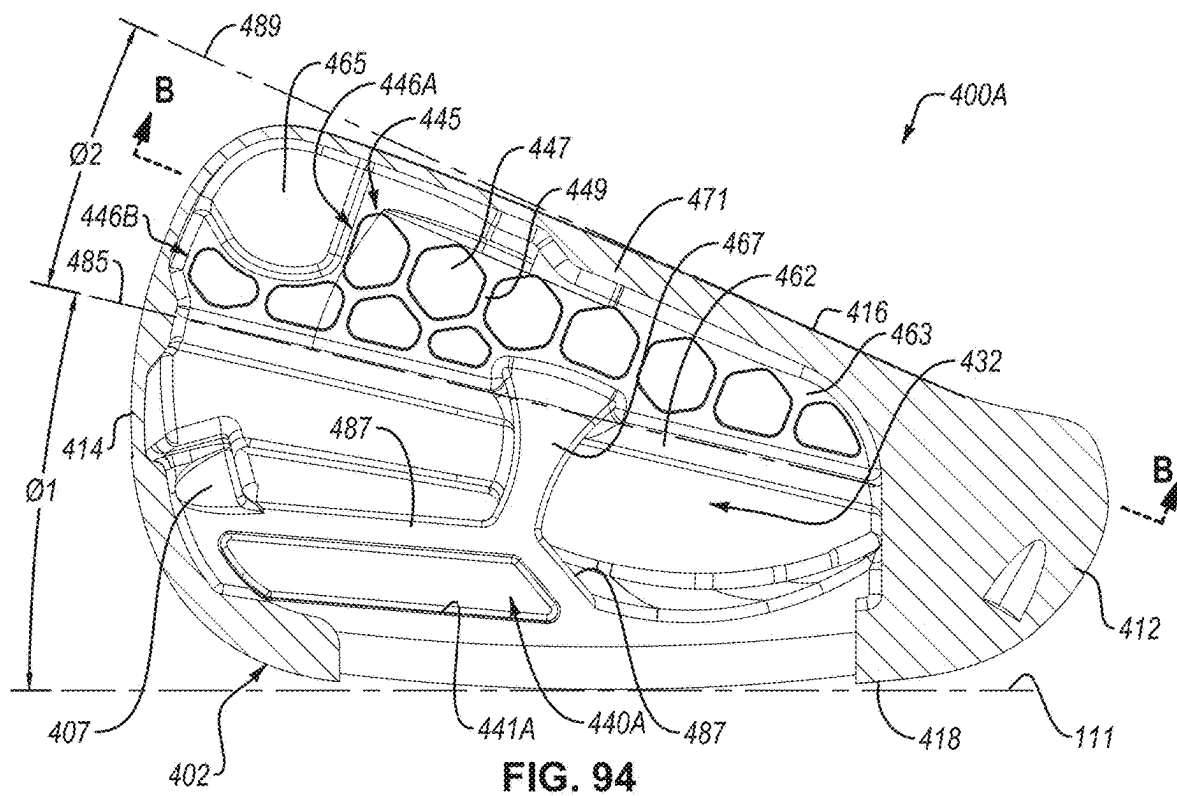


FIG. 92



FIG. 93



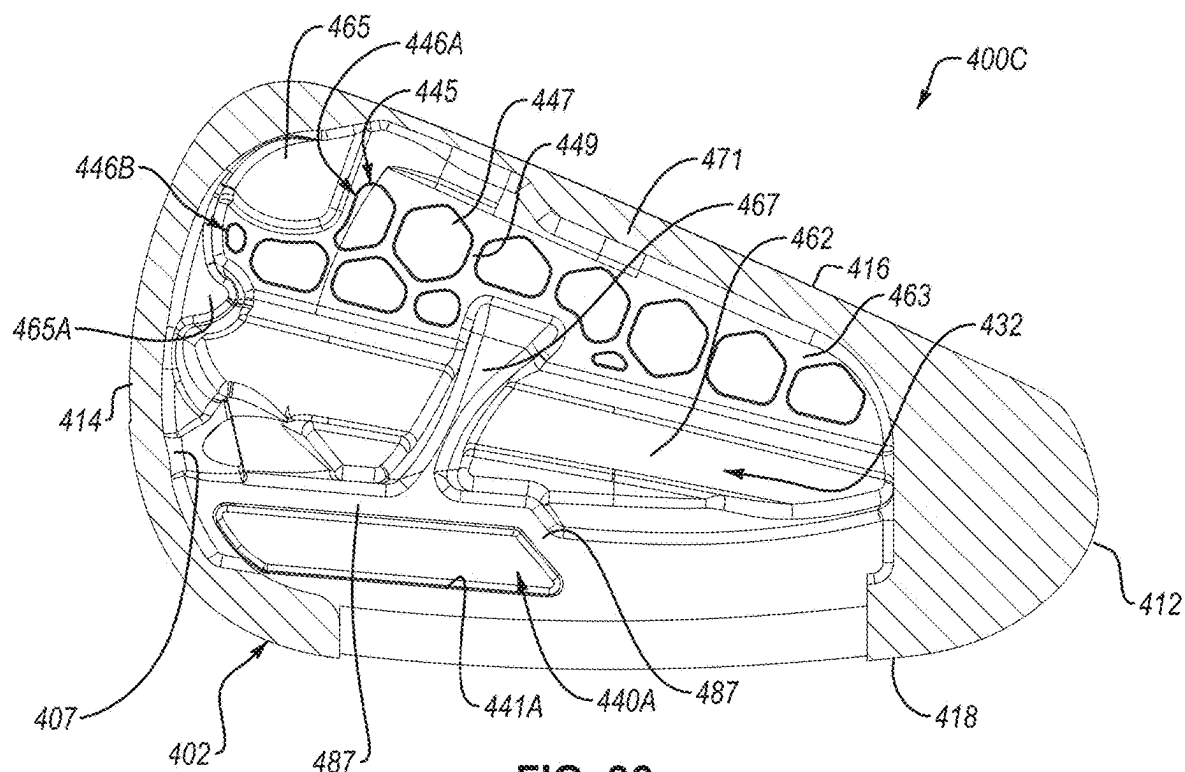


FIG. 96

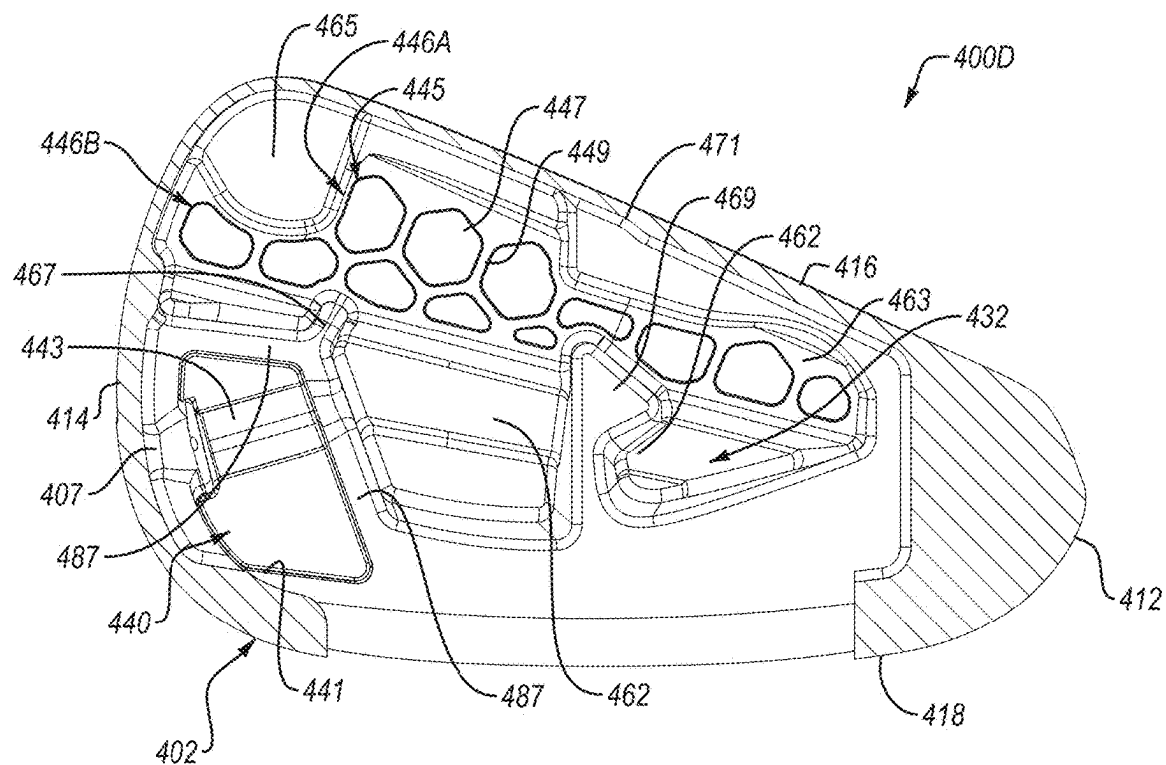
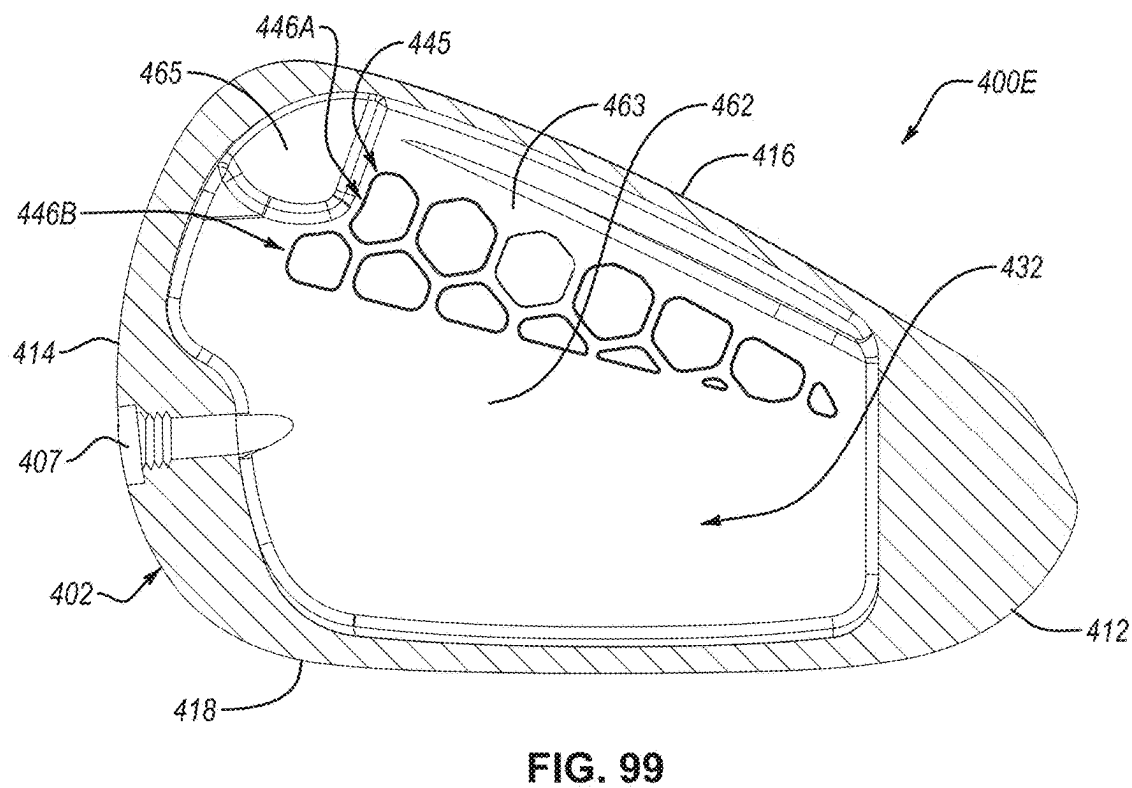
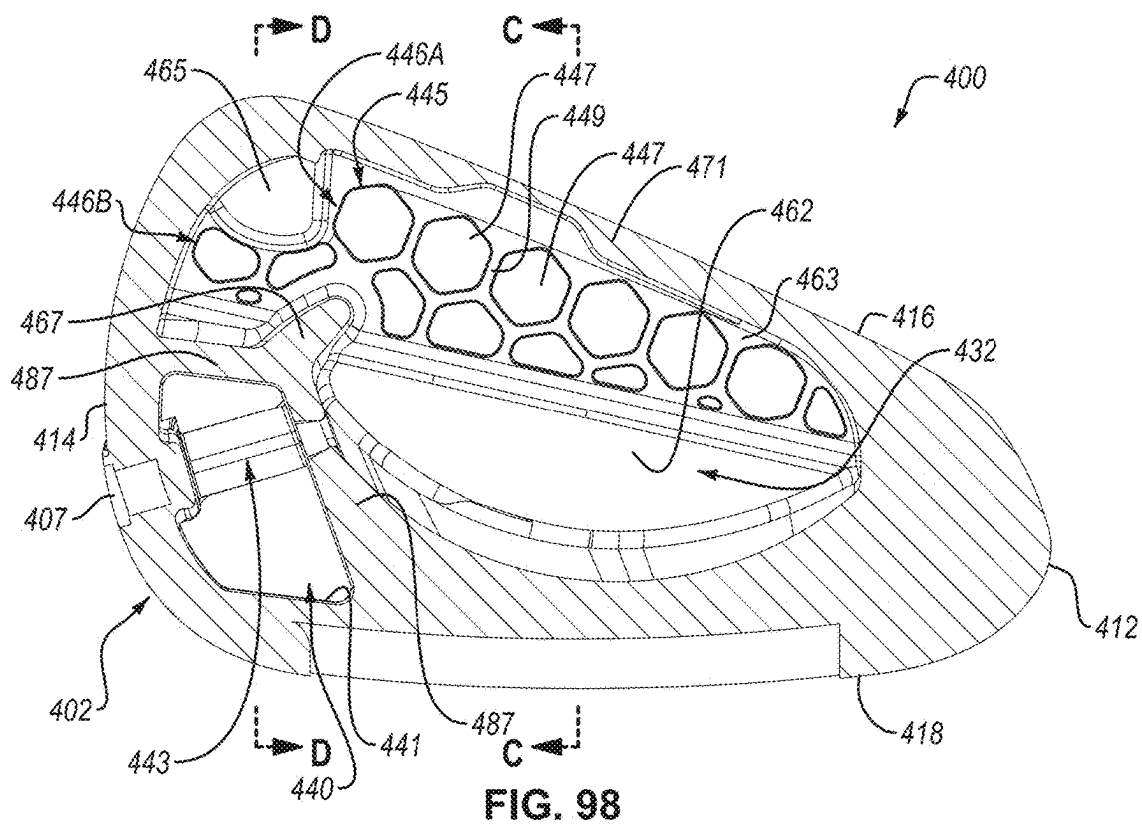


FIG. 97



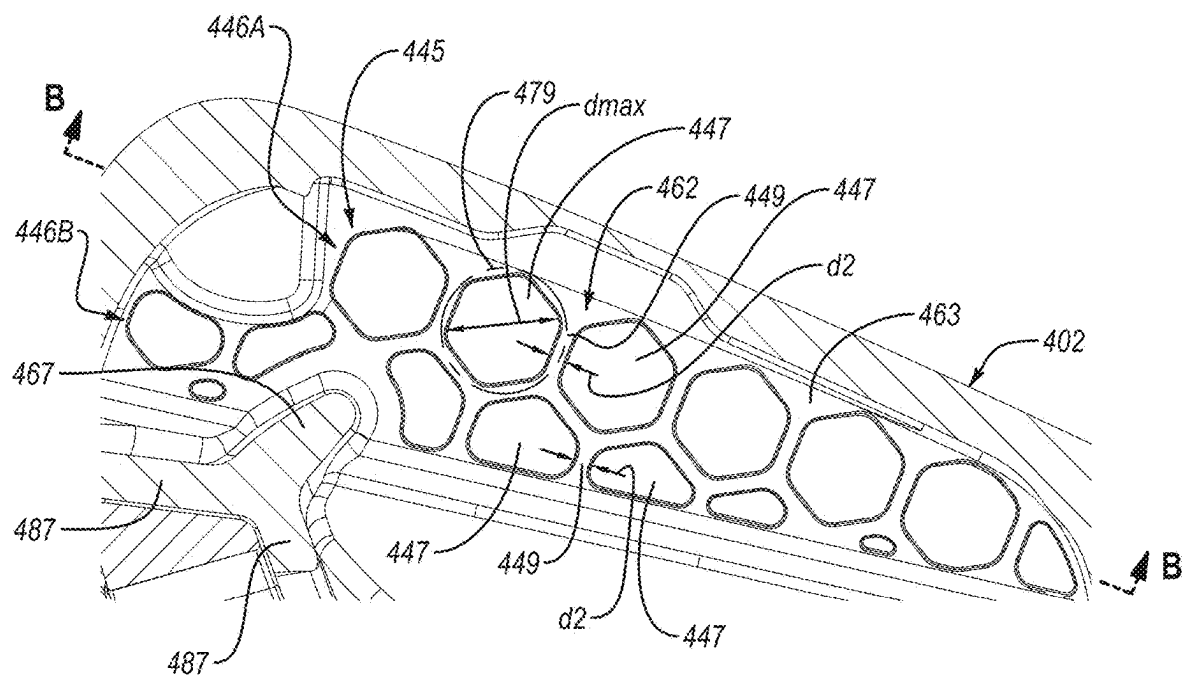


FIG. 100

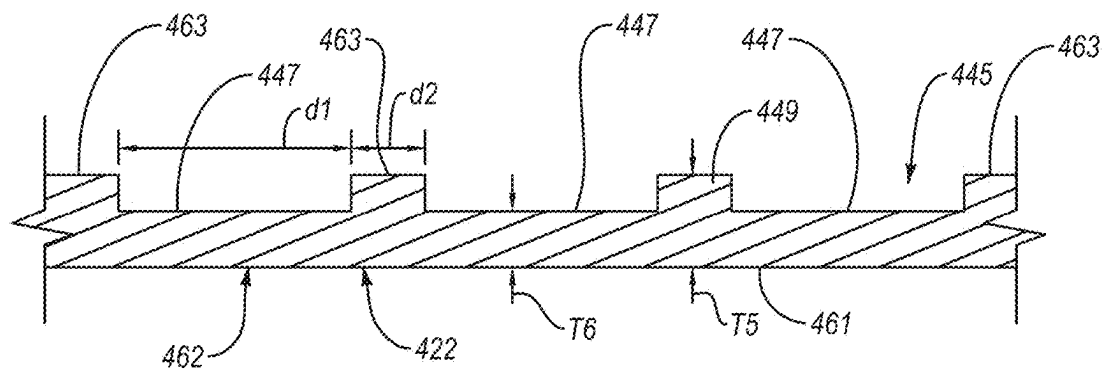


FIG. 101

800

Golf Club Head	400A	400B	400C	400D	400	400E	9-Iron	PW
Loft (degrees)	19.00	21.00	23.50	26.50	30.50	35.00	40.00	45.00
Lie (degrees)	60.50	61.00	61.50	62.00	62.50	63.00	63.50	64.00
Offset	3.50	3.40	3.10	2.90	2.70	2.40	2.10	1.80
Hosel Length (mm)	62.50	64.00	64.00	64.00	64.00	66.00	68.00	70.00
60° Hosel Length (mm)	63.70	63.40	63.10	62.80	62.50	64.30	66.00	67.70
Hosel Base Length (mm)	34.5	36.00	36.00	36.00	36.00	38.00	40.00	42.00
Topline Thickness, CAD (mm)	6.50	6.50	6.50	6.40	6.30	6.30	6.30	6.30
Topline Thickness, Visual (mm)	6.60	6.70	6.50	6.40	6.30	5.90	5.60	5.4
Face Thickness (mm)	1.60	1.60	1.60	1.50	1.50	1.70	1.70	2.4
Face Area (mm ²)	2738	2761	2790	2825	2829	2971	3028	3085
Blade Length (mm)	79.00	78.70	78.60	78.40	78.20	77.60	77.10	77.10
Face Length (mm)	74.00	73.70	73.60	73.40	73.20	72.10	71.10	70.60
Shaft Progression (mm)	33.70	33.60	33.50	33.40	33.30	33.00	33.00	33.30
Hosel Axis to PAR (CAD) (mm)	5.00	5.00	5.00	5.00	5.00	5.50	6.00	6.50
Face Height, Toe (mm)	54.70	55.10	55.40	55.70	56.10	57.40	58.40	59.2
Face Height, PAR (mm)	31.30	31.60	31.90	32.30	32.60	34.20	35.50	36.70
Sole Width, Toe (mm)	22.50	22.70	22.80	22.90	23.20	23.10	23.20	23.40
Sole Width, Heel (mm)	14.80	15.20	15.40	16.10	16.90	17.10	17.80	18.10
Leading Edge Height (mm)	2.00	2.30	2.40	2.50	2.90	3.60	4.10	5.20
First Scoreline Height (mm)	5.90	5.90	5.70	5.40	5.20	5.10	5.00	4.70
Number of Scorelines	13.00	13.00	13.00	13.00	14.00	14.00	15.00	15.00
Scoreline Length (mm)	54.10	53.90	53.70	53.50	53.30	51.70	50.80	50.30
Scoreline End to Toe (mm)	18.30	18.30	18.30	18.30	18.30	18.80	18.70	18.80
Sole Radius (mm)	216.00	216.00	216.00	216.00	216.00	216.00	216.00	216.00
MASS PROPERTIES								
Assembly, Mass, CAD (g)	240.7	246.2	253.8	259.3	265.9	275.0	283.2	290.0
Z-up (mm)	17.65	17.61	17.96	17.93	18.02	18.53	18.54	19.28
Cgx (mm)	1.9	2.0	1.5	0.8	0.5	2.4	3.5	3.6
D1 (mm)	7.3	7.7	8.0	8.6	9.5	10.2	10.9	12.6
D2 (mm)	36.3	36.1	36.7	37.2	37.4	35.6	34.7	35.1
D3 (mm)	62.8	63.9	63.5	63.5	63.2	63.3	64.6	65.7
Ixx (kg-mm ²)	47.0	49.0	53.0	50.0	55.0	60.0	65.0	77.0
Izz (kg-mm ²)	221.0	223.0	233.0	239.0	255.0	247.0	257.0	271.0

FIG. 102

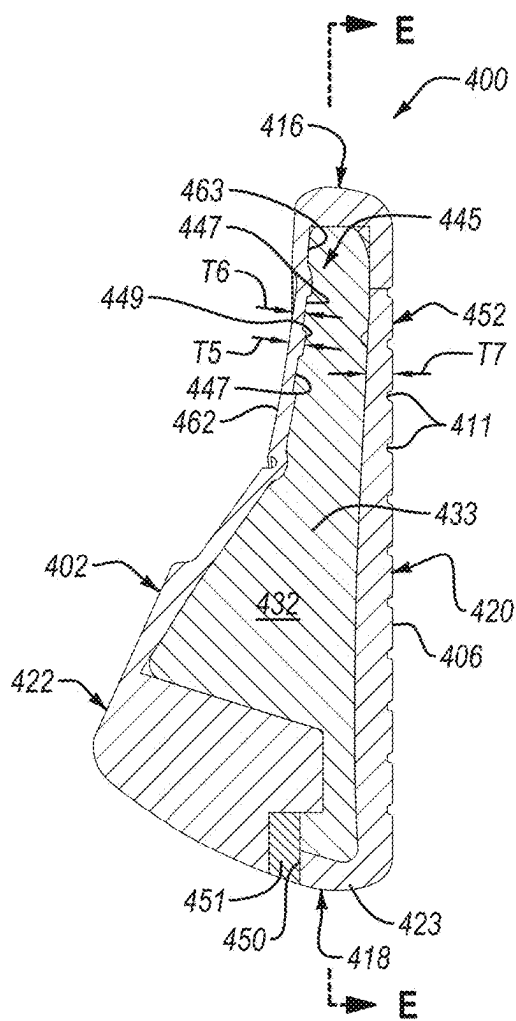


FIG. 103

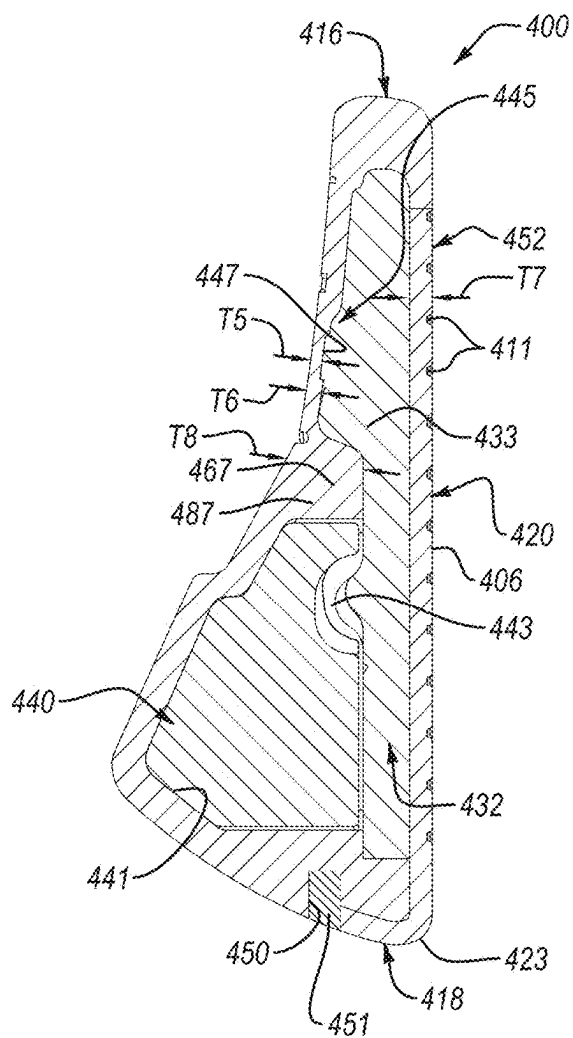


FIG. 104

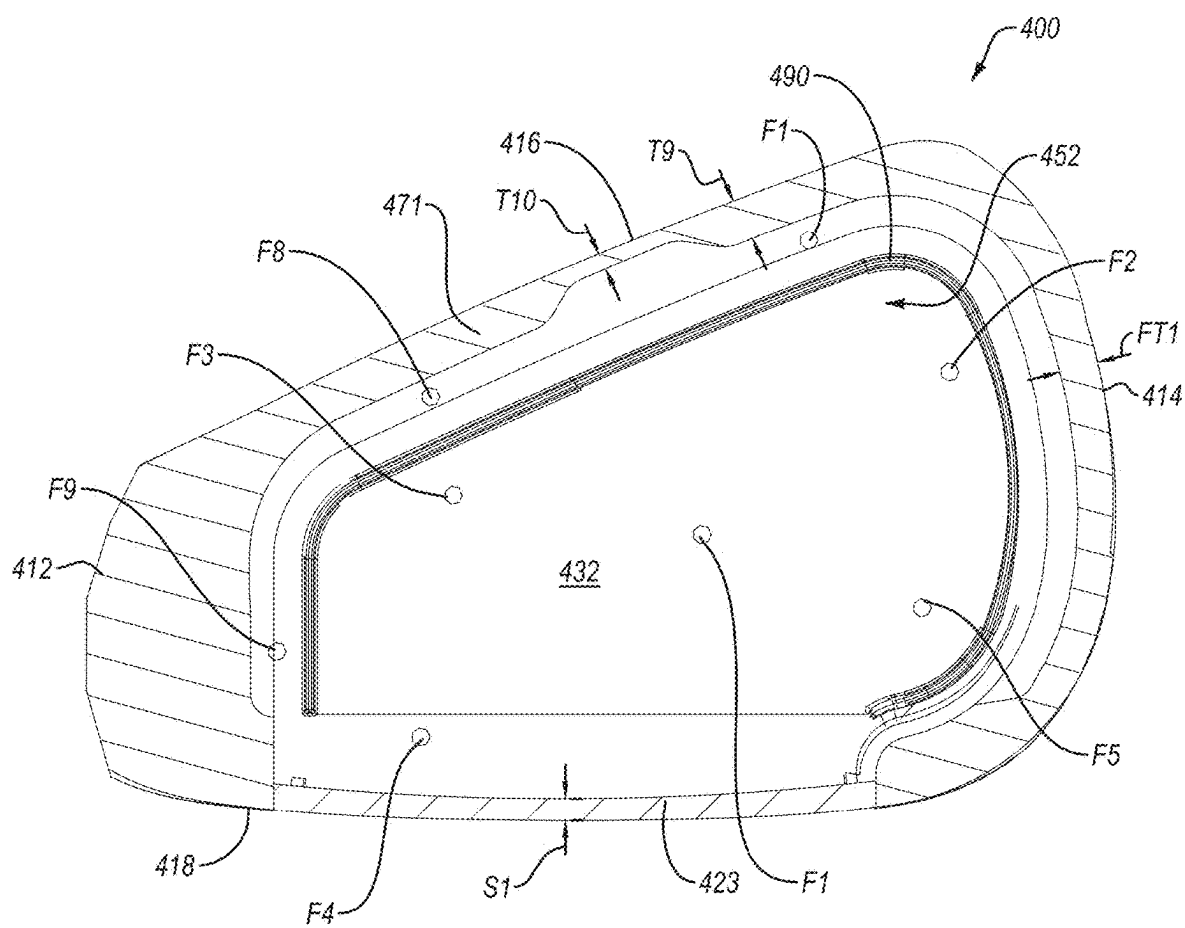


FIG. 105

900

HEAD	F1	F2	F3	F4	F5	F6	F7	F8	F9	FT1	T9	T10	T5	T6	S1
TOL	± 0.15									± 0.15	± 0.25	± 0.15	± 0.15	± 0.15	± 0.2
2	2.50	2.10	1.80	1.90	2.50		1.65			2.95	3.40	1.20	1.00	0.75	2.30
400A	2.34	1.62	1.56	1.60	3.10		1.60			1.20	3.50	1.20	1.25	1.00	2.37
400B	2.33	1.62	1.56	1.60	3.10		1.60			1.20	3.50	1.20	1.25	1.00	2.39
400C	2.31	1.62	1.56	1.60	3.10		1.60			3.20	3.50	1.20	1.00	0.75	2.49
400D	2.89	1.68	1.75	1.76	1.91		1.50			1.20	N/A	1.20	1.00	0.75	2.65
400	2.83	1.57	1.64	1.66	1.72		1.50			3.20	3.50	1.20	1.00	0.75	2.10
										TBD			1.00	0.75	
400F	3.20	1.88	1.85	1.92	1.99		1.70			4.00	N/A	1.70	1.00	0.75	2.80
9										5.10	N/A	1.70	1.00	N/A	2.83
P	3.41	2.22	2.15	2.16	3.47		2.40			N/A	N/A	N/A	1.20	N/A	3.56
A										N/A	N/A	N/A	1.00	N/A	2.80

FIG. 106

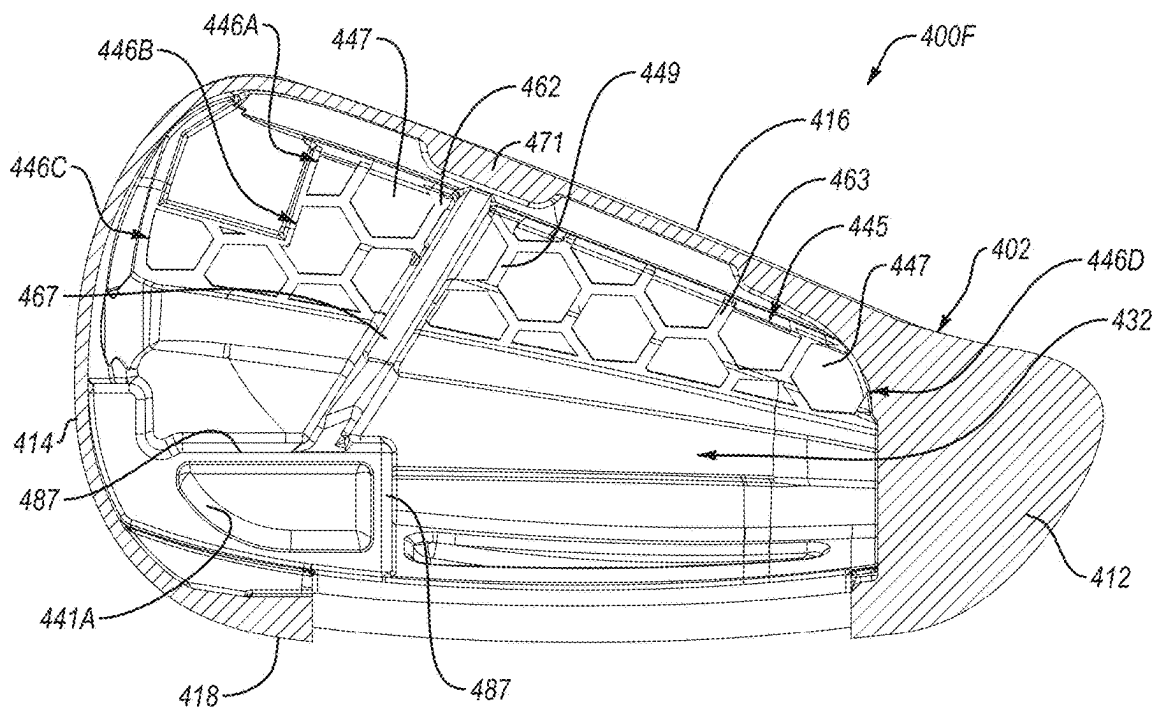


FIG. 107

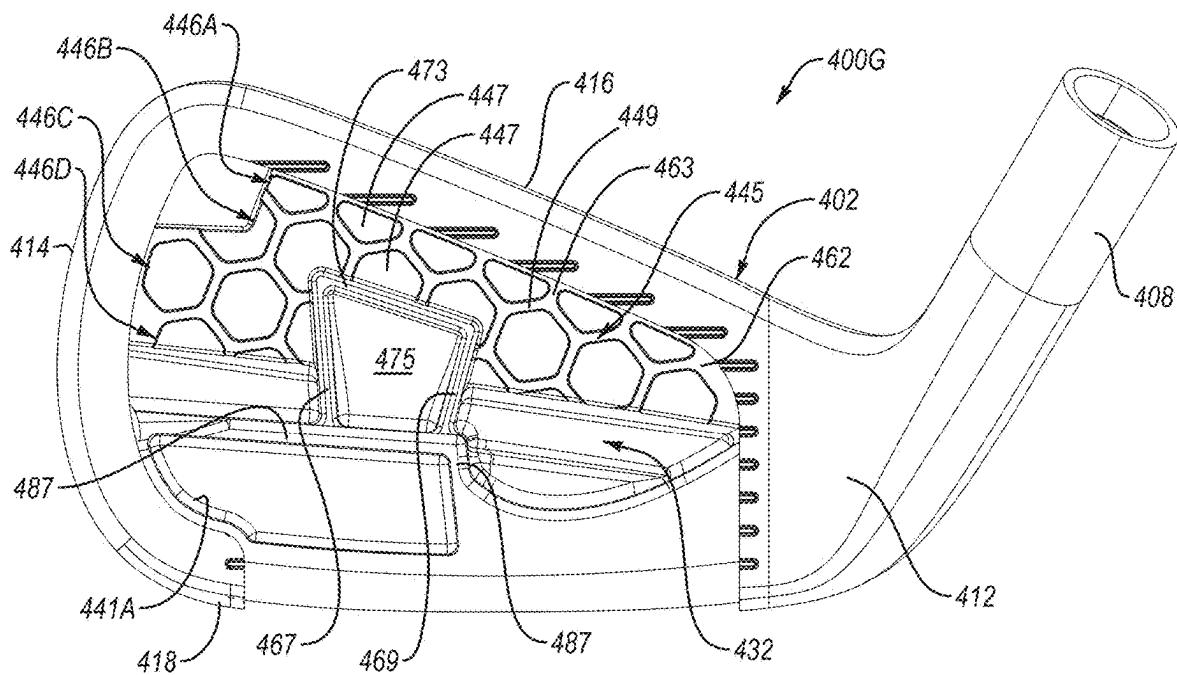
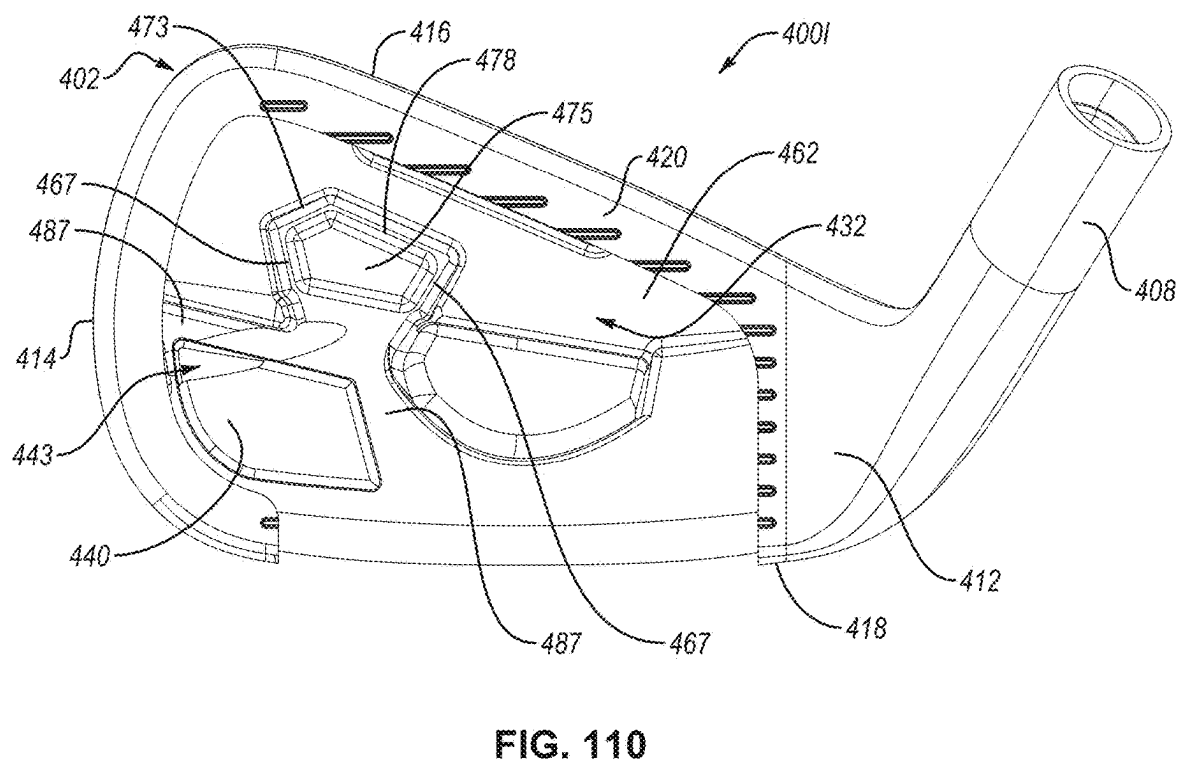
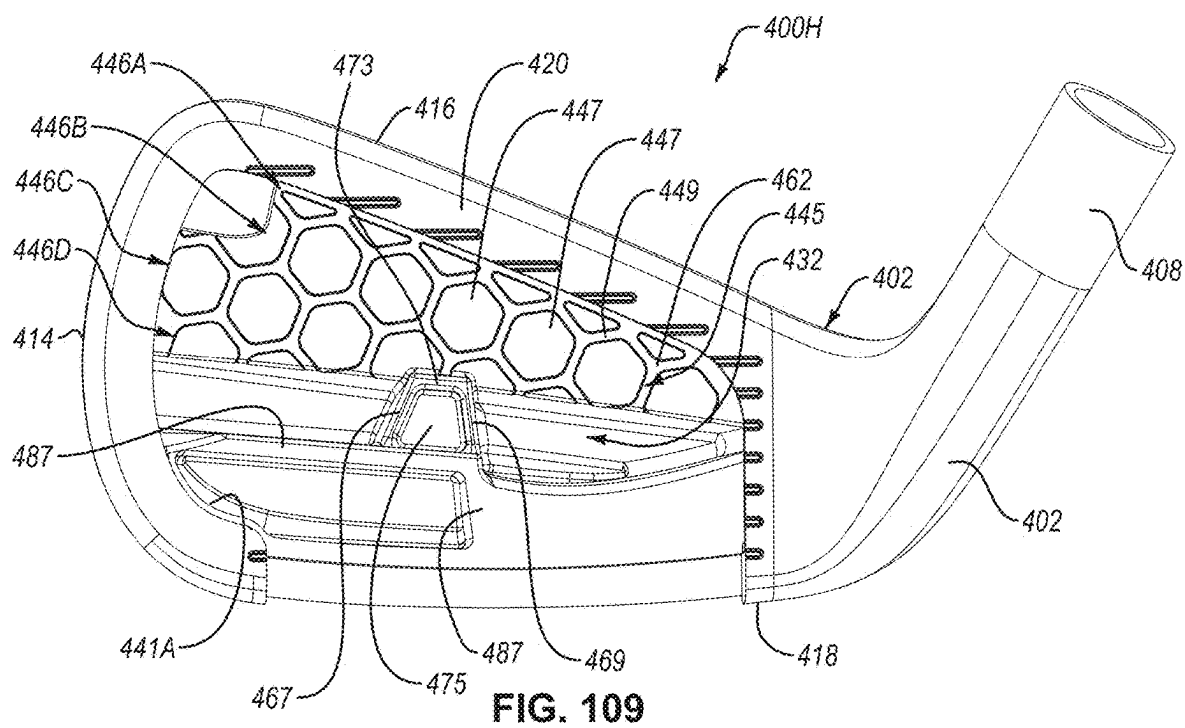


FIG. 108



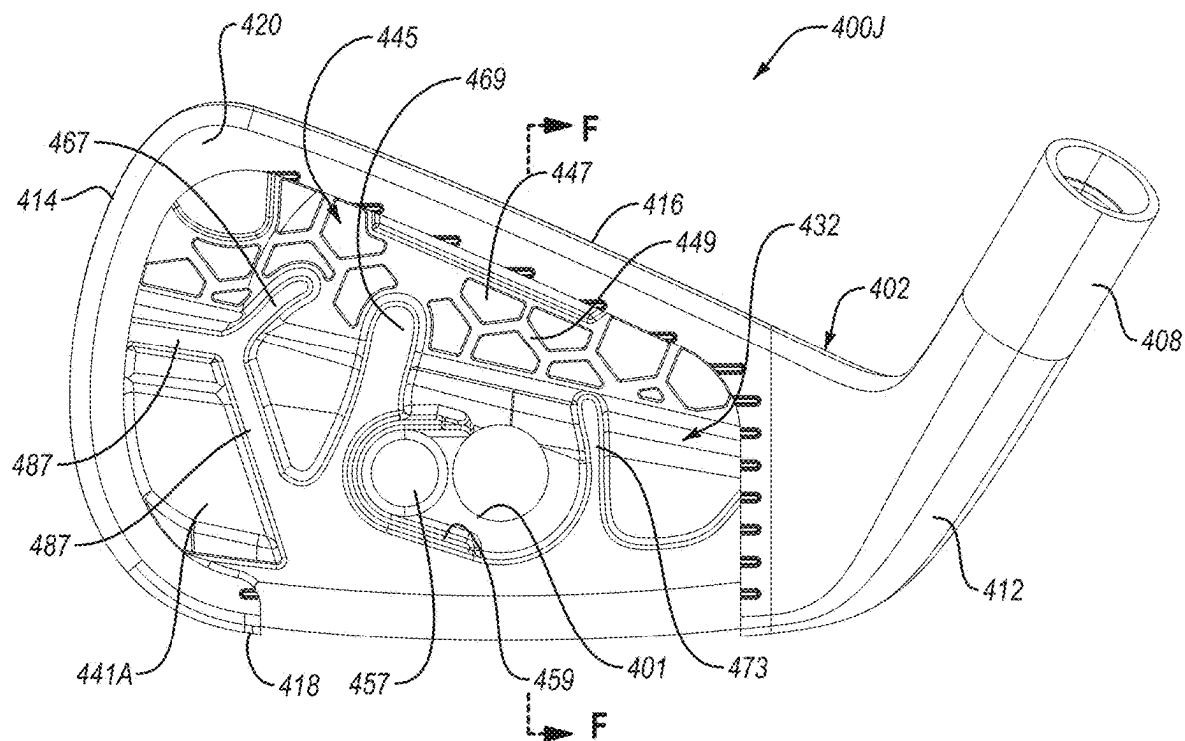


FIG. 111

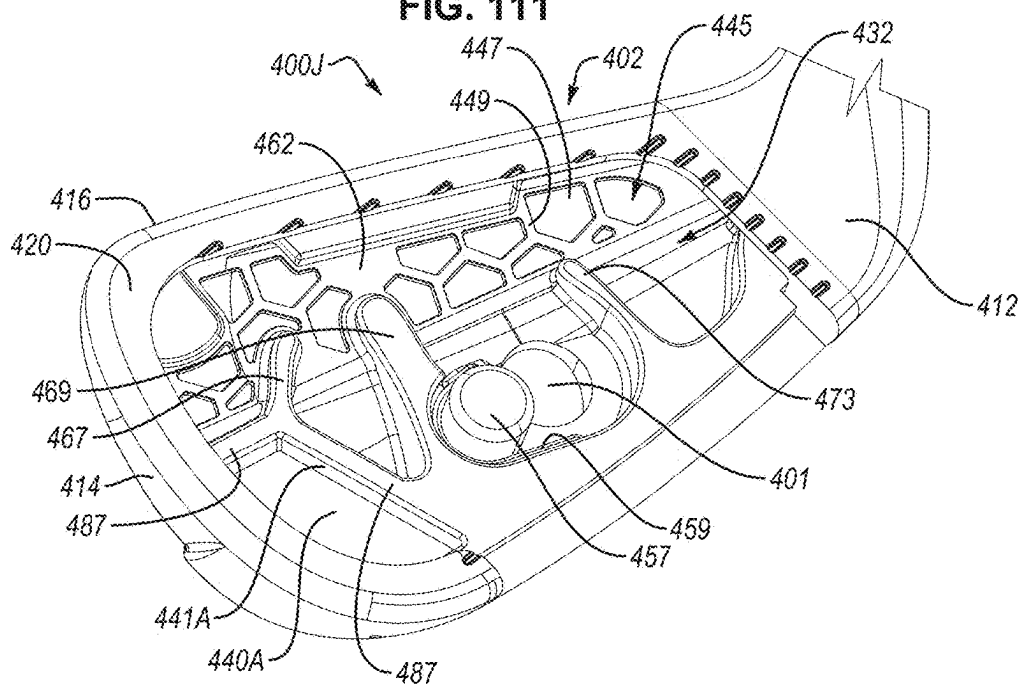


FIG. 112

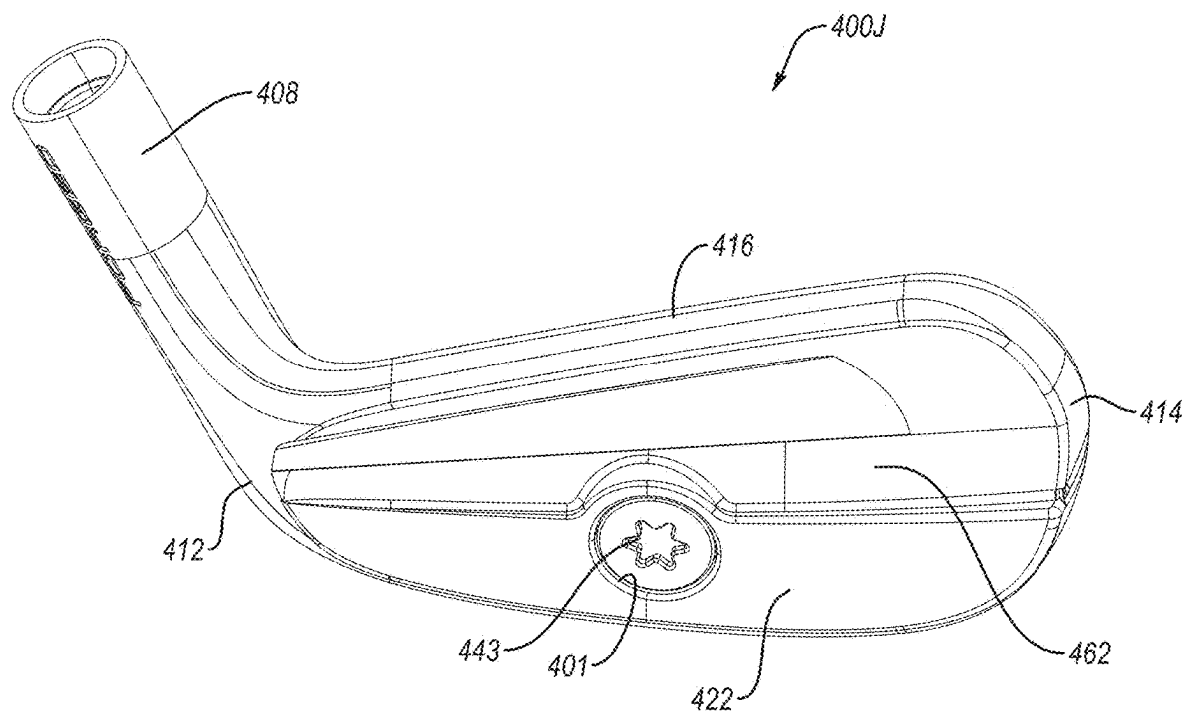


FIG. 113

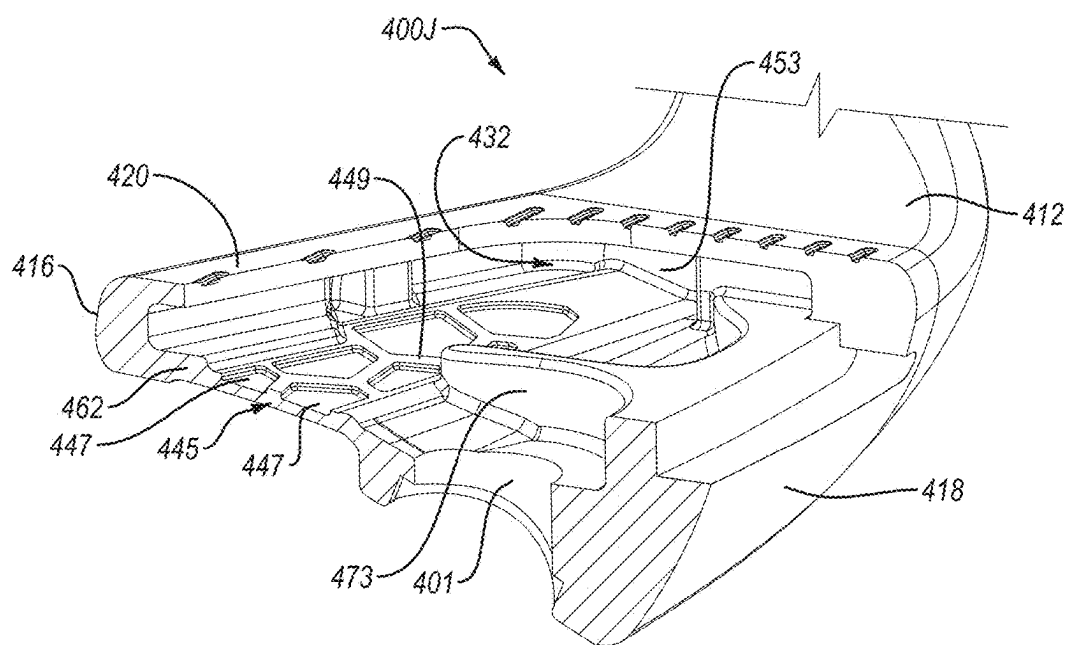


FIG. 114

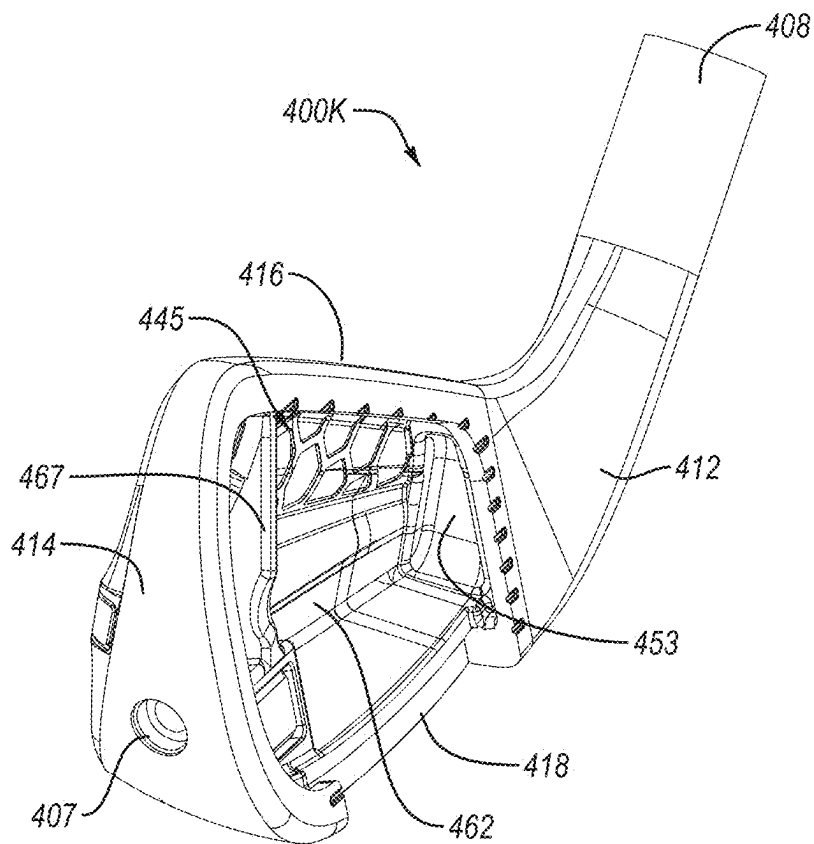


FIG. 115

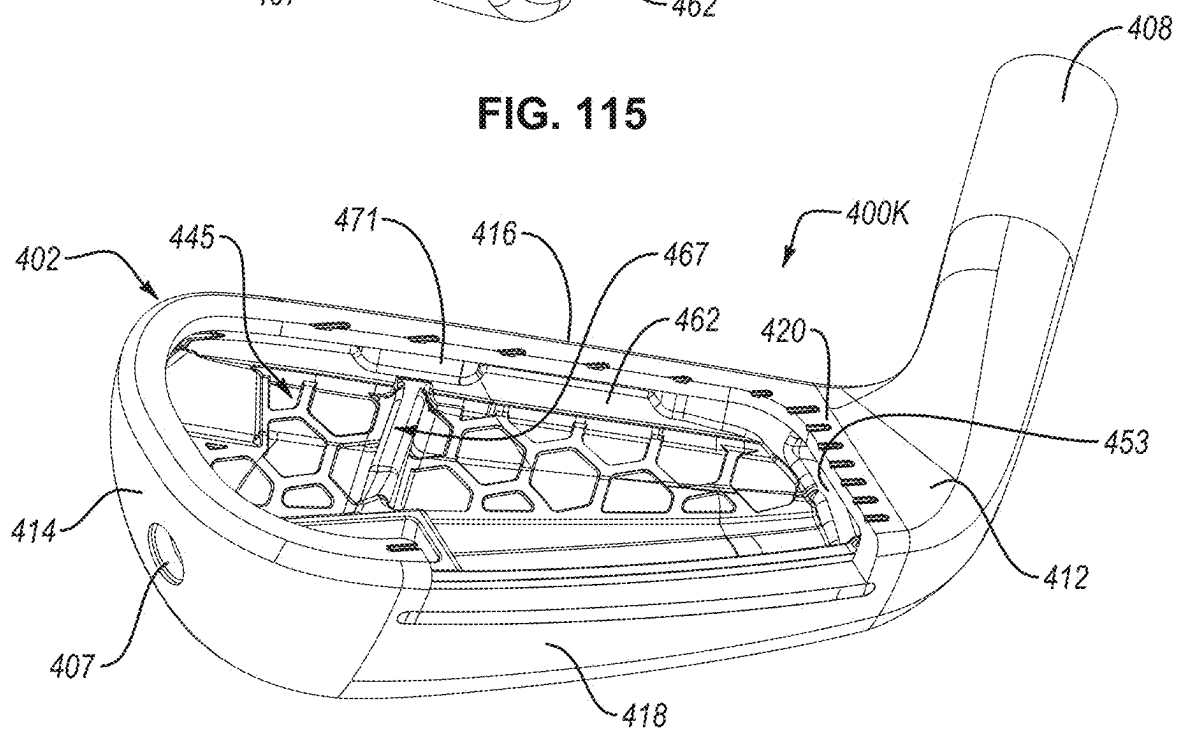


FIG. 116

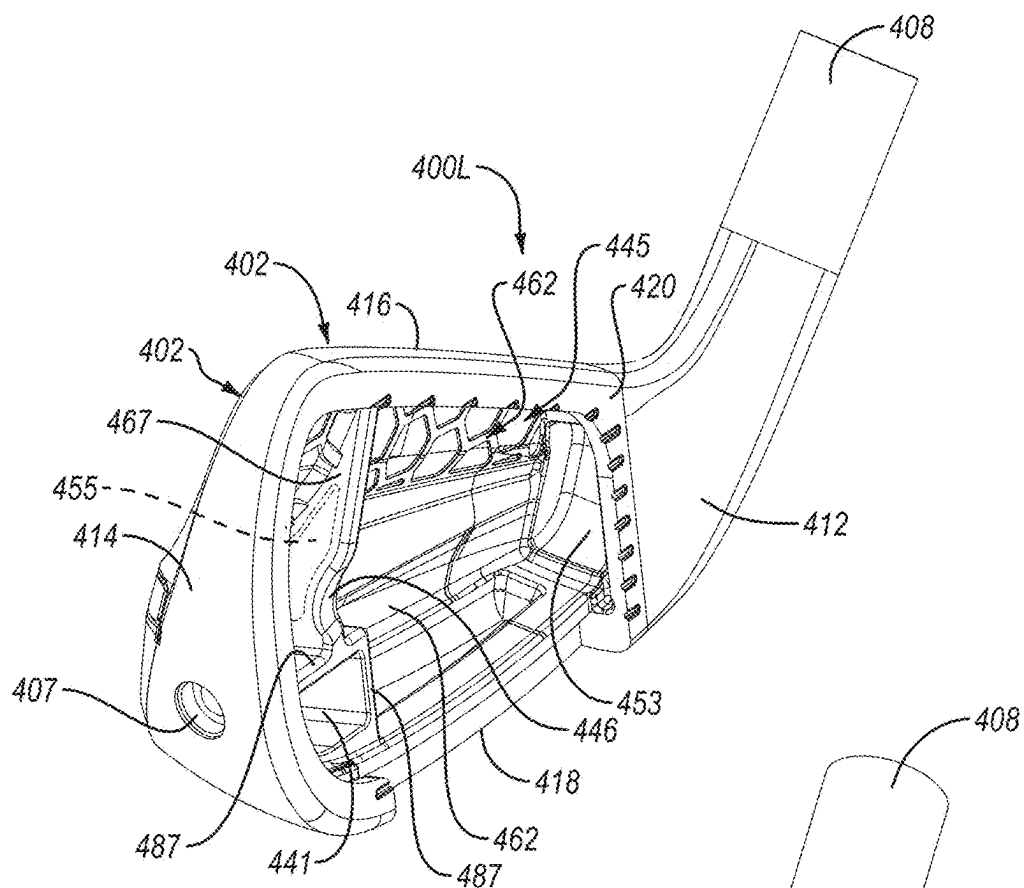


FIG. 117

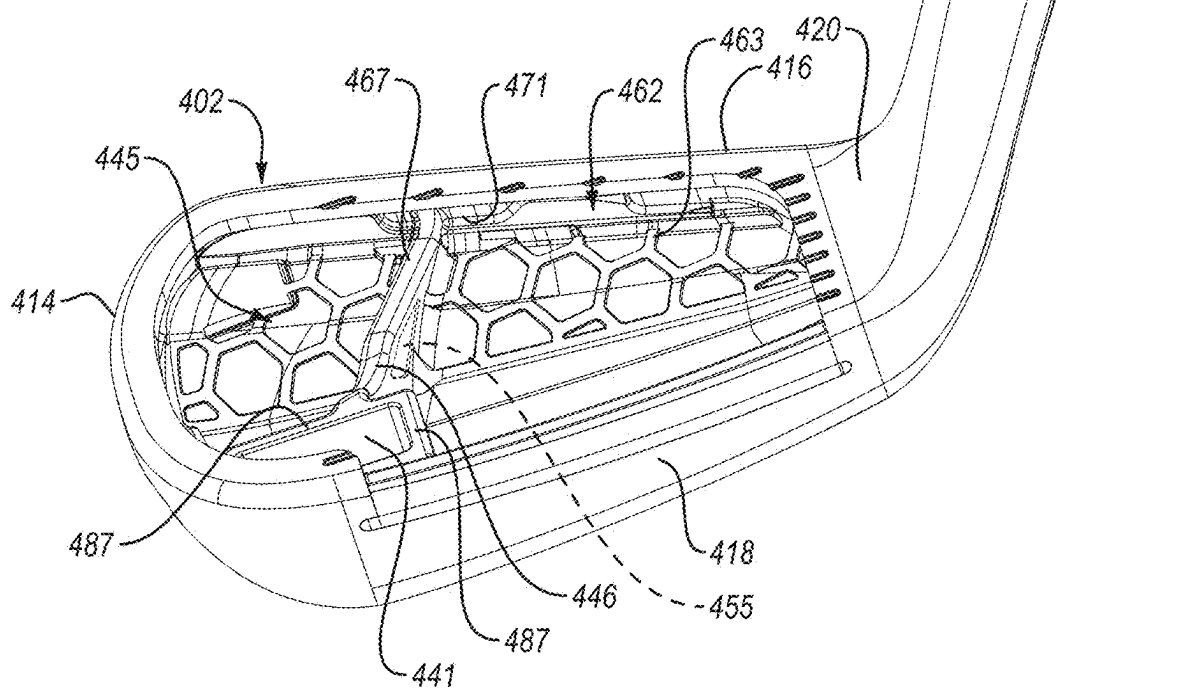


FIG. 118

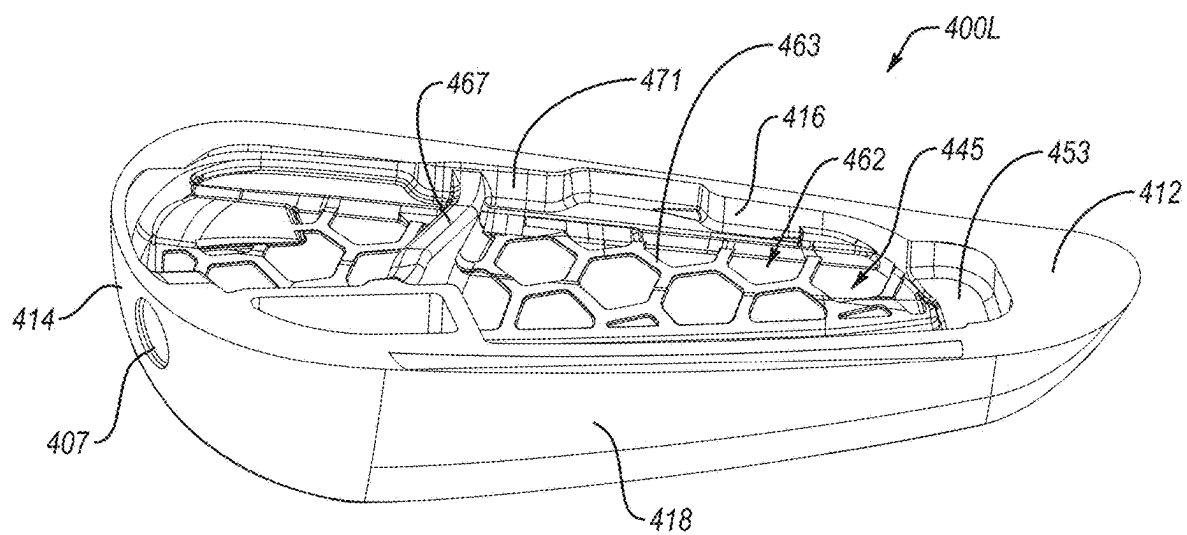


FIG. 119

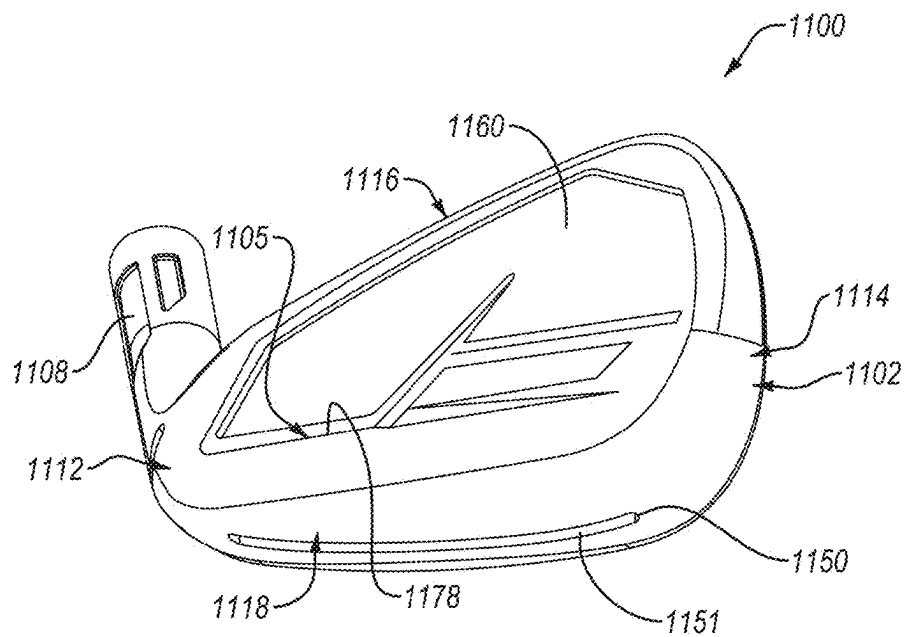


FIG. 120

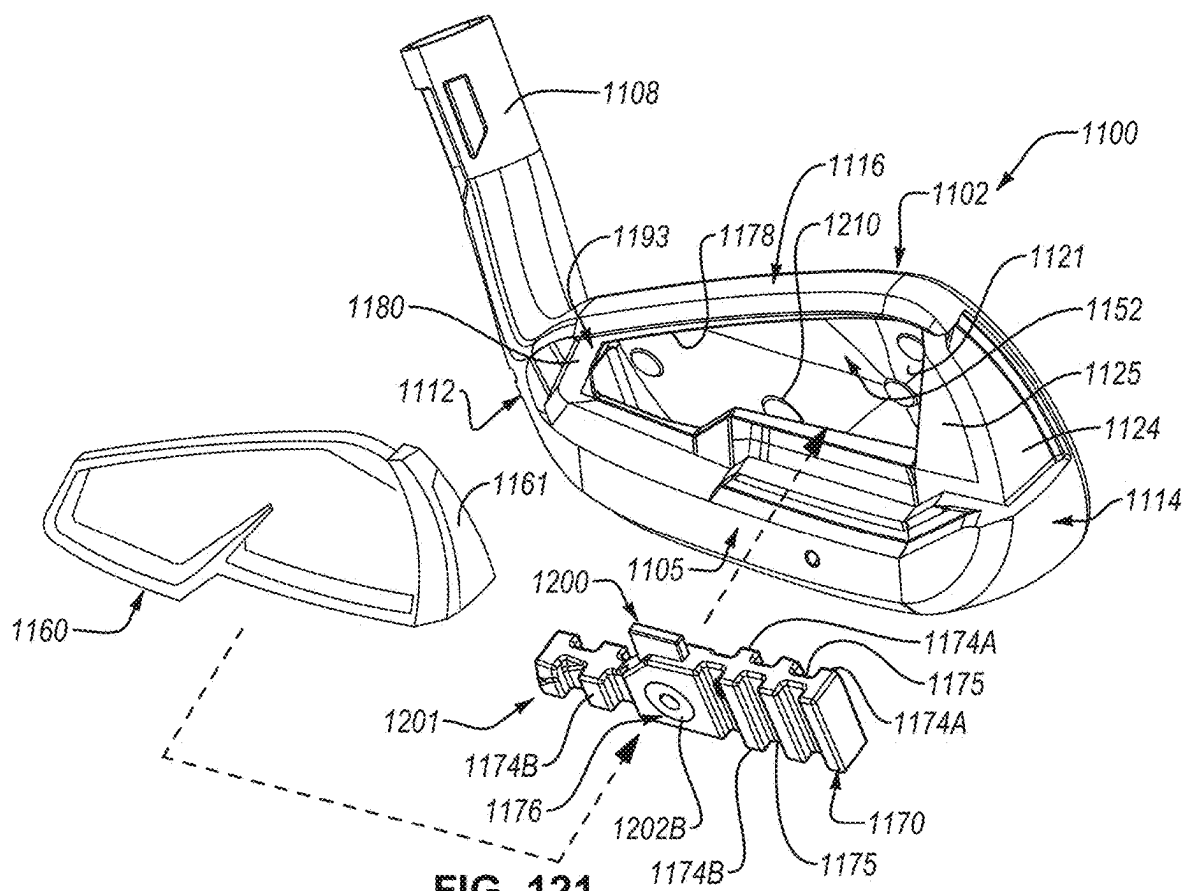
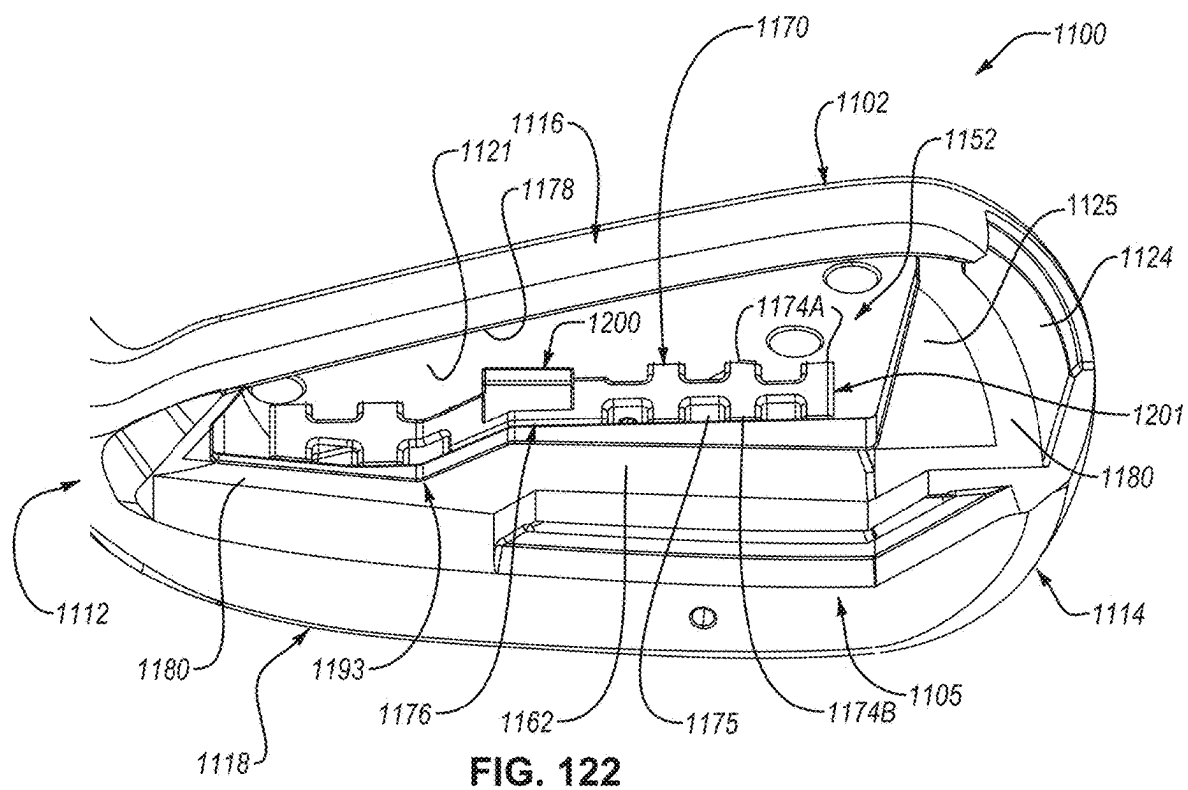
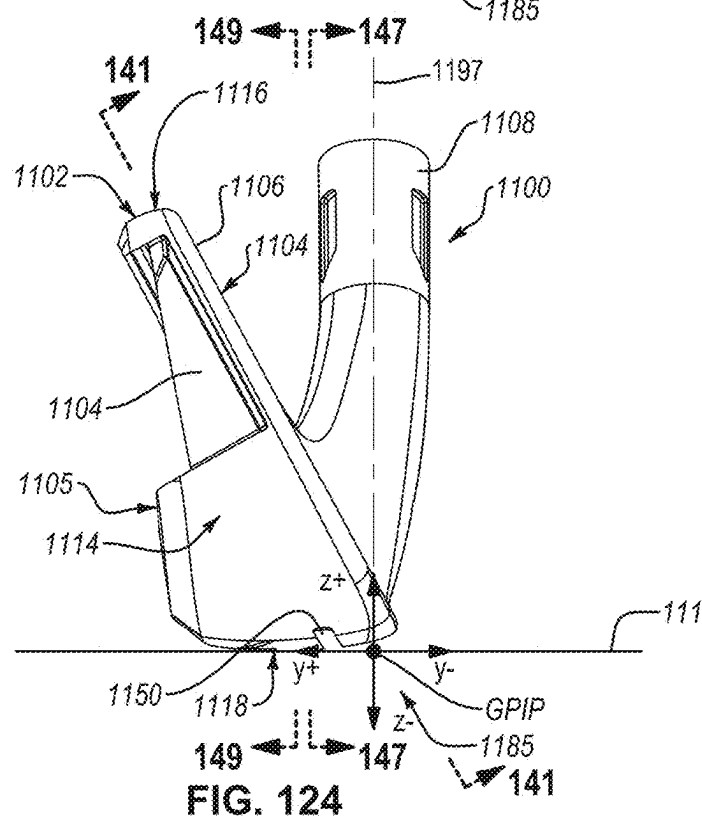
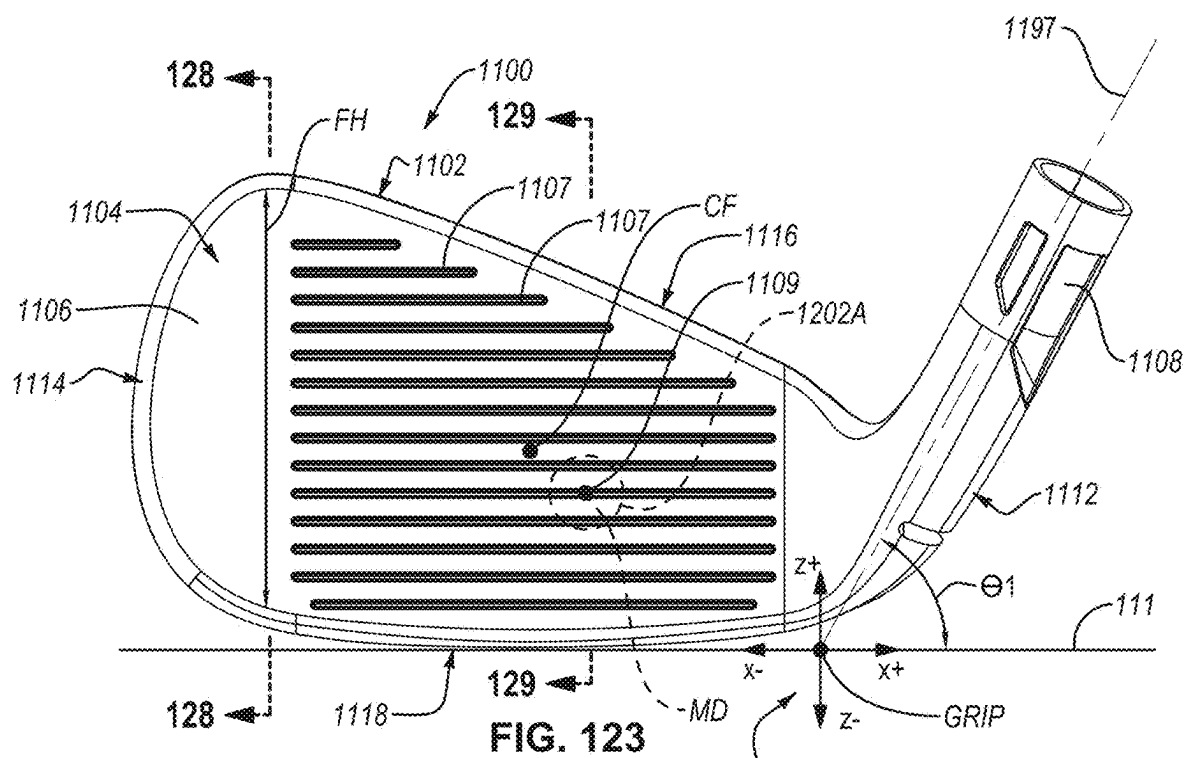


FIG. 121





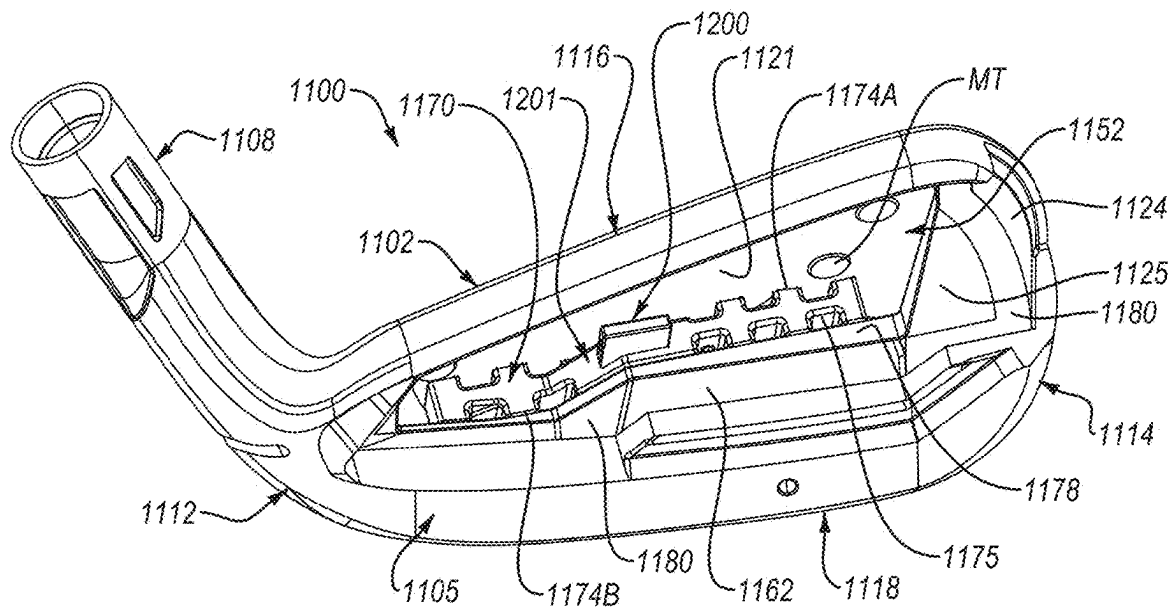


FIG. 125

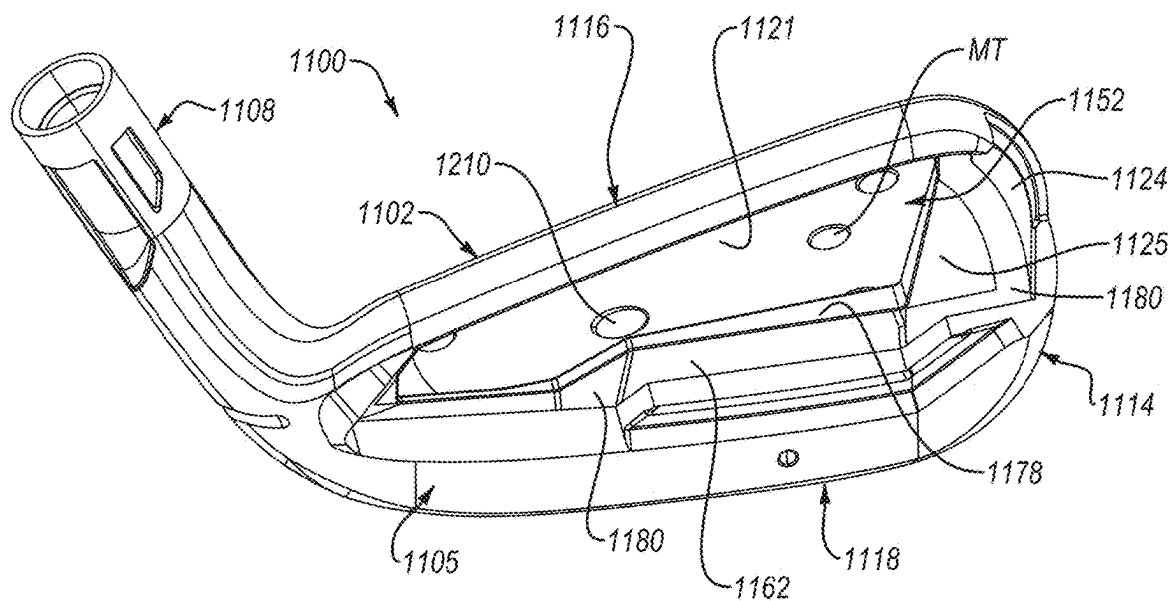


FIG. 126

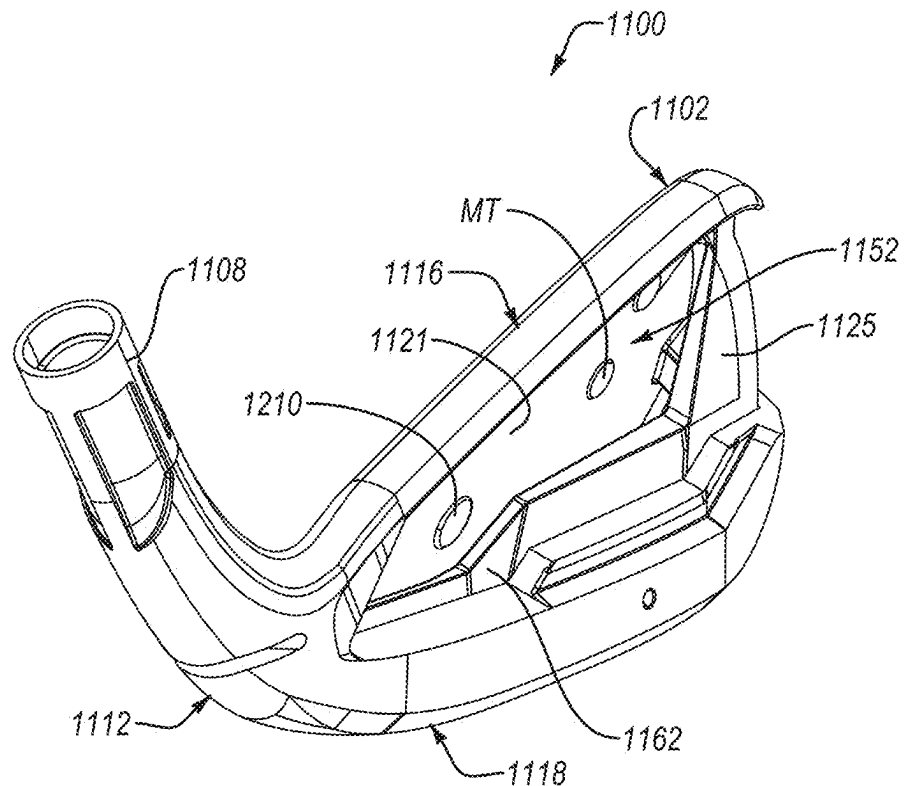


FIG. 127

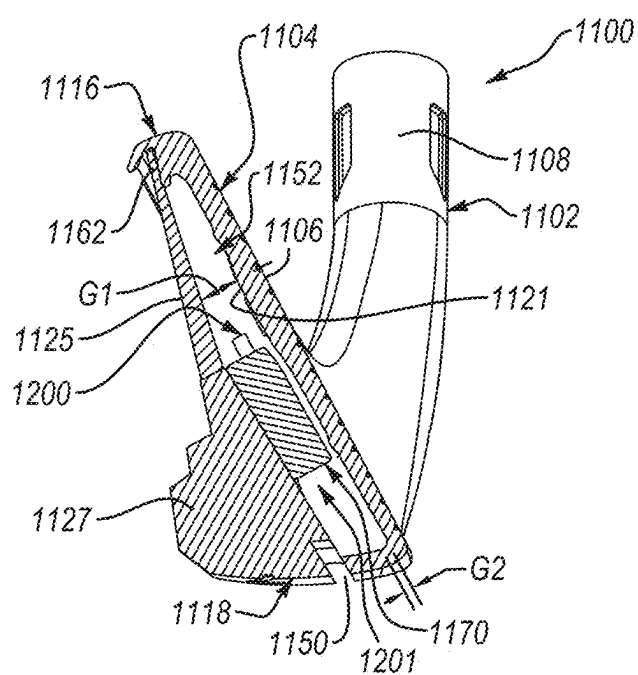
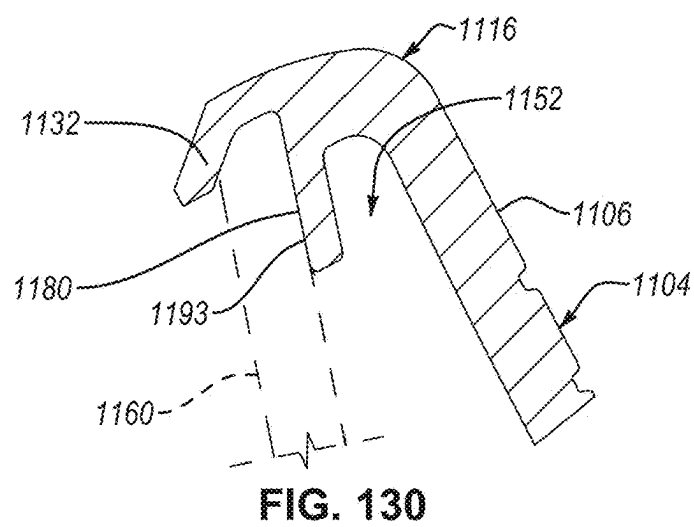
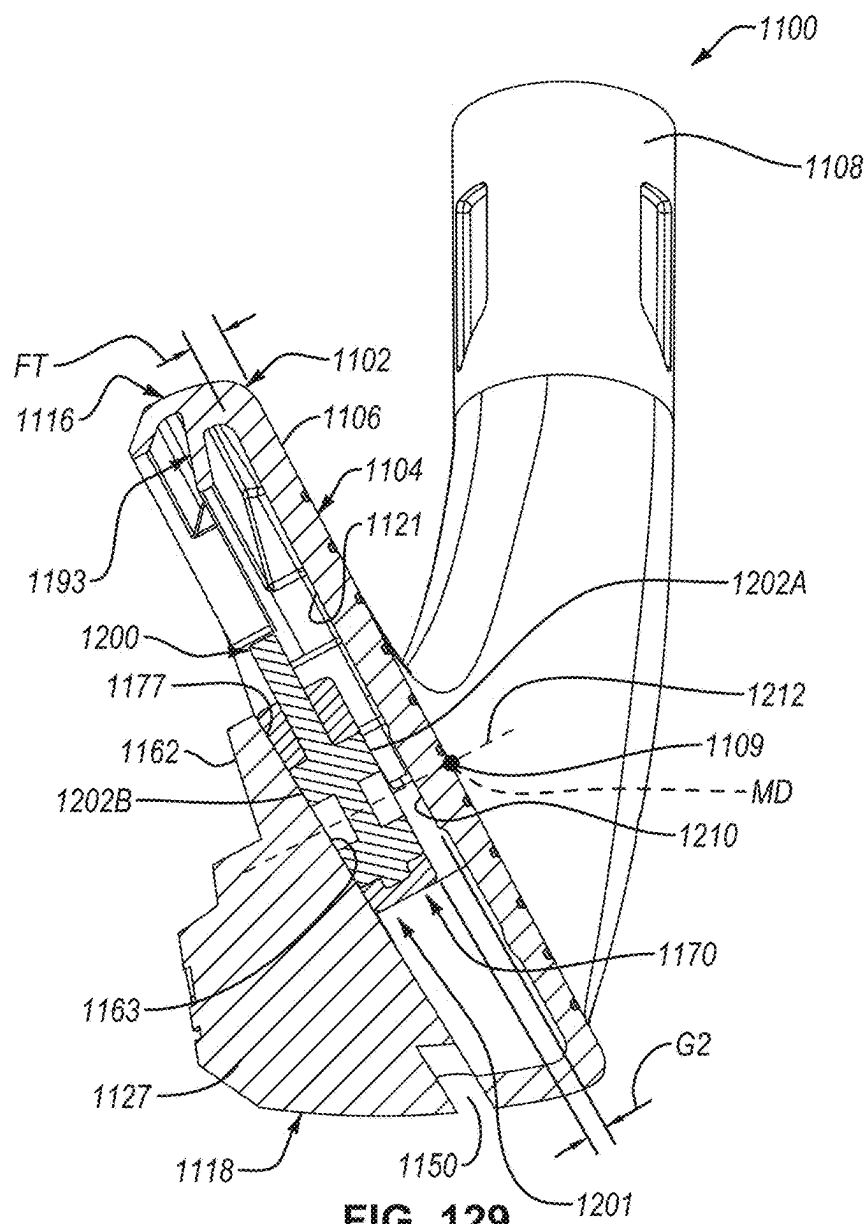


FIG. 128



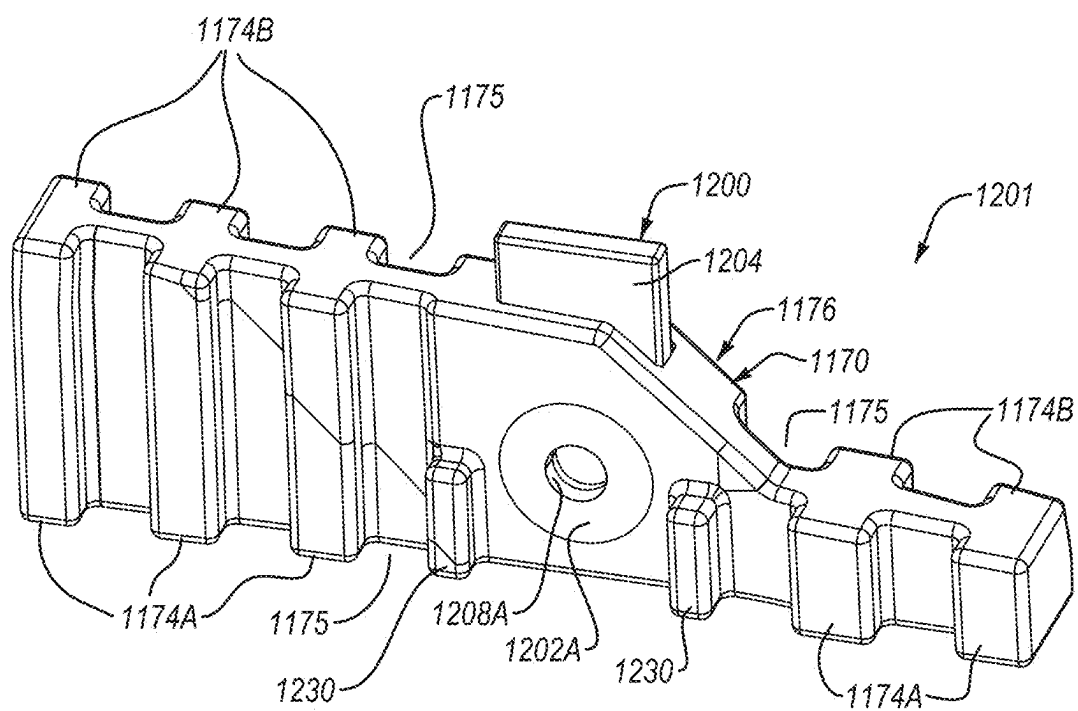


FIG. 131

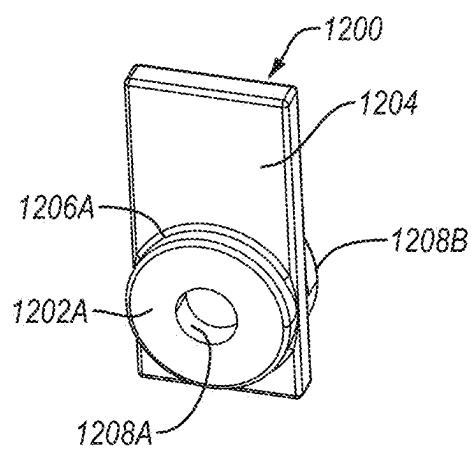
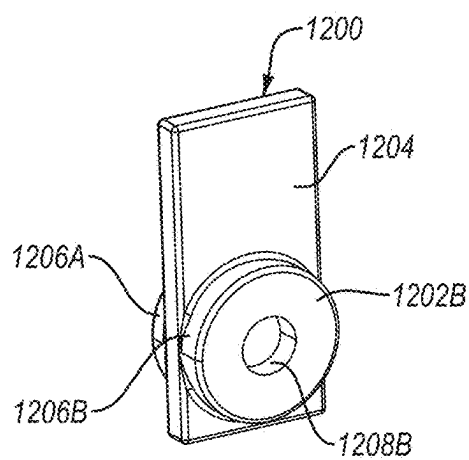
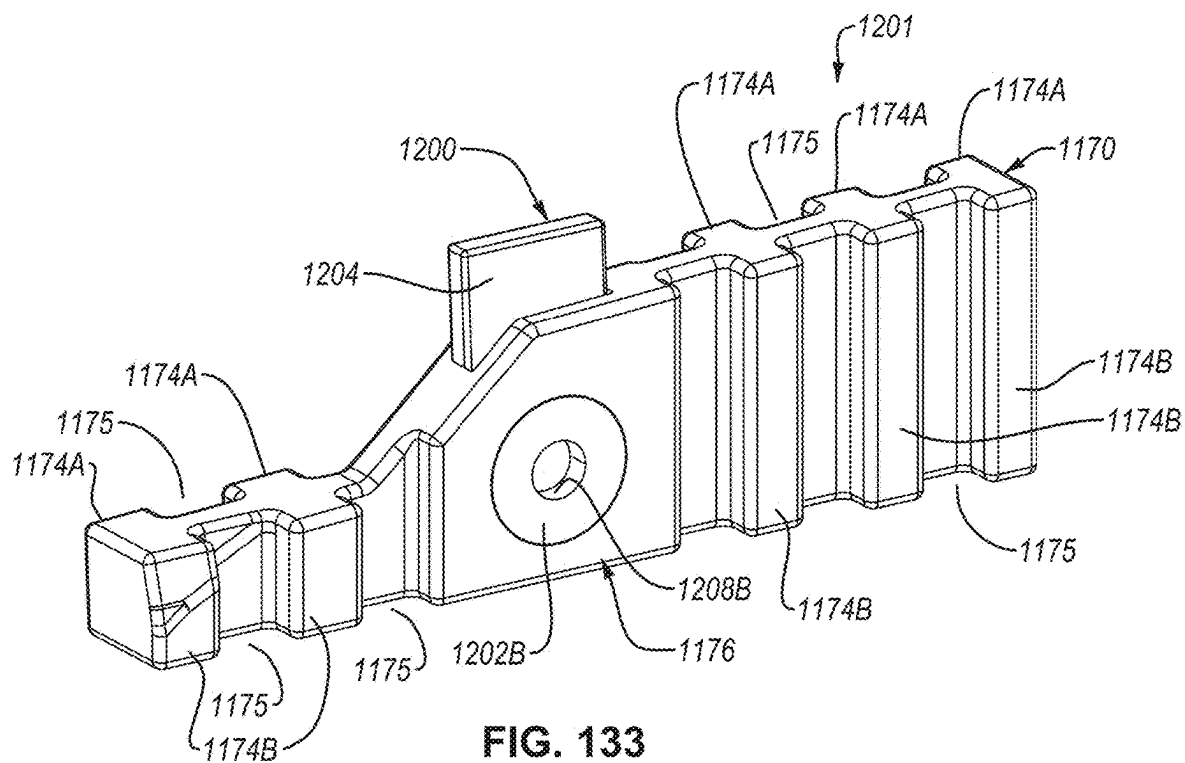
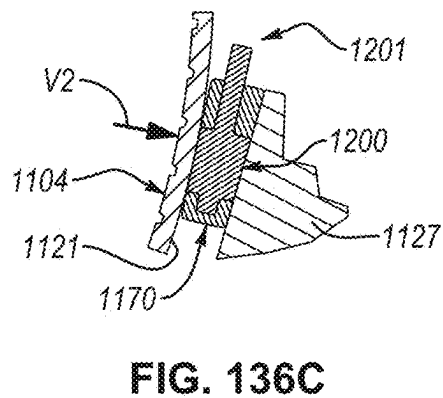
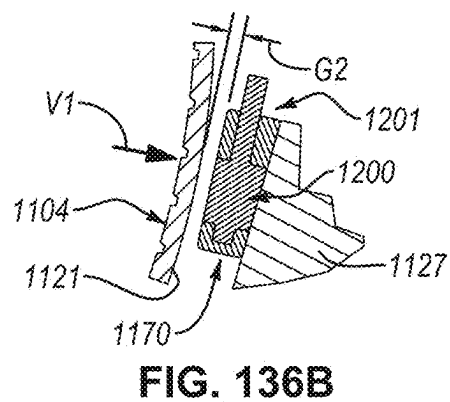
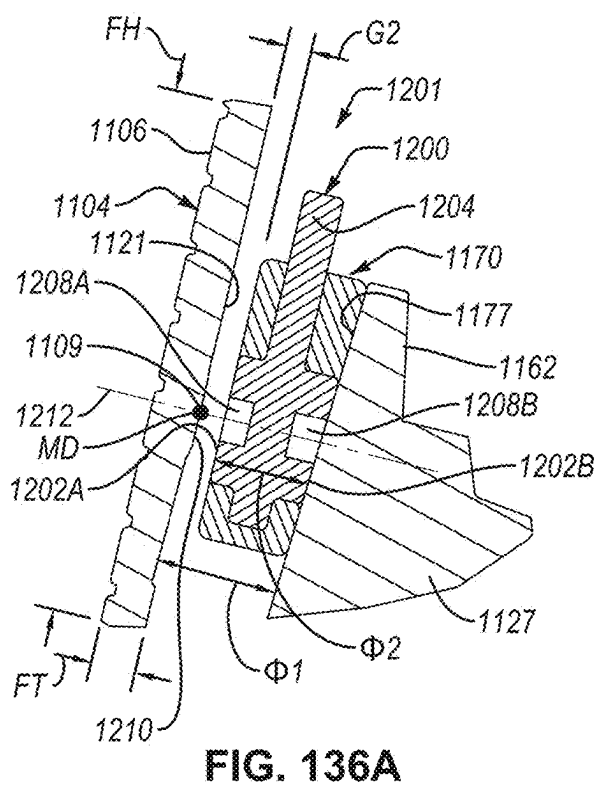
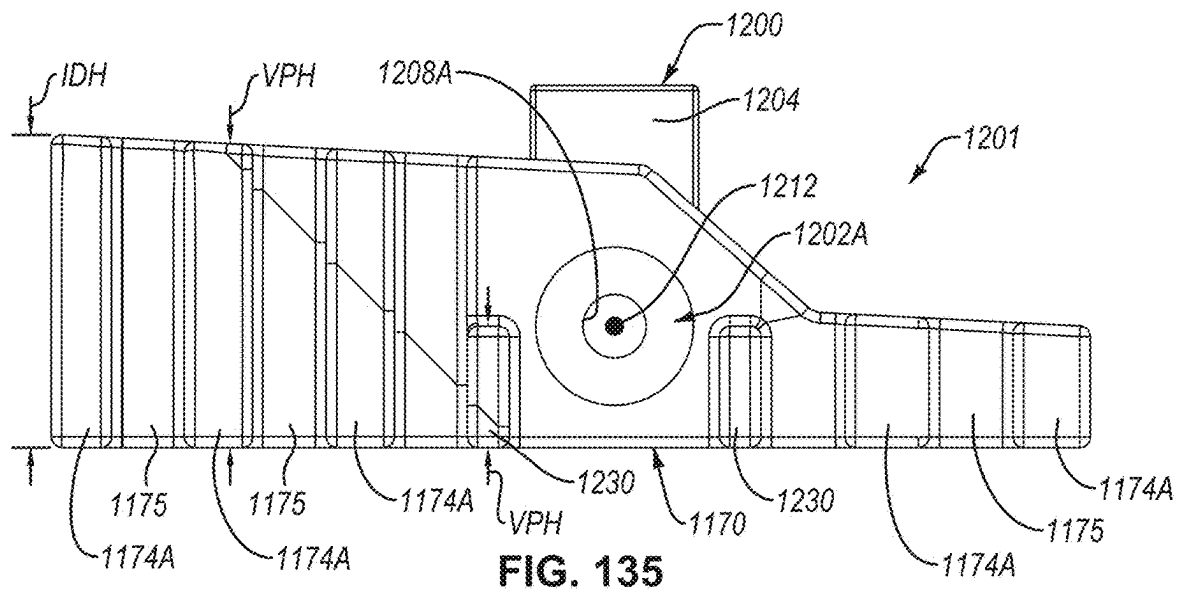
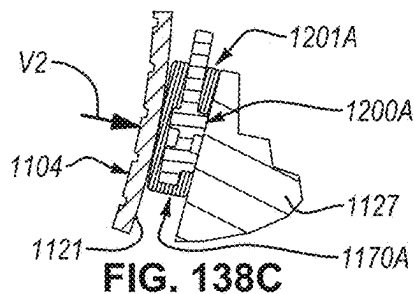
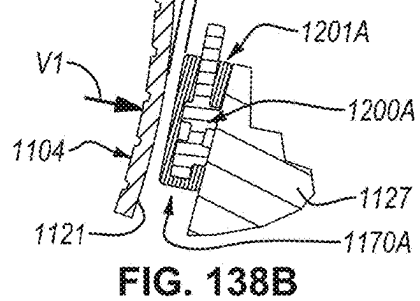
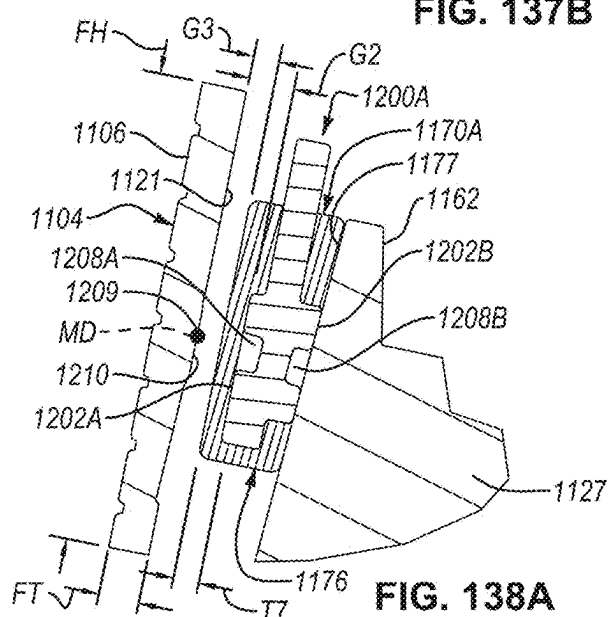
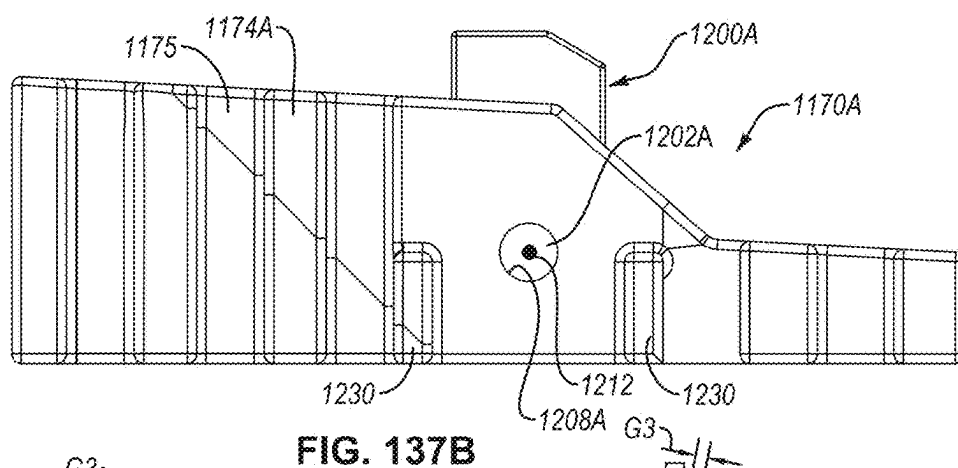
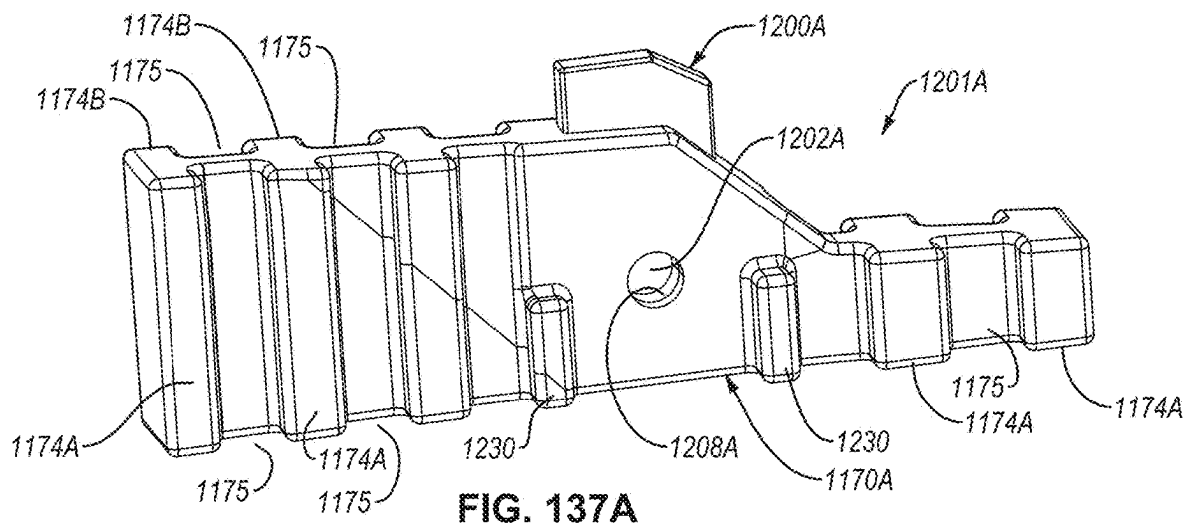


FIG. 132







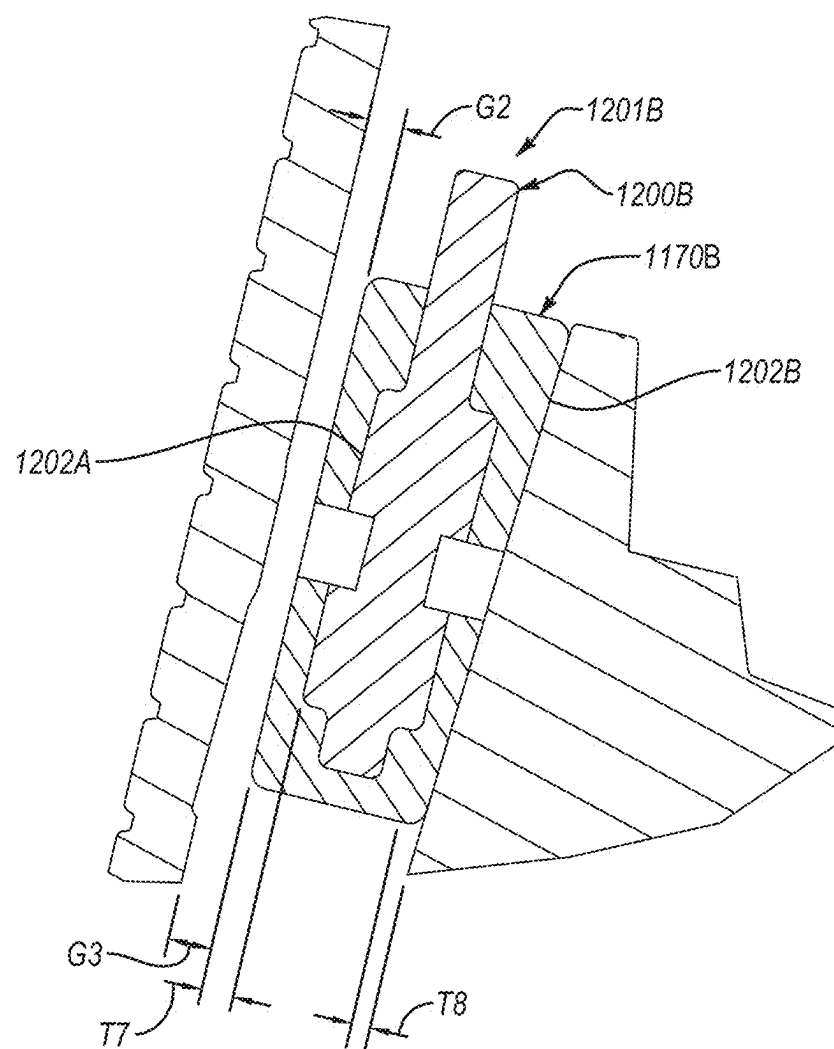


FIG. 138D

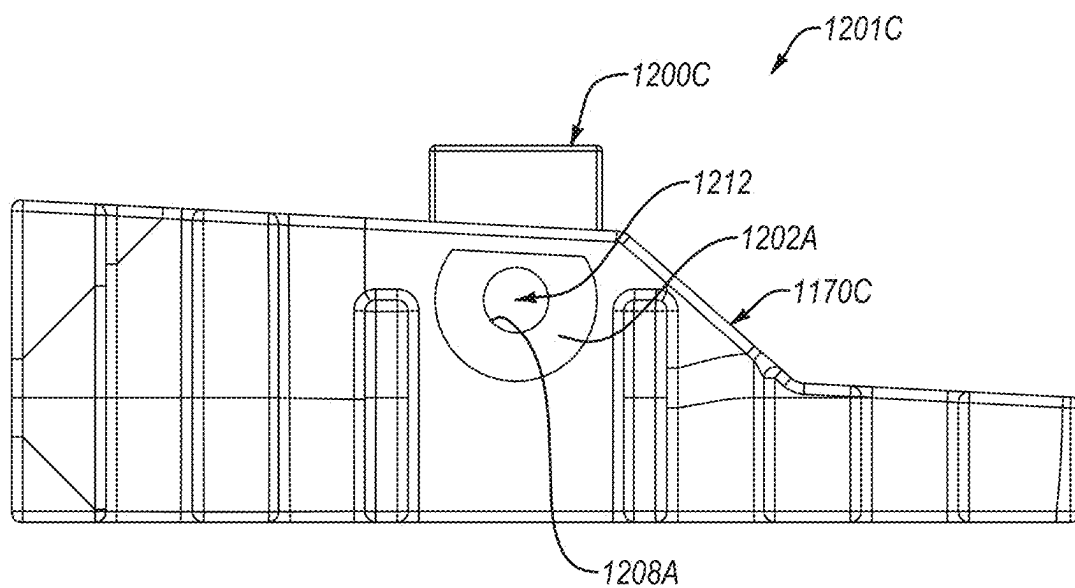


FIG. 139

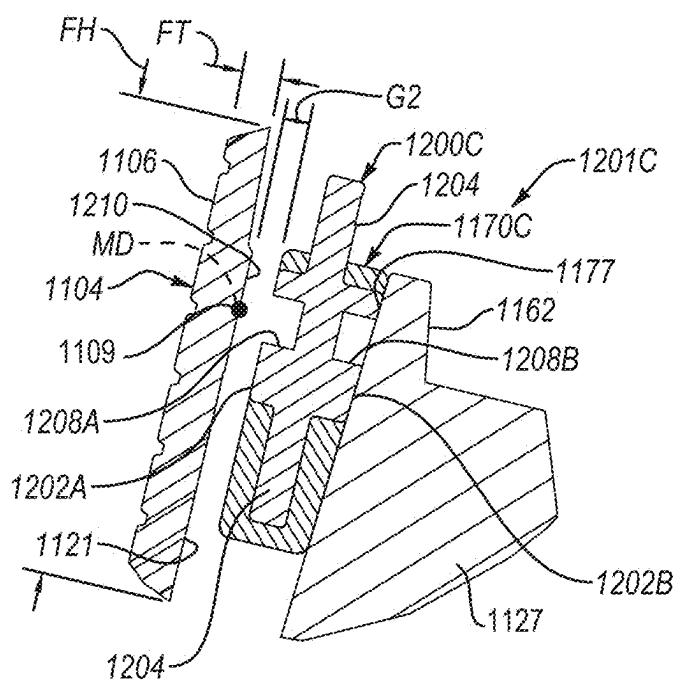


FIG. 140

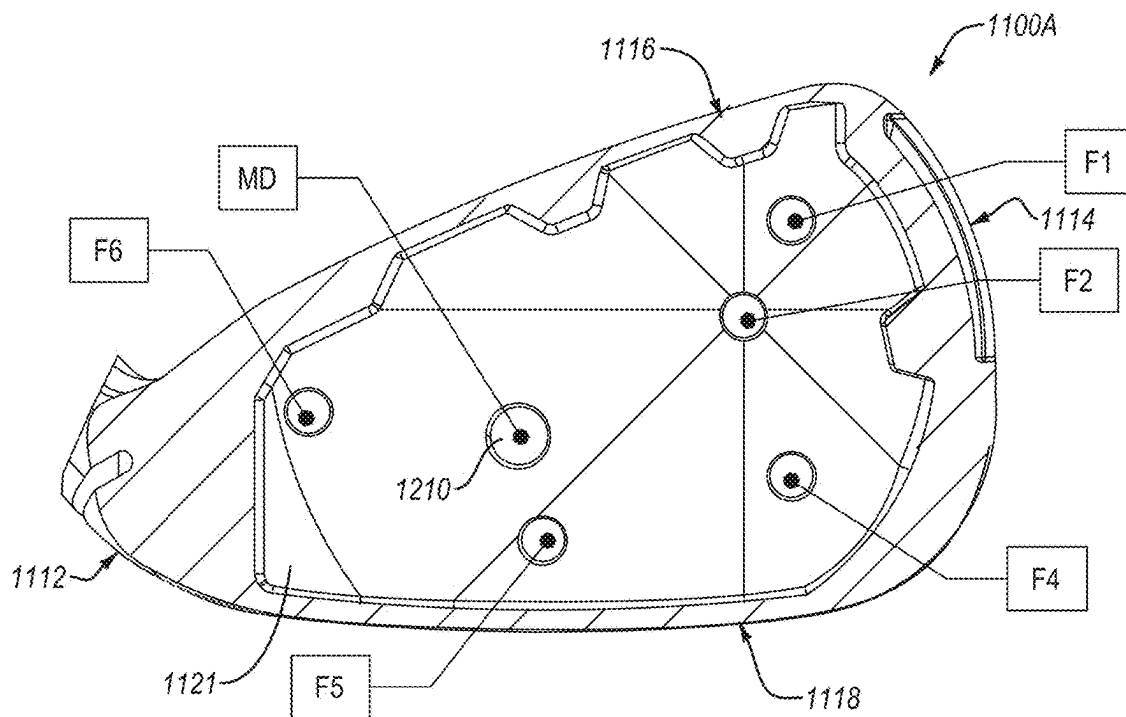


FIG. 141A

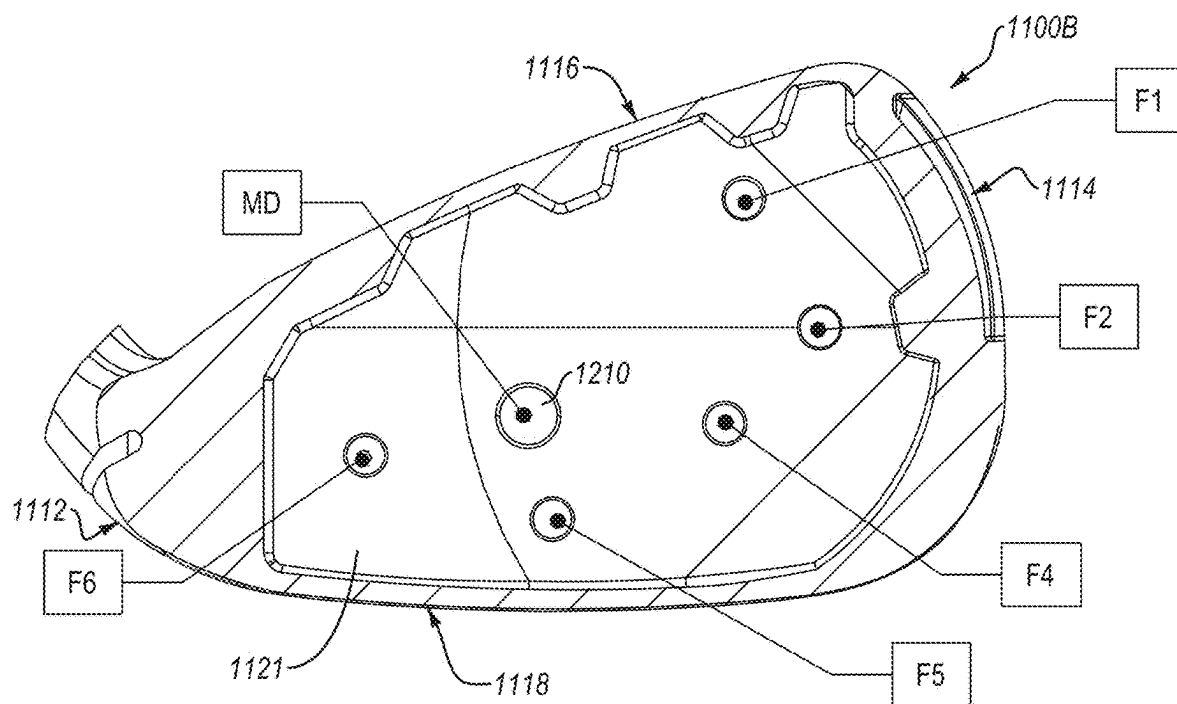


FIG. 141B

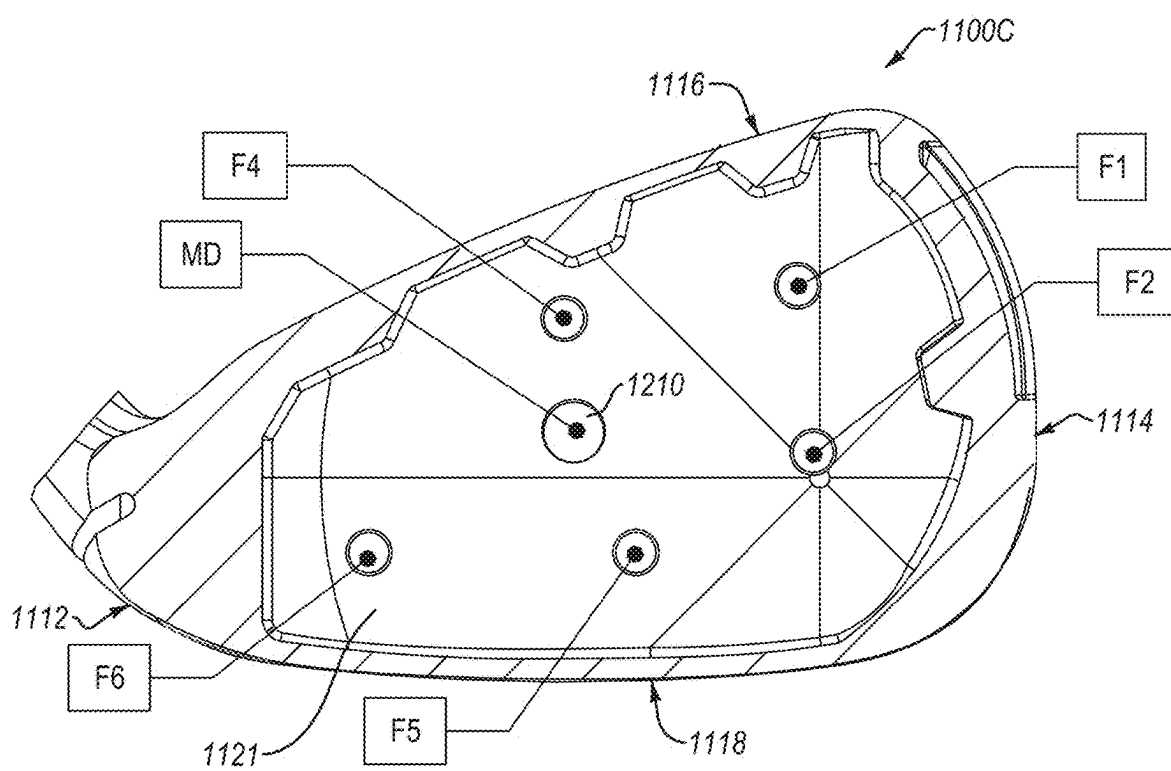
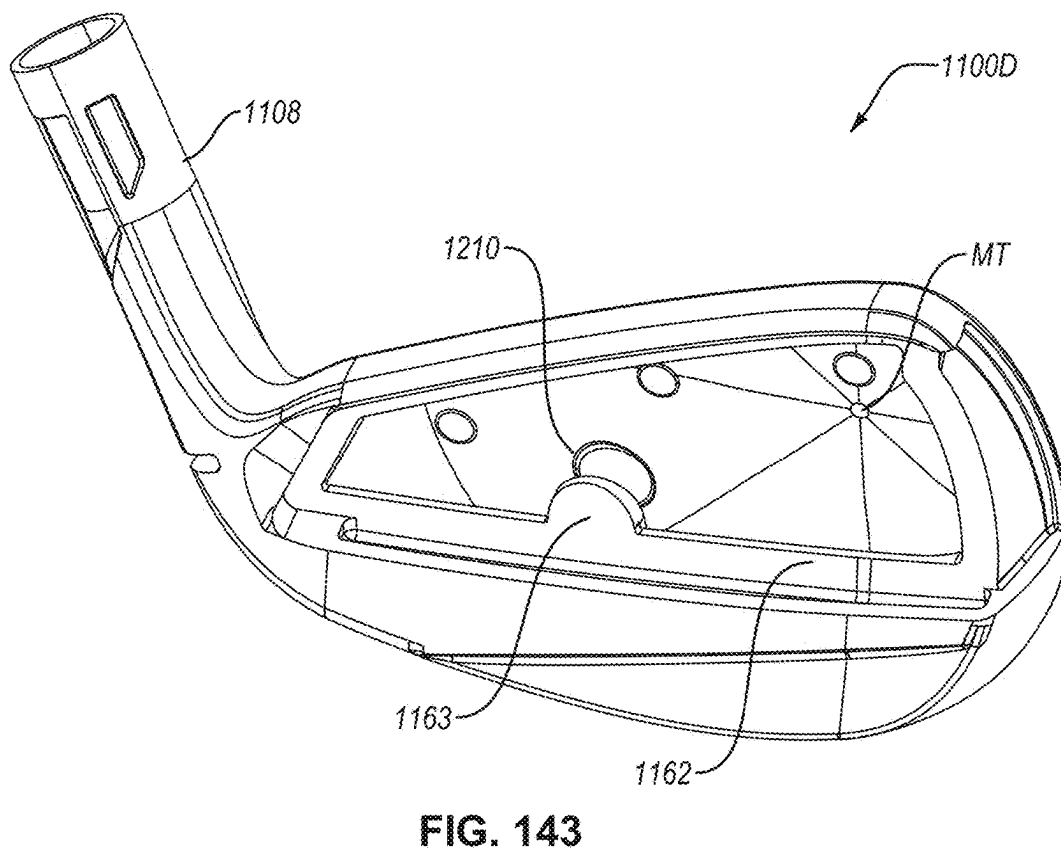
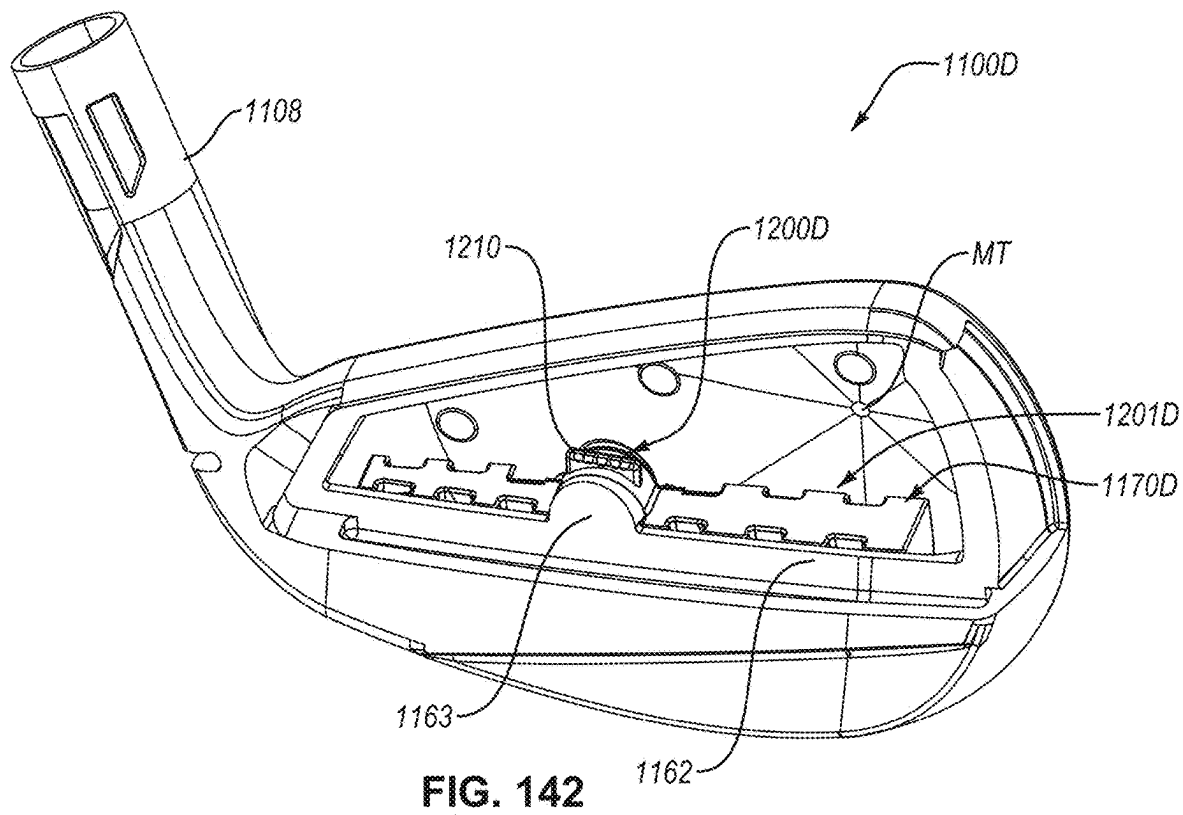


FIG. 141C



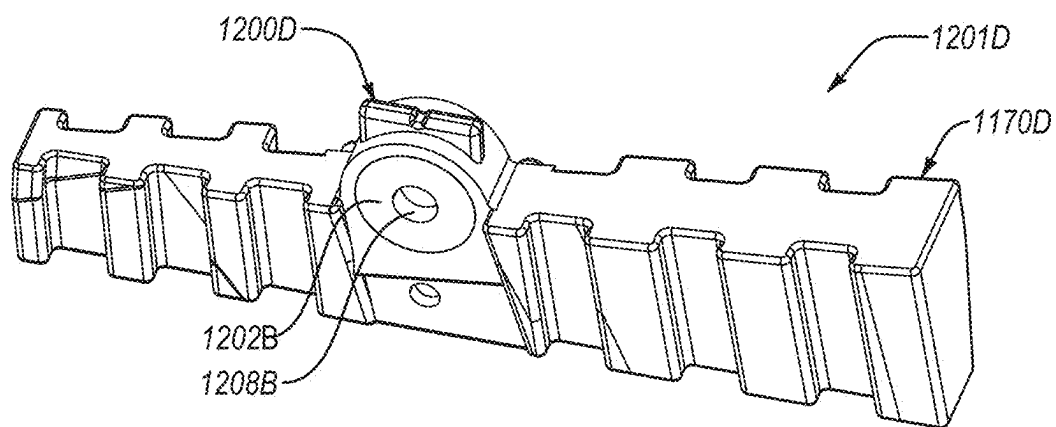


FIG. 144

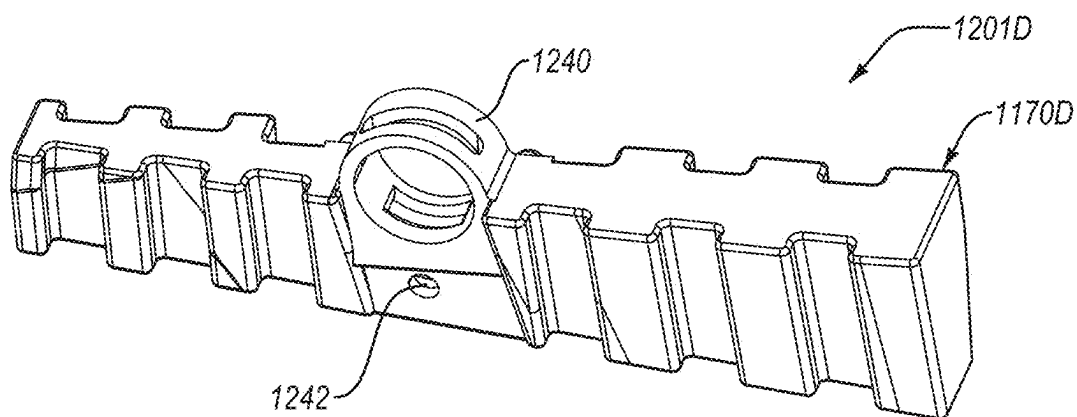


FIG. 145

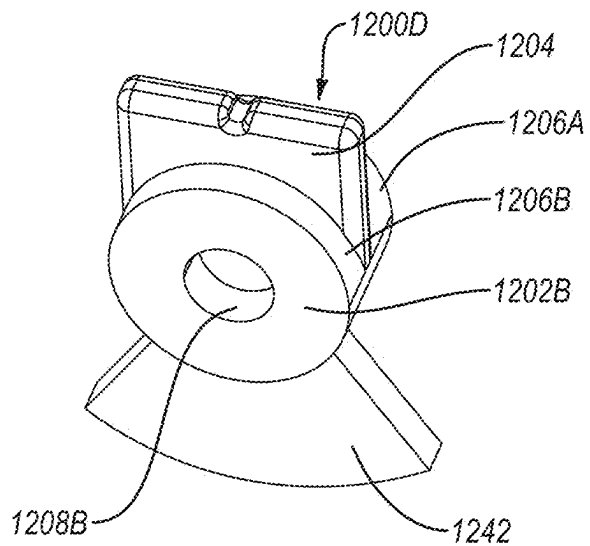


FIG. 146

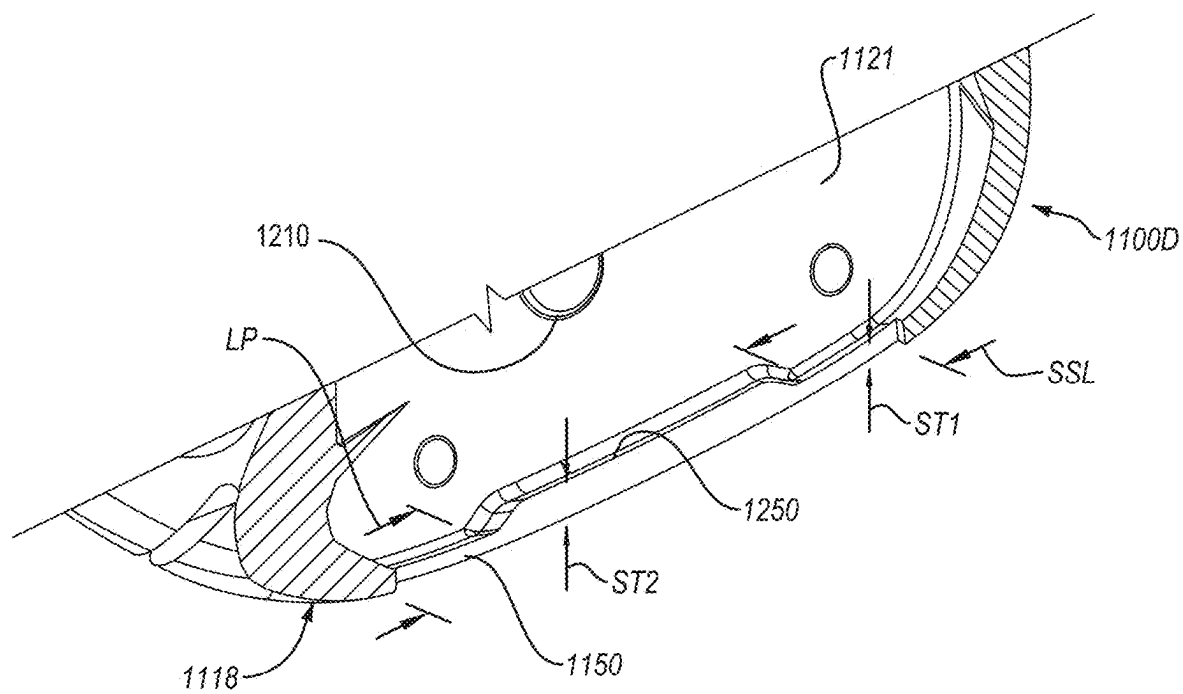


FIG. 147

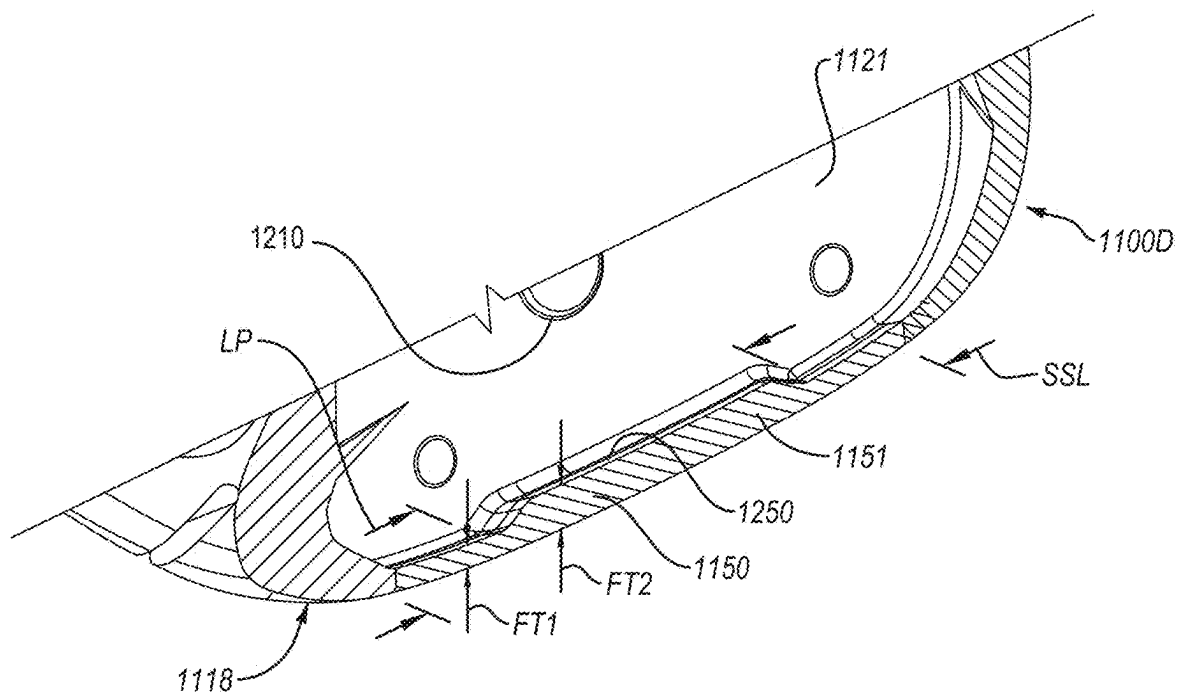


FIG. 148

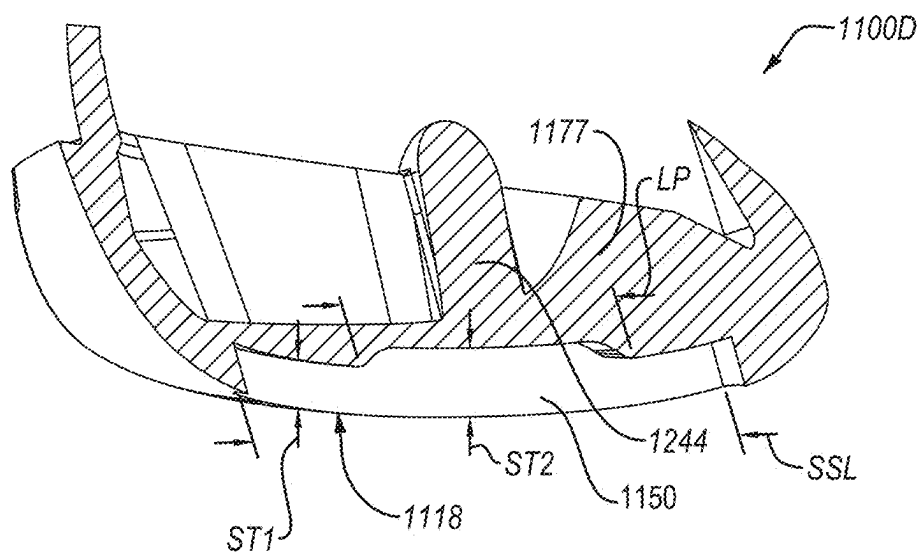


FIG. 149

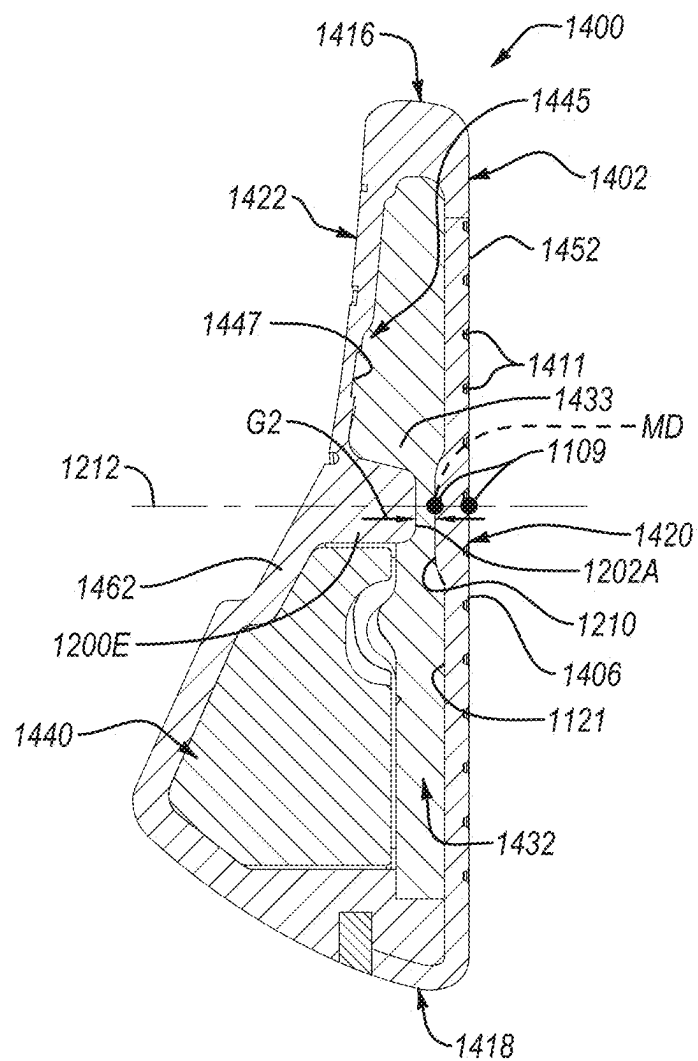


FIG. 150

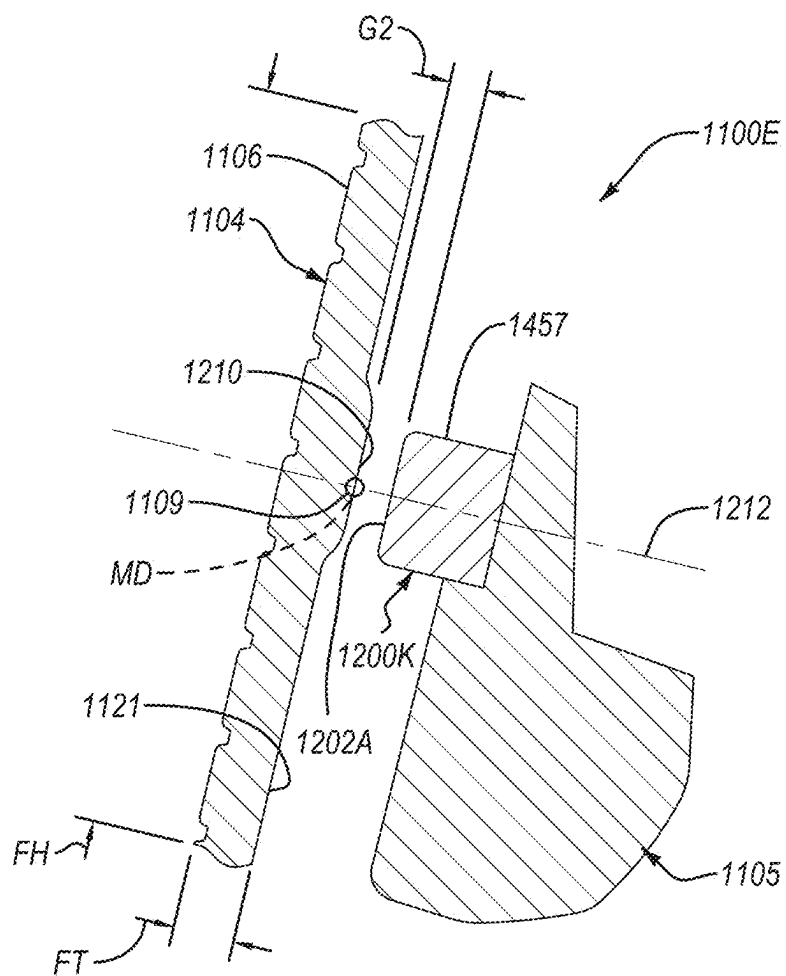


FIG. 151

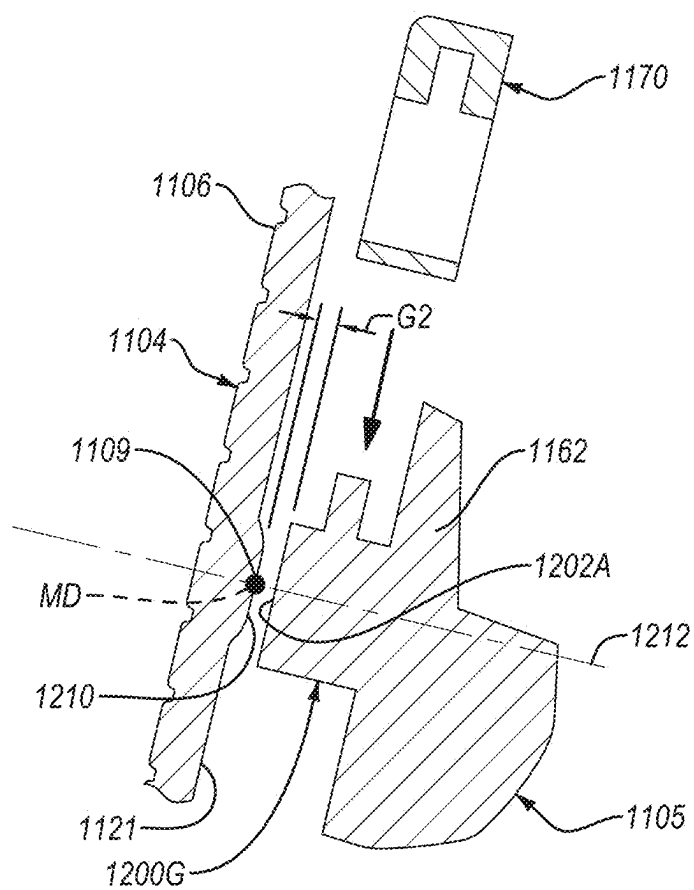


FIG. 152

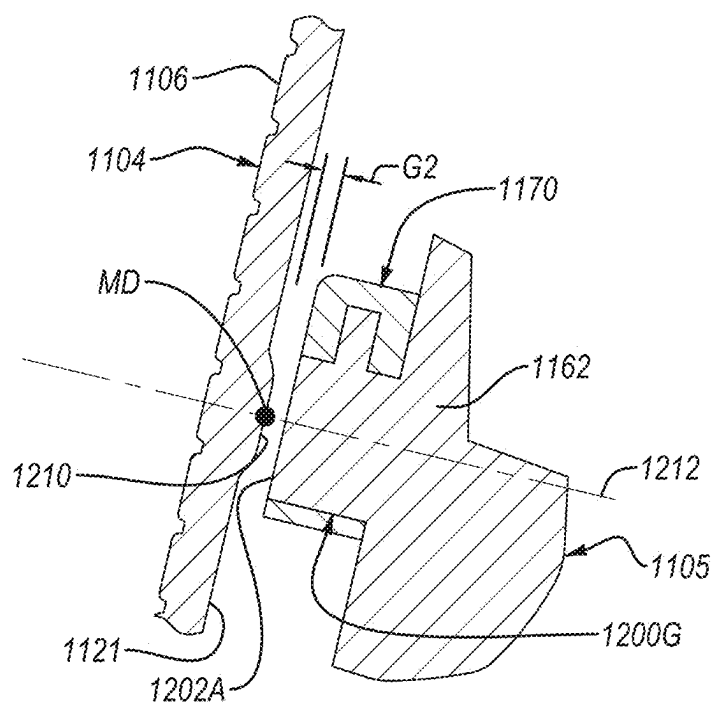


FIG. 153

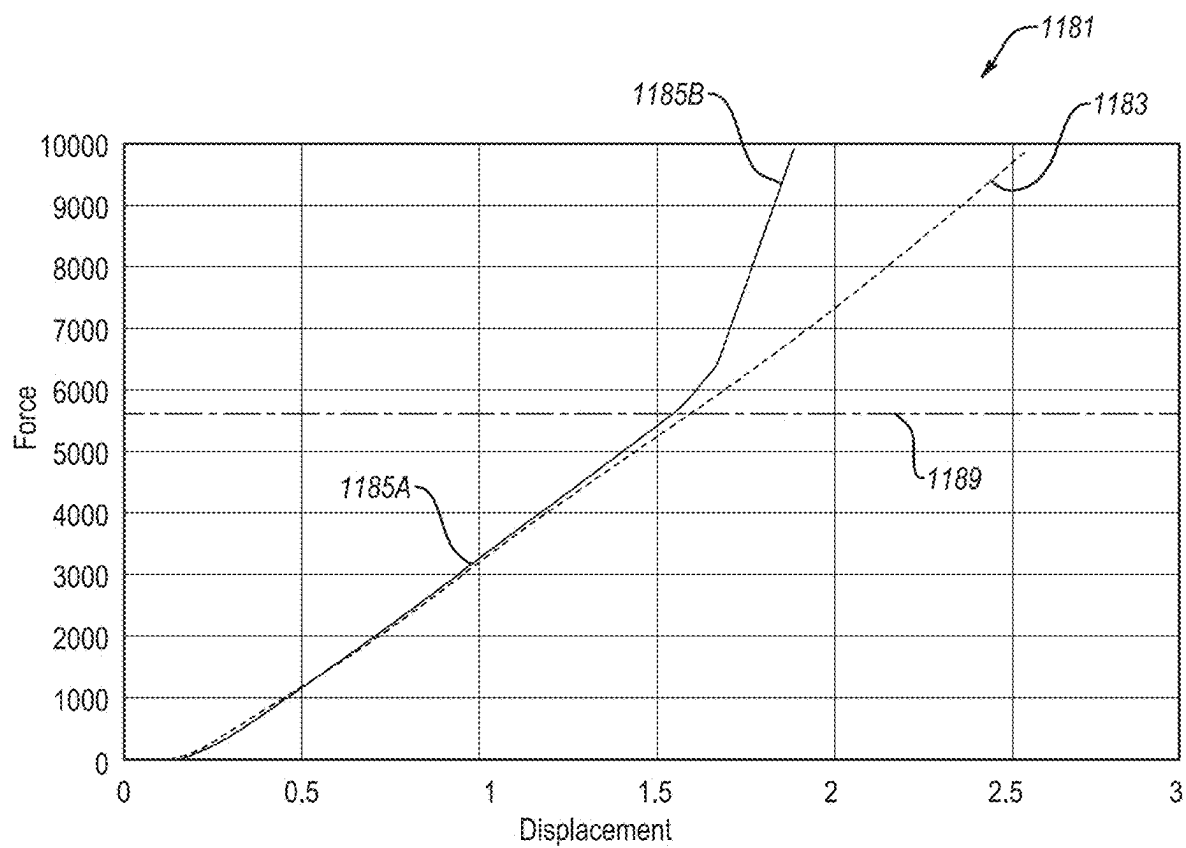


FIG. 154

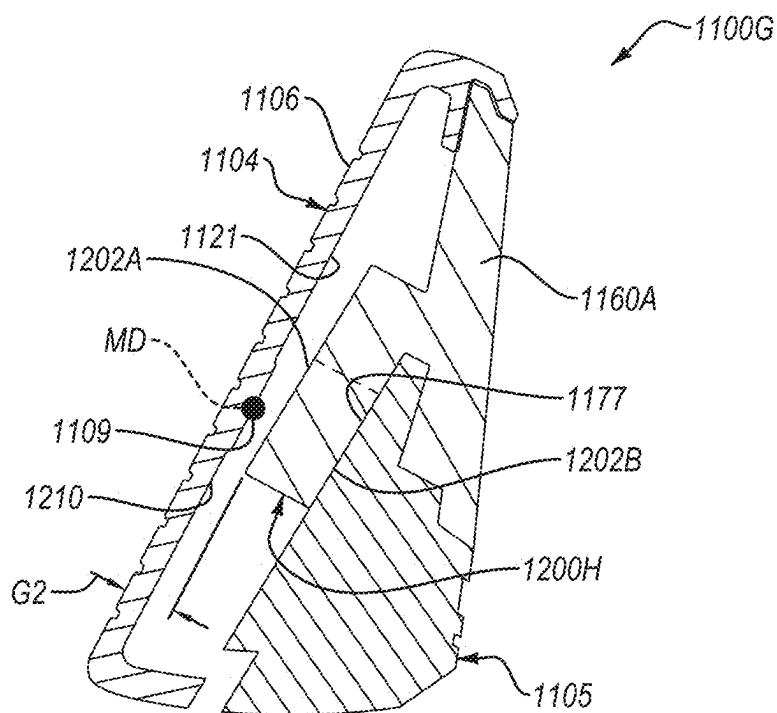


FIG. 155

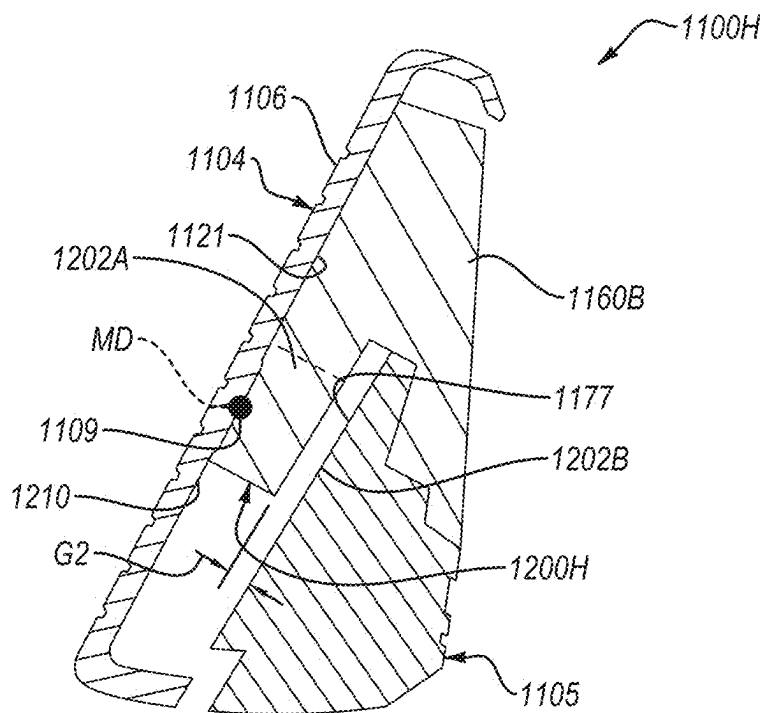


FIG. 156

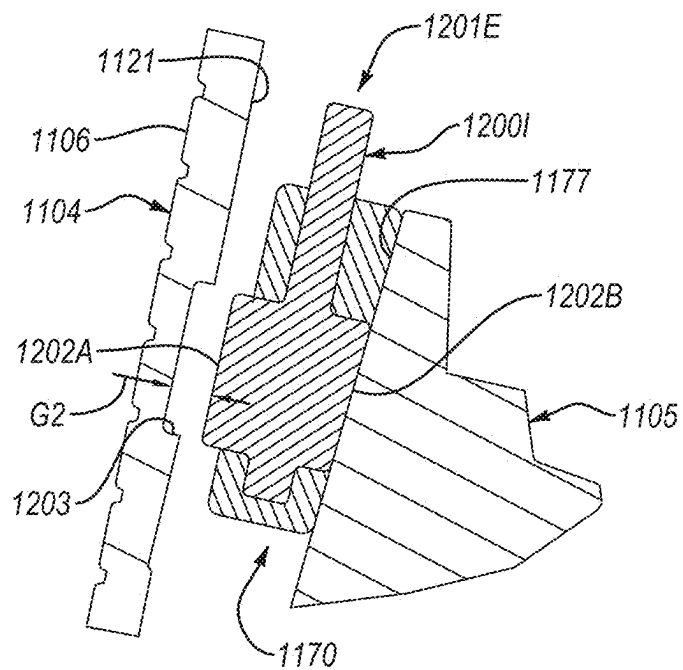


FIG. 157

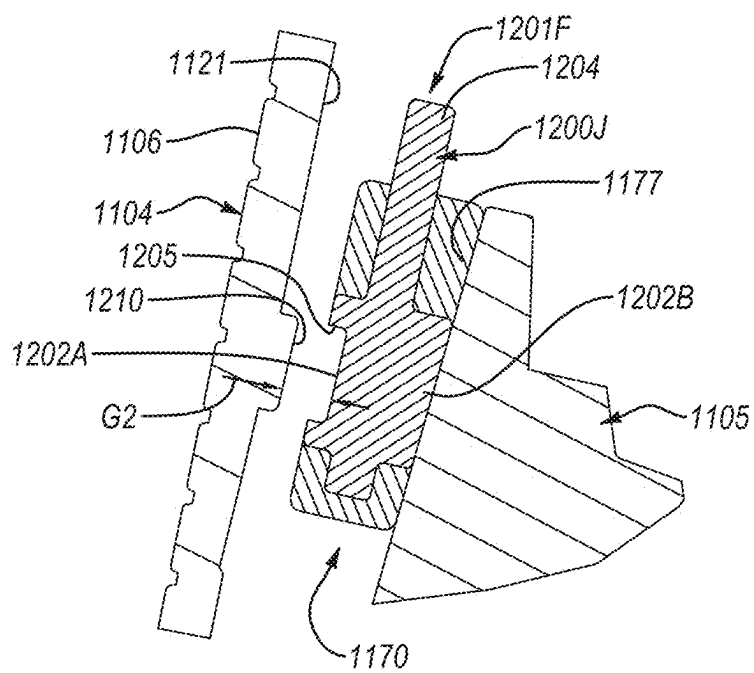


FIG. 158

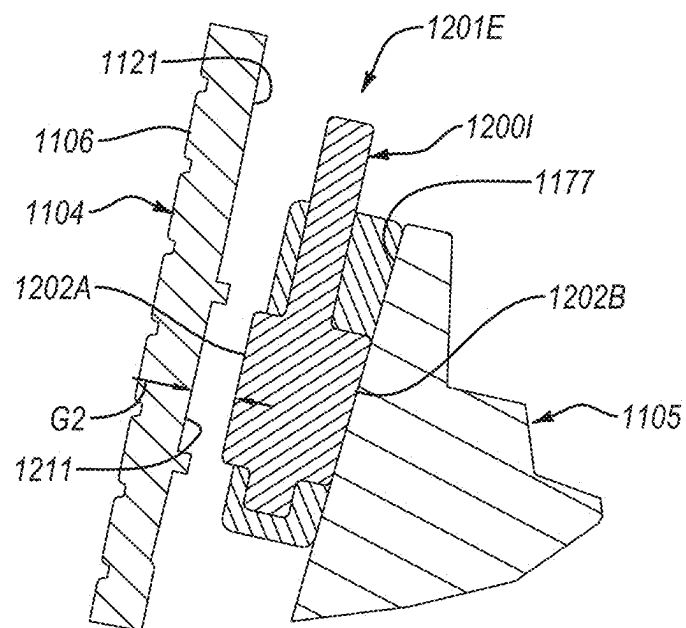


FIG. 159

1260

	Face Thickness						
Head	TOL	F1	F2	MD	F4	F5	F6
1100A	±0.10	2.90	3.25	2.35	2.37	2.14	1.82
1100B		2.67	3.50	1.80	3.30	1.89	1.88
1100C		1.84	1.80	2.30	2.00	2.62	2.05

FIG. 160

GOLF CLUB HEAD

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/616,507, filed Dec. 29, 2023, and is a continuation-in-part of U.S. patent application Ser. No. 18/506,843, filed Nov. 10, 2023, which is a continuation-in-part of U.S. patent application Ser. No. 17/889,187, filed Aug. 16, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 17/368,520, filed Jul. 6, 2021, now U.S. Pat. No. 11,458,374, which is a continuation-in-part of U.S. patent application Ser. No. 17/330,033, filed May 25, 2021, now U.S. Pat. No. 11,413,510, which is a continuation-in-part of U.S. patent application Ser. No. 17/132,541, filed Dec. 23, 2020, now U.S. Pat. No. 11,400,351, which claims the benefit of U.S. Provisional Patent Application No. 62/954,211, filed Dec. 27, 2019 and is a continuation-in-part of U.S. patent application Ser. No. 16/870,714, filed May 8, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/846,492, filed May 10, 2019, and U.S. Provisional Patent Application No. 62/954,211, filed Dec. 27, 2019, all of which are incorporated herein by reference in their entirety.

FIELD

[0002] This disclosure relates generally to golf clubs, and more particularly to the head of an iron-type golf club that reduces face deflection rates for high club head speeds.

BACKGROUND

[0003] The performance of golf equipment is continuously advancing due to the development of innovative clubs and club designs. While all clubs in a golfer's bag are important, both scratch and novice golfers rely on the performance and feel of their irons for many commonly encountered playing situations.

[0004] The strike face of golf club heads deflect when impacted by a golf ball. For conventional golf club heads, the rate of deflection of the strike face, per unit impact force, is constant across all impact forces associated with all swing speeds within the range of possible player swing speeds. Such configurations ensure face deflection characteristics of a golf club head are the same whether the golf club head is swung at slow, medium, or fast swing speeds.

SUMMARY

[0005] The subject matter of the present application has been developed in response to the present state of the art, and in particular, in response to the shortcomings of golf clubs and associated golf club heads, that have not yet been fully solved by currently available techniques. Accordingly, the subject matter of the present application has been developed to provide a golf club and golf club head that overcome at least some of the above-discussed shortcomings of prior art techniques. For example, disclosed herein are golf club heads that promote different rates of deflection of the strike face for different swing speeds. Such a configuration enables the strike face of a golf club head to deflect more for slower swing speeds and deflect less for faster swing speeds.

[0006] Disclosed herein is an iron-type golf club head. The iron-type golf club head comprises a heel portion, a toe portion, a sole portion, a top portion, a front portion com-

prising a strike face and a front interior surface opposite the strike face, a rear portion, and a discrete deflection impeding surface spaced apart from the front interior surface such that a gap, perpendicular to the strike face, is defined between the front interior surface and the discrete deflection impeding surface. The gap is between, and inclusive of, 75% and 125% of a minimum face thickness of the front portion. The preceding subject matter of this paragraph characterizes example 1 of the present disclosure.

[0007] The gap is between, and inclusive of, 1.0 millimeter (mm) and 2.0 mm. The preceding subject matter of this paragraph characterizes example 2 of the present disclosure, wherein example 2 also includes the subject matter according to example 1, above.

[0008] The gap is between, and inclusive of, 1.1 mm and 1.9 mm. The preceding subject matter of this paragraph characterizes example 3 of the present disclosure, wherein example 3 also includes the subject matter according to example 2, above.

[0009] The gap is between, and inclusive of, 1.2 mm and 1.6 mm. The preceding subject matter of this paragraph characterizes example 4 of the present disclosure, wherein example 4 also includes the subject matter according to example 3, above.

[0010] A maximum deflection of the front interior surface, when the strike face is impacted by a 10 kilonewton (kN) force, is between, and inclusive of, 155% and 185% of the gap. The preceding subject matter of this paragraph characterizes example 5 of the present disclosure, wherein example 5 also includes the subject matter according to any of examples 1-4, above.

[0011] When the strike face is impacted by a 10 kN force, the front interior surface deflects between, and inclusive of, 75% and 225% of the minimum face thickness. The preceding subject matter of this paragraph characterizes example 6 of the present disclosure, wherein example 6 also includes the subject matter according to any of examples 1-5, above.

[0012] When the strike face is impacted by a 10 kN force, the front interior surface deflects between, and inclusive of, 75% of the minimum face thickness and 75% of a maximum face thickness of the front portion. The preceding subject matter of this paragraph characterizes example 7 of the present disclosure, wherein example 7 also includes the subject matter according to any of examples 1-6, above.

[0013] A total area of the discrete deflection impeding surface is between, and inclusive of, 1% and 25% of a total area of the front interior surface. The preceding subject matter of this paragraph characterizes example 8 of the present disclosure, wherein example 8 also includes the subject matter according to any of examples 1-7, above.

[0014] The total area of the discrete deflection impeding surface is between, and inclusive of, 1% and 5% of the total area of the front interior surface. The preceding subject matter of this paragraph characterizes example 9 of the present disclosure, wherein example 9 also includes the subject matter according to example 8, above.

[0015] The total area of the discrete deflection impeding surface is between, and inclusive of, 6% and 10% of the total area of the front interior surface. The preceding subject matter of this paragraph characterizes example 10 of the present disclosure, wherein example 10 also includes the subject matter according to any of examples 8-9, above.

[0016] The total area of the discrete deflection impeding surface is between, and inclusive of, 30 mm² and 1000 mm². The preceding subject matter of this paragraph characterizes example 11 of the present disclosure, wherein example 11 also includes the subject matter according to any of examples 8-10, above.

[0017] The total area of the discrete deflection impeding surface is between, and inclusive of, 30 mm² and 100 mm². The preceding subject matter of this paragraph characterizes example 12 of the present disclosure, wherein example 12 also includes the subject matter according to example 11, above.

[0018] The total area of the discrete deflection impeding surface is between, and inclusive of, 35 mm² and 65 mm². The preceding subject matter of this paragraph characterizes example 13 of the present disclosure, wherein example 13 also includes the subject matter according to example 12, above.

[0019] The front interior surface of the front portion further comprises a discrete protrusion protruding rearwardly from an immediately adjacent portion of the front interior surface. The discrete protrusion protrudes from the immediately adjacent portion a distance that is between, and inclusive of, 5% and 30% of the minimum face thickness. A shape of the discrete protrusion matches a shape of the discrete deflection impeding surface. The preceding subject matter of this paragraph characterizes example 14 of the present disclosure, wherein example 14 also includes the subject matter according to any of examples 1-13, above.

[0020] The distance that the discrete protrusion protrudes from the immediately adjacent portion of the front interior surface is between, and inclusive of, 0.1 mm and 0.3 mm. The preceding subject matter of this paragraph characterizes example 15 of the present disclosure, wherein example 15 also includes the subject matter according to example 14, above.

[0021] The distance that the discrete protrusion protrudes from the immediately adjacent portion of the front interior surface varies across a width of the discrete protrusion. The preceding subject matter of this paragraph characterizes example 16 of the present disclosure, wherein example 16 also includes the subject matter according to any of examples 14-15, above.

[0022] The discrete protrusion comprises a recess. The recess is configured to nestably receive the discrete deflection impeding surface when the strike face is impacted by a 10 kN force. The preceding subject matter of this paragraph characterizes example 17 of the present disclosure, wherein example 17 also includes the subject matter according to any of examples 14-16, above.

[0023] The front interior surface defines a maximum deflection point of the front portion. The discrete deflection impeding surface is in line with the maximum deflection point, in a direction perpendicular to the strike face, and is spaced apart from the maximum deflection point by the gap. The preceding subject matter of this paragraph characterizes example 18 of the present disclosure, wherein example 18 also includes the subject matter according to any of examples 1-17, above.

[0024] A size of the gap, in the direction perpendicular to the strike face, is such that the discrete deflection impeding surface impedes deflection of the strike face only for impacts with the strike face above a threshold force. The preceding subject matter of this paragraph characterizes example 19 of

the present disclosure, wherein example 19 also includes the subject matter according to any of examples 1-18, above.

[0025] The modulus of elasticity of the discrete deflection impeding surface is at least at least 0.0150 gigapascals (GPa). The preceding subject matter of this paragraph characterizes example 20 of the present disclosure, wherein example 20 also includes the subject matter according to any of examples 1-19, above.

[0026] A ratio of the size of the gap to a maximum height of the strike face is at most 0.05 and at least 0.02. The preceding subject matter of this paragraph characterizes example 21 of the present disclosure, wherein example 21 also includes the subject matter according to any of examples 1-20, above.

[0027] The discrete deflection impeding surface has a circular shape. The preceding subject matter of this paragraph characterizes example 22 of the present disclosure, wherein example 22 also includes the subject matter according to any of examples 1-21, above.

[0028] The front interior surface of the front portion further comprises a discrete protrusion protruding rearwardly from immediately adjacent portions of the front interior surface. The discrete protrusion protrudes from the immediately adjacent portion of the front interior surface a distance. The discrete protrusion has a circular shape. The preceding subject matter of this paragraph characterizes example 23 of the present disclosure, wherein example 23 also includes the subject matter according to example 22 above.

[0029] The discrete deflection impeding surface has an annulus shape. The discrete protrusion has a circular shape. The preceding subject matter of this paragraph characterizes example 24 of the present disclosure, wherein example 24 also includes the subject matter according to example 23 above.

[0030] A first portion of a perimeter of the discrete deflection impeding surface is curved. A second portion of the perimeter of the discrete deflection impeding surface is flat. The preceding subject matter of this paragraph characterizes example 25 of the present disclosure, wherein example 25 also includes the subject matter according to any of examples 1-24, above.

[0031] A center of the discrete deflection impeding surface is in line with a location on the strike face, in a direction perpendicular to the strike face. The location on the strike face is no more than 10 mm toward and no more than 10 mm heelward of a geometric center of the strike face. The location on the strike face is no more than 10 mm above and no more than 10 mm below the geometric center of the strike face. The preceding subject matter of this paragraph characterizes example 26 of the present disclosure, wherein example 26 also includes the subject matter according to any of examples 1-25, above.

[0032] The rear portion further comprises a rear opening and a badge at least partially covering the rear opening. The badge defines the discrete deflection impeding surface. The preceding subject matter of this paragraph characterizes example 27 of the present disclosure, wherein example 27 also includes the subject matter according to any of examples 1-26, above.

[0033] A maximum deflection of the front interior surface, when the strike face is impacted by a 10 kN force, is between, and inclusive of, 125% and 225% of the gap. The preceding subject matter of this paragraph characterizes

example 28 of the present disclosure, wherein example 28 also includes the subject matter according to any of examples 1-27, above.

[0034] The rear interior surface comprises a recess having a shape that matches the shape of the discrete deflection impeding surface and is configured to receive the discrete deflection impeding surface when the strike face is deflected. The preceding subject matter of this paragraph characterizes example 29 of the present disclosure, wherein example 29 also includes the subject matter according to any of examples 1-28, above.

[0035] The front interior surface of the front portion further comprises a discrete protrusion protruding rearwardly from immediately adjacent portions of the front interior surface. The iron-type golf club head further comprising a recess facing the front interior surface. The recess has a shape that matches the shape of the discrete protrusion and is configured to receive the discrete protrusion when the strike face is deflected. The discrete deflection impeding surface is formed in the recess. The preceding subject matter of this paragraph characterizes example 30 of the present disclosure, wherein example 30 also includes the subject matter according to any of examples 1-29, above.

[0036] The rear portion comprises a rear wall and the discrete deflection impeding surface forms a one-piece monolithic seamless construction with the rear wall. The preceding subject matter of this paragraph characterizes example 31 of the present disclosure, wherein example 31 also includes the subject matter according to any of examples 1-30, above.

[0037] Further disclosed herein is another example of an iron-type golf club head. The iron-type golf club head comprises a heel portion, a toe portion, a sole portion, a top portion, a front portion comprising a strike face and a front interior surface opposite the strike face, a rear portion, and a discrete deflection impeding surface spaced apart from the front interior surface such that a gap is defined between the front interior surface and the discrete deflection impeding surface. A thickness of the strike face, a material of the strike face, and a size of the gap, in the direction perpendicular to the strike face, is such that a first displacement-force rate of the strike face, defined as displacement of the strike face per impact force to the strike face at a given location on the strike face for impact forces below a force threshold, is at least 30% greater than a second displacement-force rate of the strike face, defined as displacement of the strike face per impact force to the strike face at the given location on the strike face for impact forces above the force threshold. The preceding subject matter of this paragraph characterizes example 32 of the present disclosure.

[0038] The first displacement-force rate of the strike face is at least 40% greater than the second displacement-force rate of the strike face. The preceding subject matter of this paragraph characterizes example 33 of the present disclosure, wherein example 33 also includes the subject matter according to example 32, above.

[0039] The first displacement-force rate of the strike face is at least 50% greater than the second displacement-force rate of the strike face. The preceding subject matter of this paragraph characterizes example 34 of the present disclosure, wherein example 34 also includes the subject matter according to example 33, above.

[0040] The first displacement-force rate of the strike face is at least 60% greater than the second displacement-force

rate of the strike face. The preceding subject matter of this paragraph characterizes example 35 of the present disclosure, wherein example 35 also includes the subject matter according to example 34, above.

[0041] The first displacement-force rate of the strike face is at least 70% greater than the second displacement-force rate of the strike face. The preceding subject matter of this paragraph characterizes example 36 of the present disclosure, wherein example 36 also includes the subject matter according to example 35, above.

[0042] The second displacement-force rate of the strike face is constant and the second displacement-force rate of the strike face is constant. The preceding subject matter of this paragraph characterizes example 37 of the present disclosure, wherein example 37 also includes the subject matter according to any of examples 32-36, above.

[0043] The first displacement-force rate of the strike face is between, and inclusive of 0.00019 mm/N and 0.00030 mm/N, and the second displacement-force rate of the strike face is between, and inclusive of, 0.00004 mm/N and 0.00015 mm/N. The preceding subject matter of this paragraph characterizes example 38 of the present disclosure, wherein example 38 also includes the subject matter according to any of examples 32-37, above.

[0044] The force threshold is between, and inclusive of, 5,000 newtons (N) and 7,000 N. The preceding subject matter of this paragraph characterizes example 39 of the present disclosure, wherein example 39 also includes the subject matter according to any of examples 32-38, above.

[0045] The force threshold is greater than 6,000 N. The preceding subject matter of this paragraph characterizes example 40 of the present disclosure, wherein example 40 also includes the subject matter according to any of examples 32-39, above.

[0046] A first displacement of the strike face for impact forces up to the force threshold is at least 0.8 mm. A second displacement of the strike face for impact forces greater than the force threshold is at least 0.8 mm. The second displacement is less than the first displacement. The preceding subject matter of this paragraph characterizes example 41 of the present disclosure, wherein example 41 also includes the subject matter according to any of examples 32-40, above.

[0047] The size of the gap is between, and inclusive of, 1.0 millimeter and 2 mm. The preceding subject matter of this paragraph characterizes example 42 of the present disclosure, wherein example 42 also includes the subject matter according to any of examples 32-41, above.

[0048] The front interior surface defines a maximum deflection point of the front portion. The discrete deflection impeding surface is in line with the MD point, in a direction perpendicular to the strike face, and is spaced apart from the MD point by the gap. The preceding subject matter of this paragraph characterizes example 43 of the present disclosure, wherein example 43 also includes the subject matter according to any of examples 32-42, above.

[0049] A thickness of the strike face, a material of the strike face, and a size of the gap, in the direction perpendicular to the strike face, is such that the discrete deflection impeding surface impedes deflection of the strike face for first impact forces with a golf ball so that a maximum coefficient of restitution (COR) value of the strike face, when receiving the first impact forces, remains below an upper COR threshold value, the discrete deflection impeding surface does not impede deflection of the strike face for

second impact forces with the golf ball so that the COR value of the strike face, when receiving the second impact forces, is higher than the upper COR threshold value, and the second impact forces are lower than the first impact forces. The preceding subject matter of this paragraph characterizes example 44 of the present disclosure, wherein example 44 also includes the subject matter according to any of examples 32-43, above.

[0050] Additionally disclosed herein is yet another example of an iron-type golf club head. The iron-type golf club head comprises a heel portion, a toe portion, a sole portion, a top portion, a front portion comprising a strike face, made of a first material having a first modulus of elasticity, and a front interior surface opposite the strike face, a rear portion comprising a rear wall that defines a rear interior surface, and a damper assembly. The damper assembly comprises an internal damper made of a second material having a second modulus of elasticity that is less than the first modulus of elasticity and a backstop at least partially embedded in the second material of the internal damper made of a third material having a third modulus of elasticity less than the first modulus of elasticity. The backstop comprises a discrete deflection impeding surface and a rear engagement surface. The damper assembly is located between the rear interior surface of the rear wall and the front interior surface of the front portion such that the internal damper is wedged between the rear interior surface and the front interior surface, the rear engagement surface engages the rear interior surface, and the discrete deflection impeding surface is spaced apart from the front interior surface such that a gap, perpendicular to the strike face, is defined between the front interior surface and the discrete deflection impeding surface. The gap is between, and inclusive of, 75% and 125% of a minimum face thickness of the front portion. The preceding subject matter of this paragraph characterizes example 45 of the present disclosure.

[0051] The front interior surface defines a maximum deflection point of the front portion. The discrete deflection impeding surface is in line with the MD point in a direction perpendicular to the strike face and is spaced apart from the MD point by the gap. The preceding subject matter of this paragraph characterizes example 46 of the present disclosure, wherein example 46 also includes the subject matter according to example 45, above.

[0052] A size of the gap, in the direction perpendicular to the strike face, is such that the discrete deflection impeding surface impedes deflection of the strike face only for impacts with the strike face above a threshold force. The preceding subject matter of this paragraph characterizes example 47 of the present disclosure, wherein example 47 also includes the subject matter according to any of examples 45-46, above.

[0053] The third modulus of elasticity is at least 100 pounds per square inch (psi). The preceding subject matter of this paragraph characterizes example 48 of the present disclosure, wherein example 48 also includes the subject matter according to any of examples 45-47, above.

[0054] A ratio of the size of the gap to a maximum height of the strike face is at most 0.05 and at least 0.02. The preceding subject matter of this paragraph characterizes example 49 of the present disclosure, wherein example 49 also includes the subject matter according to any of examples 45-48, above.

[0055] A ratio of the first modulus of elasticity to the third modulus of elasticity is between, and inclusive of, 0.004 and

6.2. The preceding subject matter of this paragraph characterizes example 50 of the present disclosure, wherein example 50 also includes the subject matter according to any of examples 45-49, above.

[0056] A ratio of the second modulus of elasticity to the third modulus of elasticity is between, and inclusive of, 0.000000625 and 0.070. The preceding subject matter of this paragraph characterizes example 51 of the present disclosure, wherein example 51 also includes the subject matter according to example 50, above.

[0057] The backstop further comprises a base plate, a first protrusion protruding from the base plate in a first direction, and a second protrusion protruding from the base plate in a second direction, opposite the first direction. The discrete deflection impeding surface is defined by a distal end of the first protrusion. The rear engagement surface is defined by a distal end of the second protrusion. The preceding subject matter of this paragraph characterizes example 52 of the present disclosure, wherein example 52 also includes the subject matter according to any of examples 45-51, above.

[0058] The base plate comprises an upper section extending upwardly toward the top portion away from the first protrusion and the second protrusion. The base plate comprises a lower section, extending downwardly toward the sole portion away from the first protrusion and the second protrusion. At least a portion of the upper section and the lower section of the base plate is embedded in the second material of the internal damper. The preceding subject matter of this paragraph characterizes example 53 of the present disclosure, wherein example 53 also includes the subject matter according to example 52, above.

[0059] A portion of the upper section of the base plate is not embedded in the second material of the internal damper and extends upwardly toward the top portion away from the internal damper. The preceding subject matter of this paragraph characterizes example 54 of the present disclosure, wherein example 54 also includes the subject matter according to example 53, above.

[0060] An entirety of the lower section of the base plate is embedded in the second material of the internal damper. The preceding subject matter of this paragraph characterizes example 55 of the present disclosure, wherein example 55 also includes the subject matter according to example 54, above.

[0061] A width of the lower section of the base plate increases in a downwardly direction toward the sole portion. The preceding subject matter of this paragraph characterizes example 56 of the present disclosure, wherein example 56 also includes the subject matter according to example 55, above.

[0062] The discrete deflection impeding surface and the rear engagement surface are not embedded in the second material. The rear engagement surface abuts the rear interior surface. The preceding subject matter of this paragraph characterizes example 57 of the present disclosure, wherein example 57 also includes the subject matter according to any of examples 45-56, above.

[0063] The discrete deflection impeding surface and the rear engagement surface are both embedded in the second material. The rear engagement surface is spaced apart from the rear interior surface such that a thickness, perpendicular to the rear interior surface, of the second material is between the rear interior surface and the rear engagement surface. The preceding subject matter of this paragraph characterizes

example 58 of the present disclosure, wherein example 58 also includes the subject matter according to any of examples 45-57, above.

[0064] The gap is greater than the thickness. The preceding subject matter of this paragraph characterizes example 59 of the present disclosure, wherein example 59 also includes the subject matter according to example 58, above.

[0065] A summation of the gap and the thickness is between, and inclusive of, 0.9 mm and 2.0 mm. The preceding subject matter of this paragraph characterizes example 60 of the present disclosure, wherein example 60 also includes the subject matter according to example 59, above.

[0066] A ratio of a summation of the gap and the thickness to the minimum face thickness is between, and inclusive of, 0.49 and 1.48. The preceding subject matter of this paragraph characterizes example 61 of the present disclosure, wherein example 61 also includes the subject matter according to any of examples 59-60, above.

[0067] The discrete deflection impeding surface is embedded in the second material of the internal damper, such that the second material is interposed between the discrete deflection impeding surface and the front interior surface of the front portion and a second gap is defined between the second material and the front interior surface. The second gap is less than the gap. The rear engagement surface is not embedded in the second material. The preceding subject matter of this paragraph characterizes example 62 of the present disclosure, wherein example 62 also includes the subject matter according to any of examples 45-61, above.

[0068] A thickness of the second material interposed between the discrete deflection impeding surface and the front interior surface is between, and inclusive of, 0.5 mm and 1.5 mm. The preceding subject matter of this paragraph characterizes example 63 of the present disclosure, wherein example 63 also includes the subject matter according to example 62, above.

[0069] The backstop further comprises a first blind hole formed in the discrete deflection impeding surface and a second blind hole formed in the rear engagement surface. The preceding subject matter of this paragraph characterizes example 64 of the present disclosure, wherein example 64 also includes the subject matter according to any of examples 45-63, above.

[0070] The second material is a polymeric material. The third material is a metallic material. The preceding subject matter of this paragraph characterizes example 65 of the present disclosure, wherein example 65 also includes the subject matter according to any of examples 45-64, above.

[0071] The second material is a first polymeric material. The third material is a second polymeric material. The preceding subject matter of this paragraph characterizes example 66 of the present disclosure, wherein example 66 also includes the subject matter according to any of examples 45-65, above.

[0072] The internal damper comprises a plurality of vertical projections and a plurality of relief cutouts. Each one of the plurality of relief cutouts separates a corresponding two of the plurality of vertical projections. The plurality of vertical projections contact the front interior surface. First ones of the plurality of vertical projections extend an entire height of the internal damper. Second ones of the plurality of vertical projections extend along only a portion of the entire height of the internal damper. The second ones of the

plurality of vertical projections straddle the discrete deflection impeding surface. The preceding subject matter of this paragraph characterizes example 67 of the present disclosure, wherein example 67 also includes the subject matter according to any of examples 45-66, above.

[0073] A hardness of the third material is greater than hardness of the second material. The hardness of the third material is less than the hardness of the first material. The preceding subject matter of this paragraph characterizes example 68 of the present disclosure, wherein example 68 also includes the subject matter according to any of examples 45-67, above.

[0074] The hardness of the second material is less than 5 Shore D. The hardness of the third material is greater than 40 Shore D. The preceding subject matter of this paragraph characterizes example 69 of the present disclosure, wherein example 69 also includes the subject matter according to example 68, above.

[0075] The discrete deflection impeding surface is parallel to the strike face. The preceding subject matter of this paragraph characterizes example 70 of the present disclosure, wherein example 70 also includes the subject matter according to any of examples 45-69, above.

[0076] The discrete deflection impeding surface is angled relative to the rear engagement surface. The preceding subject matter of this paragraph characterizes example 71 of the present disclosure, wherein example 71 also includes the subject matter according to example 70, above.

[0077] A maximum deflection of the front interior surface, when the strike face is impacted by a 10 kN force, is between, and inclusive of, 125% and 225% of the gap. The preceding subject matter of this paragraph characterizes example 72 of the present disclosure, wherein example 72 also includes the subject matter according to any of examples 45-71, above.

[0078] The described features, structures, advantages, and/or characteristics of the subject matter of the present disclosure may be combined in any suitable manner in one or more examples and/or implementations. In the following description, numerous specific details are provided to impart a thorough understanding of examples of the subject matter of the present disclosure. One skilled in the relevant art will recognize that the subject matter of the present disclosure may be practiced without one or more of the specific features, details, components, materials, and/or methods of a particular example or implementation. In other instances, additional features and advantages may be recognized in certain examples and/or implementations that may not be present in all examples or implementations. Further, in some instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of the subject matter of the present disclosure. The features and advantages of the subject matter of the present disclosure will become more fully apparent from the following description and appended claims, or may be learned by the practice of the subject matter as set forth hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0079] In order that the advantages of the subject matter may be more readily understood, a more particular description of the subject matter briefly described above will be rendered by reference to specific examples that are illustrated in the appended drawings. Understanding that these drawings depict only typical examples of the subject matter

and are not therefore to be considered to be limiting of its scope, the subject matter will be described and explained with additional specificity and detail through the use of the drawings, in which:

[0080] FIG. 1 is a front elevation view of a golf club head, according to one or more examples of the present disclosure;

[0081] FIG. 2 is a side elevation view of the golf club head of FIG. 1, according to one or more examples of the present disclosure;

[0082] FIG. 3 is a cross-sectional side elevation view of the golf club head of FIG. 1, taken along the line 3-3 of FIG. 1, according to one or more examples of the present disclosure;

[0083] FIG. 4 is a perspective view of the golf club head of FIG. 1, from a bottom of the golf club head, according to one or more examples of the present disclosure;

[0084] FIG. 5 is a bottom plan view of the golf club head of FIG. 1, according to one or more examples of the present disclosure;

[0085] FIG. 6 is a back elevation view of the golf club head of FIG. 1, according to one or more examples of the present disclosure;

[0086] FIG. 7 is a perspective view of the golf club head of FIG. 1, from a rear-toe of the golf club head, according to one or more examples of the present disclosure;

[0087] FIG. 8 is a perspective view of the golf club head of FIG. 1, from a rear-heel of the golf club head, according to one or more examples of the present disclosure;

[0088] FIG. 9 is a perspective view of the golf club head of FIG. 1, from a bottom-rear of the golf club head, according to one or more examples of the present disclosure;

[0089] FIG. 10 is a front elevation view of a golf club head damper, according to one or more examples of the present disclosure;

[0090] FIG. 11 is a back perspective view of a golf club head badge and the damper of FIG. 10, according to one or more examples of the present disclosure;

[0091] FIG. 12 is a bottom perspective view of the golf club head badge and damper of FIG. 11, according to one or more examples of the present disclosure;

[0092] FIG. 13 is a back perspective view of a golf club head, according to one or more examples of the present disclosure;

[0093] FIG. 14 is a cross-sectional side view of a golf club head, according to one or more examples of the present disclosure;

[0094] FIG. 15 is a cross-sectional back view of a golf club head, according to one or more examples of the present disclosure;

[0095] FIG. 16 is a cross-sectional side view of a golf club head, according to one or more examples of the present disclosure;

[0096] FIG. 17 is a cross-sectional back view of a golf club head, according to one or more examples of the present disclosure;

[0097] FIG. 18 is a cross-sectional back view of a golf club head, according to one or more examples of the present disclosure;

[0098] FIG. 19 is a perspective view of a golf club head, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0099] FIG. 20 is a rear cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0100] FIG. 21 is a front elevation view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0101] FIG. 22 is a back perspective view of a golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0102] FIG. 23 is a perspective view of a golf club head, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0103] FIG. 24 is a rear perspective view of the golf club head of FIG. 23 without a shim or badge installed, according to one or more examples of the present disclosure;

[0104] FIG. 25 is a top perspective view of a golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0105] FIG. 26 is a bottom perspective view of a golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0106] FIG. 27 is a side cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0107] FIG. 28 is a side cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0108] FIG. 29A is a side cross-sectional view of the upper region of FIG. 27, according to one or more examples of the present disclosure;

[0109] FIG. 29B is a side cross-sectional view of a lower region of FIG. 27, according to one or more examples of the present disclosure;

[0110] FIG. 30 is a perspective view of the damper from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0111] FIG. 31 is a rear elevation view of the shim from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0112] FIG. 32 is a rear perspective view of the shim from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0113] FIG. 33 is a front elevation view of the shim from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0114] FIG. 34 is a front perspective view of the shim from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0115] FIG. 35 is a heelward perspective view of the shim from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0116] FIG. 36 is a toward perspective view of the shim from the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0117] FIG. 37 is a front perspective view of the shim from the golf club head of FIG. 23, according to one or more examples of the present disclosure;

[0118] FIG. 38 is a lower perspective view of the shim from the golf club head of FIG. 23, according to one or more examples of the present disclosure;

[0119] FIG. 39 is a side cross-sectional view of a golf club head according to one or more examples of the present disclosure;

[0120] FIG. 40 is an exploded view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0121] FIG. 41 is a side cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0122] FIG. 42 is a side cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0123] FIG. 43 is a top cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0124] FIG. 44 is an exploded view of a golf club head according to one or more examples of the present disclosure;

[0125] FIG. 45 includes graphical representations of a golf club head undergoing first through fourth mode frequency vibration and associated characteristics of the golf club head, according to one or more examples of the present disclosure;

[0126] FIG. 46 includes graphical representations of a golf club head undergoing first through fourth mode frequency vibration and associated characteristics of the golf club head, according to one or more examples of the present disclosure;

[0127] FIG. 47 is a rear perspective view of the golf club head of FIG. 23 with a shim or badge installed, according to one or more examples of the present disclosure;

[0128] FIG. 48 is a toe-side elevation view of the golf club head of FIG. 23, according to one or more examples of the present disclosure;

[0129] FIG. 49 is a front elevation view of the golf club head of FIG. 23, according to one or more examples of the present disclosure;

[0130] FIG. 50 is a rear perspective view of the golf club head of FIG. 23 without a shim or badge installed, according to one or more examples of the present disclosure;

[0131] FIG. 51 is a toe-side elevation view of the golf club head of FIG. 23 without a shim or badge installed, according to one or more examples of the present disclosure;

[0132] FIG. 52 is a perspective view of the golf club head of FIG. 23, according to one or more examples of the present disclosure;

[0133] FIG. 53 is a front perspective view of the shim or badge from the golf club head of FIG. 23, according to one or more examples of the present disclosure;

[0134] FIG. 54 is a rear, heel-side perspective view of a golf club head, without a shim or badge installed, according to one or more examples of the present disclosure;

[0135] FIG. 55 is a rear, toe-side perspective view of a golf club head, without a shim or badge installed, according to one or more examples of the present disclosure;

[0136] FIG. 56 is a rear, toe-side perspective view of a golf club head, with a shim or badge installed, according to one or more examples of the present disclosure;

[0137] FIG. 57 is heel-side cross-sectional view of the golf club head of FIG. 19, according to one or more examples of the present disclosure;

[0138] FIG. 58 is a front elevation view of a golf club head in accordance with the embodiments of the current disclosure;

[0139] FIG. 59 is an illustration of the central region of a golf club head in accordance with the embodiments of the current disclosure;

[0140] FIG. 60 is another illustration of the central region of a golf club head in accordance with the embodiments of the current disclosure;

[0141] FIG. 61 is another illustration of the central region of a golf club head in accordance with the embodiments of the current disclosure;

[0142] FIG. 62 is a front elevation view of a golf club head in accordance with the embodiments of the current disclosure;

[0143] FIG. 63 is a front elevation view of a golf club head in accordance with the embodiments of the current disclosure;

[0144] FIG. 64 is a perspective view of an iron-type golf club head, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0145] FIG. 65 is a cross-sectional side view of the golf club head of FIG. 64, taken along the line 14-14 of FIG. 64, according to one or more examples of the present disclosure;

[0146] FIG. 66 is a cross-sectional side view of the golf club head of FIG. 64, taken along the line 15-15 of FIG. 64, according to one or more examples of the present disclosure;

[0147] FIG. 67 is a front view of the golf club head of FIG. 64, according to one or more examples of the present disclosure;

[0148] FIG. 68 is a cross-sectional rear view of the golf club head of FIG. 64, taken along the line 68-68 of FIG. 66, according to one or more examples of the present disclosure;

[0149] FIG. 69 is a perspective view of a body of the golf club head of FIG. 64, from a top of the golf club head, according to one or more examples of the present disclosure;

[0150] FIG. 70 is a perspective view of the body of the golf club head of FIG. 64, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0151] FIG. 71 is a perspective view of an iron-type golf club head, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0152] FIG. 72 is a cross-sectional side view of the golf club head of FIG. 71, taken along the line 72-72 of FIG. 71, according to one or more examples of the present disclosure;

[0153] FIG. 73 is a front view of the golf club head of FIG. 71, according to one or more examples of the present disclosure;

[0154] FIG. 74 is a cross-sectional rear view of the golf club head of FIG. 71, taken along the line 74-74 of FIG. 72, according to one or more examples of the present disclosure;

[0155] FIG. 75 is a perspective view of the body of the golf club head of FIG. 71, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0156] FIG. 76 is a perspective view of an iron-type golf club head, from a front of the golf club head, according to one or more examples of the present disclosure;

[0157] FIG. 77 is perspective view of the golf club head of FIG. 76, from a rear of the golf club head, according to one or more examples of the present disclosure;

[0158] FIG. 78 is perspective view of the golf club head of FIG. 76, from a front and bottom of the golf club head, according to one or more examples of the present disclosure;

[0159] FIG. 79 is a front view of the golf club head of FIG. 76, according to one or more examples of the present disclosure;

[0160] FIG. 80 is perspective view of the golf club head of FIG. 76, from a front and bottom of the golf club head, and with a strike plate removed, according to one or more examples of the present disclosure;

[0161] FIG. 81 is perspective view of the golf club head of FIG. 76, from a front of the golf club head and with a strike plate removed, according to one or more examples of the present disclosure;

[0162] FIG. 82 is a first perspective view of an internal weight of an iron-type golf club head, according to one or more examples of the present disclosure;

[0163] FIG. 83 is a second perspective view of the internal weight of FIG. 82, according to one or more examples of the present disclosure;

[0164] FIG. 84 is a front view of the internal weight of FIG. 82, according to one or more examples of the present disclosure;

[0165] FIG. 85 is a side view of the internal weight of FIG. 82, according to one or more examples of the present disclosure;

[0166] FIG. 86 is perspective view of the golf club head of FIG. 76, from a front and heel of the golf club head, and with a strike plate and internal weight removed, according to one or more examples of the present disclosure;

[0167] FIG. 87 is perspective view of the golf club head of FIG. 76, from a front and toe of the golf club head, and with a strike plate and an internal weight removed, according to one or more examples of the present disclosure;

[0168] FIG. 88 is perspective view of an iron-type golf club head, from a front and heel of the golf club head, and with a strike plate removed, according to one or more examples of the present disclosure;

[0169] FIG. 89 is perspective view of the golf club head of FIG. 88, from a front and heel of the golf club head, and with a strike plate and an internal weight removed, according to one or more examples of the present disclosure;

[0170] FIG. 90 is perspective view of the golf club head of FIG. 88, from a front and toe of the golf club head, and with a strike plate removed, according to one or more examples of the present disclosure;

[0171] FIG. 91 is perspective view of the golf club head of FIG. 88, from a front and toe of the golf club head, and with a strike plate and an internal weight removed, according to one or more examples of the present disclosure;

[0172] FIG. 92 is a perspective view of an internal weight of an iron-type golf club head, according to one or more examples of the present disclosure;

[0173] FIG. 93 is a side view of the internal weight of FIG. 92, according to one or more examples of the present disclosure;

[0174] FIG. 94 is a cross-sectional front view of an iron-type golf club head, taken along a line similar to line A-A of FIG. 88, according to one or more examples of the present disclosure;

[0175] FIG. 95 is a cross-sectional front view of an iron-type golf club head, taken along a line similar to line A-A of FIG. 88, according to one or more examples of the present disclosure;

[0176] FIG. 96 is a cross-sectional front view of an iron-type golf club head, taken along a line similar to line A-A of FIG. 88, according to one or more examples of the present disclosure;

[0177] FIG. 97 is a cross-sectional front view of an iron-type golf club head, taken along a line similar to line A-A of FIG. 8, according to one or more examples of the present disclosure;

[0178] FIG. 98 is a cross-sectional front view of an iron-type golf club head, taken along a line similar to line A-A of FIG. 88, according to one or more examples of the present disclosure;

[0179] FIG. 99 is a cross-sectional front view of an iron-type golf club head, taken along a line similar to line A-A of FIG. 88, according to one or more examples of the present disclosure;

[0180] FIG. 100 is a cross-sectional front close-up view of the iron-type golf club head of FIG. 98, taken along a line similar to line A-A of FIG. 88, according to one or more examples of the present disclosure;

[0181] FIG. 101 is a cross-sectional top close-up view of an iron-type golf club head, taken along a line similar to line B-B of FIGS. 94 and 100, according to one or more examples of the present disclosure;

[0182] FIG. 102 is a chart showing values for various properties of the iron-type golf club heads of a correlated set of golf club heads, according to one or more examples of the present disclosure;

[0183] FIG. 103 is a cross-sectional side elevation view of the iron-type golf club head of FIGS. 76-81, 86, 87, and 98, taken along the line C-C of FIG. 98, according to one or more examples of the present disclosure;

[0184] FIG. 104 is a cross-sectional side elevation view of the iron-type golf club head of FIGS. 76-81, 86, 87, and 98, taken along the line D-D of FIG. 98, according to one or more examples of the present disclosure;

[0185] FIG. 105 is a cross-sectional rear elevation view of the iron-type golf club head of FIG. 103, taken along the line E-E of FIG. 103, according to one or more examples of the present disclosure;

[0186] FIG. 106 is a chart showing values for various properties of the iron-type golf club heads of a correlated set of golf club heads, according to one or more examples of the present disclosure;

[0187] FIG. 107 is a cross-sectional front elevation view of an iron-type golf club head, according to one or more examples of the present disclosure;

[0188] FIG. 108 is front elevation view of an iron-type golf club head, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0189] FIG. 109 is front elevation view of an iron-type golf club head, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0190] FIG. 110 is front elevation view of an iron-type golf club head, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0191] FIG. 111 is front elevation view of an iron-type golf club head, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0192] FIG. 112 is perspective view of the iron-type golf club head of FIG. 111, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0193] FIG. 113 is rear view of the iron-type golf club head of FIG. 111, according to one or more examples of the present disclosure;

[0194] FIG. 114 is a cross-sectional side view of the iron-type golf club head of FIG. 111, taken along the line F-F of FIG. 111, according to one or more examples of the present disclosure;

[0195] FIG. 115 is a side view of an iron-type golf club head, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0196] FIG. 116 is a bottom view of the iron-type golf club head of FIG. 115, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0197] FIG. 117 is a side view of an iron-type golf club head, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0198] FIG. 118 is a bottom view of the iron-type golf club head of FIG. 117, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0199] FIG. 119 is a cross-sectional bottom view of the iron-type golf club head of FIG. 117, shown with a strike plate removed, according to one or more examples of the present disclosure;

[0200] FIG. 120 is a perspective view of an iron-type golf club head, shown from a rear of the golf club head, according to one or more examples of the present disclosure;

[0201] FIG. 121 is an exploded perspective view of the iron-type golf club head of FIG. 120, according to one or more examples of the present disclosure;

[0202] FIG. 122 is a rear elevation view of the iron-type golf club head of FIG. 120, shown with a rear badge removed, according to one or more examples of the present disclosure;

[0203] FIG. 123 is a front elevation view of the iron-type golf club head of FIG. 120, according to one or more examples of the present disclosure;

[0204] FIG. 124 is a side elevation view of the iron-type golf club head of FIG. 120, shown with a rear badge removed, according to one or more examples of the present disclosure;

[0205] FIG. 125 is a perspective view of the iron-type golf club head of FIG. 120, shown from a rear of the golf club head and with the rear badge removed, according to one or more examples of the present disclosure;

[0206] FIG. 126 is a perspective view of the iron-type golf club head of FIG. 120, shown from a rear of the golf club head and with the rear badge and a damper assembly removed, according to one or more examples of the present disclosure;

[0207] FIG. 127 is a perspective view of the iron-type golf club head of FIG. 120, shown from a heel side of the golf club head and with the rear badge and a damper assembly removed, according to one or more examples of the present disclosure;

[0208] FIG. 128 is a cross-sectional side elevation view of the iron-type golf club head of FIG. 120, taken along the line 128-128 in FIG. 123 and shown with the rear badge removed, according to one or more examples of the present disclosure;

[0209] FIG. 129 is a cross-sectional side elevation view of the iron-type golf club head of FIG. 120, taken along the line 129-129 in FIG. 123 and shown with the rear badge removed, according to one or more examples of the present disclosure;

[0210] FIG. 130 is a cross-sectional side elevation view of a top portion of the iron-type golf club head of FIG. 120, taken along the line 129-129 in FIG. 123 and shown with the rear badge removed, according to one or more examples of the present disclosure;

[0211] FIG. 131 is a perspective view of a damper assembly of an iron-type golf club head, shown from a front of the damper assembly, according to one or more examples of the present disclosure;

[0212] FIG. 132 is a perspective view of a backstop of the damper assembly of FIG. 131, shown from a front of the backstop, according to one or more examples of the present disclosure;

[0213] FIG. 133 is a perspective view of a damper assembly of an iron-type golf club head, shown from a rear of the damper assembly, according to one or more examples of the present disclosure;

[0214] FIG. 134 is a perspective view of the backstop of the damper assembly of FIG. 131, shown from a rear of the backstop, according to one or more examples of the present disclosure;

[0215] FIG. 135 is a front elevation view of the damper assembly of FIG. 131, according to one or more examples of the present disclosure;

[0216] FIG. 136A is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with an undeflected strike face, according to one or more examples of the present disclosure;

[0217] FIG. 136B is a cross-sectional side elevation view of the iron-type golf club head of FIG. 136A, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with a partially deflected strike face, according to one or more examples of the present disclosure;

[0218] FIG. 136C is a cross-sectional side elevation view of the iron-type golf club head of FIG. 136A, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with a more deflected strike face than in FIG. 136B, according to one or more examples of the present disclosure;

[0219] FIG. 137A is a perspective view of a damper assembly of an iron-type golf club head, shown from a front of the damper assembly, according to one or more examples of the present disclosure;

[0220] FIG. 137B is a front elevation view of the damper assembly of FIG. 137A, according to one or more examples of the present disclosure;

[0221] FIG. 138A is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with an undeflected strike face, according to one or more examples of the present disclosure;

[0222] FIG. 138B is a cross-sectional side elevation view of the iron-type golf club head of FIG. 138A, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with a partially deflected strike face, according to one or more examples of the present disclosure;

[0223] FIG. 138C is a cross-sectional side elevation view of the iron-type golf club head of FIG. 138A, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with a more deflected strike face than in FIG. 138B, according to one or more examples of the present disclosure;

[0224] FIG. 138D is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with an undeflected strike face, according to one or more examples of the present disclosure;

[0225] FIG. 139 is a front elevation view of a damper assembly of an iron-type golf club head, according to one or more examples of the present disclosure;

[0226] FIG. 140 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123, providing a close-up view of the damper assembly of the iron-type golf club head, and shown with an undeflected strike face, according to one or more examples of the present disclosure;

[0227] FIG. 141A is a cross-sectional front elevation view of one example of an iron-type golf club head, taken along a line similar to the line 141-141 in FIG. 124, according to one or more examples of the present disclosure;

[0228] FIG. 141B is a cross-sectional front elevation view of a second example of an iron-type golf club head, taken along a line similar to the line 141-141 in FIG. 124, according to one or more examples of the present disclosure;

[0229] FIG. 141C is a cross-sectional front elevation view of a third example of an iron-type golf club head, taken along a line similar to the line 141-141 in FIG. 124, according to one or more examples of the present disclosure;

[0230] FIG. 142 is a perspective view of an iron-type golf club head, shown from a rear of the golf club head and with the rear badge removed, according to one or more examples of the present disclosure;

[0231] FIG. 143 is a perspective view of the iron-type golf club head of FIG. 142, shown from a rear of the golf club head and with the rear badge and a damper assembly removed, according to one or more examples of the present disclosure;

[0232] FIG. 144 is a perspective view of a damper assembly of the iron-type golf club head of FIG. 142, shown from a rear of the damper assembly, according to one or more examples of the present disclosure;

[0233] FIG. 145 is a perspective view of an internal damper of the damper assembly of FIG. 144, according to one or more examples of the present disclosure;

[0234] FIG. 146 is a perspective view of a backstop of the damper assembly of FIG. 144, according to one or more examples of the present disclosure;

[0235] FIG. 147 is a cross-sectional perspective view of an iron-type golf club head, taken along a line similar to the line 147-147 of FIG. 124 and shown with a filler material removed from a slot of the iron-type golf club head, according to one or more examples of the present disclosure;

[0236] FIG. 148 is a cross-sectional perspective view of the iron-type golf club head of FIG. 147, taken along the line similar to the line 147-147 of FIG. 124 and shown with a filler material in the slot of the iron-type golf club head, according to one or more examples of the present disclosure;

[0237] FIG. 149 is a cross-sectional perspective view of an iron-type golf club head, taken along a line similar to the line 149-149 of FIG. 124 and shown with a filler material removed from a slot of the iron-type golf club head, according to one or more examples of the present disclosure;

[0238] FIG. 150 is a cross-sectional side elevation view of an iron-type golf club head, having a hollow body construction and filled with a filler material, taken along the line 129-129 in FIG. 123, according to one or more examples of the present disclosure;

[0239] FIG. 151 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of

a backstop of the iron-type golf club head, according to one or more examples of the present disclosure;

[0240] FIG. 152 is an exploded cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of a backstop of the iron-type golf club head, according to one or more examples of the present disclosure;

[0241] FIG. 153 is a cross-sectional side elevation view of the iron-type golf club head of FIG. 153, taken along the line similar to the line 129-129 in FIG. 123 and providing a close-up view of a backstop of the iron-type golf club head, according to one or more examples of the present disclosure;

[0242] FIG. 154 is a chart showing strike face displacement-force data for two examples of an iron-type golf club head, according to one or more examples of the present disclosure;

[0243] FIG. 155 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of a backstop of the iron-type golf club head, according to one or more examples of the present disclosure;

[0244] FIG. 156 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of a backstop of the iron-type golf club head, according to one or more examples of the present disclosure;

[0245] FIG. 157 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of a damper assembly of the iron-type golf club head, according to one or more examples of the present disclosure;

[0246] FIG. 158 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of a damper assembly of the iron-type golf club head, according to one or more examples of the present disclosure;

[0247] FIG. 159 is a cross-sectional side elevation view of an iron-type golf club head, taken along a line similar to the line 129-129 in FIG. 123 and providing a close-up view of a damper assembly of the iron-type golf club head, according to one or more examples of the present disclosure; and

[0248] FIG. 160 is a chart showing values for various properties of the iron-type golf club heads of FIGS. 141A-141C, according to one or more examples of the present disclosure.

DETAILED DESCRIPTION

[0249] The following describes examples of golf club heads in the context of an iron-type golf club, but the principles, methods and designs described may be applicable in whole or in part to utility golf clubs (also known as hybrid golf clubs), metal-wood-type golf club, driver-type golf clubs, putter-type golf clubs, and the like.

[0250] The following describes exemplary embodiments of golf club heads in the context of an iron-type golf club, but the principles, methods and designs described may be applicable in whole or in part to utility golf clubs (also known as hybrid golf clubs), metal-wood-type golf clubs, driver-type golf clubs, putter-type golf clubs, and other golf clubs.

[0251] FIG. 1 illustrates one embodiment of an iron-type golf club head 100 including a body 113 having a heel portion 102, a toe portion 104, a sole portion 108, a topline portion 106, and a hosel 114. The golf club head 100 is

shown in FIG. 1 in a normal address position with the sole portion 108 resting upon a ground plane 111, which is assumed to be perfectly flat. As used herein, “normal address position” means the position of the golf club head 100 when a vector normal to a geometric center of a strike face 110 of the golf club head 100 lies substantially in a first vertical plane (i.e., a plane perpendicular to the ground plane 111), a centerline axis 115 of the hosel 114 lies substantially in a second vertical plane, and the first vertical plane and the second vertical plane substantially perpendicularly intersect. The geometric center of the strike face 110 is determined using the procedures described in the USGA “Procedure for Measuring the Flexibility of a Golf Club head,” Revision 2.0, Mar. 25, 2005. The strike face 110 is the front surface of a strike plate 109 of the golf club head 100. The strike face 110 has a rear surface 131, opposite the strike face 110 (see, e.g., FIG. 3). In some embodiments, the strike plate has a thickness that is less than 2.0 mm, such as between 1.0 mm and 1.75 mm. Additionally or alternatively, the strike plate may have an average thickness less than or equal to 2 mm, such as an average thickness between 1.0 mm and 2.0 mm, such as an average thickness between 1.25 mm and 1.75 mm. In some embodiments, the strike plate has a thickness that varies. In some embodiments, the strike plate has a thinned region coinciding and surrounding the center of the face such that the center face region of the strike plate is the thinnest region of the strike plate. In other embodiments, the strike plate has a thickened region coinciding and surrounding the center of the face such that the center face region of the strike plate is the thickest region of the strike plate. As used herein, the minimum thickness of the strike plate, at least in regions containing grooves or scorelines, is a minimum thickness of the strike plate excluding the grooves. For example, the minimum thickness of the strike plate is measured as a distance from lands or a land area, between the grooves, and a rear surface of the strike plate.

[0252] As shown in FIG. 1, a lower tangent point 291 on the outer surface of the golf club head 100, of a line 295 forming a 45° angle relative to the ground plane 111, defines a demarcation boundary between the sole portion 108 and the toe portion 104. Similarly, an upper tangent point 292 on the outer surface of the golf club head 100 of a line 299 forming a 45° angle relative to the ground plane 111 defines a demarcation boundary between the topline portion 106 and the toe portion 104. In other words, the portion of the golf club head 100 that is above and to the left (as viewed in FIG. 1) of the lower tangent point 291 and below and to the left (as viewed in FIG. 1) of the upper tangent point 292 is the toe portion 104.

[0253] The strike face 110 includes grooves 112 designed to impact and affect spin characteristics of a golf ball struck by the golf club head 100. In some embodiments, the toe portion 104 may be defined to be any portion of the golf club head 100 that is toward of the grooves 112. In some embodiments, the body 113 and the strike plate 109 of the golf club head 100 can be a single unitary cast piece, while in other embodiments, the strike plate 109 can be formed separately and be adhesively or mechanically attached to the body 113 of the golf club head 100.

[0254] FIGS. 1 and 2 show an ideal strike location 101 on the strike face 110 and respective coordinate system with the ideal strike location 101 at the origin. As used herein, the ideal strike location 101 is located on the strike face 110 and coincides with the location of the CG 127 of the golf club

head 100 along an x-axis 105 and is offset from a leading edge 179 of the golf club head 100 (defined as the midpoint of a radius connecting the sole portion 108 and the strike face 110) by a distance d , which is 16.5 mm in some implementations, along the strike face 110, as shown in FIG. 2. The x-axis 105, a y-axis 107, and a z-axis 103 intersect at the ideal strike location 101, which defines the origin of the orthogonal axes. With the golf club head 100 in the normal address position, the x-axis 105 is parallel to the ground plane 111 and is oriented perpendicular to a normal plane extending from the strike face 110 at the ideal strike location 101. The y-axis 107 is also parallel to the ground plane 111 and is perpendicular to the x-axis 105. The z-axis 103 is oriented perpendicular to the ground plane 111, and thus is perpendicular to the x-axis 105 and the y-axis 107. In addition, a z-up axis 171 can be defined as an axis perpendicular to the ground plane 111 and having an origin at the ground plane 111.

[0255] In certain embodiments, a desirable CG-y location is between about 0.25 mm to about 20 mm along the y-axis 107 toward the rear portion of the club head. Additionally, according to some embodiments, a desirable CG-z location is between about 12 mm to about 25 mm along the z-up axis 171.

[0256] The golf club head 100 may be of solid construction (also referred to as “blades” and/or “musclebacks”), hollow, cavity back, or other construction. However, in the illustrated embodiments, the golf club head 100 is depicted as having a cavity-back construction because the golf club head 100 includes an open cavity 161 behind the strike plate 109 (see, e.g., FIG. 3). FIG. 3 shows a cross-sectional side view, along the cross-section lines 3-3 of FIG. 1, of the golf club head 100.

[0257] In the embodiment shown in FIGS. 1-3, the grooves 112 are located on the strike face 110 such that they are centered along the X-axis 105 about the ideal strike location 101 (such that the ideal strike location 101 is located within the strike face 110 on an imaginary line that is both perpendicular to and that passes through the midpoint of the longest score-line groove 112). In other embodiments (not shown in the drawings), the grooves 112 may be shifted along the X-axis 105 to the toe side or the heel side relative to the ideal striking location 101, the grooves 112 may be aligned along an axis that is not parallel to the ground plane 111, the grooves 112 may have discontinuities along their lengths, or the strike face 110 may not have grooves 112. Still other shapes, alignments, and/or orientations of grooves 112 on the strike face 110 are also possible.

[0258] In reference to FIG. 1, the golf club head 100 has a sole length L_B (i.e., length of the sole) and a club head height H_{CH} (i.e., height of the golf club head 100). The sole length L_B is defined as the distance between two points 116, 117 projected onto the ground plane 111. The heel side point 116 is defined as the intersection of a projection of the hosel axis 115 onto the ground plane 111. The toe side point 117 is defined as the intersection point of the vertical projection of the lower tangent point (described above) onto the ground plane 111. Accordingly, the distance between the heel side point 116 and the toe side point 117 is the sole length L_B of the golf club head 100. The club head height H_{CH} is defined as the distance between the ground plane 111 and the uppermost point of the club head in a direction parallel to the z-up axis 171.

[0259] Referring to FIG. 2, the golf club head 100 includes a club head front-to-back depth D_{CH} defined as the distance between two points 118, 119 projected onto the ground plane 111. A forward end point 118 is defined as the intersection of the projection of the leading edge 143 onto the ground plane 111 in a direction parallel to the z-up axis 171. A rearward end point 119 is defined as the intersection of the projection of the rearward-most point of the club head onto the ground plane 111 in a direction parallel to the z-up axis 171. Accordingly, the distance between the forward end point 118 and rearward end point 119 of the golf club head 100 is the depth D_{CH} of the golf club head 100.

[0260] Referring to FIGS. 3 and 6-9, the body 113 of the golf club head 100 further includes a sole bar 135 that defines a rearward portion of the sole portion 108 of the body 113. The sole bar 135 has a relatively large thickness in relation to the strike plate 109 and other portions of the golf club head 100. Accordingly, the sole bar 135 accounts for a significant portion of the mass of the golf club head 100 and effectively shifts the CG of the golf club head 100 relatively lower and rearward. As particularly shown in FIG. 3, the sole portion 108 of the body 113 includes a forward portion 189 with a thickness less than that of the sole bar 135. The forward portion 189 is located between the sole bar 135 and the strike face 110. As described more fully below, the body 113 includes a channel 150 formed in the sole portion 108 between the sole bar 135 and the strike face 110 to effectively separate the sole bar 135 from the strike face 110. The channel 150 is located closer to the forward end point 118 than the rearward end point 119.

[0261] In certain embodiments of the golf club head 100, such as those where the strike plate 109 is separately formed and attached to the body 113, the strike plate 109 can be formed of forged maraging steel, maraging stainless steel, or precipitation-hardened (PH) stainless steel. In general, maraging steels have high strength, toughness, and malleability. Being low in carbon, maraging steels derive their strength from precipitation of inter-metallic substances other than carbon. The principle alloying element is nickel (e.g., 15% to nearly 30%). Other alloying elements producing inter-metallic precipitates in these steels include cobalt, molybdenum, and titanium. In one embodiment, the maraging steel contains 18% nickel. Maraging stainless steels have less nickel than maraging steels but include significant chromium to inhibit rust. The chromium augments hardenability despite the reduced nickel content, which ensures the steel can transform to martensite when appropriately heat-treated. In another embodiment, a maraging stainless steel C455 is utilized as the strike plate 109. In other embodiments, the strike plate 109 is a precipitation hardened stainless steel such as 17-4, 15-5, or 17-7. After forming the strike plate 109 and the body 113 of the golf club head 100, the contact surfaces of the strike plate 109 and the body 113 can be finish-machined to ensure a good interface contact surface is provided prior to welding. In some embodiments, the contact surfaces are planar for ease of finish machining and engagement.

[0262] The strike plate 109 can be forged by hot press forging using any of the described materials in a progressive series of dies. After forging, the strike plate 109 is subjected to heat-treatment. For example, 17-4 PH stainless steel forgings are heat treated by 1040° C. for 90 minutes and then solution quenched. In another example, C455 or C450

stainless steel forgings are solution heat-treated at 830° C. for 90 minutes and then quenched.

[0263] In some embodiments, the body 113 of the golf club head 100 is made from 17-4 steel. However another material such as carbon steel (e.g., 1020, 1030, 8620, or 1040 carbon steel), chrome-molybdenum steel (e.g., 4140 Cr—Mo steel), Ni—Cr—Mo steel (e.g., 8620 Ni—Cr—Mo steel), austenitic stainless steel (e.g., 304, N50, or N60 stainless steel (e.g., 410 stainless steel) can be used.

[0264] In addition to those noted above, some examples of metals and metal alloys that can be used to form the components of the parts described include, without limitation: titanium alloys (e.g., 3-2.5, 6-4, SP700, 15-3-3-3, 10-2-3, or other alpha/near alpha, alpha-beta, and beta/near beta titanium alloys), aluminum/aluminum alloys (e.g., 3000 series alloys, 5000 series alloys, 6000 series alloys, such as 6061-T6, and 7000 series alloys, such as 7075), magnesium alloys, copper alloys, and nickel alloys.

[0265] In still other embodiments, the body 113 and/or the strike plate 109 of the golf club head 100 are made from fiber-reinforced polymeric composite materials and are not required to be homogeneous. Examples of composite materials and golf club components comprising composite materials are described in U.S. Patent Application Publication No. 2011/0275451, published Nov. 10, 2011, which is incorporated herein by reference in its entirety.

[0266] The body 113 of the golf club head 100 can include various features such as weighting elements, cartridges, and/or inserts or applied bodies as used for CG placement, vibration control or damping, or acoustic control or damping. For example, U.S. Pat. No. 6,811,496, incorporated herein by reference in its entirety, discloses the attachment of mass altering pins or cartridge weighting elements.

[0267] In some embodiments, the golf club head 100 includes a flexible boundary structure (“FBS”) at one or more locations on the golf club head 100. Generally, the FBS feature is any structure that enhances the capability of an adjacent or related portion of the golf club head 100 to flex or deflect and to thereby provide a desired improvement in the performance of the golf club head 100. The FBS feature may include, in several embodiments, at least one slot, at least one channel, at least one gap, at least one thinned or weakened region, and/or at least one of any of various other structures. For example, in several embodiments, the FBS feature of the golf club head 100 is located proximate the strike face 110 of the golf club head 100 in order to enhance the deflection of the strike face 110 upon impact with a golf ball during a golf swing. The enhanced deflection of the strike face 110 may result, for example, in an increase or in a desired decrease in the coefficient of restitution (“COR”) of the golf club head 100. When the FBS feature directly affects the COR of the golf club head 100, the FBS may also be termed a COR feature. In other embodiments, the increased perimeter flexibility of the strike face 110 may cause the strike face 110 to deflect in a different location and/or different manner in comparison to the deflection that occurs upon striking a golf ball in the absence of the channel, slot, or other flexible boundary structure.

[0268] In the illustrated embodiment of the golf club head 100, the FBS feature is a channel 150 that is located on the sole portion 108 of the golf club head 100. As indicated above, the FBS feature may comprise a slot, a channel, a gap, a thinned or weakened region, or other structure. For clarity, however, the descriptions herein will be limited to

embodiments containing a channel, such as the channel 150, with it being understood that other FBS features may be used to achieve the benefits described herein.

[0269] Referring to FIG. 3, the channel 150 is formed into the sole portion 108 and extends generally parallel to and spaced rearwardly from the strike face 110. Moreover, the channel 150 is defined by a forward wall 152, a rearward wall 154, and an upper wall 156. The rearward wall 154 is a forward portion of the sole bar 135. The channel 150 includes an opening 158 defined on the sole portion 108 of the golf club head 100. The forward wall 152 further defines, in part, a first hinge region 160 located at the transition from the forward portion of the sole 108 to the forward wall 152, and a second hinge region 162 located at a transition from an upper region of the forward wall 152 to the sole bar 135. The first hinge region 160 and the second hinge region 162 are portions of the golf club head 100 that contribute to the increased deflection of the strike face 110 of the golf club head 100 due to the presence of the channel 150. In particular, the shape, size, and orientation of the first hinge region 160 and the second hinge region 162 are designed to allow these regions of the golf club head 100 to flex under the load of a golf ball impact. The flexing of the first hinge region 160 and second hinge region 162, in turn, creates additional deflection of the strike face 110.

[0270] The hosel 114 of the golf club head 100 can have any of various configurations, such as shown and described in U.S. Pat. No. 9,731,176. For example, the hosel 114 may be configured to reduce the mass of the hosel 114 and/or facilitate adjustability between a shaft and the golf club head 100. For example, the hosel 114 may include a notch 177 that facilitates flex between the hosel 114 and the body 113 of the golf club head 100.

[0271] The topline portion 106 of the golf club head 100 can have any of various configurations, such as shown and described in U.S. Pat. No. 9,731,176. For example, the topline portion 106 of the golf club head 100 may include weight reducing features to achieve a lighter weight topline. According to one embodiment shown in FIG. 9, the weight reducing features of the topline portion 106 of the golf club head 100 include a variable thickness of the top wall 169 defining the topline portion 106. More specifically, in a direction lengthwise along the topline portion 106, the thickness of the top wall 169 alternates between thicker and thinner so as to define pockets 190 between ribs 192 or pads. The pockets 190 are those portions of the top wall 169 having a thickness less than that of the portions of the top wall 169 defining the ribs 192. The pockets 190 help to reduce mass in the topline portion 106, while the ribs 192 promote strength and rigidity of the topline portion 106 and provide a location where a bridge bar 140 can be fixed to the topline portion 106 as is explained in more detail below. As shown in FIG. 9, the alternating wall thickness of the top wall 169 can extend into the toe wall forming the toe portion 104. In the illustrated embodiment, the top wall 169 includes two pockets 190 and three ribs 192. However, in other embodiments, the top wall 169 can include more or less than two pockets 190 and three ribs 192.

[0272] Referring to FIGS. 6-9, the back portion 128 of the golf club head 100 includes a bridge bar 140 that extends uprightly from the sole bar 135 to the topline portion 106. As defined herein, uprightly can be vertically or at some angle greater than zero relative to horizontal. The bridge bar 140 structurally interconnects the sole bar 135 directly with the

topline portion 106 without being interconnected directly with the strike plate 109. In other words, the bridge bar 140 is directly coupled to a top surface 157 of the sole bar 135, at a top end 144 of the bridge bar 140, and a bottom surface 159 of the topline portion 106, at a bottom end 142 of the bridge bar 140. However, the bridge bar 140 is not directly coupled to the strike plate 109. In fact, an unoccupied gap or space is present between the bridge bar 140 and the rear surface 131 of the strike plate 109. The bridge bar 140 can be made of the same above-identified materials as the body 113 of the golf club head 100. Alternatively, the bridge bar 140 can be made of a material that is different than that of the rest of the body 113. However, the material of the bridge bar 140 is substantially rigid so that the portions of the golf club head 100 coupled to the bridge bar 140 are rigidly coupled. The bridge bar 140 is non-movably or rigidly fixed to the sole bar 135 and the topline portion 106. In one embodiment, the bridge bar 140 is co-formed (e.g., via a casting technique) with the topline portion 106 and the sole bar 135 so as to form a one-piece, unitary, seamless, and monolithic, construction with the topline portion 106 and the sole bar 135. However, according to another embodiment, the bridge bar 140 is formed separately from the topline portion 106 and the sole bar 135 and attached to the topline portion 106 and the bridge bar 140 using any of various attachment techniques, such as welding, bonding, fastening, and the like. In some implementations, when attached to or formed with the topline portion 106 and the sole bar 135, the bridge bar 140 is not under compression or tension.

[0273] The bridge bar 140 spans the cavity 161, and more specifically, spans an opening 163 to the cavity 161 of the golf club head 100. The opening 163 is at the back portion 128 of the golf club head 100 and has a length L_O extending between the toe portion 104 and the heel portion 102. The bridge bar 140 also has a length L_{BB} and a width W_{BB} transverse to the length L_{BB} . The length L_{BB} of the bridge bar 140 is the maximum distance between the bottom end 142 of the bridge bar 140 and the top end 144 of the bridge bar 140. The length L_{BB} of the bridge bar 140 is less than the length L_O . The width W_{BB} of the bridge bar 140 is the minimum distance from a given point on one elongated side of the bridge bar 140 to the opposite elongated side of the bridge bar 140 in a direction substantially parallel with the x-axis 105 (e.g., heel-to-toe direction). The width W_{BB} of the bridge bar 140 is less than the length L_O of the opening 163. In one implementation, the width W_{BB} of the bridge bar 140 is less than 20% of the length L_O . According to another implementation, the width W_{BB} of the bridge bar 140 is less than 10% or 5% of the length L_O . The width W_{BB} of the bridge bar 140 can be greater at the bottom end 142 than at the top end 144 to promote a lower Z-up. Alternatively, the width W_{BB} of the bridge bar 140 can be greater at the top end 144 than at the bottom end 142 to promote a higher Z-up. In yet other implementations, the width W_{BB} of the bridge bar 140 is constant from the top end 144 to the bottom end 142. In some implementations, the length L_{BB} of the bridge bar 140 is 2-times, 3-times, or 4-times the width W_{BB} of the bridge bar 140.

[0274] Referring to FIG. 6, an areal mass of the rear portion 128 of the golf club head 100 between the topline portion 106, the sole portion 108, the toe portion 104, and the heel portion 102 is between 0.0005 g/mm² and 0.00925 g/mm², such as, for example, about 0.0037 g/mm². Generally, the areal mass of the rear portion 128 is the mass per

unit area of the area defined by the opening **163** to the cavity **161**. In some implementations, the area of the opening **163** is about 1,600 mm².

[0275] In some embodiments, the golf club head may include a topline portion weight reduction zone that includes weight reducing features that yield a mass per unit length within the topline portion weight reduction zone of between about 0.09 g/mm to about 0.40 g/mm, such as between about 0.09 g/mm to about 0.35 g/mm, such as between about 0.09 g/mm to about 0.30 g/mm, such as between about 0.09 g/mm to about 0.25 g/mm, such as between about 0.09 g/mm to about 0.20 g/mm, or such as between about 0.09 g/mm to about 0.17 g/mm. In some embodiments, the topline portion weight reduction zone yields a mass per unit length within the weight reduction zone less than about 0.25 g/mm, such as less than about 0.20 g/mm, such as less than about 0.17 g/mm, such as less than about 0.15 g/mm, or such as less than about 0.10 g/mm. The golf club head has a topline portion made from a metallic material having a density between about 7,700 kg/m³ and about 8,100 kg/m³, e.g., steel. If a different density material is selected for the topline construction that could either increase or decrease the mass per unit length values. The weight reducing features may be applied over a topline length of at least 10 mm, such as at least 20 mm, such as at least 30 mm, such as at least 40 mm, such as at least 45 mm, such as at least 50 mm, such as at least 55 mm, or such as at least 60 mm.

[0276] Additional and different golf club head features may be included in one or more embodiments. For example, additional golf club head features are described in U.S. Pat. Nos. 10,406,410, 10,155,143, 9,731,176, 9,597,562, 9,044,653, 8,932,150, 8,535,177, and 8,088,025, which are incorporated by reference herein in their entireties. Additional and different golf club head features are also described in U.S. Patent Application Publication No. 2018/0117425, published May 3, 2018, which is incorporated by reference herein in its entirety. Additional and different golf club head features are also described in U.S. Patent Publication No. 2019/0381370, published Dec. 19, 2019, which is incorporated by reference herein in its entirety.

Coefficient of Restitution and Characteristic Time

[0277] As used herein, the terms “coefficient of restitution,” “COR,” “relative coefficient of restitution,” “relative COR,” “characteristic time,” and “CT” are defined according to the following. The coefficient of restitution (COR) of an iron club head is measured according to procedures described by the USGA Rules of Golf as specified in the “Interim Procedure for Measuring the Coefficient of Restitution of an Iron Club head Relative to a Baseline Plate,” Revision 1.2, Nov. 30, 2005 (hereinafter “the USGA COR Procedure”). Specifically, a COR value for a baseline calibration plate is first determined, then a COR value for an iron club head is determined using golf balls from the same dozen(s) used in the baseline plate calibration. The measured calibration plate COR value is then subtracted from the measured iron club head COR to obtain the “relative COR” of the iron club head. The COR values are obtained by striking the strike face of the golf club head with a golf ball traveling at a speed of about 133 ft/s.

[0278] To illustrate by way of an example: following the USGA COR Procedure, a given set of golf balls may produce a measured COR value for a baseline calibration plate of 0.845. Using the same set of golf balls, an iron club

head may produce a measured COR value of 0.825. In this example, the relative COR for the iron club head is 0.825–0.845=–0.020. This iron club head has a COR that is 0.020 lower than the COR of the baseline calibration plate, or a relative COR of –0.020.

[0279] The characteristic time (CT) is the contact time between a metal mass attached to a pendulum that strikes the face center of the golf club head at a low speed under conditions prescribed by the USGA club conformance standards.

Damper and Badge Structures

[0280] As manufacturers of iron-type golf club heads design cavity-back club heads for a high moment of inertia (MOI), low center of gravity (CG), and other characteristics, acoustic and vibration dampers may be provided to counteract unpleasant sounds and vibration frequencies produced by features of the club heads, such as resulting from thin toplines, thin strike faces, and other club head characteristics. Heel-to-toe badges and/or dampers may be provided such that unpleasant sounds and vibration frequencies are dampened, while maintaining acceptable COR and CT values for the strike face. Heel-to-toe badges and/or dampers may also be provided with relief cutouts (also referred to as channels and grooves, such as to provide projection or ribs on the damper) to maintain COR and CT values of the strike face, improve COR and CT values for off-center strikes, and to provide for a larger “sweet-spot” on the strike face.

[0281] FIG. 10 illustrates one embodiment of a damper **280** of an iron-type golf club head. The damper **280** includes one or more relief cutouts **281a-281g** on front surface **284** that reduce the surface area of the damper **280** that contacts a rear surface of the strike face. Any number of relief cutouts may be provided. The damper **280** includes one or more projections **282a-282h** on front surface **284** that contact the rear surface of the strike face. Any number of projections may be provided. The number of projections may correspond with the number of relief cutouts. For example, as depicted in FIG. 10, damper **280** has one more projection than relief cutout, such that the damper **280** contacts the rear surface of the strike face on both sides of each relief cutout. In another embodiment, the damper **280** may have fewer projections than relief cutouts. In yet another embodiment, the damper **280** may have an equal number of projections and relief cutouts.

[0282] In one or more embodiments, the width and shape of each of the relief cutouts **281a-281g** and each of the projections **282a-282h** may differ in order to provide different damping characteristics of the damper **280** (e.g., sound and feel) and different performance characteristics at different locations across the strike face (e.g., CT and COR). For example, wide relief cutouts may be provided in the damper **280** near the ideal strike location (e.g., location **101** in FIG. 1) to retain more COR while still benefiting from sound and feel across the strike face. In another example, narrow relief cutouts may be provided in the damper **280** at the ideal strike location to provide for better sound and feel at the expense of reduced performance characteristics. In yet another example, uniform cutouts may be provided in the damper **280** to provide for a balance between sound and feel with performance characteristics.

[0283] In one or more embodiments, the relief cutout widths may provide for zones of contact by the projections of the damper. For example, in a damper with wider pro-

jections near the ideal strike location of the strike face, the damper will provide for better damping near the ideal strike location and will maintain a greater percentage of COR and CT near the heel and toe locations of the strike face. By maintaining a greater percentage of COR and CT near the heel and toe locations of the strike face, a perceived “sweet spot” of the strike face can be enlarged, providing for more consistent COR and CT across the strike face, resulting in consistent ball speeds resulting from impact across the strike face.

[0284] To provide for adequate sound and vibration damping, and to meet other club head specifications, the amount of surface area that the damper contacts the strike face determines the level of damping provided by the damper and impacts the performance specifications of the club head. For example, the damper need not be compressed to provide for damping. For example, the damper may move with the strike face, while still providing for sound and vibration damping. However, in some embodiments, the damper is compressed by the strike face. For example, a strike face may flex up to about 1.5 mm. In embodiments where the damper **280** is compressed, the damper may be compressed up to about 0.3 mm, up to about 0.6 mm, up to about 1.0 mm, up to about 1.5 mm, or up to another distance.

[0285] The damper **280** can be described by a projection ratio of the surface area of the projections contacting the strike face to a surface area of a projected area of the entire damper **280** (i.e., a combined surface area of the projections and the relief cutouts). In one or more embodiments, the projection ratio is no more than about 25%, between about 25% and 50%, or another percentage. In some embodiments, the surface area of the entire damper **280** is more than about 2 times the surface area of the projections, such as about 2.3 times (i.e., $542 \text{ mm}^2/235 \text{ mm}^2$), about 2.2 times (i.e., $712 \text{ mm}^2/325 \text{ mm}^2$), or about 1.8 times (i.e., $722 \text{ mm}^2/396 \text{ mm}^2$). Dampers with other ratios may be provided. For example, a numerically higher projection ratio (e.g., about 50%) may provide for increased vibration and sound damping at the expense of performance characteristics. Likewise, a numerically lower projection ratio (e.g., about 25%) may provide for increased performance characteristics at the expense of vibration and sound damping.

[0286] As depicted in FIG. 10, the damper **280** may include alternating projections **282a-282h** and relief cutouts **281a-281g**. The alternating projections **282a-282h** and relief cutouts **281a-281g** reduces the surface area of the projected surface of the damper **280** from contacting a rear surface of the strike face. By providing the relief cutouts **281a-281g** in the damper **280**, flexibility of the strike face can be maintained when compared to a solid damper (i.e., a damper without relief). In one embodiment, when compared to a solid damper that reduces COR of a strike face by about 5 points, a damper with relief cutouts may reduce COR of the strike face by only about 2.5 points. In another embodiment, when compared to a solid damper, a damper with relief cutouts may reduce COR of the strike face by 4 points less than the solid damper.

[0287] The damper **280** may be provided in any shape suitable to fit within the cavity and provide for vibration and sound damping. In one or more embodiments, the damper **280** may be provided with a tapered profile that reaches a peak height adjacent to a toeside of the damper. For example, the damper **280** may have a length of about 75 mm measured from the heel portion to the toe portion, a toeside

height of about 16 mm, and heelside height of about 10 mm. In another example, the toeside height is no less than twice the heelside height. Other measurements may be provided, such as a length of greater than 40 mm measured from the heel portion to the toe portion, greater than 50 mm measured from the heel portion to the toe portion, greater than 60 mm measured from the heel portion to the toe portion, greater than 70 mm measured from the heel portion to the toe portion, or another length.

[0288] In one or more embodiments, the golf club head may include strike face of a golf club head may include localized stiffened regions, variable thickness regions, or inverted cone technology (ICT) regions located on the strike face at a location that surrounds or that is adjacent to the ideal striking location of the strike face. In these embodiments, additional features may be provided by the damper **280** to accommodate for the localized stiffened regions, variable thickness regions, or ICT regions. For example, the damper **280** may include a cutout **283** provided to receive and/or contact a portion of the strike face corresponding to a localized stiffened region, a variable thickness region, or an ICT region. As such, the cutout **283** is provided to match a shape of the region, such as a circular region, an elliptical region, or another shape of the region. In one example, the cutout **283** receives, but does not contact, at least a portion of the of a rear surface of the localized stiffened region, variable thickness region, or ICT region. In another example, the cutout **283** receives and is in contact with at least a portion of the rear surface of the localized stiffened region, variable thickness region, or ICT region. In this example, the damper contacts less than about 50% of the rear surface area, less than about 40%, or another portion of the rear surface area.

[0289] In one or more embodiments, the damper **280** is provided in lieu of localized stiffened regions, variable thickness regions, or ICT regions located on the strike face. For example, the damper **280** may be provided with characteristics that stiffen a localized region of the strike face more than surrounding regions of the strike face, such as to increase the durability of the club head strike face, to increase the area of the strike face that produces high CT and/or COR, or a combination of these reasons. To stiffen a localized region of the strike face, relief cutouts may be provided adjacent to the localized region, resulting in a stiffened local region and one or more flexible adjacent regions. Additional and different relief cutouts may be provided to effectuate localized stiffened regions of the strike face using the damper **280**.

[0290] In one or more embodiments, additional relief cutouts may be provided on any surface of the damper **280**, such as a top surface **285**, an intermediate surface **286**, a rear surface **287**, or another surface, such as depicted in FIG. 11. For example, the additional relief cutouts may be provided for weight savings, water drainage from the cavity, ease of damper installation, aesthetic characteristics, and to provide other performance benefits.

[0291] In one or more embodiments, relief cutouts on the front surface **284** and/or the intermediate surface **286** of the damper **280** provide for a volume and mass savings compared to a damper without relief cutouts. In one example, a damper without relief cutouts is 7589 mm^3 with a mass of 9.9 g. Providing relief cutouts on the front surface **284** reduces the volume of the damper to 7278 mm^3 and reduces the mass to 9.5 g, providing a 4.1% mass savings. Providing

relief cutouts on the front surface **284** and the intermediate surface **286** reduces the volume of the damper to 6628 mm³ and reduces the mass to 8.6 g, providing a 12.7% mass savings. In another example, another damper without relief cutouts is 5976 mm³ with a mass of 7.8 g. Providing relief cutouts on the front surface **284** reduces the volume of the damper to 5608 mm³ and reduces the mass to 7.3 g, providing a 6.1% mass savings. Providing relief cutouts on the front surface **284** and the intermediate surface **286** reduces the volume of the damper to 4847 mm³ and reduces the mass to 6.3 g, providing an 18.7% mass savings.

[0292] FIGS. 11-12 illustrate additional views of one embodiment of a damper **280** of an iron-type golf club head. The damper **280** includes a top surface **285**, an intermediate rear surface **286**, and a rear surface **287**. Additional and different surfaces may be provided.

[0293] In one or more embodiments, relief cutouts are provided in the top surface **285** of the damper **280**. For example, one or more relief cutouts **281a-281g** on front surface **284** (depicted in FIG. 10) may extend to the top surface **285**. The relief cutouts provided in the top surface **285** may allow for water trapped in front of the damper **280** to drain from the cavity. The relief cutouts provided in the top surface **285** may also provide for aesthetic benefits, such as allowing the damper to be more pleasing to the golfer and to blend into the feature lines of the golf club head. The relief cutouts provided in the top surface **285** may also provide for weight savings and may add to the flexibility of the damper for ease of installation into the cavity. Any number of relief cutouts may be provided in the top surface **285**.

[0294] In one or more embodiments, relief cutouts are also provided in the intermediate rear surface **286** of the damper **280**. The relief cutouts provided in the intermediate rear surface **286** may also provide for weight savings and may add to the flexibility of the damper for ease of installation into the cavity. Any number of relief cutouts may be provided in the intermediate rear surface **286**. Projections may also be provided in the intermediate rear surface **286** for contact with a rear portion and/or a sole bar of the club head. In an example, uniform projections and uniform relief cutouts are provided in the intermediate rear surface **286**. In this example, the intermediate rear surface **286** includes the same number of projections as the front surface **284**. In another example, the intermediate rear surface **286** includes more projections than the front surface **284**. In another example, the intermediate rear surface **286** includes fewer projections than the front surface **284**.

[0295] FIG. 11 also illustrates one embodiment of a badge **288** of an iron-type golf club head. The badge **288** may be positioned above the damper **280** within the cavity of the club head. For example, the badge **288** may be adhesively secured or otherwise mechanically attached or connected to the rear surface of the strike face. The badge **288** may be provided in any shape. For example, the badge **288** may be provided in a tapered shape, with a peak height adjacent to the toeside of the badge. The badge **288** may provide additional vibration and sound damping, as well as serve aesthetic purposes within the cavity. In one or more embodiments, the damper **280** extends a greater distance from heel to toe than the badge **288**.

[0296] In some embodiments, the damper **280** is provided with a pattern or other relief on the front surface **284** that reduces the surface area of the damper **280** that contacts a

rear surface of the strike face. Any type of relief may be provided that reduces the surface area of the front surface of the damper that contacts the rear surface of the strike face. For example, the damper **280** may be provided with a honeycomb pattern, a cross-cut pattern, a nubbin pattern, pattern, another pattern, or a pattern inversion. The pattern and/or other relief may be symmetrical across the front surface of the damper, or the pattern may vary across the front surface. The pattern and/or other relief provides that less than 100% of the front surface of the damper contact the rear surface of the strike face, such as 20% to 80% of the projected area of the front surface of the damper contacting the rear surface of the strike face.

[0297] Additional and different golf club badge and/or damper features may be included in one or more embodiments. For example, additional golf club badge and/or damper features are described in U.S. Pat. Nos. 10,427,018, 9,937,395, and 8,920,261, which are incorporated by reference herein in their entireties.

Damper Materials

[0298] A variety of materials and manufacturing processes may be used in providing the damper **280**. In one or more embodiments, the damper **280** is a combination of Santoprene and Hybrar. For example, using different ratios of Santoprene to Hybrar, the durometer of the damper **280** may be manipulated to provide for different damping characteristics, such as interference, dampening, and stiffening properties. In one embodiment, a ratio of about 85% Santoprene to about 15% Hybrar is used. In another embodiment, a ratio of at least about 80% Santoprene to about 10% Hybrar is used. Other ratios may be used.

[0299] Examples of materials that may be suitable for use as a damper structure include, without limitation: viscoelastic elastomers; vinyl copolymers with or without inorganic fillers; polyvinyl acetate with or without mineral fillers such as barium sulfate; acrylics; polyesters; polyurethanes; polyethers; polyamides; polybutadienes; polystyrenes; polyisoprenes; polyethylenes; polyolefins; styrene/isoprene block copolymers; hydrogenated styrenic thermoplastic elastomers; metallized polyesters; metallized acrylics; epoxies; epoxy and graphite composites; natural and synthetic rubbers; piezoelectric ceramics; thermoset and thermoplastic rubbers; foamed polymers; ionomers; low-density fiber glass; bitumen; silicone; and mixtures thereof. The metallized polyesters and acrylics can comprise aluminum as the metal. Commercially available materials include resilient polymeric materials such as Scotchweld™ (e.g., DP-105™) and Scotchdamp™ from 3M, Sorbothane™ from Sorbothane, Inc., DYAD™ and GP™ from Soundcoat Company Inc., Dynamat™ from Dynamat Control of North America, Inc., NoViFlex™ Sylomer™ from Pole Star Maritime Group, LLC, Isoplast™ from The Dow Chemical Company, Legetolex™ from Piqua Technologies, Inc., and Hybrar™ from the Kuraray Co., Ltd.

[0300] In some embodiments, the filler material may have a modulus of elasticity ranging from about 0.001 GPa to about 25 GPa, and a durometer ranging from about 5 to about 95 on a Shore D scale. In other examples, gels or liquids can be used, and softer materials which are better characterized on a Shore A or other scale can be used. The Shore D hardness on a polymer is measured in accordance with the ASTM (American Society for Testing and Materials) test D2240.

[0301] In some embodiments, the damper material may have a density of about 0.95 g/cc to about 1.75 g/cc, or about 1 g/cc. The damper material may have a hardness of about 10 to about 70 shore A hardness. In certain embodiments, a shore A hardness of about 40 or less is preferred. In certain embodiments, a shore D hardness of up to about 40 or less is preferred.

[0302] In some embodiments, the damper material may have a density between about 0.16 g/cc and about 0.19 g/cc or between about 0.03 g/cc and about 0.19 g/cc. In certain embodiments, the density of the damper material is in the range of about 0.03 g/cc to about 0.2 g/cc, or about 0.04-0.10 g/cc. The density of the damper material may impact the COR, durability, strength, and damping characteristics of the club head. In general, a lower density material will have less of an impact on the COR of a club head. The damper material may have a hardness range of about 15-85 Shore OO hardness or about 80 Shore OO hardness or less.

[0303] In one or more embodiments, the damper 280 may be provided with different durometers across a length of the damper 280. For example, the damper 280 may be co-molded using different materials with different durometers, masses, densities, colors, and/or other material properties. In one embodiment, the damper 280 may be provided with a softer durometer adjacent to the ideal striking location of the strike face than adjacent to the heel and toe portions. In another embodiment, the damper 280 may be provided with a harder durometer adjacent to the ideal striking location of the strike face than adjacent to the heel and toe portions. In these examples, the different material properties used to co-mold the damper 280 may provide for better performance and appearance.

[0304] Additional and different damper materials and manufacturing processes can be used in one or more embodiments. For example, additional damper materials and manufacturing processes are described in U.S. Pat. Nos. 10,427,018, 9,937,395, 9,044,653, 8,920,261, and 8,088,025, which are incorporated by reference herein in their entireties. For example, the damper 280 may be manufactured at least in part of rubber, silicone, elastomer, another relatively low modulus material, metal, another material, or any combination thereof.

Club Head and Damper Interaction

[0305] FIG. 13 illustrates one embodiment of the damper 280 positioned within the cavity 161 of a golf club head 100. For example, the damper 280 is inserted from a toeside of the club head 100 into the cavity 161. Likewise, a badge 288 (not depicted) may also be inserted from the toeside of the golf club head and affixed within the cavity 161. In one or more embodiments, the damper 280 is positioned low in the cavity 161 below an upper edge of the rear portion 128 (i.e., below the cavity opening line). For example, the damper 280 is positioned about 1 mm below an upper edge of the upper edge of the rear portion 128. The damper may also be positioned below the badge 288.

[0306] As discussed above, in one or more embodiments, the damper 280 may include relief cutouts on one or more surfaces of the damper 280 which allow water to drain out of the cavity 161 from below and around the damper 280. For example, if the club head 100 is submerged in a water bucket, such as for cleaning, the relief cutouts allow water to drain from the cavity 161. In testing embodiments of the damper 280, a club head 100 without the relief cutouts

retained 1.2 g of water. In contrast, a club head 100 with the relief cutouts retained only 0.3 g of water.

[0307] FIG. 14 illustrates a cross-section view of one embodiment of the damper 280 positioned within the cavity 161 of a golf club head 100. The front surface 284 of the damper 280 contacts a rear surface of the strike face 110. The intermediate surface 286 and the rear surface 287 of the damper 280 each contact the rear portion 128 and/or the sole bar 135. As depicted in FIG. 14, the damper 280 contacts the strike face 110, the rear portion 128 and/or the sole bar 135 at varying heights within the cavity 161. Further, channel 150 may be rearward intermediate surface 286.

[0308] In one or more embodiments, a badge 288 may also be positioned within the cavity 161. As depicted in FIG. 14, the badge 288 is positioned above the damper 280 and separated from the damper 280. For example, the damper 280 and the badge 288 may be separated by about 1 mm or another distance. In another embodiment, the badge 288 is positioned above of and in contact with the damper 280. In this embodiment, the badge 288 may lock the damper in place within the cavity 161. The badge 288 may be an ABS plastic or another material, secured within the cavity to the rear surface of the strike face 110 by an adhesive or tape. In one example, the badge is secured by tape with a thickness of about 1.1 mm, providing additional vibration and sound damping of the strike face 110. In some embodiments, the damper 280 extends rearward of the badge 288.

[0309] FIG. 15 illustrates another cross-section view of one embodiment of the damper 280 positioned within the cavity 161 of a golf club head 100. The heel portion 102 of the club head 100 includes a negative heel tab 196 for receiving the heel tab 293 of the damper 280. The toe portion 104 of the club head 100 includes a negative toe tab 195 for receiving the toe tab 294 of the damper 280. During installation, the damper 280 may be inserted into the cavity 161 and locked into place using the toe tab 294 and the heel tab 293. The club head 100 may also include a center tab 191 for further securing the damper 280 within the cavity 161.

[0310] As depicted in FIG. 15, a portion of the negative toe tab 195 overlaps a portion of the damper 280 when the damper 280 is positioned within the cavity 161. Likewise, a portion of the negative heel tab 196 overlaps a portion of the damper 280 when the damper 280 is positioned within the cavity 161. In one or more embodiments, the top edge of each of the negative toe tab 195, the center tab 191, and the negative heel tab 196 are substantially colinear.

[0311] In one or more embodiments, the damper 280 may be positioned in contact with a "donut" (not depicted in FIG. 15) of the strike face 110. For example, the damper 280 may be positioned in contact with a lower portion of the "donut," such as below the peak of the "donut." In some embodiments, the "donut" further secures the damper within the cavity 161.

[0312] In one or more embodiments, the damper 280 may be positioned in the cavity 161 and secured with an interference fit between the damper 280 and the body 113. For example, the damper 280 may be under compression when it is positioned within the cavity 161, such as at least 0.2 mm of compression, 0.4 mm of compression, 0.6 mm of compression, or another length of compression. In an embodiment, the front surface 284 of the damper 280 is compressed by at least 0.2 mm by the strike face 110 and the rear surface 287 is compressed by at least 0.2 mm by the rear portion

128. In another embodiment, the damper **280** is preloaded by about 0.6 mm by the damper **280** contacting the body **113**. **[0313]** FIG. **16** illustrates a cross-section view of another embodiment of the damper **280** positioned within the cavity **161** of a golf club head **100**. The front surface **284** of the damper **280** contacts a rear surface of the strike face **110**. The intermediate surface **286** and the rear surface **287** of the damper **280** each contact the rear portion **128** and/or the sole bar **135**. As depicted in FIG. **16**, the damper **280** contacts the strike face **110**, the rear portion **128** and/or the sole bar **135** at varying heights within the cavity **161**. Further, channel **150** may be rearward intermediate surface **286**.

[0314] FIG. **17** illustrates another cross-section view of one embodiment of the damper **280** positioned within the cavity **161** of a golf club head **100**. The heel portion **102** of the club head **100** includes a negative heel tab **196** for receiving the heel tab **293** of the damper **280**. The toe portion **104** of the club head **100** includes a negative toe tab **195** for receiving the toe tab **294** of the damper **280**. During installation, the damper **280** may be inserted into the cavity **161** and locked into place using the toe tab **294** and the heel tab **293**. The club head **100** may also include a center tab **191** for further securing the damper **280** within the cavity **161**.

[0315] As depicted in FIG. **17**, a portion of the negative toe tab **195** overlaps a portion of the damper **280** when the damper **280** is positioned within the cavity **161**. Likewise, a portion of the negative heel tab **196** overlaps a portion of the damper **280** when the damper **280** is positioned within the cavity **161**. In one or more embodiments, the top edge of each of the negative toe tab **195**, the center tab **191**, and the negative heel tab **196** are not substantially colinear.

Localized Stiffened Regions and Inverted Cone Technology

[0316] In one or more embodiments, the strike face of a golf club head may include localized stiffened regions, variable thickness regions, or inverted cone technology (ICT) regions located on the strike face at a location that surrounds or that is adjacent to the ideal striking location of the strike face. The aforementioned regions may also be referred to as a “donut” or a “thickened central region.” The regions may be circular, elliptical, or another shape. For example, the localized stiffened region may include an area of the strike face that has increased stiffness due to being relatively thicker than a surrounding region, due to being constructed of a material having a higher Young’s Modulus (E) value than a surrounding region, and/or a combination of these factors. Localized stiffened regions may be included on a strike face for one or more reasons, such as to increase the durability of the club head strike face, to increase the area of the strike face that produces high CT and/or COR, or a combination of these reasons.

[0317] Examples of localized stiffened regions, variable thickness configurations, and inverted cone technology regions are described in U.S. Pat. Nos. 6,800,038, 6,824, 475, 6,904,663, 6,997,820, and 9,597,562, which are incorporated by reference herein in their entireties. For example, ICT regions may include symmetrical “donut” shaped areas of increased thickness that are located within the unsupported face region. In some embodiments, the ICT regions are centered on the ideal striking location of the strike face. In other embodiments, the ICT regions are centered heelward of the ideal striking location of the strike face, such as to stiffen the heel side of the strike face and to add flexibility

to the toe side of the strike face, such as to reduce lateral dispersion (e.g., a draw bias) produced by the golf club head.

[0318] In some embodiments, the ICT region(s) include(s) an outer span and an inner span that are substantially concentric about a center of the ICT regions. For example, the outer span may have a diameter of between about 15 mm and about 25 mm, or at least about 20 mm. In other embodiments, the outer span may have a diameter greater than about 25 mm, such as about 25-35 mm, about 35-45 mm, or more than about 45 mm. The inner span of the ICT region may represent the thickest portion of the unsupported face region. In certain embodiments, the inner diameter may be between about 5 mm and about 15 mm, or at least about 10 mm.

[0319] In other embodiments, the localized stiffened region comprises a stiffened region (e.g., a localized region having increased thickness in relation to its surrounding regions) having a shape and size other than those described above for the inverted cone regions. The shape may be geometric (e.g., triangular, square, trapezoidal, etc.) or irregular. For these embodiments, a center of gravity of the localized stiffened region (CG_{LSR}) may be determined by defining a boundary for the localized stiffened region and calculating or otherwise determining the center of gravity of the defined region. An area, volume, and other measurements of the localized stiffened region are also suitable for measurement upon defining the appropriate boundary.

Club Head Measurements

[0320] FIG. **18** illustrates club head measurements that may apply to one or more embodiments, including club head **100**, club head **300**, or another club head. In one or more embodiments the golf club head **300**, as shown in FIG. **18**, the internal cavity **361** is partially or entirely filled with a filler material and/or a damper, such as a non-metal filler material of a thermoplastic material, a thermoset material, or another material. In other embodiments, the internal cavity **361** is not filled with a filler material and remains an unfilled or partially filled hollow cavity within the club head. In other embodiments, such as the club head **100**, as shown in FIG. **1**, the cavity **161** is not closed by a back wall and remains unfilled or partially filled with a filler material and/or a damper. In some embodiments, the golf club head **300** may include a face insert **310** that wraps from the face into the crown, topline, rear portion, and/or sole, such as in a face to crown to rear transition region **321** and/or a face to sole transition region **309**.

[0321] Referring back to FIG. **18**, club head **300** includes a sole bar **339**. A maximum sole bar height $H_{solebar}$ measured as the distance perpendicular from the ground plane (GP) to a top edge of the sole bar **339** when the golf club head is in proper address position on the ground plane, may be between 7.5 and 8 mm, between 6 mm and 9 mm, between 8 mm and 10 mm, between 9 and 12 mm, between 11 mm and 15 mm, or another distance.

[0322] FIG. **18** also shows the thicknesses of various portions of the golf club head **300**. The golf club head **300** has a topline thickness $T_{topline}$, a minimum face thickness $T_{facemin}$, a maximum face thickness $T_{facemax}$, a sole wrap thickness $T_{solewrap}$, a sole thickness T_{sole} , and a rear thickness T_{rear} . The topline thickness $T_{topline}$ is the minimum thickness of the wall of the body defining the top portion of the body of the golf club head. The minimum face thickness $T_{facemin}$ is the minimum thickness of the wall or plate of the

body defining the strike face of the body of the golf club head. The maximum face thickness $T_{facemax}$ is the maximum thickness of the wall or plate of the body defining the strike face of the body of the golf club head. The sole wrap thickness $T_{solewrap}$ is the minimum thickness of the wall of the body defining the transition between the strike face and the sole portion of the body of the golf club head. The sole thickness T_{sole} is the minimum thickness of the wall of the body defining the sole portion of the body of the golf club head. The rear thickness T_{rear} is the minimum thickness of the wall of the body defining the rear portion of the body or the rear panel of the golf club head.

[0323] In one or more embodiments, the topline thickness $T_{topline}$ is between 1 mm and 3 mm, inclusive (e.g., between 1.4 mm and 1.8 mm, inclusive), the minimum face thickness $T_{facemin}$ is between 2.1 mm and 2.4 mm, inclusive, the maximum face thickness $T_{facemax}$ (typically at center face or an ideal strike location **301**) is between 3.1 mm and 4.0 mm, inclusive, the sole wrap thickness $T_{solewrap}$ is between 1.2 mm and 3.3 mm, inclusive (e.g., between 1.5 mm and 2.8 mm, inclusive), the sole thickness T_{sole} is between 1.2 mm and 3.3 mm, inclusive (e.g., between 1.7 mm and 2.75 mm, inclusive), and/or the rear thickness T_{rear} is between 1 mm and 3 mm, inclusive (e.g., between 1.2 mm and 1.8 mm, inclusive). In certain embodiments, a ratio of the sole wrap thickness $T_{solewrap}$ to the maximum face thickness $T_{facemax}$ is between 0.40 and 0.75, inclusive, a ratio of the sole wrap thickness $T_{solewrap}$ to the maximum face thickness $T_{facemax}$ is between 0.4 and 0.75, inclusive (e.g., between 0.44 and 0.64, inclusive, or between 0.49 and 0.62, inclusive), a ratio of the topline thickness $T_{topline}$ to the maximum face thickness $T_{facemax}$ is between 0.4 and 1.0, inclusive (e.g., between 0.44 and 0.64, inclusive, or between 0.49 and 0.62, inclusive), and/or a ratio of the sole wrap thickness $T_{solewrap}$ to the maximum sole bar height $H_{solebar}$ is between 0.05 and 0.21, inclusive (e.g., between 0.07 and 0.15, inclusive). In certain embodiments, a ratio of a minimum thickness in the face to sole transition region **309** to $T_{facemax}$ is between 0.40 and 0.75, inclusive (e.g., between 0.44 and 0.64, preferably between 0.49 and 0.62), and a ratio of the minimum face thickness $T_{facemin}$ to the face to crown to rear transition region **321** (excluding the weld bead) is between 0.40 and 1.0, inclusive (e.g. between 0.44 and 0.64, preferably between 0.49 and 0.62).

[0324] In one or more embodiments, the strike face may be welded to the body (e.g., a cast body), defining the cavity behind the strike face and forward of the rear portion, such as by welding a strike plate welded to a face opening on the body. In some embodiments, the strike face is manufactured with a forging process and the body is manufactured with a casting process. The welded strike face may include an undercut portion that wraps underneath the cavity and forms part of the sole portion. The undercut portion of the topline portion may include a minimum topline thickness, such as 1 mm, 1.1 mm, 1.2 mm, 1.3 mm, 1.4 mm, less than 1.5 mm, or another thickness. In an embodiment, the minimum topline thickness is between 1.4 mm and 1.8 mm, 1.3 mm and 1.9 mm, 1 mm and 2.5 mm, or another thickness. The welded strike face may include an undercut portion that wraps above the cavity and forms part of the topline portion. The undercut portion of the sole portion may include a minimum sole thickness, such as 1.25 mm, 1.4 mm, 1.55 mm, less than 1.6 mm, or another thickness. In an embodiment, the minimum sole thickness is between 1.6 mm and 2

mm, 1.5 mm and 2.2 mm, 1 mm and 3 mm, or another thickness. In some embodiments, the strike face is integrally cast or forged with the body. In some embodiments, the body and the strike face form a one-piece, unitary, monolithic construction.

[0325] The golf club head may be described with respect to a coordinate system defined with respect to an ideal striking location. The ideal striking location defines the origin of a coordinate system in which an x-axis is tangential to the strike face at the ideal striking location and is parallel to a ground plane when the body is in a normal address position, a y-axis extends perpendicular to the x-axis and is also parallel to the ground plane, and a z-axis extends perpendicular to the ground plane, wherein a positive x-axis extends toward the heel portion from the origin, a positive y-axis extends rearwardly from the origin, and a positive z-axis extends upwardly from the origin.

[0326] The golf club head may also be described with respect to a central region of the golf club head. For example, the body may be described with respect to a central region defined by a location on the x-axis, such as $-25 \text{ mm} < x < 25 \text{ mm}$, $-20 \text{ mm} < x < 20 \text{ mm}$, $-15 \text{ mm} < x < 15 \text{ mm}$, $-30 \text{ mm} < x < 30 \text{ mm}$, or another location. In some embodiments, the aforementioned measurements and other features may be described with respect to the central region, such as maximum face thickness $T_{facemax}$ of 3.5 mm within the central region of the face. In some embodiments, the damper may be described with respect to the central region, such as having a length from the heel portion to the toe portion of between 80% to 150% of the length of the central region, between 30% to 200% of the length of the central region, or between other percentages. In one example, defining a central region at $-25 \text{ mm} < x < 25 \text{ mm}$ has a length of 50 mm. In this example, providing a damper having a length of 75 mm from the heel portion to the toe portion results in the damper being 150% of the length of the central region.

[0327] The golf club head may also be described with respect to other characteristics of the golf club head, such as a face length measured from the par line to the toe portion ending at approximately the Z-up location of the club head. In another example, the golf club head may be described with respect to the score lines of the face, such as from a heelward score line location to a toward score line location. In yet another example, the golf club head may be described by a blade length measured from a point on the surface of the club head on the toe side that is furthest from the ideal striking location on the x-axis to a point a point on the surface of the club head on the heel side that is furthest from the ideal striking location on the x-axis.

Additional Club Head Structure

[0328] FIG. 19 illustrates one embodiment of an iron-type golf club head **100** including a body **113** having a heel portion **102**, a toe portion **104**, a sole portion **108**, a topline portion **106**, a rear portion **128**, and a hosel **114**. The golf club head **100** is manufactured with a cavity **161** (not depicted in FIG. 19), and a shim or badge **188** is adhered, bonded, or welded to the body **113** to produce a cap-back iron, giving the appearance of a hollow-body iron. In this way, the golf club **100** can be manufactured with the performance benefits of a game improvement iron, while providing the appearance of a blade, player's iron, and/or a hollow-body iron.

[0329] For example, a cap-back iron can capitalize on the performance benefits of a low CG, cavity-back iron, and the sound and feel benefits of a hollow-body iron. For example, by using a lightweight and rigid shim or badge 188 to close a cavity opening 163 in the cavity 161, the golf club head can provide increased stiffness in the topline portion 106, while maintaining a low CG. Various shim or badge 188 arrangements and materials can be used, and a filler material and/or damper 180 can be included within the cavity 161 to improve sound and feel, while minimizing loss in COR.

[0330] In some embodiments, the club head 100 is manufactured using as a unitary cast body 113. In these embodiments, the heel portion 102, toe portion 104, sole portion 108, topline portion 106, rear portion 128, front portion (not depicted in FIG. 19 and including strike face 110), and hosel 114 are cast as a single body 113. A separately formed shim or badge 188 is then received at least in part by the body 113, such as by the topline portion 106 and the rear portion 128. In some embodiments, the club head 100 includes an upper ledge 193 (not depicted in FIG. 19) and a lower ledge 194 (not depicted in FIG. 19) configured to receive the shim or badge 188. In some embodiments, at least a portion of the rear surface of the strike face 110 can be machined or chemical etched before installing the shim or badge 188, such as to finish the surface to increase durability and/or to machine variable face thicknesses across the strike face 110. For example, in embodiments where the strike face 110 is cast from Ti as part of a unitary cast body 113, the rear surface of the strike face can be machined or chemical etched to remove the potentially brittle alpha case layer from the strike face.

[0331] The shim or badge 188 is separately formed from and affixed to the unitary cast body 113. For example, the shim or badge 188 can be bonded to exterior of club head (i.e., not bladder molded or co-molded) as a separately formed piece.

[0332] The shim or badge 188 is configured to close a cavity opening 163 in the cavity 161 and to form, enclose, or otherwise define an internal cavity. The volume of the internal cavity can be between about 1 cc and about 50 cc, and preferably between between 5 cc to 20 cc. In some embodiments, the volume of the internal cavity is between about 5 cc and about 30 cc, or between about 8 cc and about 20 cc. For the purposes of measuring the internal cavity volume herein, the shim or badge 188 is assumed to be removed and an imaginary continuous wall or substantially back wall is utilized to calculate the internal cavity volume.

[0333] The club head 100 can have an external water-displaced club head volume between about 15 cc and about 150 cc, preferably between 30 cc and 75 cc, preferably between 35 cc and 65 cc, more preferably between about 40 cc and about 55 cc. A water-displaced volume is the volume of water displaced when placing the fully manufactured club head 100 into a water bath and measuring the volume of water displaced by the club head 100. The water-displaced volume differs from the material volume of the club head 100, as the water-displaced volume can be larger than the material volume, such as due to including the enclosed internal cavity and/or other hollow features of the club head. In some embodiments, the external water-displaced club head volume can be between about 30 cc and about 90 cc, between about 30 cc and about 70 cc, between about 30 cc

and about 55 cc, between about 45 cc and about 100 cc, between about 55 cc and about 95 cc, or between about 70 cc and about 95 cc.

[0334] A ratio of the internal cavity volume to external water displaced club head volume can be between about 0.05 and about 0.5, between 0.1 and 0.4, preferably between 0.14 and 0.385. In some embodiments, the ratio of the internal cavity volume to external water displaced club head volume can be between 0.20 and 0.35, or between 0.23 and 0.30.

[0335] In some embodiments, the club head 100 is manufactured by casting or forging a body 113 without the strike face 110 and/or strike face 110. In these embodiments, the strike face 110 and/or strike face 110 can be welded or otherwise attached to the body 113. In some embodiments, at least part of the strike face 110 and/or strike face 110 wraps one or more of the heel portion 102, toe portion 104, sole portion 108, and/or topline portion 106. For example, the body 113 can be cast from a steel alloy (e.g., carbon steel with a modulus of elasticity of about 200 GPa) and the strike face 110 and/or strike face 110 can be cast or forged from higher strength steel alloy (e.g., stainless steel 17-4 with a modulus of elasticity of about 210 GPa or 4140 with a modulus of elasticity of about 205 GPa), from a titanium alloy (e.g., with a modulus of elasticity between 110 GPa and 120 GPa), or manufactured from another material. Examples of golf club head constructions are disclosed in U.S. Pat. No. 10,543,409, filed Dec. 29, 2016, issued Jan. 28, 2020, and U.S. Pat. No. 10,625,126, filed Sep. 15, 2017, issued Apr. 21, 2020, which are incorporated herein by reference in their entirety.

[0336] In some embodiments, the club head 100 is manufactured with an unfinished, raw surface material. In some embodiments, the club head 100 has a finished surface material, such as with a satin finish, a physical vapor deposition (PVD) coating, a quench polish quench (QPQ) coating, or another finish. In some embodiments, a color can be embedded into the club head 100 material before casting, forging, or another process. In these embodiments, the embedded color gives the club head 100 an appearance of having a finish applied, while allowing the color to last longer than a coating or another finish applied during manufacturing.

[0337] The club head 100 can have a Zup between about 10 mm and about 20 mm, more preferably less than 19 mm, more preferably less than 18 mm, more preferably less than 17 mm, more preferably less than 16 mm. As used herein, “Zup” means the CG z-axis location determined according to this above ground coordinate system. Zup generally refers to the height of the CG above the ground plane as measured along the z-axis. In some embodiments, the club head 100 has a CG location (without the shim) between about 17 mm and about 18 mm above the ground plane, or between about 15 mm and about 18 mm above the ground plane.

[0338] The club head 100 can have a moment of inertia (MOI) about the CGz (also referred to as “Izz”) of between about 180 kg-mm² and about 290 kg-mm², preferably between 205 kg-mm² and 255 kg-mm², a MOI about the CGx (also referred to as “Ixx”) of between about 40 kg-mm² and about 75 kg-mm², preferably between 50 kg-mm² and 60 kg-mm², and a MOI about the CGy (also referred to as “Iyy”) of between about 240 kg-mm² and about 300 kg-mm², preferably between 260 kg-mm² and 280 kg-mm². For example, by placing discretionary weight at the toe can

increase the MOI of the golf club resulting in a golf club that resists twisting and is thereby easier to hit straight even on mishits.

[0339] FIG. 20 illustrates cross-sectional back view of the golf club head 100. Numerals 2001, 2003, 2005, 2007, 2007, 2009, and 2011 refer to features of club head 100. The features of club head 100 may also be applicable to club heads 300, 500, and 600. As depicted, the heel portion 102, toe portion 104, sole portion 108, and/or topline portion 106 can include thinned regions. The thinned regions can redistribute discretionary weight within the club head 100. For example, including thinned region 2001 in the topline portion 106 can allow discretionary weight to be redistributed low, such as to lower the center of gravity of the golf club head 100. Targeted thick regions, such as thickened regions 2003, 2005, can be included to retain stiffness in the topline portion 106, such as to maintain acoustic frequencies, producing a better sound and feel of the golf club head 100. Likewise, thinned regions 2007, 2009 and a thickened region 2011 can be included the toe portion 102. For example, the thinned region 2001 can be between about 0.8 mm and about 1.4 mm, preferably between about 0.95 mm and about 1.25 mm. The thinned region 2007 can be between about 0.8 mm and about 2.5 mm, preferably between about 1.95 mm and about 2.25 mm, or between about 0.95 mm and about 1.25 mm.

[0340] The strike face 110 can include a donut 145 (also referred to as a thickened central region, localized stiffened regions, variable thickness regions, or inverted cone technology (ICT)). The center of the donut 145 can be the location of a peak thickness of the strike face 110. For example, a peak or maximum thickness of the donut 145 can be between about 2.5 mm and about 3.5 mm, preferably between about 2.75 mm and about 3.25 mm, more preferably between about 2.9 mm and about 3.1 mm. The strike face 110 can have a minimum or off-peak thickness of the donut 145 can be between about 1.4 mm and about 2.6 mm, preferably between about 1.55 mm and about 2.35 mm, more preferably between about 1.70 mm and about 2.2 mm.

[0341] The position of the donut 145 relative to a geometric center of the strike face 110 can be different for one or more irons within a set of club heads. For example, a set of club heads may include a selection of club heads, designated based on having different lofts of the strike face 110 at address, typically including numbered irons (e.g., 1-9 irons) and/or wedges (e.g., PW, AW, GW, and LW). The geometric center of the strike face 110 is determined using the procedures described in the USGA "Procedure for Measuring the Flexibility of a Golf Club head," Revision 2.0, Mar. 25, 2005.

[0342] For example, in longer irons with less loft (e.g., typically designated with numerically lower numbers), the position of the donut 145 can be lower and more toward relative to the geometric center of the strike face 110. In shorter irons (e.g., typically designated with numerically higher number) and wedges, the position of the donut 145 can be higher and more heelward relative to the geometric center of the strike face 110. The location of the donut 145 relative to a geometric center of the strike face 110 can influence localized flexibility of the strike face 110 and can influence launch conditions. For example, shifting the donut 145 can stiffen heelward locations the strike face 110 and can add flexibility to toward locations on the strike face 110. Further, shifting the donut 145 upward, downward,

toward, and heelward can influence launch conditions, such as impart a draw bias, fade bias, or to otherwise reduce lateral dispersion produced by the golf club head.

[0343] FIG. 21 a front elevation view of the golf club head 100 showing a peak/maximum and minimum/off-peak thicknesses of the strike face 110 of club head 100, measured at locations on the strike face 110 without grooves and/or scoring lines. Numerals 2101, 2103, 2105, 2107, 2109 refer to features of club head 100. The features of club head 100 may also be applicable to club heads 300, 500, and 600.

[0344] The strike face 110 has a peak or maximum thickness, such as at a center of donut 145, between about 2.5 mm and about 3.5 mm, preferably between about 2.75 mm and about 3.25 mm, more preferably between about 2.9 mm and about 3.1 mm. The strike face 110 has a minimum or off-peak thickness of the donut 145 can be between about 1.4 mm and about 2.6 mm, preferably between about 1.55 mm and about 2.35 mm, more preferably between about 1.70 mm and about 2.2 mm. The maximum face thickness may not be aligned with the geometric center of the face, such as when the donut 145 is shifted lower and toward to create a draw bias, such as in longer irons (e.g., 1-7 irons). In some embodiments, the donut 145 can be centered higher in short irons and wedges, and the donut 145 can be centered lower in middle and long irons.

[0345] For example, the minimum or off-peak thicknesses 2101, 2103, 2105, 2107, 2109 can vary based on iron loft. For example, for long irons with lofts between about 16 degrees and about 25 degrees (e.g., 1-5 irons), the off-peak thicknesses 2101, 2103, 2105, 2107, 2109 are preferably between about 1.6 mm and 1.9 mm, and a peak thickness between about 2.95 mm and about 3.25 mm. For example, for mid irons with lofts between about 21.5 degrees and about 32.5 degrees (e.g., 6-7 irons), the off-peak thicknesses 2101, 2103, 2105, 2107, 2109 are preferably between about 1.55 mm and 1.85 mm, and a peak thickness between about 2.9 mm and about 3.2 mm. For example, for short irons and wedges with lofts between about 28.5 degrees and about 54 degrees (e.g., 8 iron-AW), the off-peak thicknesses 2101, 2103, 2105, 2107, 2109 are preferably between about 1.95 mm and 2.25 mm, and a peak thickness between about 2.7 mm and about 3.05 mm. For example, for wedges with lofts between about 49 degrees and about 65 degrees (e.g., SW-LW), the off-peak thicknesses 2101, 2103, 2105, 2107, 2109 are preferably between about 1.6 mm and 1.9 mm, and a peak thickness between about 2.85 and about 3.15.

[0346] The strike face 110 of the golf club head 100 has coefficient of restitution (COR) change value between -0.015 and +0.008, the COR change value being defined as a difference between a measured COR value of the strike face 110 and a calibration plate COR value. In some embodiments, the damper 280 and/or filler material reduces the COR of the golf club head by no more than 0.010. A characteristic time (CT) at a geometric center of the strike face 110 is at least 250 microseconds. In some embodiments, the strike face 110 is made from a titanium alloy and a maximum thickness of less than 3.9 millimeters, inclusive. The strike face 110, excluding grooves, has a minimum thickness between 1.5 millimeters and 2.6 millimeters. The strike face 110 is a first titanium alloy and the body is a second titanium alloy, and the first titanium alloy is different than the second titanium alloy.

[0347] In some embodiments, the strike face **110** is a titanium alloy and the body **113** is a steel alloy. For example, the body can be a carbon steel with a modulus of elasticity of about 200 GPa and the face can be a higher strength titanium or steel alloy (e.g., stainless (17-4) with a modulus of elasticity of about 210 GPa, **4140** with a modulus of elasticity of about 205 GPa, or a Ti alloy with a modulus of elasticity between 110 GPa and 120 GPa).

[0348] In some embodiments, club heads within a set can have bodies **113** and/or strike faces **109** of different alloys. For example, longer irons can have bodies **113** and/or strike faces **109** of a first alloy (e.g., 3-8 irons using 450 SS with a modulus of elasticity of about 190-220 GPa), middle and short irons can have bodies **113** and/or strike faces **109** of a second alloy (e.g., 9 iron-AW using 17-4 PH SS with a modulus of elasticity of about 190-210 GPa), and short irons and wedges can have bodies **113** and/or strike faces **109** of a third alloy (SW-LW using 431 SS with a modulus of elasticity of about 180-200 GPa). Additional and different alloys can be used for different irons and wedges. In some embodiments, the club heads can be cast using alloys with a yield strength between 250 MPa and 1000 MPa, preferably greater than 500 MPa. Preferably, the iron-type club heads having a loft between 16 degrees and 33 degrees are formed from a material having a higher modulus of elasticity than the iron-type club heads having a loft greater than 33 degrees. Preferably, the iron-type club heads having a loft between 16 degrees and 33 degrees are formed from a material having a nickel content of at least 5% by weight and a Copper content of no more than 2% by weight.

[0349] In some embodiments, short irons and/or wedges can be manufactured using a different alloy and can have a thicker face than mid and long irons. In some embodiments, club heads with lofts greater 40 degrees can be manufactured using a different alloy (e.g., 17-4 PH SS) than club heads with lofts below 40 degrees (e.g., 450 SS). In some embodiments, a relatively stronger alloy may be required to cast ledges **193**, **194** for receiving the shim or badge **188**. In embodiments without ledges **193**, **194**, a relatively weaker alloy may be used.

[0350] In some embodiments, the club head **100** has a blade length between about 75 mm and about 86.5 mm, preferably between 77.5 mm and 84 mm. In some embodiments, the club head **100** has a topline width between about 5.5 mm and about 11 mm, preferably between 7 mm and 9 mm. In some embodiments, the club head **100** has a toeward face height between about 52 mm and about 68 mm, preferably between 54 mm and 66 mm. In some embodiments, the club head **100** has a PAR face height between about 28 mm and about 43 mm, preferably between 30 mm and 41 mm. In some embodiments, the club head **100** has a hosel to PAR width between about 4 mm and about 8 mm, preferably between 5 mm and 7 mm.

[0351] FIG. 22 illustrates a back perspective view of the golf club head **100** showing an upper ledge **193** and a lower ledge **194** configured to receive the shim or badge **188** (not depicted in FIG. 22). Numerals **2201** and **2203** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The shim or badge **188** can close the cavity opening **163**, enclosing and defining an internal cavity. The body **113** includes a heel portion **102**, a toe portion **104**, a sole portion **108**, a topline portion **106**, a rear portion **128**, and a hosel **114**. For example, the sole portion **108** extends rearwardly from a

lower end of the strike face **110** to a lower end of the rear portion **128**. A sole bar **135** can define a rearward portion of the sole portion **108**. A cavity **161** can be defined by a region of the body **113** rearward of the strike face **110**, forward of the rear portion **128**, above the sole portion **108**, and below the topline portion **106**.

[0352] The upper ledge **193** can be formed at least as part of the topline portion **106** and the lower ledge **194** can be formed at least as part of the rear portion **128**. In some embodiments, the upper ledge **193** is formed at least as part of both the topline portion **106** and the rear portion **128**. In some embodiments, the lower ledge **194** is formed at least as part of both the topline portion **106** and the rear portion **128**.

[0353] The shim or badge **188** (not depicted in FIG. 22) can be received at least in part by the upper ledge **193** and the lower ledge **194**. The shim or badge **188** is configured to close an opening **163** in the cavity **161**, enclosing an internal cavity volume. The upper ledge **193** and the lower ledge **194** can be planar or non-planar, and are shaped to receive at least a portion of the shim or badge **188** with a corresponding planar or non-planar shape.

[0354] In some embodiments, the ledges **193**, **194** can be discontinuous, such as provided as a one or more partial ledges and/or a series of tabs forming a discontinuous ledge. In some embodiments, a sealing wiper can be provided around shim or badge **188** to prevent water from intruding into the cavity **161**. The sealing wiper can be a gasket or another material provided around shim, such as to seal a discontinuous ledge.

[0355] For example, the upper ledge **193** has an upper ledge width **2201** with a width between about 0.5 mm and about 4.0 mm, preferably 3.25 mm, and a thickness between about 0.5 mm and about 1.5 mm, preferably about 1.0 mm. The lower ledge **194** has a lower ledge width **2203** has a width between about 0.1 mm and about 3.0 mm, preferably about 2.25 mm, and a thickness between about 0.8 mm and about 2 mm, preferably about 1.3 mm. In some embodiments, the width and thickness of the upper ledge **193** and/or lower ledge **194** are minimized to allow additional discretionary weight to be relocated in the club head **100**, such as lower in the club head **100**. In some embodiments, the upper ledge **193** is wider than the lower ledge **194** to provide additional structural support for the topline portion **106**, such as to improve feel, sound, and to better support the strike face **110**. The shim has an area as projected onto the strike face of between about 1200 mm² and about 2000 mm², more preferably between 1500 mm² and 1750 mm².

[0356] According to the embodiment depicted in FIG. 22, the upper ledge **193** extends from in a general heel-to-toe direction from the heel portion **102** to the toe portion **104** and across the topline portion **106**, such as from the lower heelside of the cavity opening **163** to the toeside of the cavity opening **163**, such as forming an upper edge, heelward edge, and toeward edge of the cavity opening **163**. The lower ledge **194** extends in a general heel-to-toe direction across the rear portion **128**, such as from the lower heelside of the cavity opening **163** to the lower toeside of the cavity opening **163**, such as forming a lower edge of the cavity opening **163**. In some embodiments, the upper ledge **193** can have an area between about 75 mm² and about 750 mm², preferably between 200 mm² and 500 mm². The lower ledge **194** can have an area between about 25 mm² and about 250 mm², preferably between 100 mm² and 300 mm². A total

ledge area of the upper and lower ledges **193**, **194**, as projected onto the strike face **110**, can be relatively small compared to an area of the cavity opening **163**. For example, the total ledge area can be between about 100 mm² and about 1000 mm², preferably between about 300 mm² and about 800 mm².

[0357] The area of the cavity opening **163**, as projected onto the strike face **110**, can be between about 800 mm² and about 2500 mm², preferably between 1200 mm² and 2000 mm², more preferably between 800 mm² and 1400 mm² or more preferably between 300 mm² and about 800 mm². For example, a ratio of the total ledge area to the area of the cavity opening **163** can be between about 4% and about 55%, preferably between 30% and 45%.

[0358] The total ledge area of the upper and lower ledges **193**, **194**, as projected onto the strike face **110**, can also be relatively small compared to an area of the shim or badge **188**, as projected onto the strike face **110**. For example, a ratio of the total ledge area to the area of the shim or badge **188** can be between about 15% and about 63%, preferably between 25% and 40%. A ratio the area of the cavity opening **163**, as projected onto the strike face **110**, to the area of the shim or badge **188**, as projected onto the strike face **110**, is at least about 50%, 53%, 56%, 59%, 62%, 65%, 68%, 71%, and no more than about 100%.

[0359] In some embodiments, the upper ledge **193** and/or lower ledge **194** can be eliminated, and the shim or badge **188** can be received at least in part by the topline portion **106** and/or rear portion **128**. For example, the shim or badge **188** can be bonded directly to a surface of the topline portion **106** and/or rear portion **128**. In another example, the topline portion **106** and/or the rear portion **128** can include a notch, slot, channel, or groove for receiving at least a portion of the shim or badge **188**. In this example, the shim or badge **188** can first hook into the topline portion **106** or the rear portion **128**, then the shim or badge **188** can be rotated and bonded to the rear portion **128** or the topline portion **106**, respectively.

[0360] FIG. 23 illustrates another embodiment of an iron-type golf club head **500** including a body **113** having a heel portion **102**, a toe portion **104**, a sole portion **108**, a topline portion **106**, a rear portion **128**, and a hosel **114**. The golf club head **500** is manufactured with a cavity **161** (not depicted in FIG. 23), and a shim or badge **188** is adhered, bonded, or welded to the body **113** to produce a cap-back iron, giving the appearance of a hollow-body iron. In this embodiment, the shim or badge **188** wraps into at least a portion of the toe portion **104**. In some embodiments, the shim or badge **188** also wraps into at least a portion of the heel portion **102**, toe portion **104**, sole portion **108**, topline portion **106**, and/or rear portion **128**. Various shim or badge **188** arrangements and materials can be used, and a filler material and/or damper **180** can be included within the cavity **161** to improve sound and feel, while minimizing loss in COR.

[0361] Although golf club heads **100**, **500** can have different shims **188**, other design elements of the golf club heads **100**, **500** can be used interchangeably between the embodiments. For example, the dimensions, material properties, and other design elements that are discussed with respect to golf club head **100** can be incorporated into the club head **500**, and vice versa. For example, both club heads **100**, **500** can be configured to receive a damper **180**, **280**

and/or a filler material within an internal cavity defined by affixing a shim or badge **188** to the golf club head **100**, **500**.

[0362] FIG. 24 illustrates the iron-type golf club head **500** without the shim or badge **188** installed. In some embodiments, in addition to the club head **500** including an upper ledge **193** and a lower ledge **194** configured to receive the shim or badge **188**, the club head **500** can also include a toeside ledge **125** in the toe portion **104** for receiving at least a portion of the shim or badge **188** in the toe portion **104**. In these embodiments, at least a portion of the shim or badge **188** is received in and/or enclosing a toeside cavity **124**.

[0363] In some embodiments, a damper **280** is installed in the cavity **161** before installing the shim or badge **188**. In some embodiments, the damper **280** is received entirely within the lower undercut region **164**, which is defined within the cavity **161** rearward of the strike face **110**, forward of the sole bar **135**, and above the sole portion **108**. In some embodiments, at least a portion of the damper **280** is received within the lower undercut region **164**. In some embodiments, a filler material (e.g., a foam or another material) can be injected into the cavity **161** after installing the shim or badge **188**.

[0364] FIG. 25 illustrates a top perspective view of a golf club head **100** showing topline portion **106** and hosel **114**. Numerals **2501**, **2503**, and **2505** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The topline portion **106** can have a topline width, measured at various locations **2501**, **2503**, **2505** across the topline portion **106**, between about 5 mm and about 10 mm, preferably between 7 mm and 9 mm. In some embodiment the topline width varies at the locations **2501**, **2503**, **2505**. In some embodiments, longer irons in a set can have a wider topline width than shorter irons. For example, short irons and wedges (e.g., 9 iron-LW) can have a topline width between about 7.15 mm and about 7.65 mm, mid irons (e.g., 8 iron) can have a topline width between about 7.55 mm and about 8.05 mm, and long irons (e.g., 4-7 iron) can have a topline width between about 7.75 mm and about 8.25 mm. The aforementioned dimensions are also applicable to golf club heads **300**, **500**, and **600**.

[0365] In some embodiments, a weight reducing feature can be used to selectively reduce the wall thickness around the hosel **114**, such as for freeing up discretionary weight in the club head **100**. For example, the weight reducing features removing weight from the hosel **114** can be used to remove mass from the hosel **114** wall thickness. The weight reducing feature can remove at least 1 g, such as at least 2 g, such as at least 3 g, such as at least 4 g of mass from the hosel. In the design shown, about 4 g was removed from the hosel **114** and reallocated to lower in the club head, resulting in a downward Zup shift of about 0.6 mm while maintaining the same overall head weight. The flute design shown can use flutes on the front side, rear side, and underside of the hosel **114**, making the flutes less noticeable from address. By employing weight reducing features on the side and/or underside of the hosel, the golf club head can have a traditional look, while providing the performance benefits of weight reducing features and weight redistribution in the golf club head. For example, U.S. Pat. No. 10,265,587, incorporated herein by reference in its entirety, discloses additional details on weight reducing features.

[0366] In some embodiments, variable length hosels can be used within a set of irons. For example, shorter hosels can

be used to redistribute mass lower in the club head **100**. In some embodiments, a peak hosel height can be less than a peak toe height relative to ground plane when club head is at address.

[0367] FIG. 26 illustrates a bottom perspective view of a golf club head **100** showing a hosel **114**, a channel **150** and a weld point **2607**. Numerals **2601**, **2603**, **2605**, and **2607** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The hosel **114** includes a weight reducing feature can be used to selectively reduce the wall thickness around the hosel **114**. The flute design shown can use flutes on the front side, rear side, and underside of the hosel **114**, making the flutes more noticeable from below. By employing weight reducing features on the side and/or underside of the hosel, the golf club head can have a traditional look, while providing the performance benefits of weight reducing features and weight redistribution in the golf club head.

[0368] The channel **150** can have a channel width **2601** between 1.5 mm and 2.5 mm, preferably between 1.85 mm and 2.15 mm. The channel **150** can have a channel length **2603** between about 55 mm and about 70 mm, preferably between 63.85 mm and 64.15 mm. A channel setback **2605** from the leading edge between about 5 mm and about 20 mm, preferably between about 5 mm and about 9 mm, more preferably between 6 mm and 8 mm, more preferably between 6.35 mm and 7.35 mm. In embodiments with strike faces **109** welded to the body **113**, a weld point **2607** can be offset from the leading edge, such as by the channel setback **2605**.

[0369] FIG. 27 is a side cross-sectional view of the golf club head **100** showing a lower undercut region **164** in lower region **29B** and an upper undercut region **165** in upper region **29A**. Numerals **2701**, **2703**, and **2705** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The channel **150** has a width **2601** and a channel depth **2701** beyond the sole portion **108**. The channel depth **2701** beyond the sole portion can be between about 1.0 mm and about 3.0 mm, preferably between 1.5 mm and 2.5 mm, preferably between 1.85 mm and 2.15 mm. The sole portion **108** has a sole thickness **2705** of between about 1.5 mm and about 3 mm, more preferably between 1.85 mm and 2.35 mm. A total channel depth can be a combination of the sole thickness **2705** and the channel depth **2701** beyond the sole portion **108**. A topline thickness **2703** of the topline portion **106** can be between about 0.5 mm and about 2 mm, more preferably between 0.95 mm and 1.25 mm.

[0370] The sole bar **135** has a height, measured as the distance perpendicular from the ground plane (GP) to a top edge of the sole bar **135** when the golf club head is in proper address position on the ground plane. For example, the sole bar height can be between about 7.5 mm and about 35 mm, preferably between 10 mm and 30 mm, more preferably 15 mm and 26 mm. In some embodiments, the sole bar **135** can have a peak height between about 10 mm and about 30 mm, preferably between 15 mm and 26 mm. The sole bar **135** can have an off-peak height between about 7.5 mm and about 26 mm, preferably between 7.5 mm and 15 mm. A ratio of the sole bar height to the sole thickness **2705** can be between about 2:1 and about 20:1, more preferably 5:1, 6:1, 10:1, or 15:1. A ratio of the sole thickness **2705** to the sole bar height can be between about 1:25 and about 1:2.5, preferably between 1:14 and 1:7.

[0371] FIG. 28 is a side cross-sectional view of the golf club head **100** of FIG. 19 showing the topline portion **106**, the sole portion **108**, the strike face **110**, the sole bar **135**, the upper ledge **193**, the lower ledge **194**, the lower undercut region **164** and the upper undercut region **165**. Numerals **2801**, **2803**, **2805**, and **2807** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**.

[0372] The lower undercut region **164** is defined within the cavity rearward of the strike face **110**, forward of the sole bar **135**, and above the sole portion **108**. The lower undercut region **164** can be forward of the lower ledge **194**. For example, the lower ledge **194** can extend above the sole bar **135** to further define the lower undercut region **164**. An upper undercut region **165** is defined within the cavity rearward of the strike face **110**, and below the topline portion **106**. The upper undercut region **165** can be forward of the upper ledge **193**. For example, upper ledge **193** can extend below the topline portion **106** to further define the upper undercut region **165** forward of an upper ledge **193**. In various embodiments, the upper ledge **193** can extend inward toward the strike face **110**, outward away from the strike face **110**, or downward parallel with the strike face **110**.

[0373] The upper undercut region **165** can be defined at least in part by the upper ledge **193**, and includes an upper undercut width **2801** and an upper undercut depth **2805**. The upper undercut width **2801** can be between about 1.5 mm and about 7.5 mm, preferably between 2 mm and 6.5 mm, more preferably about 2.75 mm. The upper undercut depth **2805** can be between about 3 mm and about 15 mm, preferably between 4 mm and 13 mm, more preferably about 5 mm. A ratio of the upper undercut depth **2805** to the upper undercut width **2801** is at least 1.25, preferably at least 1.5, preferably at least 1.75. For example, an upper undercut depth **2805** can be 5 mm and upper undercut width **2801** as 2.75 mm, resulting in a ratio of about 1.8. The upper undercut width **2801** and the upper undercut depth **2805** is measured at a cross-section taken at the geometric center face or at a scoreline midline. Alternatively, the upper undercut depth **2805** is measured in a cross-section through 5 mm toward or 5 mm heelward of the geometric center face in the y-z plane.

[0374] The lower undercut region **164** can be defined at least in part by the lower ledge **194**, and includes a lower undercut width **2803** and a lower undercut depth **2807**. The lower undercut width **2803** can be between about 2 mm and about 15 mm, preferably between 4 mm and 6 mm. The lower undercut depth **2807** can be between about 10 mm and about 30 mm, preferably between 11 mm and 26 mm. The lower undercut width **2803** and the lower undercut depth **2807** is measured at a cross-section taken at the geometric center face or at a scoreline midline.

[0375] In some embodiments, the lower undercut depth **2807** is greater than the upper undercut depth **2806**, such as having a ratio of at least 2:1, preferably 2.5:1, more preferably 3:1.

[0376] In some embodiments, in order to cast a unitary body **113** without metal defects, a ratio of an undercut width to undercut depth should not exceed about 1:3.5. For example, to cast the golf club head **100** as a single piece (i.e., a unitary casting), the ratio of undercut width to undercut depth should not be greater than about 1:3.5 or 1:3.6 to allow for ample space for wax injection pickouts within the

undercut. The ratio of the lower undercut width **2803** to the lower undercut depth **2807** can be between about 1:4.0 and about 1:2.0, preferably between about 1:3.5 and about 1:2.5. Table 1 below provides examples of lower undercut widths **2803**, lower undercut depths **2807**, and corresponding ratios:

TABLE 1

Exemplary Lower Undercut Ratios			
Example No.	Lower Undercut Width	Lower Undercut Depth	Ratio
1	6.5 mm	17 mm	1:2.6
2	5.25 mm	19 mm	1:3.6
3	4.5 mm	15.3 mm	1:3.4
4	4.7 mm	16.9 mm	1:3.6
5	5.2 mm	17.9 mm	1:3.4
6	7.5 mm	26 mm	1:3.5

[0377] In embodiments where the club head **100** comprises a strike face **110** welded to the body, and in embodiments where the lower undercut region **164** and/or the upper undercut region **165** are machined in the club head **100**, the ratio of width to depth of an undercut can be less than 25-28%.

[0378] FIG. 29A is a side cross-sectional view of the upper region **29A** of FIG. 27. Numerals **2901** and **2903** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The upper region **29A** includes the upper undercut region **165**. The upper undercut region **165** is at least in part defined by the upper ledge **193**. The upper ledge **193** has an upper ledge width **2901** is between about 0.5 mm and about 4.0 mm, preferably 3.25 mm, and an upper ledge thickness **2903** between about 0.5 mm and about 1.5 mm, preferably about 1.0 mm. The topline portion **106** has a topline thickness **2703** is between about 0.5 mm and about 2 mm, more preferably between 0.95 mm and 1.25 mm.

[0379] The upper undercut region **165** can be defined as a cavity formed rearward of the strike face **110**, below the topline portion **106**, forward of the upper ledge **193**, heelward of the toe portion **104**, and toward of the heel portion **102**. In some embodiments, the upper undercut region **165** can be defined as a cavity formed rearward of the strike face **110**, forward of and below the topline portion **106**, heelward of the toe portion **104**, and toward of the heel portion **102**.

[0380] FIG. 29B is a side cross-sectional view of the lower region **29B** of FIG. 27. Numerals **2905** and **2907** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The lower region **29B** includes the lower ledge **194**. The lower ledge **194** has a lower ledge width **2905** is between about 0.1 mm and about 3.0 mm, preferably about 2.25 mm, and a lower ledge thickness **2907** is between about 0.8 mm and about 2 mm, preferably about 1.3 mm.

[0381] Referring back to FIG. 28, the lower undercut region **164** is at least in part defined by the lower ledge **194**. For example, the lower undercut region **164** can be defined as a cavity formed rearward of the strike face **110**, forward of the lower ledge **194** and the sole bar **135**, heelward of the toe portion **104**, and toward of the heel portion **102**. In some embodiments, lower undercut region **164** can be defined as a cavity formed rearward of the strike face **110**,

forward of the sole bar **135**, heelward of the toe portion **104**, and toward of the heel portion **102**.

Damper and/or Filler Materials

[0382] FIG. 30 is a perspective view of a damper **280** from the golf club head **100** of FIG. 19. The damper **280** includes one or more projections **282**. For example, when the damper **280** is installed, each of the projections **282** can make contact with a rear surface of the strike face **110** or a front surface of the sole bar **135**. The damper **280** also includes one or more relief cutouts **281**, such as between the projections **282**, which do not contact the rear surface of the strike face **110** or the front surface of the sole bar **135**.

[0383] In some embodiments, the damper **280** is a combination of a combination of Santoprene and Hybrar, such as with a hybrar content between about 10% and about 40%, more particularly 15% or 30%. Other materials can also be used. The damper **280** can also be co-molded using different materials with different durometers, masses, densities, colors, and/or other material properties. In some embodiments, using a damper **280** can lower the CG when compared to using a filler material. Additional weighted materials can also be included in the damper **280**, such as to further lower CG of the golf club head, such as using weight plugs or inserts made from a Tungsten alloy, another alloy, or another material.

[0384] In some embodiments, a damper **280** and/or a filler material is only used in a subset of clubs within a set. For example, some club heads **100** can provide adequate sound and feel without a damper **280** and/or a filler material. In this example, only long and mid irons (e.g., 2-8 irons) include a damper **280** and/or a filler material. Short irons and wedges (e.g., 9 iron-LW) can be manufactured without a damper **280** or a filler material. In these embodiments, each club head **100** within a set can be manufactured with or without the damper **280** and/or the filler material based on the sound and feel characteristics independent to each club head **100**.

[0385] In some embodiments, a filler material can be used in place of the damper **280**. In other embodiments, a filler material can be used in conjunction with the damper **280**. For example, a foam, hot melt, epoxy, adhesive, liquified thermoplastic, or another material can be injected into the club head **100** filling or partially filling the cavity **161**. In some embodiments, the filler material is heated past melting point and injected into the club head **100**.

[0386] In some embodiments, the filler material is used to secure the damper **280** in place during installation, such as using hot melt, epoxy, adhesive, or another filler material. In some embodiments, a filler material can be injected into the club head **100** to make minor changes to the weight of the club head **100**, such as to adjust the club head for proper swing weight, to account for manufacturing variances between club heads, and to achieved a desired weight of each head. In these embodiments, the club head weight can be increased between about 0.5 grams and about 5 grams, preferably up to 2 grams.

Shim Structure and Materials

[0387] FIG. 31 is a rear elevation view of the shim or badge **188** from the golf club head of FIG. 19. The shim or badge **188** is manufactured from a light weight, stiff material (s), which may provide additional support for the topline portion **106** to provide better sound and feel. The shim or

badge **188** may dampen vibrations and sounds. Examples of such shims, badges, and inserts are disclosed in U.S. Pat. No. 8,920,261, which is incorporated by reference herein in its entirety. Additionally, the shim or badge **188** can also be used for decorative purposes and/or for indicating the manufacturer name, logo, trademark, or the like.

[0388] The shim or badge **188** can be manufactured from one or more materials. The shim or badge **188** may be made from any suitable material that provides a desired stiffness and mass to achieve one or more desired performance characteristics. In some embodiments, shim or badge **188** is co-molded or otherwise formed from multiple materials. For example, the shim or badge **188** can be formed from one or more of ABS (acrylonitrile-butadiene-styrene) plastic, a composite (e.g., true carbon or another material), a metal or metal alloy (e.g., titanium, aluminum, steel, tungsten, nickel, cobalt, an alloy including one or more of these materials, or another alloy), one or more of various polymers (e.g., ABS plastic, nylon, and/or polycarbonate), a fiber-reinforced polymer material, an elastomer or a viscoelastic material (e.g., rubber or any of various synthetic elastomers, such as polyurethane, a thermoplastic or thermoset material polymer, or silicone), any combination of these materials, or another material. In some embodiments, the shim or badge **188** can be formed from a first material (e.g., ABS plastic) with a second material (e.g., aluminum) inlaid into the first material.

[0389] The average thickness of the shim or badge **188** can be between about 0.5 mm and about 6 mm. A relatively thicker shim or badge **188** (e.g., average thickness of about 3 mm) may be more effective than a thinner shim or badge **188** (e.g., average thickness of about 1 mm).

[0390] The shim or badge **188** can have an average density (i.e., mass divided by water-displaced volume) that is lower than the body **113**, such as between about 0.5 g/cc and about 20 g/cc, preferably between 1 g/cc and 2 g/cc, between 3 g/cc and 4 g/cc, or between 4 g/cc and 5 g/cc. A thinner shim or badge **188** can be used with a tighter material stack-up, increasing the density and durability of the shim or badge **188**. The shim or badge **188** can have a mass between about 2.5 grams and about 15 grams, preferably between 2.5 grams and 10 grams, more preferably between 2.5 grams and 9 grams. A ratio of the average density to the mass can be between about 0.033 l/cc and about 8 l/cc, preferably between 0.08 l/cc and 0.8 l/cc, more preferably between 0.15 l/cc and 0.375 l/cc. The material density of the shim or badge **188**, defined by the mass of the shim or badge **188** divided by the volume of the shim or badge **188**, can be less than 7.8 g/cc, preferably between 1 g/cc and 2 g/cc, more preferably between 1.0 g/cc and 1.5 g/cc.

[0391] The shim or badge **188** can have an area weight (e.g., average thickness divided by average density) of between about 0.0065 cm⁴/g and about 1.2 cm⁴/g. The mass and thickness of the shim or badge **188** can vary within a set of club heads **100**. For example, shorter irons and wedges have relatively thicker and heavier shims or badges **188** than mid and long irons.

[0392] FIG. 32 is a rear perspective view of the shim or badge **188** from the golf club head of FIG. 19. Numerals **3201**, **3203** and **3205** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. The shim or badge **188** can be three-dimensional and non-planar. A rear surface of the shim or badge **188** can include one or more three-dimensional

features, such as ridges, depressions, ledges, lips, valleys, inlays, channels, slots, cavities, and other features. The three-dimensional features on the rear surface the shim or badge **188** can confer aesthetic and performance benefits to the club head **100**.

[0393] For example, the three-dimensional features on the rear surface the shim or badge **188** can correspond to features of the golf club head **100**, such as to give the appearance of a hollow body iron. In other examples, the three-dimensional features on the rear surface the shim or badge **188** can reduce the weight of at least a portion of the shim or badge **188**, such as to redistribute discretionary weight lower in the club head **100**. In further examples, the three-dimensional features on the rear surface the shim or badge **188** can increase structural stability of the shim and/or badge **188**, and can provide additional support the topline portion **106**, and can provide other performance benefits to the golf club head **100**, such as altering sound and feel characteristics of the golf club head **100**.

[0394] In some embodiments, the shim or badge **188** can include a ridge **3201**, a channel **3203**, a depression **3205**. Given the three-dimensional features of the shim or badge **188**, the projected area can be less than a surface area of one or more surfaces of the shim or badge **188**. The shim or badge **188** has an area as projected onto the strike face of between about 1200 mm² and about 2000 mm², more preferably between 1500 mm² and 1750 mm².

[0395] FIG. 33 is a front elevation view of the shim or badge **188** from the golf club head of FIG. 19. Numerals **3301**, **3303** and **3305** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. A front surface of the shim or badge **188** can have one or more three-dimensional features, such as ridges, depressions, ledges, lips, valleys, inlays, channels, slots, cavities, and other features. The three-dimensional features on the front surface the shim or badge **188** can performance benefits to the club head **100**, such as weight reduction and redistribution, increasing structural stability, altering sound and feel characteristics, and providing other performance benefits to the golf club head **100**.

[0396] The shim or badge **188** can have a ledge **3303** used for installing the shim or badge **188** onto the golf club head **100**. In some embodiments, the width **3301** of the ledge **3303** is between about 0.5 mm and 5.0 mm, more preferably between 0.5 mm to 3.5 mm, more preferably between 1.0 mm and 3.0 mm, more preferably between 1.0 mm and 2.0 mm, more preferably between 1.25 mm and 1.75 mm. In some embodiments, the ledge width **3301** is variable, such as with a wider or narrower width on one or more of an upper portion, lower portion, toward portion, heelward portion, and/or another portion of the ledge **3303**. In some embodiments, a ledge width **3301** less than 1 mm can negatively impact durability of the shim or badge **188**, such as when an ABS plastic is used.

[0397] FIG. 34 a front perspective view of the shim or badge **188** from the golf club head of FIG. 19. Numeral **3401** refers to a feature of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. In some embodiments, the ledge **3303** extends around the perimeter of the shim or badge **188**. In other embodiments, the ledge **3303** is discontinuous, such as with the ledge **3303** separated into one or more of an upper ledge portion, a lower ledge portion, a toward ledge portion, a heelward ledge portion, and/or another ledge portion. Support ridges **3305**

can also be provided to stiffen and provide structural support for the shim or badge **188** and the topline portion **106**.

[0398] The ledge **3303** can be defined by a center thickened region **3401**. In some embodiments, the center thickened region **3401** is configured to fit within and close a cavity opening **163** in the cavity **161**. In some embodiments, the center thickened region **3401** is configured to fit over and close a cavity opening **163** in the cavity **161**. In some embodiments, the ledge **3303** can receive a portion of the club head **100** during installation. In this example, the shape of the ledge **3303** can correspond to the upper ledge **193** and the lower ledge **194** of the club head **100**.

[0399] The ledge **3303** can be non-planar in one or more of the upper portion, lower portion, toeward portion, heelward portion, and/or another portion of the ledge **3303**. For example, the ledge **3303** can be convex, concave, wavy, rounded, or provided with another non-planar surface.

[0400] FIG. **35** is a heelward perspective view of the shim or badge **188** from the golf club head of FIG. **19**. Numerals **3501** and **3503** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. In some embodiments, the shim or badge thickness, as measured from the front surface to the rear surface of the shim or badge **188**, can vary from the upper portion to the lower portion of the shim or badge **188**. For example, an upper thickness **3501** of the shim or badge **188** is different from the lower thickness **3503** of the shim or badge **188**. In some embodiments, the shim or badge **188** is thickest in the lower portion of the shim or badge **188**, such as near to or at the bottom of the badge, and the shim or badge **188** is thinnest in the upper portion of the shim or badge **188**, such as near to or at the top of the badge.

[0401] FIG. **35** also depicts the ledge **3303** and the ledge width **3301** discussed above with respect to FIG. **33**. The ledge **3303** can extend around the perimeter of the shim or badge **188** and can provide a bonding surface between the shim or badge **188** and golf club head.

[0402] In some embodiments, a ratio of the upper thickness **3501** to the lower thickness **3503** can be between about 150% and about 500%, more preferably at least 150%, 200%, 250%, or 300%. Likewise, a ratio of the thinnest portion to the thickest portion of the shim or badge **188** can also be between about 150% and about 500%, more preferably at least 150%, 200%, 250%, or 300%.

[0403] In some embodiments, the shim or badge **188** has a minimum thickness between about 0.5 mm and about 3 mm, preferably between 0.5 mm and 1.5 mm. In some embodiments, the shim or badge **188** has a maximum thickness between about 0.75 mm and about 17 mm, preferably between 3 mm and 13 mm.

[0404] FIG. **36** is a toeward perspective view of the shim or badge **188** from the golf club head of FIG. **19**. Numerals **3601** and **3603** refer to features of club head **100**. The features of club head **100** may also be applicable to club heads **300**, **500**, and **600**. In some embodiments, the shim or badge **188** has a maximum depth **3601** between about 5 mm and about 20 mm, preferably less than 16 mm, and more preferably less than 15 mm. In some embodiments, the shim or badge **188** has a minimum depth **3603** between about 1 mm and about 6 mm, preferably at least 2 mm, more preferably at least 2.5 mm.

[0405] FIG. **37** is a front perspective view of the shim or badge **188** from the golf club head **500** of FIG. **23**. Numeral **3701** refers to a feature of club head **500**. The features of

club head **100** may also be applicable to club heads **100**, **300**, and **600**. In this embodiment, the shim or badge **188** is configured to wrap into at least a portion of the toe portion **104**. For example, the shim or badge **188** has a toewrap portion **3701**, such as to be received by or enclosing the toeside cavity **124** of the golf club head **500**. In some embodiments, the toewrap portion **3701** is separated from the center thickened region **3401** by a channel or slot for receiving at least a portion of the toeside ledge **125** in the toe portion **104** of the golf club head **500**. In this embodiment, additional discretionary mass can be freed up in the toe portion and redistributed in the body, such as to further lower Zup. For example, high density steel in the toe portion can be replaced with the lower density material of the shim.

[0406] FIG. **38** is a lower perspective view of the shim or badge **188** from the golf club head of FIG. **23**. In some embodiments, the shim or badge **188** has a ledge **3303**. In some embodiments, the ledge **3303** of the shim or badge **188** is configured to match a profile of the sole bar **135**, the upper ledge **193**, the lower ledge **194**, or another feature of the golf club head **500**.

Rear Fascia, Shim, Plate, or Badge

[0407] Exemplary club head structures, including a rear fascia, plate, or badge, are described in U.S. patent application Ser. No. 16,870,714, filed May 8, 2020, titled "IRON-TYPE GOLF CLUB HEAD," which is incorporated herein by reference in its entirety.

[0408] According to some examples of the golf club head **100**, as shown in FIG. **39**, the body **113** of the golf club head **100** has a cavity-back configuration and the golf club head **100** further includes a rear fascia **188**, shim, rear plate, or badge, coupled to the back portion **129** of the body **113**. As used herein, the terms rear fascia, shim, rear plate, and badge can be used interchangeably. The rear fascia **188** encloses the internal cavity **147** by covering, at the back portion **129** of the body **113**, the plate opening **176**. Accordingly, the rear fascia **188**, in effect, converts the cavity-back configuration of the golf club head **100** into more of a hollow-body configuration. As will be explained in more detail, enclosing the internal cavity **147** with the rear fascia **188** allows a filler material (e.g., filler material **233** of FIG. **65**) and/or damper to retainably occupy at least a portion of the internal cavity **147**. The filler material **233** and/or damper can include organic and/or inorganic materials. In some examples, the filler material **233** and/or damper does not contain glass bubbles or inorganic solids.

[0409] As depicted in FIG. **39**, the rear fascia **188** can bond to a surface without a pronounced ledge. For example, the upper edge of the rear fascia **188** can bond directly to the topline portion **106**. Likewise, the lower edge of the rear fascia **188** can bond directly to the back portion **129**. In some embodiments, the rear fascia **188** does not bond to a ledge of the topline portion **106** or back portion **129**, such as one or more substantially vertical ledges (e.g., approximately 90 degrees with respect to the ground plane at address). In some embodiments, the rear fascia **188** bonds to a first surface on the topline portion **106** and a second surface on the back portion **129**. In some embodiments, the first surface and the second surface are not parallel surfaces, the surfaces are transverse to each other, or the surfaces are at an angle to each other, such as an angle between 25 degrees and 90 degrees to each other.

[0410] The rear fascia **188** is made from one or more of the polymeric materials described herein, in some examples, and adhered or bonded to the body **113**. In other examples, the rear fascia **188** is made from one or more of the metallic materials described herein and adhered, bonded, or welded to the body **113**. The rear fascia **188** can have a density ranging from about 0.9 g/cc to about 5 g/cc. Moreover, the rear fascia **188** may be a plastic, a carbon fiber composite material, a titanium alloy, or an aluminum alloy. In certain embodiments, where the rear fascia **188** is made of aluminum, the rear fascia **188** may be anodized to have various colors such as red, blue, yellow, or purple.

[0411] The golf club head **100** disclosed herein may have an external head volume equal to the volumetric displacement of the golf club head **100**. For example, the golf club head **100** of the present application can be configured to have a head volume between about 15 cm³ and about 150 cm³. In more particular embodiments, the head volume may be between about 30 cm³ and about 90 cm³. In yet more specific embodiments, the head volume may be between about 30 cm³ and about 70 cm³, between about 30 cm³ and about 55 cm³, between about 45 cm³ and about 100 cm³, between about 55 cm³ and about 95 cm³, or between about 70 cm³ and about 95 cm³. The golf club head **100** may have a total mass between about 230 g and about 300 g.

[0412] In some embodiments, the volume of the internal cavity is between about 1 cm³ and about 50 cm³, between about 5 cm³ and about 30 cm³, or between about 8 cc and about 20 cc. For the purposes of measuring the internal cavity volume herein, the aperture is assumed to be removed and an imaginary continuous wall or substantially back wall is utilized to calculate the internal cavity volume.

[0413] In some embodiments, the mass of the filler material **233**, and/or the damper, divided by the external head volume is between about 0.08 g/cm³ and about 0.23 g/cm³, between about 0.11 g/cm³ and about 0.19 g/cm³, or between about 0.12 g/cm³ and about 0.16 g/cm³. For example, in some embodiments, the mass of the filler material **233** and/or damper may be about 5.5 grams and the external head volume may be about 50 cm³ resulting in a ratio of about 0.11 g/cm³.

[0414] In some embodiments, the density of the filler material **233** and/or the damper, after it is fully formed and/or positioned within the internal cavity **147**, is at least 0.21 g/cc, such as between about 0.21 g/cc and about 0.71 g/cc or between about 0.22 g/cc and about 0.49 g/cc. In certain embodiments, the density of the filler material **233** and/or the damper is in the range of about 0.22 g/cc to about 0.71 g/cc, or between about 0.35 g/cc and 0.60 g/cc. The density of the filler material **233** and/or the damper impacts the COR, durability, strength, and filling capacity of the club head. In general, a lower density material will have less of an impact on the COR of a club head. The density of the filler material **233** and/or the damper is the density after the filler material **233** and/or the damper is fully formed and/or positioned within and enclosed by the internal cavity **147**.

[0415] During development of the golf club head **100**, use of a lower density filler material and/or damper having a density less than 0.21 g/cc was investigated, but the lower density did not meet certain sound performance criteria. This resulted in using a filler material **233** and/or the damper having a density of at least 0.21 g/cc to meet sound performance criteria.

[0416] In one embodiment, the filler material **233** and/or the damper has a minor impact on the coefficient of restitution (herein “COR”) as measured according to the United States Golf Association (USGA) rules set forth in the Procedure for Measuring the Velocity Ratio of a Club Head for Conformance to Rule 4-1e, Appendix II Revision 2 Feb. 8, 1999, herein incorporated by reference in its entirety.

[0417] Table 2 below provides examples of the COR change relative to a calibration plate of multiple club heads of the construction described herein both a filled and unfilled state. The calibration plate dimensions and weight are described in section 4.0 of the Procedure for Measuring the Velocity Ratio of a Club Head for Conformance to Rule 4-1e.

[0418] Due to the slight variability between different calibration plates, the values described below are described in terms of a change in COR relative to a calibration plate base value. For example, if a calibration plate has a 0.831 COR value, Example 1 for an un-filled head has a COR value of -0.019 less than 0.831 which would give Example 1 (Un-filled) a COR value of 0.812. The change in COR for a given head relative to a calibration plate is accurate and highly repeatable.

TABLE 2

COR Values Relative to a Calibration Plate			
Example No.	Unfilled COR Relative to Calibration Plate	Filled COR Relative to Calibration Plate	COR Change Between Filled and Unfilled
1	-0.019	-0.022	-0.003
2	-0.003	-0.005	-0.002
3	-0.006	-0.010	-0.004
4	-0.006	-0.017	-0.011
5	-0.026	-0.028	-0.002
6	-0.007	-0.017	-0.01
7	-0.013	-0.019	-0.006
8	-0.007	-0.007	0.000
9	-0.012	-0.014	-0.002
10	-0.020	-0.022	-0.002
Average	-0.0119	-0.022	-0.002

[0419] Table 2 illustrates that before the filler material **233** and/or the damper is introduced into the cavity **147** of the golf club head **100**, an Unfilled COR drop off relative to the calibration plate (or first COR drop off value) is between 0 and -0.05, between 0 and -0.03, between -0.00001 and -0.03, between -0.00001 and -0.025, between -0.00001 and -0.02, between -0.00001 and -0.015, between -0.00001 and -0.01, or between -0.00001 and -0.005. In one embodiment, the average COR drop off or loss relative to the calibration plate for a plurality of Unfilled COR golf club heads **100**, within a set of irons, is between 0 and -0.05, between 0 and -0.03, between -0.00001 and -0.03, between -0.00001 and -0.025, between -0.00001 and -0.02, between -0.00001 and -0.015, or between -0.00001 and -0.01.

[0420] Table 2 further illustrates that after the filler material **233** and/or the damper is introduced into the cavity **147** of golf club head **100**, a Filled COR drop off relative to the calibration plate (or second COR drop off value) is more than the Unfilled COR drop off relative to the calibration plate. In other words, the addition of the filler material **233** and/or the damper in the Filled COR golf club heads slows the ball speed (Vout-Velocity Out) after rebounding from the

face by a small amount relative to the rebounding ball velocity of the Unfilled COR heads. In some embodiments shown in Table 2, the COR drop off or loss relative to the calibration plate for a Filled COR golf club head is between 0 and -0.05, between 0 and -0.03, between -0.00001 and -0.03, between -0.00001 and -0.025, between -0.00001 and -0.02, between -0.00001 and -0.015, between -0.00001 and -0.01, or between -0.00001 and -0.005. In one embodiment, the average COR drop off or loss relative to the calibration plate for a plurality of Filled COR golf club head within a set of irons is between 0 and -0.05, between 0 and -0.03, between -0.00001 and -0.03, between -0.00001 and -0.025, between -0.00001 and -0.02, between -0.00001 and -0.015, between -0.00001 and -0.01, or between -0.00001 and -0.005.

[0421] However, the amount of COR loss or drop off for a Filled COR head is minimized when compared to other constructions and filler materials. The last column of Table 2 illustrates a COR change between the Unfilled and Filled golf club heads which are calculated by subtracting the Unfilled COR from the Filled COR table columns. The change in COR (COR change value) between the Filled and Unfilled club heads is between 0 and -0.1, between 0 and -0.05, between 0 and -0.04, between 0 and -0.03, between 0 and -0.025, between 0 and -0.02, between 0 and -0.015, between 0 and -0.01, between 0 and -0.009, between 0 and -0.008, between 0 and -0.007, between 0 and -0.006, between 0 and -0.005, between 0 and -0.004, between 0 and -0.003, or between 0 and -0.002. Remarkably, one club head was able to achieve a change in COR of zero between a filled and unfilled golf club head. In other words, no change in COR between the Filled and Unfilled club head state. In some embodiments, the COR change value is greater than -0.1, greater than -0.05, greater than -0.04, greater than -0.03, greater than -0.02, greater than -0.01, greater than -0.009, greater than -0.008, greater than -0.007, greater than -0.006, greater than -0.005, greater than -0.004, or greater than -0.003. In certain examples, the filler material in the internal cavity reduces the COR by no more than 0.025 or 0.010.

[0422] In some embodiments, at least one, two, three, or four golf clubs out of an iron golf club set has a change in COR between the Filled and Unfilled states of between 0 and -0.1, between 0 and -0.05, between 0 and -0.04, between 0 and -0.03, between 0 and -0.02, between 0 and -0.01, between 0 and -0.009, between 0 and -0.008, between 0 and -0.007, between 0 and -0.006, between 0 and -0.005, between 0 and -0.004, between 0 and -0.003, or between 0 and -0.002.

[0423] In yet other embodiments, at least one pair or two pair of iron golf clubs in the set have a change in COR between the Filled and Unfilled states of between 0 and -0.1, between 0 and -0.05, between 0 and -0.04, between 0 and -0.03, between 0 and -0.02, between 0 and -0.01, between 0 and -0.009, between 0 and -0.008, between 0 and -0.007, between 0 and -0.006, between 0 and -0.005, between 0 and -0.004, between 0 and -0.003, or between 0 and -0.002.

[0424] In other embodiments, an average of a plurality of iron golf clubs in the set has a change in COR between the Filled and Unfilled states of between 0 and -0.1, between 0 and -0.05, between 0 and -0.04, between 0 and -0.03, between 0 and -0.02, between 0 and -0.01, between 0 and -0.009, between 0 and -0.008, between 0 and -0.007,

between 0 and -0.006, between 0 and -0.005, between 0 and -0.004, between 0 and -0.003, or between 0 and -0.002.

[0425] The filler material 233 and/or the damper fills the cavity 147 located above the sole slot 126. A recess or depression in the filler material 233 and/or the damper engages with the thickened portion of the strike plate. In some embodiments, the filler material 233 and/or the damper is a two-part polyurethane foam that is a thermoset and is flexible after it is cured. In one embodiment, the two-part polyurethane foam is any methylene diphenyl diisocyanate (a class of polyurethane prepolymer) or silicone based flexible or rigid polyurethane foam.

Shim Mass Per Unit Length

[0426] Exemplary club head structures are described in U.S. Pat. No. 10,493,336, titled "IRON-TYPE GOLF CLUB HEAD," which is incorporated herein by reference in its entirety.

[0427] Referring to FIG. 19, an areal mass of the shim or badge 188 of the golf club head 100 between the rear portion 128, the topline portion 106, the sole portion 108, the toe portion 104, and the heel portion 102 is between 0.0005 g/mm² and 0.00925 g/mm², such as, for example, about 0.0037 g/mm². Generally, the areal mass of the shim or badge 188 is the mass per unit area of the area defined by the opening 163 to the cavity 161 (see FIG. 22). In some implementations, the area of the opening 163 is about 1,600 mm².

[0428] In some embodiments, the shim or badge 188 has a mass per unit length of between about 0.09 g/mm and about 0.40 g/mm, such as between about 0.09 g/mm and about 0.35 g/mm, such as between about 0.09 g/mm and about 0.30 g/mm, such as between about 0.09 g/mm and about 0.25 g/mm, such as between about 0.09 g/mm and about 0.20 g/mm, such as between about 0.09 g/mm and about 0.17 g/mm, or such as between about 0.1 g/mm and about 0.2 g/mm. In some embodiments, the shim or badge 188 has a mass per unit length less than about 0.25 g/mm, such as less than about 0.20 g/mm, such as less than about 0.17 g/mm, such as less than about 0.15 g/mm, such as less than about 0.10 g/mm. In one implementation, the shim or badge 188 has a mass per unit length of 0.16 g/mm.

Club Head, Damper, Filler Material, and Shim Interaction

[0429] FIG. 40 is an exploded view of the golf club head 100 showing the body 113, the damper 280 and the shim or badge 188. In some embodiments, a unitary cast body 113 is provided. A unitary cast body is manufactured by casting the strike face 110 and the strike face 110 with the body 113 as a single piece. In other embodiments, the body 113 is cast separately from the strike face 110 and/or the strike face 110, and the strike face 110 and/or the strike face 110 is welded to the body 113.

[0430] After the body 113 is manufactured, the damper 280 can be installed within the cavity 161 of the body 113. In some embodiments, an adhesive, an epoxy, and/or a hotmelt is used to install the damper 280 within the cavity. For example, an adhesive can be applied to the damper 280 before installation and/or a hotmelt can be injected into the cavity 161 after the damper 280 has been installed. In some embodiments, hotmelt can be injected into the toeside of the cavity 161. In some embodiments, an adhesive can be

applied to a rear surface of the damper 280, such as to bond the rear surface of the damper 280 to the sole bar 135 or rear portion 128.

[0431] After the damper 280 is installed in the body 113, the shim or badge 188 can be installed on the body 113, enclosing at least a portion of the cavity 161 to define or form an internal cavity. In some embodiments, the shim or badge 188 can be installed using a tape, such as an industrial strength double-sided tape (e.g., DC2000 series 0.8 mm 3M Very High Bond (VHB) or 1.1 mm 3M VHB tape), an adhesive, an epoxy, a weld, a screw(s), or another fastener (s). In some embodiments, a tape is used rather than screws, clamps, or other fasteners to improve aesthetics of the club head. In some embodiments, at least a portion of the shim or badge 188 snaps in place, such as using a friction fit. After installation, the force required to remove the shim or badge 188 can be between about 20 kilogram-force (kgf) and about 50 kgf, more preferably between 25 kgf and 35 kgf. In some embodiments, a sealing wiper is installed around shim to help prevent water intrusion, such as when a discontinuous ledge is used.

[0432] After installing the damper 280 to the body 113, the club head 100 has the appearance of a hollow body iron. The shim or badge 188 seals the cavity 161, such as preventing water from entering the cavity 161. In some embodiments, no portion of the shim or badge 188 contacts the strike face 110. In some embodiments, no structure attached to the badge or shim or badge 188 contacts the strike face 110. In some embodiments, at least a portion of the shim protrudes forward of one or more of the ledges 193, 194 and toward the strike face 110. For example, at least a portion of the cavity 161 separates the shim or badge 188 from the strike face 110.

[0433] An assembled club head weight can be between about 200 grams and about 350 grams, more preferably between 230 grams and 305 grams. A combined weight of damper 280 and shim or badge 188 can be between about 8 g and about 20 g, preferably less than about 13 g, more preferably less than 12 g. In some embodiments, the combined weight of damper 280 and shim or badge 188 can be between about 0.2% and about 10% of the assembled club head weight, preferably between 2.6% and 8.7%, more preferably less than about 5%.

[0434] FIG. 41 is a side cross-sectional view of the golf club head 100. Numerals 4101, 4103, 4105, 4107, 4121, 4123, 4125, and 4127 refer to features of club head 100. The features of club head 100 may also be applicable to club heads 300, 500, and 600. The golf club head 100, as assembled, includes a sole portion 108, a topline portion 106, a rear portion 128, strike face 110, a strike face 110, a sole bar 135, a damper 280, and a shim or badge 188.

[0435] The golf club head 100 includes an upper undercut region 165. In some embodiments, no part of the damper 280 or the shim or badge 188 is within the upper undercut region 165. In some embodiments using a filler material, no filler material is within the upper undercut region 165.

[0436] The golf club head 100 includes a lower undercut region 164. In some embodiments, the damper 280 is installed entirely within the lower undercut region 164. In some embodiments, at least a portion of the damper 280 is installed partially within the lower undercut region 164, thus the damper extends above an opening of the lower undercut region 164 defined by a line perpendicular to the strike face 110 and extending to the upper most point of the lower ledge

194. In some embodiments, the damper 280 does not contact the sole portion 108 and does not entirely fill the lower undercut region 164. The damper 280 can fill a portion of the cavity 161. In some embodiments, the damper 280 fills between about 5% and about 70% of the cavity 161, preferably between 5% and 50%, preferably between 20% and 50%, preferably between 5% and 20%, preferably between 50% and 70%.

[0437] The golf club head 100 may include installation surfaces 4101, 4103, 4105, 4107 for receiving at least a portion of the shim or badge 188. Likewise, the shim or badge 188 can include corresponding installation surfaces 4121, 4123, 4125, and 4127 for receiving at least a portion of the club head 100. In some embodiments, the shim or badge 188 is adhered, taped, bonded, welded, or otherwise affixed to the body 113 between installation surfaces 4101, 4103, 4105, 4107 and installation surfaces 4121, 4123, 4125, and 4127. In some embodiments, the shim or badge 188 is installed using a tape between the installation surfaces 4123, 4125 and the installation surfaces 4103, 4105, respectively. In some embodiments, the tape separates the body 113 from the shim or badge 188. The separation can be between about 0.5 mm and about 1.5 mm, preferably between 0.8 mm and 1.1 mm. In some embodiments, the shim or badge 188 does not contact any portion of the strike face 110 or the strike face 110. For example, when installed, the shim or badge 188 can be up to 10 mm from the strike face 110, such as between 0.1 mm and 10 mm, preferably between 0.1 mm and 5 mm, alternatively between 2 mm and 7 mm. In some embodiments, the shim or badge 188 extends within the cavity 161 and contacts at least a portion of the strike face 110 and/or the strike face 110.

[0438] When compared to using a bridge bar 140 (e.g., depicted in FIG. 6), the shim or badge 188 can allow the club head 100 to have a lower center of gravity (CG). For example, by manufacturing the shim or badge 188 from a light weight, stiff material(s), the shim or badge or rear fascia 188 can provide support for the topline portion 106, such as to provide better sound and feel, while allowing additional discretionary weight be positioned lower in the golf club head 100. Thus, using a shim or badge 188 can allow the golf club head 100 to achieve similar modes for sound and feel, while conferring additional performance benefits achieved by freeing up additional discretionary weight.

[0439] A coefficient of restitution (COR) of the golf club head 100 can be affected by installation of the damper 280 and/or the shim or badge 188. For example, installing the damper 280 and/or a filler material can reduce the COR by between about 1 and about 4 points, preferably no more than 3 points, more preferably no more than 2 points. Installing the shim or badge 188 (e.g., such as a shim or badge 188 that does not contact a rear surface of the strike face and stiffens the topline portion 106) can increase COR by between about 1 and about 6 points, preferably by at least 1 point, more preferably by at least 2 points. Installing the shim or badge 188 with the damper 280 can minimize or negate the loss of COR caused by the damper 280, and in some cases can increase COR for the strike face. For example, installing the shim or badge 188 with the damper 280 can affect COR by between a loss of about 2 points and a gain of about 6 points.

TABLE 3

COR Values Relative to a Calibration Plate			
Example No.	COR Relative to Calibration Plate Without Shim and Without Damper	COR Relative to Calibration Plate With Shim and With Damper	COR Change Between Without Shim and Damper and with Shim and Damper
1	-0.004	-0.004	-0.000
2	-0.002	-0.004	-0.002
3	-0.004	-0.003	0.001
4	-0.004	-0.004	-0.000
5	-0.003	-0.004	-0.001
Average	-0.0034	-0.0038	-0.0004
6	0.000	-0.010	-0.010
7	-0.004	-0.009	-0.005
8	0.000	-0.011	-0.011
9	-0.003	-0.007	-0.004
10	-0.005	-0.009	-0.004
Average	-0.0024	-0.0092	-0.0068
11	-0.001	-0.004	-0.003
12	-0.001	-0.006	-0.005
13	-0.003	-0.007	-0.004
14	-0.005	-0.008	-0.003
15	-0.002	-0.002	0.000
Average	-0.0024	-0.0054	-0.003
16	-0.004	-0.010	-0.006
17	-0.004	-0.009	-0.005
18	-0.004	-0.008	-0.004
19	0.000	-0.005	-0.005
20	-0.005	-0.008	-0.003
Average	-0.0034	-0.008	-0.0046

[0440] Table 3 illustrates the results of COR testing on four different iron embodiments. Examples 1-5 are results for a first 4 iron embodiment. Examples 1-5 show that adding a shim and damper can reduce COR by less than 1 point (i.e., 0.4 points). Examples 6-10 are results for a second 4 iron embodiment. Examples 6-10 show that adding a shim and damper can reduce COR by over 6 points (i.e., 6.8 points). Examples 11-15 are results for a first 7 iron embodiment. Examples 11-15 show that adding a shim and damper can reduce COR by an average of 3 points. Examples 16-20 are results for a second 7 iron embodiment. Examples 16-20 show that adding a shim and damper can reduce COR by an average of 4.6 points. In some embodiments, installing a damper and a shim results in a COR change value of no more than -0.011 compared to a club head without the badge and damper installed.

[0441] As used herein, a COR change value of 0.001 is considered a change value of 1 point and a negative sign means a decrease in COR. If no sign is present, then that represents an increase. For example, Example No. 3 shows an initial COR value of -0.004 without a shim or damper and a value of -0.003 including a shim and damper for a positive COR change value of 0.001 or a 1 point change in COR (i.e., COR increased).

[0442] FIG. 42 is a side cross-sectional view of the golf club head 100, showing a cross-section through the Y-Z plane through a geometric center of the strike face 110, with the club head at zero loft (depicted as cross-section 42-42 in FIG. 21). Numerals 4201, 4203, 4205, 4207, 4209, 4211, and 4213 refer to features of club head 100. The features of club head 100 may also be applicable to club heads 300, 500, and 600. The club head 100 has an upper undercut depth 4201, a lower undercut depth 4203, and a club head section

height 4205. In some embodiments, no portion of shim or badge 188 extends into upper undercut region 165 or the lower undercut region 164.

[0443] An upper portion 4207 of the lower undercut region 164 is at least partial defined by an upper surface 4209 of the lower ledge 194. In some embodiments, the geometric center of the strike face 110 is located above the upper portion 4207 of the lower undercut region 164. In some embodiments, the lower undercut region 164 does not extend beyond the geometric center of the strike face 110.

[0444] A lower portion 4211 of the upper undercut region 165 is at least partial defined by a lower surface 4213 of the lower ledge 193. In some embodiments, the geometric center of the strike face 110 is located below the lower portion 4211 of the upper undercut region 165. In some embodiments, the upper undercut region 165 does not extend beyond the geometric center of the strike face 110.

[0445] In some embodiments, the upper undercut depth 4201 is between about 2 mm and about 10 mm, preferably at least 3 mm, more preferably less than the lower undercut depth 4203, more preferably less than a maximum depth of the lower undercut depth 4203. In some embodiments, the upper undercut depth 4201 is between about 25% and about 50% of the lower undercut depth 4203, preferably between 30% and 40% of the lower undercut depth 4203. In some embodiments, the upper undercut depth 4201 is between about 10% and about 25% of the club head section height 4205, preferably between 13% and 18% of the club head section height 4205, more preferably at least 5% of the club head section height 4205.

[0446] In some embodiments, the lower undercut depth 4203 is less than 50% of the club head section height 4205, more preferably between 30% and 50% of the club head section height 4205, more preferably between 38% and 43% of the club head section height 4205.

[0447] In some embodiments, the lower undercut depth 4203 is at least 2 times the upper undercut depth 4201, preferably at least 2.5 times the upper undercut depth 4201.

[0448] FIG. 43 is a top cross-sectional view of the golf club head 100, showing the body 113 including locating or interlocking features 4301, 4303. Numerals 4301 and 4303 refer to features of club head 100. The features of club head 100 may also be applicable to club heads 300, 500, and 600. In some embodiments, the body 113 includes one or more locating or interlocking features 4301, 4303 that engages the damper 280 during installation. In some embodiments, there is a toeside locating or interlocking feature 4301 and a heelside locating or interlocking feature 4303. In some embodiments, the damper 280 is installed by first positioning the damper 280 in an upper position within the cavity 161, then is moved into a lower position within the cavity 161, engaging one or more of the locating or interlocking features 4301, 4303.

[0449] FIG. 44 is an exploded view of the golf club head 600, showing the body 113 including a shim or badge 188, a fill port 4403 and a screw 4401. Numerals 4401 and 4403 refer to features of club head 600. The features of club head 100 may also be applicable to club heads 100, 300, and 500. In some embodiments, after the shim or badge 188 is installed onto the body 113, a filler material can be injected into the body 113 through the fill port 4403. After the filler material is injected into the body 113, the screw 4401 can be installed in the fill port 4403. In some embodiments, the shim or badge 188 can prevent the filler material from

leaving the body **113** and can also to achieve a desired aesthetic and further dampening. In some embodiments, the filler material completely fills the cavity **161**. In some embodiments, the filler material only partially fills the cavity **161**, such as between 25% and 75% of the cavity **161**, preferably less than 50% of the cavity **161**.

Club Head Sound and Feel

[0450] Exemplary club head structures for acoustic mode altering and dampening are described in U.S. Pat. No. 10,493,336, titled “IRON-TYPE GOLF CLUB HEAD,” which is incorporated herein by reference in its entirety.

[0451] The sound generated by a golf club is based on the rate, or frequency, at which the golf club head vibrates and the duration of the vibration upon impact with a golf ball. Generally, for iron-type golf clubs, a desired first mode frequency is generally above 2000 Hz, such as around 3,000 Hz and preferably greater than 3,200 Hz. Additionally, the duration of the first mode frequency is important because a longer duration may feel like a golf ball was poorly struck, which results in less confidence for the golfer even when the golf ball was well struck. Generally, for iron-type golf club heads, a desired first mode frequency duration is generally less than 10 ms and preferably less than 7 ms.

[0452] In some embodiments, the golf club head **100** has a COR between about 0.5 and about 1.0 (e.g., greater than about 0.79, such as greater than about 0.8) and a Z-up less than about 18 mm, preferably less than 17 mm, more preferably less than 16 mm. In some examples, the golf club head **100** has a first mode frequency between about 3,000 Hertz (Hz) and 4,000 Hz and a fourth mode frequency between about 5,000 Hz and about 7,000 Hz, preferably a first mode frequency between 3,394 Hz and 3,912 Hz and a fourth mode frequency between 5,443 Hz and 6,625 Hz. In these examples, the golf club head **100** has a first mode frequency duration between about 5 milliseconds (ms) and about 9 ms and a fourth mode frequency duration between about 2.5 ms and about 4.5 ms, preferably a first mode frequency duration between about 5.4 ms and about 8.9 ms and a fourth mode frequency duration of about 3.1 ms and about 3.9 ms.

[0453] FIGS. 45-46 provide graphical representations of a golf club head undergoing first through fourth mode frequency vibration and associated characteristics of the golf club head. In some embodiments, such as for a 4 iron, includes a first mode frequency of 3,318 Hz with a first mode frequency duration of 4.8 ms, a second mode frequency of 3,863 Hz with a second mode frequency duration of 5 ms, a third mode frequency of 4,647 Hz with a third mode frequency duration of 2.4 ms, and a fourth mode frequency of 6,050 Hz with a fourth mode frequency duration of 11.6 ms. In some embodiments, such as for a 7 iron, includes a first mode frequency of 3,431 Hz with a first mode frequency duration of 7 ms, a second mode frequency of 4,088 Hz with a second mode frequency duration of 4 ms, a third mode frequency of 4,389 Hz with a third mode frequency duration of 2.8 ms, and a fourth mode frequency of 5,716 Hz with a fourth mode frequency duration of 10 ms.

[0454] Although the foregoing discussion cites features related to golf club head **100** and its variations (e.g. **300**, **500**, **600**), the many design parameters discussed above substantially apply to all golf club heads **100**, **300**, **500**, and **600** due to the common features of the club heads. With that in mind, in some embodiments of the golf clubs described

herein, the location, position or orientation of features of the golf club head, such as the golf club head **100**, **300**, **500**, and **600**, can be referenced in relation to fixed reference points, e.g., a golf club head origin, other feature locations or feature angular orientations. In some instances, the features of club heads **100**, **300**, **500**, and **600** discussed above are referred to by numerals corresponding to their figure numbers (e.g., FIGS. 1-46) and can be applicable to all golf club heads **100**, **300**, **500**, and **600**. Features from **100**, **300**, **500**, and **600** can be used between embodiments. For example, each of golf club heads **100**, **300**, **500**, and **600** can be provided with or without a damper and/or a filler material.

Toewrap Badge Structure

[0455] As club heads continue to relocate discretionary weight low and rearward, it can become more difficult to remove additional mass from high on an iron club head body (i.e., above the center of gravity or Zup) and relocate the mass low on the club head body in order to lower the center of gravity of the club head. In some embodiments, removing too much mass in the central region of the topline portion of the club head can negatively impact the sound, feel, and aesthetics of the club head, and can also compromise durability of the club head body due to stress and deflection caused by removing too much weight from the topline portion.

[0456] Referring back to FIGS. 23, 24, 37, and 38, and as depicted in FIG. 47, the club head **500** can include a body **113** having a heel portion **102**, a toe portion **104**, a sole portion **108**, a topline portion **106**, a rear portion **128**, a strike face **110** (not depicted in FIG. 47), and a hosel **114**.

[0457] The club head portions can be described with respect to an x-axis, y-axis, and z-axis. An x-axis can be defined being tangent to the strike face at the origin and parallel to a ground plane. The x-axis extends in a positive direction from the origin heelward to the heel portion **102** of the club head body and in a negative direction toward from the origin to the toe portion **104** of the club head body. The y-axis intersects the origin and is parallel to the ground plane. The y-axis is orthogonal to the x-axis and extends in a positive direction rearward from the origin to the rear portion **128** of the club head body. The z-axis intersects the origin and is orthogonal to the x-axis, the y-axis, and the ground plane. The z-axis extends in a positive direction from the origin upward to the topline portion **106** of the club head body and in a negative direction from the origin downward to the sole portion **108** of the club head body.

[0458] The heel portion **102** is defined as the portion of the golf club head extending to and including the hosel portion **114** (i.e., the club shaft receiving portion) from a y-z plane passing through the origin. For example, the heel portion extends heelward from a scoreline mid-plane SLmid. The scoreline mid-plane SLmid is a plane defined at the midpoint of the longest scoreline on the strike face **110**, normal to the strike face **110** and normal to the ground plane GP when the golf club is in a zero-loft address position. The toe portion **104** is defined as the portion of the golf club head extending from the y-z plane in a direction opposite the heel portion. For example, the toe portion **104** extends toward from the scoreline mid-plane SLmid.

[0459] The sole portion **108** portion is defined as the portion of the golf club extending to and including the sole of the golf club head from an x-y plane passing through the origin. The sole portion **108** extends downwards from to an

address mid-plane ML, defined 20 mm above and parallel to the ground plane GP, to a lowest point of the club head (i.e., the sole), located at the ground plane GP, when the golf club is in a zero-loft address position.

[0460] The topline portion 106 portion is defined as the portion of the golf club extending to and including the topline of the golf club head from an x-y plane passing through the origin. The topline portion 106 extends upwards from the address mid-plane ML, defined 20 mm above and parallel to the ground plane GP, to a highest point of the club head (i.e., the topline) when the golf club is at a zero-loft address position.

[0461] The rear portion 128 is defined as the portion of the golf club extending to and including the sole bar of the golf club head from an x-z plane passing through the origin. The rear portion 128 extends rearward from the rear surface of the strike face 110 to a rearward-most point of the club head when the golf club is at a zero-loft address position.

[0462] The strike face 110 is defined as the portion of the golf club extending to and including the strike face of the golf club head from an x-z plane passing through the origin. The strike face 110 extends forward from the rear surface of the strike face 110 to a forward-most point of the club head when the golf club is at a zero-loft address position.

[0463] The body 113 can be a unitary cast body having the strike face 110 cast as a single piece with the other portions of the body. Alternatively, one or more of the portions of the body can be manufactured separately and attached to the body 113. For example, the strike face 110 can be welded to the body 113. Other portions of the club head body 113 can also be welded or otherwise attached to the body 113, such as at least a portion of the sole portion 108 and/or the topline portion 106, for example. In some embodiments, the strike face 110 can wrap into the sole portion 108 and/or the topline portion 106.

[0464] The body 113 also includes a hosel portion 114. The hosel portion 114 can include one or more weight reducing features to remove mass from the hosel portion 114, as discussed herein. For example, selectively reducing a wall thickness around the hosel portion 114 can allow for discretionary mass to be relocated to the rear portion 128 of the club head 500, for example.

[0465] As discussed herein, the strike face 110 (not depicted in FIG. 47) has a strike face 110, which can have a variable face thickness profile with a minimum face thickness no less than 1.0 mm and a maximum face thickness no more than 3.5 mm. The variable thickness profile can be provided symmetrically (e.g., with a “donut” shaped area of increased thickness located within the unsupported strike face) or asymmetrically (e.g., with at least one transition region between a thicker region and a thinner region within the unsupported strike face).

[0466] A shim or badge 188 can be formed separately from the body 113 and attached to the body 113. The shim or badge 188 can be received at least in part by the body 113. For example, as depicted in FIG. 47, the shim or badge 188 is received by the body 113 within the rear portion 128 and within the toe portion 104. The shim or badge 188 can be received below the topline portion 106 and above the sole bar 135. In this embodiment, the shim or badge 188 in part forms the outermost surface of the rear portion 128 and the toe portion 104. The body 113 also in part forms the outermost surface of the rear portion 128 and toe portion

104, such as above and below the badge. The body 113 also extends heelward of the shim or badge 188.

[0467] The shim or badge 188 can be formed from one or more materials. For example, the shim or badge 188 can be formed of a lower density material than the body 113. The shim or badge 188 can also be formed from a combination of materials, such as a polymer, a composite, a metal, and/or another material. In some embodiments, the shim or badge 188 can be a multi-material shim formed from a first material having a first density between about 0.5 g/cc and about 2 g/cc and a second material having a second density between about 1.5 g/cc and about 10 g/cc. For example, the first material can be a polymer material and the second material can be a metal or a composite material. In other embodiments, a first material can be a polymer material, a second material can be a composite material, and a third material can be a metal.

[0468] The iron-type golf club head 500 is provided with a weight reduction zone 175 located in the toe portion 104 of the club head 500. The weight reduction zone 175 can include one or more weight reduction features, such as a mass reduction in the toe portion 104 and the badge or shim or badge 188 extending into the weight reduction zone 175 in the toe portion 104. The weight features in the weight reduction zone can reduce between 0.5 g and 4.0 g from the toe portion 104, more preferably between 0.7 g and 3 g, more preferably at least 0.9 g. The weight reduction zone 175 can extend between about 5 mm and 55 mm above the ground plane, preferably between about 10 mm and 45 mm above the ground plane when the club head is in a zero-loft address position. In some embodiments, the weight reduction zone 175 can extend from the sole (e.g., between about 0 mm and about 5 mm above the ground plane) upward. In some embodiments, the weight reduction zone can extend from the topline downward. The weight reduction zone 175 can have a length between about 5 mm and about 15 mm as measured on a plane parallel to the z-axis, such as between about 5 mm and about 10 mm, such as between about 10 mm and about 15 mm. In some embodiments, the weight reduction zone can have a length between about 15 mm and about 55 mm as measured on a plane parallel to the z-axis, such as between about 25 mm and about 45 mm.

[0469] The weight reduction features can shift a center of gravity z-axis location (Zup) by 0.5 mm toward a ground plane, such as between about 0.25 mm and about 4 mm toward the ground plane. In some embodiments, the club head can have a center of gravity z-axis location (Zup) between about 12 mm and about 19 mm above a ground plane, such as between about 13 and about 18 mm, such as between about 14 mm and about 17 mm, preferably no more than 18 mm, more preferably no more than 17.5 mm, and more preferably no more than 17 mm.

[0470] The toe portion the shim or badge 188 replaces high density material in the toe portion of the body (i.e., between about 2.5 g/cc and about 20 g/cc) with a lower density material of the toe portion of the shim or badge 188 (i.e., between about 0.5 g/cc and about 2 g/cc). The shim or badge 188 can wrap from a rear portion 128 of the body into the toe portion 104 of the body 113 to create a multi-material toe portion of the body. The multi-material toe portion can include a first material having a first density between about 2.5 g/cc and about 20 g/cc, and a second material having a second density between about 0.5 g/cc and about 2 g/cc.

Mass removal in the high toe-region of the body allows for lower of the center-of gravity.

[0471] The shim or badge 188 includes a toe-to-rear-portion transition region 178. In some embodiments, the toe-to-rear-portion transition region 178 can form an edge as the shim or badge 188 wraps from the toe portion 104 to the rear portion 128. In some embodiments, the edge can be beveled, creating a ribbon between the rear portion 128 and toe portion 104. In other embodiments, the toe-to-rear-portion transition region 178 can be rounded between the rear portion 128 and toe portion 104. The body 113 also includes a toe-to-topline-portion transition region 181 and a toe-to-sole-portion transition region 182. In some embodiments, transition regions 181, 182 can be rounded between the toe portion 104, the topline portion 106, and/or the sole portion 108. In other embodiments, the transition regions 181, 182 can be provided with an edge, such as a beveled edge. Additional and different features can define the transition regions 178, 181, 182.

[0472] FIG. 48 depicts a toe view of the club head 500 at zero loft. To orient the club head 500 into the toe view, the club head 500 is first oriented in a zero-loft address position. The zero-loft address position has the club head 500 soled on a ground plane and rotated such that a vertical axis tangent to a face plane FP and normal to ground plane GP. The club head is then rotated 90-degrees from a face-on view about a vertical axis counter-clockwise, resulting in a view of the toe portion 104. To orient the club head 500 in a rear view (not depicted in FIG. 48), the club head 500 is rotated another 90-degrees about a vertical axis counter-clockwise (i.e., 180-degrees from the face-on view), resulting in a view of the rear portion 128.

[0473] As depicted in FIG. 48, the shim or badge 188 can extend into the toe portion 104, in part forming an outermost surface of the toe portion 104 when received by the body 113. The outermost surface of the toe portion 104 is defined by the toe view of the club head discussed above. The shim or badge 188 can also form at least part of an outermost surface of the rear portion 128 when received by the body 113. The outermost surface of the rear portion 128 is defined by the rear view of the club head discussed above. In some embodiments, the shim or badge 188 extends into the toe portion 104 by wrapping from the toe portion 104 onto the rear portion 128 to connect at least a portion of the outermost surface of the toe portion 104 and a portion of the outermost surface of the rear portion 128.

[0474] The shim or badge 188 can extend into at least a portion of the toe portion 104 to form a non-continuous, multi-material toe portion 104. For example, the shim or badge 188 can be formed from a polymer material, or a combination of different materials, and the body 113 above and below the shim or badge 188 can be formed from a metal, such as part of a cast metal body 113.

[0475] In some embodiments, the forward-most portion of the shim or badge 188 in the toe portion 104, shown by leading edge line LE, extends beyond a forward-most portion of the shim or badge 188 in the rear portion 128, such as when positioned in the toe view of the club head. The forward-most portion of the shim or badge 188 in the toe portion 104, shown by leading edge line LE, does not extend beyond the face plane line FP. In some embodiments, the face plane line FP and the leading edge line LE are separated by between about 0.5 mm and about 5 mm. Further, in some

embodiments, a gap is positioned between the forward-most portion of the shim or badge 188 in the toe portion 104 and the toe portion 104.

[0476] In some embodiments, the forward-most portion of the shim or badge 188 in the toe portion 104, shown by leading edge line LE, is substantially parallel to the strike face 110, shown by face plane line FP. An upper-most edge of the toe portion of the badge, shown by the upper edge line UP, and a lower-most edge of the toe portion of the badge, shown by the lower edge line LP, may be substantially perpendicular to the strike face 110.

[0477] In some embodiments, the width W1 from the leading edge line LE and the first trailing edge line TE1 is between about 2 mm and about 6 mm, preferably between about 4 mm and about 5 mm. In some embodiments, the width W2 from the leading edge line LE and the second trailing edge line TE2 is between about 10 mm and about 14 mm, preferably between about 11 mm and about 12 mm. In some embodiments, the width W3 from the face plane line FP and the first trailing edge line TE1 is between about 3 mm and about 8 mm, preferably between about 5 mm and about 6 mm. In some embodiments, the width W4 from the face plane line FP and the second trailing edge line TE2 is between about 11.5 mm and about 15.5 mm, preferably between about 12.5 mm and about 13.5 mm.

[0478] In some embodiments, the height H1 from ground plane line GP to the lower edge line LP as measured along the z-axis is between about 10 mm and about 20 mm, preferably between about 12 mm and about 18 mm. In some embodiments, the height H1 from ground plane line GP to the lower edge line LP as measured along the z-axis is within 2 mm of Zup or between Zup-2 mm and Zup+2 mm, preferably Zup±1.5 mm, even more preferably Zup±1 mm. Removing mass above Zup and then redistributing it lower in the club head is preferred, which is a reason some embodiments may have height H1 within 2 mm of Zup. In some embodiments, the height H2 from the lower edge line LP to the upper edge line UP as measured along the z-axis is between about 10 mm and about 30 mm, preferably between about 14 mm and about 25 mm. In some embodiments, the height H3 from the upper edge line UP to a topline plane line TOP as measured along the z-axis is between about 1 mm and about 15 mm, preferably between about 3 mm and about 13 mm. In some embodiments, the height H3 can be eliminated and the shim or badge 188 can extend directly from the topline downward. In some embodiments, the height H1 can be eliminated and the shim or badge 188 can extend directly from the sole upward. In some embodiments, the height H2 can be the entire height of the club head.

[0479] In some embodiments, the height H1 may range from $0.9 \cdot Z_{up}$ to $1.1 \cdot Z_{up}$, and the height H2 may range from $0.7 \cdot Z_{up}$ to $1.3 \cdot Z_{up}$.

[0480] FIG. 49 is a front elevation view of the golf club head 500 (i.e., oriented in a face-on view). FIG. 49 depicts the toeward and heelward boundaries of the scorelines. For example, the scorelines extend toeward up to toeward line SLt and heelward up to heelward line SLh. The scorelines end just before the par line PL. The par line PL is at the transition point between the flat strike face 110 and the organically shaped region that attaches the club head body 113 to the hosel 114 (i.e., the location of a blend of the hosel 114 into the planar strike face 110). The scoreline mid-plane SLmid is a plane defined at the midpoint of the longest

scoreline on the strike face **110**, normal to the strike face **110** and normal to the ground plane GP when the golf club is in a zero-loft address position. The scoreline mid-plane bisects the longest scoreline.

[0481] The club head **500** has a projected area between the scorelines (i.e., between toward line SLt and heelward line SLh) that is projected onto a plane tangent to the face plane between about 1300 mm² and about 2700 mm², such as between about 1400 mm² and about 2100 mm². In some embodiments, a projected area of shim or badge **188** that is projected onto a plane tangent to the face plane is greater than total area of the face within scorelines projected onto the plane tangent to the face plane (i.e., bounded by the heelward-most scoreline SLh, the toward-most scoreline SLt, the upward-most scoreline, and the lower-most scoreline).

[0482] Referring back to FIG. 47, the shim or badge **188** can extend heelward of the scorelines (i.e., heelward of heelward line SLh) and/or heelward of the par line PL. The shim or badge **188** can also extend toward of the scorelines (i.e., toward of toward line SLt). For example, a total length of the badge from a first end to a second end (in a heel-to-toe direction parallel to the ground plane) can be greater than a total length from a par line PL to the toward-most portion of the toe portion denoted by line TP (i.e., PL to TP). In some embodiments, a total length from a heelward-most scoreline (i.e., SLh) to the toward-most portion of the toe portion (i.e., TP) is less than a total length of the shim or badge **188**.

[0483] FIG. 50 is a rear perspective view of the club head **500** without the shim or badge **188** installed. The toe portion **104** includes a beam **132** with a toeside ledge **125** for receiving at least a portion of the shim or badge **188**. The beam **132** can also provide structural support for the topline portion **106** when mass is removed from the toe portion **104**. In some embodiments, the toeside ledge **125** can connect the upper ledge **193** and the lower ledge **194**. In other embodiments, the toeside ledge **125** is only connected to one of the upper ledge **193** or the lower ledge **194**. In other embodiments, the toeside ledge **125** does not connect the upper ledge **193** or the lower ledge **194**.

[0484] In some embodiments, the toe portion **104** extends toward of the beam **132**, and the shim or badge **188** wraps around the beam **132** and forward toward the strike face **110**. In other embodiments, the beam **132** provides a toward peripheral surface of the toe portion **104**, and the shim or badge **188** does not extend beyond or toward of the of the beam **132**. In some embodiments, the shim or badge **188** wraps around both a toward and a heelward side of the beam **132** and forward toward the strike face **110** on both sides of the beam **132**.

[0485] The beam **132** can have one or more relief sections **133** to further reduce discretionary mass above the center of gravity of the club head **500**. By providing relief sections **133** in the beam, additional discretionary mass can be relocated while still providing stiffness to support the badge or shim or badge **188**, the topline portion **106**, and the toe portion **104**. In some embodiments, the relief sections **133** extend only partially through the beam as depicted in FIG. 50. In other embodiments, the relief section **133** extend entirely through the beam **132** to the cavity **161**. In some embodiments, the sections **133** are filled with a filler material.

[0486] FIG. 51 is a front elevation view of the golf club head **500** (i.e., oriented in a toe view at zero-loft) without the shim or badge **188** installed. The toeside ledge **125** extends below the topline portion **106** and above the sole bar **135**. In some embodiments, the toeside ledge **125** connects the upper ledge **193** and the lower ledge **194**. In some embodiments, the relief sections **133** are at least 20% of the toward surface of the beam **132**, such as between about 20% and about 60% of the toward surface of the beam **132**. The toward surface of the beam **132** can be defined by the club head at zero-degrees loft and rotated 90 degrees counter-clockwise about a vertical axis tangent to a face plane and normal to a ground plane.

[0487] As depicted in FIG. 51, the beam **132** can have a minimum beam depth that is less than a minimum thickness of the topline portion **106**. The beam **132** can also have a maximum beam depth that is less than a minimum thickness of the sole bar **135**.

[0488] The beam **132** extends between the shim or badge **188** and the strike face **110**. The shim or badge **188** is received at least in part by the upper ledge **193**, the lower ledge **194**, and the toeside ledge **125**. In some embodiments, the shim or badge **188** can close an opening in the cavity and to enclose an internal cavity volume, such as between 5 cc and 20 cc. Alternatively, the shim or badge **188** can be provided within the cavity of a cavity-back iron.

[0489] The shim or badge **188** is received at least in part by the body **113** below the topline portion **106**. In this embodiment, the shim or badge **188** does not form or extend into any portion of the topline portion **106**. For example, an outermost surface of the topline portion **106** can be formed from a metal. For example, outermost surface of the topline portion **106** can be defined by a topline view of the club head at zero-degrees loft and rotated 90 degrees about a horizontal axis tangent to the face plane and parallel to the ground plane.

[0490] FIG. 52 is a perspective view of the club head **500** depicting three surface areas A1, A2, A3, each depicted with a different cross-hatching. The rear portion of the shim or badge **188** can have a surface area A1 of at least 1,400 mm² and no more than 5,000 mm², such as between about 1,400 mm² and about 2,100 mm², such as between about 1,750 mm² and about 1,950 mm², such as between 2,000 mm² and 4,000 mm², such as between 3,000 mm² and 4,500 mm². The surface area A1 is the area projected onto a plane parallel to the rear view discussed herein. The toe portion of the shim or badge **188** can have a surface area A2 of at least 100 mm² and no more than 400 mm², such as between about 100 mm² and about 250 mm², such as between about 200 mm² and 400 mm², such as between 200 mm² and 350 mm², such as between about 130 mm² and about 180 mm². The toe portion **104** of the body **113** above and below shim or badge **188** can have a surface area A3 of at least 500 mm², such as between about 500 mm² and about 850 mm², such as between about 600 mm² and about 750 mm². The surface areas A2, A3 are the areas projected onto a toe plane, defined as a plane perpendicular to a strike face of the club head and perpendicular to a ground plane, when the club head is in a zero loft orientation on the ground plane. The surface area A2 is greater than a surface area of the outermost surface of the toe portion above the shim or badge **188**, as projected onto the toe plane.

[0491] FIG. 53 is a perspective view of the shim or badge **188** depicting surface areas A4, A5, each depicted with a

different cross-hatching. For example, the shim or badge 188 can have a ledge 3303 used for installing the shim or badge 188 onto the golf club head 500. The ledge 3303 surrounds an inner portion 3307 of the shim or badge 188. The inner portion of the shim or badge 188 can be inserted into the cavity of the club head 500 when the shim or badge 188 is installed. The inner portion of the shim or badge 188 can have a surface area A4 of at least 700 mm², such as between about 700 mm² and about 1,600 mm², such as between about 900 mm² and about 1,400 mm². The ledge 3303 can have a surface area A5 of at least 400 mm², such as between about 400 mm² and about 1,000 mm², such as between about 550 mm² and about 750 mm².

[0492] As depicted in FIG. 53, the shim or badge 188 has a variable thickness and with a three-dimensional outer surface including a toewrap portion 3701. The inner portion 3307 of the shim or badge 188 can be three-dimensional and can protrude into the opening in the cavity of the club head 500. The toewrap portion 3701 can extend beyond all other exterior surfaces of the badge and toward the strike face 110. For example, the toewrap portion 3701 can extend beyond the inner portion 3307 proximate to the strike face 110 of the club head 500. As such, the toewrap portion 3701 can extend forward than any other portion of the shim or badge 188 when installed and the club oriented in normal address and zero-loft positions.

[0493] In some embodiments, the toewrap portion 3701 creates an angle with respect to the rear portion 128 and/or outermost surface of the shim or badge 188. For example, the toewrap portion 3701 can form an angle with respect to the rear portion 128 of the shim or badge 188. For example, the angle can be greater than about 40 degrees, such as between about 40 degrees and about 120, such as between about 60 degrees and about 100 degrees, such as about 80 degrees, about 90 degrees, about 100 degrees, or about 110 degrees. As such, the shim or badge 188 can wrap from the toe portion 104 onto the rear portion 128 forming at least a 40-degree angle as measured between the outermost surface of the toe portion 104 and the outermost surface of the rear portion 128.

[0494] In some embodiments, no portion of the shim or badge 188 directly contacts the strike face 110, such as in a hollow-body iron. In these embodiments, at least a portion of the cavity can separate the shim or badge 188 from the strike face 110. In other embodiments, a portion of the shim or badge 188 can directly contact the strike face 110, such as in a cavity-back iron. For example, toewrap portion 3701 of the shim or badge 188 can extend rearward away from the strike face 110 in the toe portion 104 in a cavity-back iron.

[0495] FIG. 54 depicts another embodiment of the club head 500, which can include a body 113 having a heel portion 102, a toe portion 104, a sole portion 108, a topline portion 106, a rear portion 128, a strike face 110 (not depicted), and a hosel 114. As discussed herein, a damper 280 can be installed within a cavity in the body 113. Alternatively or additionally, a filler material can be injected or otherwise included within the cavity in the body 113.

[0496] A sole bar can define a rearward portion of the sole portion, and a cavity can be defined by a region of the body rearward of the strike face, forward of the sole bar, above the sole, and below the topline. A lower undercut region can be defined within the cavity rearward of the strike face, forward of the sole bar, and above the sole. A lower ledge can extend above the sole bar to further define the lower undercut

region. An upper undercut region can be defined within the cavity rearward of the strike face, forward of an upper ledge and below the topline. The upper ledge can extend below the topline.

[0497] In this embodiment, no beam 132 is provided to support the shim or badge 188. Instead of including a beam 132, a recessed area 130 is provided in the toe portion 104 for supporting the shim or badge 188. For example, by hollowing out the inside the toe portion 104 and forward of the toeside ledge 125, resulting in the recessed area 130, discretionary mass can be removed and relocated lower in the body 113, while providing the toeside ledge 125 for supporting the shim or badge 188. By omitting the beam 132, the support structure for the shim or badge 188 does not need to contact the rear surface of the strike face 110, resulting a larger unsupported area of the strike face 110. The toeside ledge 125 can extend heelward from the toe portion 104 to provide support for the badge or shim or badge 188.

[0498] In some embodiments, the toeside ledge 125 can connect with the upper ledge 193 and/or the lower ledge 194. The lower ledge 193 can have a variable surface area as projected onto a plane substantially parallel to a plane tangent to the lower ledge 193. For example, a lower edge of the lower ledge 193 can be rounded and an upper edge of the lower ledge 193 can be substantially straight. Accordingly, a midpoint of the lower ledge has a greater projected surface area than the endpoints of the lower ledge proximate to the toe and the heel of the club head. In this embodiment, the lower ledge 193 is tapered at each end.

[0499] FIG. 55 depicts a toeward view of an embodiment of the club head 500, without the shim or badge 188 installed. As discussed above, additional discretionary mass can be relocated by omitting the beam 132 and providing a toeside ledge 125 directly in the toeside area of the toe portion. In some embodiments, the toeside area of the toe portion can include another recessed area 130 provided in the outside surface of the toe portion 104. The additional recessed area 130 can allow for more discretionary weight to be relocated lower in the body 113 and to allow for the shim or badge 188 to wrap into the toe portion 104 and sit substantially flush with the areas of the body 113 above and below the shim or badge 188 (as depicted in FIG. 56).

[0500] As depicted in FIG. 55, the toeside ledge 125 can largely follow the shape of the toe portion, such as having an organically rounded profile. As such, when the shim or badge 188 is installed, the club head 500 gives the appearance of a hollow iron. The damper 280 can be installed into the cavity of the body 113 prior to attaching the shim or badge 188. As discussed herein, the shim or badge 188 can include relief portions to reduce contact between the damper 180 and the strike face 110, while improving acoustics and feel of the club head 500.

[0501] FIG. 56 depicts a toeward view of the club head 500, with the shim or badge 188 installed. As depicted, the shim or badge 188 wraps from the rear of the body 113 into the toe portion 104 and toward the strike face 110. The shim or badge 188 can have a three-dimensional external surface, such as including ledges, indentions, and other features that can organically flow with the shape of the body 113. In some embodiments, a chamfered edge 137 can be provided between the shim or badge 188 and the strike face 110, such as to provide for a designed gap between the strike face 110 and the shim or badge 188.

[0502] By increasing the size of the shim or badge 188, additional discretionary weight can be relocated low in the body 113. In some embodiments, the shim or badge 188 can extend from slightly below the topline to the sole bar 135, such as to an upper edge of the sole bar 135. In some embodiments, the shim or badge 188 can extend from topline downward toward the sole portion 108. In some embodiments, the shim or badge can extend into the sole bar 135, such as below an upper edge of the sole bar 135.

[0503] FIG. 57 is a cross-section along line 57 in FIG. 54. As depicted in FIG. 57, the badge or shim or badge 188 can be three-dimensional, and can be installed into the body 113 without contacting the strike face 110. The shim or badge 188 can be installed forming a portion of the rear portion 128 and the sole bar 135. The shim or badge 188 can extend from underneath the topline to above at least a portion of the rear portion 128 and the sole bar 135. Material from the toe portion 104 can be removed, increasing the size of the cavity within the body 113 and increasing the unsupported area of the strike face 110.

Central Regions, Weighted COR, and Club Head Structures

[0504] Exemplary central regions, COR weighting factors and values, weighted COR, balance point COR, COR area, club head testing for weighted COR, CT tuning, and club head structures for increasing COR values are described in U.S. patent application Ser. No. 17/171,656, filed Feb. 9, 2021, which is incorporated herein by reference in its entirety.

[0505] Examples of iron-type, fairway wood-type, driver wood-type, driving iron-type, and hybrid-type club head structures for increasing COR values are described in U.S. patent application Ser. No. 17/191,617, filed Mar. 3, 2021, U.S. patent application Ser. No. 16/673,701, filed Nov. 4, 2019, U.S. patent application Ser. No. 17/107,462, filed Nov. 30, 2020, U.S. patent application Ser. No. 17/003,610, filed Aug. 26, 2020, U.S. patent application Ser. No. 17/107,447, filed Nov. 30, 2020, U.S. Pat. No. 9,975,018, filed Feb. 8, 2017, U.S. patent application Ser. No. 16/866,927, filed May 5, 2020, U.S. patent application Ser. No. 17/110,112, filed Dec. 2, 2020, U.S. patent application Ser. No. 17/105,234, filed Nov. 25, 2020, U.S. patent application Ser. No. 16/795,266, filed Feb. 19, 2020, U.S. patent application Ser. No. 17/131,539, filed Dec. 22, 2020, U.S. patent application Ser. No. 17/198,030, filed Mar. 10, 2021, U.S. patent application Ser. No. 16/875,802, filed May 15, 2020, U.S. patent application Ser. No. 16/990,666, filed Aug. 11, 2020, which are incorporated herein by reference in their entireties.

Central Regions

[0506] In various embodiments, central regions and striking locations can be selected for weighted COR, such as based at least in part on the type of golf club head. For example, historical data (e.g., real shot data points) can indicate that different types of golf club heads (e.g., iron-type, hybrid-type, wood-type, etc.) are typically struck at different locations on the strike face. For example, iron-type golf club heads typically strike golf balls off of the ground more often than off of a tee, such as when compared to driver wood-type club heads. Further, when iron-type golf club heads strike golf balls off of a tee, the golf ball is often teed lower than when teeing a golf ball for a driver wood-type

golf club head. Likewise, iron-type golf club heads typically strike golf balls with a steeper angle of attack, while driver wood-type golf club heads typically strike golf balls with a shallower angle of attack, and in some cases with a positive angle of attack. Likewise, hybrid-type and fairway wood-type club heads often strike golf balls off of the ground and off of a lower tee than driver wood-type golf club heads. Taken together, real shot data points for different types of golf club heads can indicate that the different types of golf club heads often strike the golf ball at different locations between the types of heads. For example, iron-type, hybrid-type, and fairway wood-type golf club heads often strike the golf ball lower on the face compared to some driver wood-type golf club heads. Using this data for different types of golf club heads, different central regions, striking locations, and COR weighting factors can be chosen based on the unique strike patterns for the particular golf club head type (e.g., different patterns between irons and woods), as well as different lofts within a golf club head type (e.g., different patterns between short and long irons).

[0507] In addition to differences between golf club head types, historical data can also indicate that differences in striking patterns exist between different groups of golfers. For example, low handicap golfers have more consistent striking patterns, as well as often striking the golf club low in the heel and high in the toe, and generally lower on the face. Higher handicap golfers have more erratic striking patterns, and often strike the golf ball high on the face. Different styles of golf swings can also result in different striking patterns. For example, some golfers have steeper angles of attack (e.g., so-called diggers) relative to other golfers with shallower angles of attack (e.g., so-called pickers), and can be grouped based on their relative angles of attack. Likewise, golfers can be grouped based on relative swing speeds (e.g., driver swing speeds: (1) less than 95 mph; (2) 95 mph to 105 mph; and (3) greater than 105 mph). Using this additional data, different central regions, striking locations, and COR weighting factors can be chosen based on the unique strike patterns for different groups of golfers and the particular golf club head type.

[0508] Further, in various embodiments, additional and different central regions can be used, such as with additional or fewer striking locations. In some embodiments, fewer striking locations can be used to simply design and/or manufacturing processes for the club head, such as with a tradeoff of incorporating fewer real shot data points on the strike face. In other embodiments, additional striking locations can be used to incorporate data for additional real shot data points on the strike face. For example, using three striking locations (e.g., FIG. 60) can include at least about 38% of real shots. In another example, using five striking locations (not depicted) can include at least about 62% of real shots. In another example, using eight striking location (e.g., FIG. 61) can include at least about 85% of real shots. Symmetric or asymmetric striking locations can also be selected based on the historical shot data. In some embodiments, the central region 120A is centered on a geometric center of the strike face 110. Alternatively, the central region 120A can be centered on a point located at a mid-point of the longest scoreline on the strike face and 20.5 mm above the ground plane when the golf club head is at a normal address position.

[0509] In the embodiment depicted in FIGS. 58-59, the central region 120A is defined for a cavity back iron-type

golf club head **100**. In other embodiments, the central region **120A** can be defined for other iron-type golf club heads, including blade irons, muscle back irons, hollow irons, and other iron-types. In other embodiments, the central region **120A** can be defined for wood-type club heads, hybrid or utility-type club heads, or other golf club heads. For example, in the embodiment depicted in FIG. 60, the central region **120A** is defined for a wood-type (e.g., FIG. 63) or a hybrid-type (e.g., FIG. 62) golf club head.

[0510] FIG. 59 illustrates a front elevation view of another golf club head **100** with striking locations **101**, **102A**, **103A**, **104A**, **105A**, **106A**, **107A** within a central region **120A** positioned on the strike face **110**. For example, the strike or strike face **110** can include the central region **120A** centered on a geometric center of the strike face **110**. In some embodiments, the central region **120A** is defined with the club head **100** at zero-degrees loft and the central region is positioned on a face plane normal to a ground plane. In some embodiments, the central region **120A** is centered on a different location on the face, such as the location of the club head center of gravity (CG) projected onto the strike face **110** or another location. The central region **120A** can be defined by a 36 millimeter (mm) by 18 mm rectangular area centered on the strike face **110**. The central region can be elongated in a heel-to-toe direction, such as tangential to the face **110** and parallel to a ground plane (GP). In some embodiments, the central region **120A** is elongated at an angle with respect to the GP, such as elongated at a 45-degree angle to GP and extending from low-to-high in a heel-to-toe direction or in another direction. In some embodiments, the central region **120A** can be defined by a larger or smaller rectangular area, defined by a different shape, such as a circular region, an octagonal region, a square region, a diamond shaped region, or another in another shape.

[0511] The central region **120A** can be used to define a central region coordinate system. For example, the central region coordinate system can be defined by the 36 millimeter (mm) by 18 mm rectangular area centered on the geometric center of the strike face. In this example, the central region coordinate system is defined with the club head at zero-degrees loft and positioned on a face plane normal to a ground plane. The central region coordinate system can be elongated in a heel-to-toe direction, and can include a central region x-axis being tangent to the strike face at the origin and parallel to a ground plane. The x-axis extends in a positive direction from the origin to the heel portion of the club head body. The central region coordinate system can also include a central region y-axis intersecting the origin being perpendicular to the ground plane and orthogonal to the x-axis. The y-axis extends in a positive direction from the origin to the topline portion of the club head body. Locations in the central region coordinate system can be referred to with x-axis and y-axis coordinates with a “cr” subscript, such as (x_{cr} , y_{cr}).

[0512] FIG. 59 illustrates the central region **120A** depicted in FIG. 58. For example, the central region **120A** includes striking locations **101**, **102A**, **103A**, **104A**, **105A**, **106A**, **107A** for a right-handed golf club head. The central region **120A** includes a first striking location **101** positioned 9 mm below the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (0, -9). The central region **120A** includes a second striking location **102A** positioned 9 mm toe-ward of the geometric center of the strike

face **110** corresponding to an (x, y) coordinate of (-9, 0). The central region **120A** includes a third striking location **103A** positioned at the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (0, 0). The central region **120A** includes a fourth striking location **104A** positioned 9 mm toe-ward of and 9 mm below the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (-9, -9). The central region **120A** includes a fifth striking location **105A** positioned 9 mm heel-ward of and 9 mm below the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (9, -9). The central region **120A** includes a sixth striking location **106A** positioned 18 mm toe-ward of the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (-18, 0). The central region **120A** includes a seventh striking location **107A** positioned 9 mm heel-ward of the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (0, -9). The above coordinates are provided in a 1 mm scale, but other scales can be used.

[0513] FIG. 60 illustrates another embodiment of a central region **120A**. The central region **120A** can be defined by a 20 millimeter (mm) by 10 mm rectangular area centered on the strike face **110**. The central region can be elongated in a heel-to-toe direction, such as tangential to the face **110** and parallel to a ground plane (GP). For example, the central region **120A** includes striking locations **101**, **102A**, **103A** for a right-handed golf club head. The central region **120A** includes a first striking location **101** at the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (0, 0). The central region **120A** includes a second striking location **102A** positioned 10 mm toe-ward of and 5 mm above the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (-10, 5). The central region **120A** includes a third striking location **103A** positioned 10 mm heel-ward of and 5 mm below the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (10, -5). The above coordinates are provided in a 1 mm scale, but other scales can be used.

[0514] FIG. 61 illustrates the central region **120A** depicted in FIG. 58. The central region **120A** can be defined by a 48 millimeter (mm) by 24 mm rectangular area centered on the strike face **110**. The central region can be elongated in a heel-to-toe direction, such as tangential to the face **110** and parallel to a ground plane (GP). For example, the central region **120A** includes striking locations **101**, **102A**, **103A**, **104A**, **105A**, **106A**, **107A**, **108A** for a right-handed golf club head. The central region **120A** includes a first striking location **101** positioned at the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (0, 0). The central region **120A** includes a second striking location **102A** positioned 12 mm toe-ward of the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (-12, 0). The central region **120A** includes a third striking location **103A** positioned 12 mm heel-ward of the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (12, 0). The central region **120A** includes a fourth striking location **104A** positioned 12 mm toe-ward of and 12 mm above the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (-12, 12). The central region **120A** includes a fifth striking location **105A** positioned 12 mm above the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (0, 12). The central region **120A** includes a sixth striking location **106A** positioned 12 mm below the geometric center of the strike

face **110** corresponding to an (x, y) coordinate of (0, -12). The central region **120A** includes a seventh striking location **107A** positioned 24 mm toe-ward of the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (-24, 0). The central region **120A** includes an eighth striking location **108A** positioned 12 mm heel-ward of and 12 mm below the geometric center of the strike face **110** corresponding to an (x, y) coordinate of (12, -12). The above coordinates are provided in a 1 mm scale, but other scales can be used.

COR Weighting Factors, COR Values, and COR Drop Off Values

[0515] Each striking location has a weighting factor and a COR value. The weighting factors can be selected based on historical data on the impact locations where golfers most often impact the golf ball on the strike face **110**. To selectively increase or optimize COR at likely impact locations on the strike face of the golf club heads, weighting factors are selected for each of the striking locations. The weighting factors and COR values are then used to calculate a weighted COR value for the golf club head. COR values are tested with the golf club head in a zero-loft address position. In some embodiments, the COR values for the striking locations can be between about 0.650 and about 0.900, such as between about 0.700 and about 0.840, such as between about 0.710 and about 0.850. In some embodiments, the weighted COR value can be between about 0.740 and about 0.800, such as between about 0.780 and about 0.790.

[0516] COR values can also be expressed as COR changes relative to a calibration plate used during COR testing. The calibration plate dimensions and weight are described in section 4.0 of the Procedure for Measuring the Velocity Ratio of a Club Head for Conformance to Rule 4-1e. Due to the slight variability between different calibration plates, difference different golf balls, and other testing variabilities, the COR values can be described in terms of a change in COR relative to a calibration plate base value established during testing. For example, if a tested calibration plate has a 0.831 COR value, a 0.844 COR value, or another COR value, measuring a change in COR for a given head relative to the tested calibration plate is accurate and highly repeatable. The change in COR relative to the calibration plate can be described as a COR drop off relative to the calibration plate. For example, COR drop off values can be calculated by subtracting a measured COR value of the calibration plate from a COR value measured at the respective coordinate of a striking location to determine a respective drop off value for the location. In some embodiments, the COR drop off value for a particular striking location can be between about -0.150 and about 0.050, preferably between about -0.140 and about 0.000. In some embodiments, the weighted COR drop off value can be between about -0.104 and about -0.044, such as between about -0.064 and about -0.054.

[0517] For example, Table 4 includes exemplary values for an embodiment of an iron-type golf club head. In this example, a COR drop off value for location **101** can be between about -0.100 and about -0.130, for location **102A** can be between about 0.000 and about -0.090, for location **103A** can be between about 0.040 and about -0.050, for **104** can be between about -0.100 and about -0.200, for location **105A** can be between about -0.090 and about -0.160, for **106** can be between about -0.100 and about -0.170, and for

location **107A** can be between about 0.000 and about -0.090. In this example, a weighted COR can be between about 0.740 and about 0.800, such as about 0.759.

TABLE 4

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, -9)	0.2347	0.730	-0.114
102 (-9, 0)	0.1935	0.804	-0.040
103 (0, 0)	0.1715	0.840	-0.004
104 (-9, -9)	0.1518	0.701	-0.143
105 (9, -9)	0.1230	0.717	-0.127
106 (-18, 0)	0.0740	0.707	-0.137
107 (9, 0)	0.0515	0.804	-0.040

[0518] The exemplary weighting factors in table 4 can be applicable for a club head that is typically struck relatively lower on the face (e.g., a 7 iron vs. a 4 iron) and/or applicable for players that typically strike the club head relatively lower on the face. Alternatively, different weighting factors can be used for club heads that are typically struck relatively higher on the face (e.g., a 4 iron vs. a 7 iron) and/or are applicable for players that typically strike the club head relatively higher on the face. For example, location **101** (0, -9) can have a weighting factor of about 0.1390, location **102A** (-9, 0) can have a weighting factor of about 0.2520, location **103A** (0, 0) can have a weighting factor of about 0.2770, location **104A** (-9, -9) can have a weighting factor of about 0.0700, location **105** (9, -9) can have a weighting factor of about 0.0890, location **106A** (-18, 0) can have a weighting factor of about 0.0740, and location **107A** (9, 0) can have a weighting factor of about 0.0980. The exemplary weighting factors and COR values described herein can be applicable to any club head, including any iron within a set of iron-type club heads.

[0519] In some embodiments, an iron-type club head (e.g., a 7 iron, a 4 iron, or another iron) can have a first COR drop off value between -0.090 and -0.130, a second COR drop off value is between 0.000 and -0.090, a third COR drop off value is between 0.010 and -0.010, a fourth COR drop off value is between -0.100 and -0.200, a fifth COR value is between -0.090 and -0.160, a sixth COR value is between -0.100 and -0.170, and a seventh COR value is between 0.000 and -0.090.

[0520] In some embodiments, an iron-type club head (e.g., a 7 iron, a 4 iron, or another iron) can have a first COR drop off value is between -0.100 and -0.130, a second COR drop off value is between -0.020 and -0.040, a third COR drop off value is between 0.006 and -0.006, a fourth COR drop off value is between -0.130 and -0.160, a fifth COR value is between -0.115 and -0.135, a sixth COR value is between -0.110 and -0.135, and a seventh COR value is between -0.010 and -0.040.

[0521] In another embodiment, Table 5 includes exemplary values for a wood-type golf club head (e.g., a fairway wood). In this example, using three (3) striking locations can incorporate historical data for approximately 38% of real shots. Further, in this example, the fairway wood can be a 15-degree fairway wood with a weighted COR of 0.804 and an unweighted COR of 0.801, resulting in a change (i.e., a delta) of 0.003.

TABLE 5

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.812	-0.032
102 (-10, 5)	0.3796	0.800	-0.044
103 (10, -5)	0.1673	0.790	-0.054

[0522] In another embodiment, Table 6 includes exemplary values for another wood-type golf club head using three (3) striking locations. In this example, the fairway wood can be a 15-degree fairway wood with a weighted COR of 0.807 and an unweighted COR of 0.799, resulting in a change of 0.008.

TABLE 6

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.823	-0.021
102 (-10, 5)	0.3796	0.805	-0.039
103 (10, -5)	0.1673	0.770	-0.074

[0523] In another embodiment, Table 7 includes exemplary values for another wood-type golf club head using three (3) striking locations. In this example, the fairway wood can be a 15-degree fairway wood with a weighted COR of 0.781 and an unweighted COR of 0.778, resulting in a change of 0.003.

TABLE 7

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.791	-0.053
102 (-10, 5)	0.3796	0.776	-0.068
103 (10, -5)	0.1673	0.766	-0.078

[0524] In another embodiment, Table 8 includes exemplary values for another wood-type golf club head using three (3) striking locations. In this example, the fairway wood can be a 15-degree fairway wood with a weighted COR of 0.789 and an unweighted COR of 0.785, resulting in a change of 0.004.

TABLE 8

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.802	-0.042
102A (-10, 5)	0.3796	0.780	-0.064
103A (10, -5)	0.1673	0.773	-0.071

[0525] In another embodiment, Table 9 includes exemplary values for another wood-type golf club head using three (3) striking locations. In this example, the fairway wood can be a 15-degree fairway wood with a weighted COR of 0.793 and an unweighted COR of 0.789, resulting in a change of 0.004.

TABLE 9

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.816	-0.028
102A (-10, 5)	0.3796	0.771	-0.073
103A (10, -5)	0.1673	0.782	-0.062

[0526] In another embodiment, Table 10 includes exemplary values for a wood-type golf club head (e.g., a driver). In this example, using eight (8) striking locations can incorporate historical data for approximately 85% of real shots. In this example, the wood-type club head can be a 9-degree driver with a weighted COR of 0.803 and an unweighted COR of 0.793, resulting in a change of 0.010.

TABLE 10

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.3107	0.823	-0.021
102A (-12, 0)	0.2261	0.805	-0.039
103A (12, 0)	0.1083	0.77	-0.074
104A (-12, 12)	0.1046	0.799	-0.045
105A (0, 12)	0.0957	0.813	-0.031
106A (0, -12)	0.0742	0.787	-0.057
107A (-24, 0)	0.0417	0.78	-0.064
108A (12, -12)	0.0388	0.772	-0.072

[0527] In another embodiment, Table 11 includes exemplary values for another wood-type golf club head using eight (8) striking locations. In this example, the wood-type club head can be a 9-degree driver with a weighted COR of 0.814 and an unweighted COR of 0.805, resulting in a change of 0.009.

TABLE 11

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.3107	0.833	0.011
102A (-12, 0)	0.2261	0.815	0.029
103A (12, 0)	0.1083	0.78	0.064
104A (-12, 12)	0.1046	0.809	0.035
105A (0, 12)	0.0957	0.818	0.026
106A (0, -12)	0.0742	0.804	0.04
107A (-24, 0)	0.0417	0.795	0.049
108A (12, -12)	0.0388	0.782	0.062

[0528] In another embodiment, Table 12 includes exemplary values for a wood-type golf club head (e.g., a fairway wood). In this example, using five (5) striking locations can incorporate historical data for approximately 62% of real shots. In this embodiment, the historical data dictates the striking locations chosen, resulting in asymmetric striking locations being included in the Table 12 (e.g., three locations toe-ward and only one location heel-ward of the origin). In this example, the wood-type club head can be a 15-degree fairway wood with a weighted COR of 0.813 and an unweighted COR of 0.812, resulting in a change of 0.001.

TABLE 12

Striking Location	COR Weighting Factor	Shots Captured	COR Value	COR Dropoff Value
101 (−3.2, 1.4)	0.2631	33,090 (16%)	0.817	−0.027
102A (0, 0)	0.2219	27,908 (14%)	0.823	−0.021
103A (−11.4, 3.7)	0.1935	24,339 (12%)	0.803	−0.041
104A (4.6, −3.3)	0.1664	20,940 (10%)	0.809	−0.035
105A (−6.7, −5.4)	0.1550	19,496 (10%)	0.807	−0.037

[0529] In another embodiment, Table 13 includes exemplary values for a wood-type golf club head using five (5) striking locations. In this example, the wood-type club head can be a 15-degree fairway wood with a weighted COR of 0.804 and an unweighted COR of 0.803, resulting in a change of 0.001.

TABLE 13

Striking Location	COR Weighting Factor	Shots Captured	COR Value	COR Dropoff Value
101 (−3.2, 1.4)	0.2631	33,090 (16%)	0.807	0.037
102A (0, 0)	0.2219	27,908 (14%)	0.812	0.032
103A (−11.4, 3.7)	0.1935	24,339 (12%)	0.797	0.047
104A (4.6, −3.3)	0.1664	20,940 (10%)	0.803	0.041
105A (−6.7, −5.4)	0.1550	19,496 (10%)	0.797	0.047

[0530] In another embodiment, Table 14 includes exemplary values for a wood-type golf club head using six (6) striking locations. In this example, the wood-type club head can be a 15-degree fairway wood, such as with a steel face welded to the body, with a weighted COR of 0.802 and an unweighted COR of 0.798, resulting in a change of 0.004.

TABLE 14

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.3000	0.814	0.030
102A (0, 2.2)	0.2000	0.816	0.028
103A (0, 5)	0.1250	0.811	0.033
104A (0, −5)	0.1250	0.793	0.051
105A (−12.7, 0)	0.1250	0.771	0.073
106A (12.7, 0)	0.1250	0.781	0.063

[0531] In another embodiment, Table 15 includes exemplary values for a wood-type golf club head (e.g., a fairway wood). In this embodiment, the historical data also dictates the striking locations chosen, resulting in asymmetric striking locations being included in the Table 15 (e.g., four locations toe-ward origin, one location heel-ward of the origin, and no locations at the origin). In this example, the wood-type club head can be a 15-degree fairway wood with a weighted COR of 0.810 and an unweighted COR of 0.810, resulting in a change of 0.000.

TABLE 15

Striking Location	COR Weighting Factor	Shots Captured	COR Value	COR Dropoff Value
101 (−3.84, 2.42)	0.2262	6,136	0.812	−0.032
102A (−0.45, 0.25)	0.2124	5,761	0.819	−0.025

TABLE 15-continued

Striking Location	COR Weighting Factor	Shots Captured	COR Value	COR Dropoff Value
103A (−7.30, 1.49)	0.2085	5,656	0.807	−0.037
104A (−2.46, −3.15)	0.1817	4,930	0.805	−0.039
105A (3.38, −0.89)	0.1712	4,643	0.805	−0.039

[0532] In another embodiment, Table 16 includes exemplary values for a wood-type golf club head with asymmetric striking locations being included. In this example, the wood-type club head can be a 15-degree fairway wood with a weighted COR of 0.804 and an unweighted COR of 0.803, resulting in a change of 0.001.

TABLE 16

Striking Location	COR Weighting Factor	Shots Captured	COR Value	COR Dropoff Value
101 (−3.84, 2.42)	0.2262	6,136	0.808	−0.036
102A (−0.45, 0.25)	0.2124	5,761	0.809	−0.035
103A (−7.30, 1.49)	0.2085	5,656	0.793	−0.051
104A (−2.46, −3.15)	0.1817	4,930	0.804	−0.040
105A (3.38, −0.89)	0.1712	4,643	0.803	−0.041

[0533] In another embodiment, Table 17 includes exemplary values for a hybrid-type golf club head using three (3) striking locations. In this example, the hybrid-type club head can be a 19-degree hybrid with a weighted COR of 0.789 and an unweighted COR of 0.786, resulting in a change of 0.003.

TABLE 17

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.797	−0.047
102A (−10, 5)	0.3796	0.785	−0.059
103A (10, −5)	0.1673	0.775	−0.069

[0534] In another embodiment, Table 18 includes exemplary values for a hybrid-type golf club head using three (3) striking locations. In this example, the hybrid-type club head can be a 19-degree hybrid with a weighted COR of 0.792 and an unweighted COR of 0.784, resulting in a change of 0.008.

TABLE 18

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.808	−0.036
102A (−10, 5)	0.3796	0.790	−0.054
103A (10, −5)	0.1673	0.755	−0.089

[0535] In another embodiment, Table 19 includes exemplary values for a hybrid-type golf club head using three (3) striking locations. In this example, the hybrid-type club head can be a 19-degree hybrid, such as with a cast face, with a weighted COR of 0.766 and an unweighted COR of 0.763, resulting in a change of 0.003.

TABLE 19

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.776	-0.068
102A (-10, 5)	0.3796	0.761	-0.083
103A (10, -5)	0.1673	0.751	-0.093

[0536] In another embodiment, Table 20 includes exemplary values for a hybrid-type golf club head using three (3) striking locations. In this example, the hybrid-type club head can be a 19-degree hybrid with a weighted COR of 0.774 and an unweighted COR of 0.770, resulting in a change of 0.004.

TABLE 20

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.787	-0.057
102A (-10, 5)	0.3796	0.765	-0.079
103A (10, -5)	0.1673	0.758	-0.086

[0537] In another embodiment, Table 21 includes exemplary values for a hybrid-type golf club head using three (3) striking locations. In this example, the hybrid-type club head can be a 19-degree hybrid with a weighted COR of 0.797 and an unweighted COR of 0.789, resulting in a change of 0.008.

TABLE 21

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.813	-0.031
102A (-10, 5)	0.3796	0.795	-0.049
103A (10, -5)	0.1673	0.760	-0.084

[0538] In another embodiment, Table 22 includes exemplary values for a hybrid-type golf club head using three (3) striking locations. In this example, the hybrid-type club head can be a 19-degree hybrid with a weighted COR of 0.802 and an unweighted COR of 0.794, resulting in a change of 0.008.

TABLE 22

Striking Location	COR Weighting Factor	COR Value	COR Dropoff Value
101 (0, 0)	0.4531	0.818	-0.026
102A (-10, 5)	0.3796	0.800	-0.044
103A (10, -5)	0.1673	0.765	-0.079

[0539] In some embodiments, the strike face 110 can have a COR area between about 100 mm² and about 300 mm², such as between about 150 mm² and about 200 mm², or between about 85 mm² and about 125 mm², such as between about 95 mm² and about 115 mm². In these embodiments, the COR area is the area of the strike face 110 defined by locations on the strike face 110 with a COR drop off value above -0.045, such as above -0.044. In some embodiments, the COR area is the area of the strike face 110 defined by locations on the strike face 110 with a COR value of 0.790, 0.800, or COR another value.

Head Structures for Increasing COR Values

[0540] In some embodiments, such as depicted in FIG. 58, the club head 100 includes a body 113 having a heel portion 102, a toe portion 104, a topline portion 106, a rear portion 128, a strike face 110 comprising a strike face 110, a sole portion 108 extending rearwardly from a lower end of the strike face 110 to a lower portion of the rear portion 128. The strike face 110 includes a geometric center defining an origin of a coordinate system when the club head is at a normal address position. For example, the coordinate system includes: an x-axis being tangent to the strike face at the origin and parallel to a ground plane; a y-axis intersecting the origin being parallel to the ground plane and orthogonal to the x-axis; and a z-axis intersecting the origin being orthogonal to both the x-axis and the y-axis. The x-axis extends in a positive direction from the origin to the heel portion of the club head body, the y-axis extends in a positive direction from the origin to the rear portion of the club head body, and the z-axis extends in a positive direction from the origin to the topline portion of the club head body.

[0541] The heel portion 102 is defined as the portion of the golf club head extending to and including the hosel portion 114 (i.e., the club shaft receiving portion) from a y-z plane passing through the origin. For example, the heel portion 102 extends heelward from a scoreline mid-plane. The scoreline mid-plane is a plane defined at the midpoint of the longest scoreline on the strike face 110, normal to the strike face 110 and normal to the ground plane when the golf club is in a zero-loft address position. The toe portion 104 is defined as the portion of the golf club head extending from the y-z plane in a direction opposite the heel portion 102. For example, the toe portion 104 extends toward from the scoreline mid-plane.

[0542] The sole portion 108 portion is defined as the portion of the golf club extending to and including the sole of the golf club head from an x-y plane passing through the origin. The sole portion 108 extends downwards from to an address mid-plane defined 20 mm above and parallel to the ground plane GP, to a lowest point of the club head (i.e., the sole), located at the ground plane, when the golf club is in a zero-loft address position. The topline portion 106 portion is defined as the portion of the golf club extending to and including the topline of the golf club head from an x-y plane passing through the origin. The topline portion 106 extends upwards from the address mid-plane, defined 20 mm above and parallel to the ground plane, to a highest point of the club head (e.g., the topline) when the golf club is at a zero-loft address position.

[0543] The rear portion 128 is defined as the portion of the golf club extending to and including the sole bar of the golf club head from an x-z plane passing through the origin. The rear portion 128 extends rearward from the rear surface of the strike face 110 to a rearward-most point of the club head when the golf club is at a zero-loft address position. The strike face 110 is defined as the portion of the golf club extending to and including the strike face of the golf club head from an x-z plane passing through the origin. The strike face 110 extends forward from the rear surface of the strike face 110 to a forward-most point of the club head when the golf club is at a zero-loft address position.

[0544] In some embodiments, the heel portion 102 extends towards, and includes, the golf club shaft receiving portion (e.g., the hosel portion 114) from a y-z plane passing through the origin, and the toe portion 104 can be defined as the

portion of the club head extending from the y-z plane in a direction opposite the heel portion **102**. In some embodiments, a sole bar can define a rearward portion of the sole portion **108**. In some embodiments, a cavity can be defined by a region of the body **113** rearward of the strike face **110**, forward of the rear portion **128**, above the sole portion **108**, and below the topline portion **106**.

[0545] In some embodiments, the club head body can be a unitary cast body. A unitary cast body is manufactured by casting the body **113** with the strike face **110**. In other embodiments, the body **113** and the strike face **110** can be cast or forged separately. In some of these embodiments, the strike face **110** is welded to the body **113**. For example, the club head can be a hollow body iron with a forged strike face **110** that is welded to a cast body **113**. In some embodiments, the club head has a center of gravity z-axis location (Z_{up}) between 10 mm and 20 mm above a ground plane, such as less than 19 mm, less than 18 mm, less than 17 mm, or less than 16 mm.

[0546] One or more club head features can be manipulated to increase COR and CT at different locations across the strike face **110**. For example, applicable club head features can be found in U.S. patent application Ser. No. 17/132,520, filed Dec. 23, 2020, which is incorporated by reference herein in its entirety. For example, a shim or badge can be received at least in part by the body to create the appearance of a hollow-body iron. The shim or badge can be configured to close an opening in the cavity and to enclose an internal cavity volume between 5 cc and 20 cc. In some embodiments, no portion of the shim or badge directly contacts the strike face, allowing the unsupported area of the strike face **110** to flex without being restricted by the shim or badge.

[0547] In some embodiments, the shim or badge includes a first layer of acrylonitrile-butadiene-styrene (ABS) plastic and a second layer of very high bond (VHB) tape. The VHB tape can have a thickness between 0.5 mm and 1.5 mm and can dampen vibrations of the club head. For example, the VHB tape can be applied directly to the topline portion **106** and can dampen some vibrations directly at the source of those vibrations at the topline. By applying damping at the propagation location of the vibrations, the vibrations can be dampened at the source, reducing vibrations that can excite other modes in the iron at other locations.

[0548] In some embodiments, a damper can be positioned within the internal cavity and can extend from the heel portion **102** to the toe portion **104**. In some embodiments, the front surface of the damper can include one or more relief portions, and the front surface of the damper can contact a rear surface of the strike face **110** (e.g., the strike face **110**) between the one or more relief portions. In some embodiments, the strike face **110** comprises an unrestricted face area extending above the damper and below the topline portion **106**. In some embodiments, the club head can be configured to receive a filler material within the internal cavity, such as through a filler port in the toe portion **104**. The filler material can extend from the heel portion **102** to the toe portion **104**.

[0549] Depending on the type of club head (e.g., iron-type, hybrid-type, wood-type, etc.), the club head can have a head height between about 25 mm and about 60 mm, such as less than about 46 mm, as measured with the club head in a normal address position. An iron-type club head can have a volume between about 10 cc and about 120 cc, such as between about 30 cc and about 100 cc, such as between

about 40 cc and about 90 cc, such as between about 50 cc and about 80 cc, such as between about 60 cc and about 80 cc. In various embodiments, the iron-type club head can include a projected face area between about 2,900 mm² and about 3,400 mm², such as between about 3,000 mm² and about 3,200 mm², such as between about 3,100 mm² and about 3,200 mm². A wood-type club head (e.g., a fairway wood) can have a volume between about 120 cc and about 240 cc, and a projected face area between about 1,800 mm² and 2,500 mm², such as between about 2,000 mm² and about 2,300 mm². A hybrid-type club head can have a volume between about 60 cc and about 150 cc, and a projected face area between about 2,000 mm² and 3,000 mm², such as between about 2,200 mm² and about 2,800 mm².

[0550] In some embodiments, an unsupported area of the strike face **110** can be increased, resulting in higher COR and CT values. For example, by removing material from the heel portion **102**, the toe portion **104**, the topline portion **106**, and/or the sole portion **108**, the unsupported face area can be increased by between about 1% and about 12%, such as between 4% and 10%, such as about 6%. In some embodiments, material is removed from low in the toe portion **104** and/or low in the heel portion **102**, resulting in an increased unsupported area of the strike face **110** toward the perimeter of the club head. In some embodiments, the strike face includes an unsupported face area between about 2300 mm² and about 3500 mm², such as between about 2500 mm² and about 3200 mm², such as between about 2700 mm² and about 3000 mm², such as between about 2600 mm² and about 2800 mm².

[0551] In some embodiments, the strike face **110** can include variable thickness regions that surround or are adjacent to an ideal striking location of the strike face **110**. For example, the variable thickness regions can include a minimum thickness of the strike face no less than 1.4 mm and a maximum thickness that is greater than the minimum thickness and that is no more than 3.4 mm. As discussed herein, the variable face thickness profile can be non-symmetrical, such as incorporating one or more blend zones, off-sets, elliptical and/or other profile shapes, and other non-symmetrical features. In some embodiments, the variable face thickness profile can be offset toe-ward of the geometric center of the strike face. In some embodiments, the variable face thickness profile can include at least one transition region (e.g., a blend zone) between a thicker region and a thinner region of the strike face **110**. CT over 259 CT. In some embodiments, the club head has a characteristic time (CT) greater than 257 microseconds, such as greater than 259 microseconds, and such as less than 300 microseconds.

[0552] In some embodiments, the strike face does not include a bulge and roll profile, such as an iron-type club head with a substantially flat strike face **110**. In other embodiments, such as in a hybrid-type or wood-type club head, the strike face **110** includes a bulge and roll profile, such as with a bulge radius greater than 500 mm and less than 1.5 inches in a front to back direction along the y-axis.

[0553] In some embodiments, the club head face thickness can vary depending on the type of club head (e.g., iron-type, hybrid-type, wood-type, and other club head types). For example, a fairway wood-type club head can have a face thickness between about 1 mm and about 3.1 mm, such as between about 1.4 mm and about 2.9 mm, such as between

about 1.55 mm and about 2.75 mm. For example, a hybrid-type club head can have a face thickness between about 1.0 mm and about 3.5 mm, such as between about 1.7 mm and about 2.5 mm, such as between about 1.75 mm and about 2.25 mm. Additional and different face thicknesses can be provided.

Additional Features

[0554] In some embodiments, the badge wraps from a toe portion to a rear portion of the golf club head. In some embodiments, the golf club head is a cavity back iron.

[0555] In some embodiments, the club head includes a transition region that transitions from the toe portion to the rear portion, and at least a portion of the transition region is formed of a material having a density between about 1.0 g/cc and about 3.0 g/cc.

[0556] In some embodiments, the transition region that transitions from the toe portion to the rear portion is formed by a badge that is separately formed from the club head body and is attached to the body. The badge can be formed from a low-density material, such that a mass of the badge divided by a volume of the badge is between about 1 g/cc and about 3 g/cc.

[0557] In some embodiments, a length of the transition region that transitions from the toe portion to the rear portion formed by the badge is at least 10 mm, more preferably at least 12.5 mm, more preferably at least 15 mm, more preferably at least 17.5 mm, and no more than 25 mm. The length of the transition region can be defined in an up-down direction along the Z-axis when the club head is in a zero-loft orientation.

[0558] In some embodiments, at least a first portion of the badge on a toe portion has a width greater than 3 mm, more preferably greater than 4 mm, more preferably greater than 5 mm, more preferably greater than 6 mm, and less than 15 mm, and at least a second portion of the badge on a toe portion has a width greater than 9 mm, more preferably greater than 10 mm, more preferably greater than 11 mm, more preferably greater than 12 mm, and less than 25 mm.

[0559] In some embodiments, the badge comprises a toe portion, wherein the toe portion of the badge is tapered from a top portion of the badge to a bottom portion of the badge such that a top portion width is less than a bottom portion width of the badge on the toe portion.

[0560] In some embodiments, at least a portion of the badge extends above and below the balance point of the club head as measured relative to the Z-axis when the club head is in a zero-loft orientation.

[0561] In some embodiments, at least a portion of the badge extends above and below the Zup point or the center of gravity of the golf club head as measured relative to the Z-axis when the club head is in a zero-loft orientation.

[0562] In some embodiments, at least a portion of the toe portion located above the badge is formed of a metal and at least a portion of the toe portion located below the badge is formed of a metal. In these embodiments, portions of the body adjacent to the badge are formed from a metal.

[0563] In some embodiments, a toe-to-topline transition region of the golf club head is formed of metal.

[0564] In some embodiments, a toe-to-sole transition region of the golf club head is formed of metal.

[0565] In some embodiments, at least a portion of the toe portion in-between the toe-to-topline transition region and

in-between the toe-to-sole transition region is formed of a non-metal material having a density between about 1 g/cc and about 3 g/cc.

[0566] In some embodiments, the badge wraps from a rear portion of the club head onto a toe portion of the club head, and further wraps from a rear portion of the club head onto a topline portion of the club head. The topline portion can be formed at least in part by the badge and the toe portion can be formed at least in part by the badge. In various embodiments, a topline portion of the badge and a toe portion of the back can be connected or separated by a portion of the body of the club head (i.e., not connected).

[0567] In some embodiments, at least a portion of the badge on the toe portion extends above and below Zup.

[0568] In some embodiments, with the club head at zero-loft orientation, the badge forms at least 30% of the outer surface area of the toe portion above a midplane of the club head. The midplane is halfway between an uppermost portion of the toe portion and a lowermost toe portion of the club head. More preferably, the badge can form at least 35% of the outer surface area of the toe portion above a midplane, more preferably at least 37% of the outer surface area of the toe portion above a midplane, more preferably at least 39% of the outer surface area of the toe portion above a midplane, more preferably at least 41% of the outer surface area of the toe portion above a midplane, more preferably at least 43% of the outer surface area of the toe portion above a midplane, and no more than 65% of the outer surface area of the toe portion above a midplane.

[0569] In some embodiments, a combined outermost surface area of the badge, as projected onto a rear plane, defined as a plane perpendicular to the toe plane and perpendicular to the ground plane, when the club head is in the zero loft orientation on the ground plane, or as projected onto the rear plane and onto the toe plane, is greater than an entire area of the face between scorelines formed in the face. The surface area of the face between scorelines is defined as the surface area in-between a heel-most portion of the scorelines and a toe-most portion of the scorelines, and is further defined as a surface area of the face between the scorelines that is projected onto a front plane, defined as a plane parallel to the rear plane, when the club head is in the zero loft orientation on the ground plane.

[0570] In some embodiments, the club head has a flat face projected area, excluding the scoreline grooves within the flat face projected area, and a badge surface area is between about 85% and about 125% of the flat face area. Accordingly, in some embodiments, the badge can have a projected surface area that is larger than the flat face projected surface area located between the grooves of the face.

[0571] In some embodiments, the flat face area is measured as if the face lacks scoreline grooves (i.e., has no grooves milled into the face).

[0572] In some embodiments, the badge forms at least part of a toe portion of the club head, at least part of a topline portion of the club head, at least part of a rear portion of the club head, and includes transition regions in between the rear portion and the toe portion, the rear portion and the topline portion, and the top line portion and the toe portion.

[0573] In some embodiments, the badge extends further heelward than the heelward-most scorelines and/or farther toward than the toward-most scorelines.

[0574] In some embodiments, a total length of the badge from a first end to a second end is greater than a total length

from a par line (i.e., the transition from a flat face surface to a curved surface proximate heel) to the toward-most portion of the toe portion.

[0575] In some embodiments, a total length from a heelward-most scoreline to the toward-most portion of the toe portion is less than a total length of the badge.

[0576] In some embodiments, an area of the toe portion of the badge, projected onto the toe plane when the club head is in the zero loft orientation on the ground plane, is at least 15%, or more preferably, at least 17%, of the total area of the toe portion, excluding the hosel that is projected onto the toe plane when the club head is in the zero loft orientation on the ground plane. In some embodiments, the projected area of the toe portion is at least 100 mm² when viewed from a toe view.

[0577] In some embodiments, the projected area of the toe portion of badge, when viewed from a toe view, is at least 5% of the projected area of the back portion of the badge, which view from a rear view, more preferably at least 7% of the projected area of the back portion of the badge.

[0578] In some embodiments, the area of badge is greater than total area of the face within scorelines (i.e., bounded by the heelward-most scoreline, the toward-most scoreline, the upward-most scoreline, and the lower-most scoreline).

[0579] According to alternative examples, as shown in FIGS. 64-70, a golf club head 200 includes at least some features similar to the features of the golf club head 100. For example, the golf club head 200 is a hollow-body-type golf club head that includes a body 202, which defines a toe portion 214, a heel portion 212, a top portion 216 (e.g., topline portion), a sole portion 218 (e.g., bottom portion), and a rear portion 222 of the golf club head 200. The body 202 also defines a face opening 277 of a front portion 220 of the golf club head 200. Additionally, the body 202 includes a hosel 208 that extends from the heel portion 212 of the body 202. The hosel 208 can be configured to receive an adjustable head-shaft connection system to help adjust the launch characteristics of the golf club head 200.

[0580] The front portion 220 includes a strike face 206 designed to impact a golf ball during a normal golf swing. The strike face 206 includes grooves 211 and a leading edge 209. The rear portion 222 includes a rear wall 262 opposite the strike face 206. The rear wall 262 is co-formed with the rest of the body 202 such that the body 202 form a one-piece, seamless, and unitary monolithic construction. In one example, the rear wall 262 has a thickness between, and inclusive of, 1 mm and 2 mm, such as between, and inclusive of, 1.07 mm and 1.4 mm, or between, and inclusive of, 1.1 mm and 1.4 mm. The body 202 further includes an internal cavity 232 that is defined as the space between the toe portion 214, the heel portion 212, the top portion 216, the sole portion 218, the front portion 220, and the rear portion 222.

[0581] At least a majority, if not all, of the strike face 206 of the golf club head 200 is formed separately from the body 202 and attached to the body 202, such as via a weld. More specifically, the strike face 206 is defined by a strike plate 252 that is separately formed from the body 202 and attached to the body 202 at the front portion 220. The strike plate 252 has a variable thickness T_{face} . In some examples, a maximum thickness of the strike plate 252 (excluding grooves and not including thickness or weld guide 290 (e.g., a maximum thickness of the strike plate 252, within a central region of the strike plate 252 identified by a rectangle

centered on a geometric center of the strike face 206 and having a length of 20 millimeters (mm) and a height of 20 mm) is between, and inclusive of, 2.25 millimeters (mm) and 2.40 mm, is between, and inclusive of, 2.1 mm and 2.8 mm, is between, and inclusive of, 2.1 mm and 3.0 mm, is between, and inclusive of, 2.0 mm and 3.0 mm, such as between, and inclusive of, 2.25 mm and 2.4 mm, or between, and inclusive of, 2.25 mm and 2.55 mm. In these examples, or alternative examples, a minimum thickness of the strike plate 252 (excluding grooves) is between, and inclusive of, 1.2 mm and 1.9 mm, such as between, and inclusive of, 1.45 mm and 1.75 mm, between, and inclusive of, 1.1 mm and 1.60 mm, or between, and inclusive of, 1.45 mm and 1.60 mm.

[0582] The body 202 is configured to mate with a portion of an outer peripheral edge of the strike plate 252, which is welded to the body 202 via a peripheral weld 253. The face opening 277 is defined between the toe portion 214, the heel portion 212, the top portion 216, and the sole portion 218 of the golf club head 200. Generally, the face opening 277 receives the strike plate 252 and helps to secure the strike plate 252 to the body 202. The face opening 277 extends entirely through the front portion 220 and is open to the internal cavity 232 of the golf club head 200. When attached to the body 202, the strike plate 252 in effect covers the face opening 277, thus helping to enclose the internal cavity 232 and form the hollow body of the golf club head 200.

[0583] In the illustrated examples, the peripheral weld 253 is peripherally discontinuous (extends about less than all of the outer periphery of the strike plate 252 such that at least one portion of the outer peripheral edge of the strike plate 252 is not welded to the body 202), however in other examples the peripheral weld may be continuous, and in other examples the strike plate 252 may not include a sole wrap portion 223 and may have a continuous peripheral weld. For example, as shown in FIGS. 64-67, the strike plate 252 includes a sole wrap portion 223 that is not welded to the body 202. More specifically, an outer peripheral edge, or perimeter, of the strike plate 252 defined along the sole wrap portion 223 of the strike plate 252 is not welded to the body 202. The sole wrap portion 223 effectively wraps around a face-to-sole transition region including a leading edge 209 and under the strike face 206 to define a portion of the bottom or the sole portion 218 of the golf club head 200. Accordingly, the sole wrap portion 223 is angled relative to the strike face 206. In some examples, the sole wrap portion 223 does not wrap around (e.g., is not vertically underneath, when the strike face 206 is perpendicular to a ground plane) an internal shelf of the sole portion 218.

[0584] Referring to FIG. 67, in some examples, the strike plate 252 also includes a jog feature 270 (e.g., notch) on a toward side of the strike plate 252. The jog feature 270 defines a transition from the peripheral edge of the sole wrap portion 223 to the peripheral edge of the toe portion of the strike plate 252. The jog feature 270 promotes a part of the strike plate 252 extending further toward than the sole wrap portion 223. Accordingly, the jog feature 270 defines an intersection between the sole wrap portion 223 and the part of the strike plate 252 that extends toward of the sole wrap portion 223. The jog feature 270 is defined by an inward-facing surface 245.

[0585] Not only is the outer peripheral edge of the sole wrap portion 223 of the strike plate 252 not welded to the body 202, but the outer peripheral edge 235 of the sole wrap

portion **223** is spaced apart from the body **202** such that a gap is defined between the outer peripheral edge **235** of the strike plate **252** and the body **202**. The gap defines a sole slot **250** of the golf club head **200**. Generally, the sole slot **250** is a groove or channel formed in a sole of the golf club head **200**. The sole slot **250** is elongated in a lengthwise direction, substantially parallel to the strike face **206**. In the illustrated examples, the sole slot **250** is a through-slot, or a slot that is open on a sole portion side of the sole slot **250** and open on an internal cavity side or interior side of the sole slot **250**. In some examples, the sole slot **250** is filled with a filler material **251**, which can be the same type of material as the filler material of the golf club head **100**.

[0586] Referring to FIG. **68**, according to certain examples, the strike plate **252** includes a weld guide **290** protruding from an internal surface **259** of the strike plate **252**. The weld guide **290** extends along the outer peripheral edge of the strike plate **252** at a location offset from the outer peripheral edge. The weld guide **290** provides a visual guide for locating an ideal weld bead and helps an inspector determine the quality of the peripheral weld **253** by comparing the peripheral weld **253** to the weld guide **290** using an X-Ray machine.

[0587] The body **202** further includes a weight port **241** at the rear portion **222** of the golf club head **200**. The weight port **241** is configured to receive and retain an external weight **243** of the golf club head **200**. In some examples, the external weight **243** is selectively insertable into and releasable from the weight port **241**, such as when a player desired to change the mass characteristics of the golf club head **200** by replacing one external weight with another external weight having a different mass. For example, the weight port **241** can include internal threads and the external weight **243** can include external threads, which threadably engage the internal threads of the weight port **241** to retain the external weight **243** in the weight port **241**. The external weight **243** can be released from the weight port **241** by engaging a release tool with a socket of the external weight **243** and rotating the release tool to rotate and release the external weight **243**. In other examples, the external weight **243** is not selectively releasable from the weight port **241** and is permanently retained in the weight port **241** via an adhesive or via a welding technique. The external weight **243** may be made of the same material as the body or it may be made of a material that is different (e.g., more dense or less dense material e.g. 1 g/cc to 19 g/cc) than the material of the body **202**. In one example, the body **202** is made of steel and the external weight **243** is made of tungsten, wherein references to materials such as steel and tungsten, here and throughout this disclosure, include alloys of such materials, including any of the alloys disclosed herein without limitation aluminum alloys, magnesium alloys, titanium alloys, steel alloys, and tungsten alloys. In one embodiment the density of the weight is at least 20% greater than the density of the body, and in further embodiments at least 30%, 40%, 50%, or 60%. In a further embodiment the density of the weight is no more than 300% greater than the density of the body, and in further embodiments no more than 275%, 250%, 225%, 200%, 175%, 150%, 125%, or 100%.

[0588] In some instances, an important characteristic may be the mass of the external weight **243** and the mass of any filler materials or internal dampers placed within the cavity, and these combined masses may be related to the external or internal volume of golf club head. Volume may be measured

using a water displacement method, or by weighing the amount of water the internal cavity can hold in both instances the external weight is in place. Accordingly, the mass of the external weight plus the mass of the filler material or internal dampers divided by the volume of the club head may be an important parameter to consider. Preferably, a ratio of the sum of the mass of external weight in grams and mass of filler material in grams to the external volume of the golf club head in cubic centimeters (cc) falls in the following inclusive ranges between 0.050 and 0.425 g/cc, more preferably between 0.240 and 0.350 g/cc, and even more preferably between 0.133 and 0.230 g/cc. A sample calculation is mass of foam filler material of 4 grams plus mass of weight of 11 grams divided by a club head volume of 60 cc is 0.25 g/cc (15 g/60 cc). Similarly, in some examples a ratio of the sum of the mass of external weight in grams and mass of filler material in grams to the internal cavity volume (e.g. internal cavity **232** volume) of the golf club head in (cc) falls in the following inclusive ranges between 0.175 and 1.425 g/cc, preferably is between 0.300 and 1.350 g/cc, more preferably is between 0.480 and 1.250 g/cc, and even more preferably is between 0.300 and 0.750.

[0589] The following discussion and the above discussion generally applies to the external weight **243** and any other external weights discussed herein. In general, the external weight **243** may have several purposes including, but not limited to, the following 1) to plug an opening to the interior cavity to prevent debris, water, etc. from entering the cavity, 2) to adjust swing weighting, 3) to place or adjust center of gravity (CG) of the golf club head. The external weight may be an assembly (e.g. two or more components (e.g. three or more components)). For longer length golf clubs (i.e. those with longer shafts) a lower mass external weight may be desirable to achieve a certain swing weight, or for senior, junior, or even ladies clubs, it may be desirable to have a lower swing weight in which case the mass of the external weight can also be decreased. Alternatively, the mass of the external weight may be increased to achieve a heavier swing weight as desired by an individual player or to achieve a target swing weight. For those that prefer to play shorter length shafts, to achieve the same target swing weight, the mass of the head must be increased and an increasing the mass of the external weight **243** is one way of achieving this.

[0590] The mass of the external weight may be varied by varying the material density (e.g. densities in the range 1 g/cc to 19 g/cc may be used for the external weight, but preferably the external weight has a density between 7 g/cc and 19 g/cc) or by varying other parameters of the external weight, such as the length of the shaft portion that enters the interior cavity. As shown in FIG. **65**, there is a lot of room within the interior cavity **232** such that the external weight **243** could extend much further into the interior cavity **232** than is shown, including up to, and potentially contacting, the rear surface of the face e.g. internal surface **259**. Generally, it is desirable for the length of the shaft to extend only so far into the interior cavity (e.g. not beyond a plane parallel to the face and passing through the transition region from the rear portion **222** to top portion **216** when the club head is in a zero loft position with a square face angle (see e.g. FIG. **48** for proper club head orientation to understand this description)). Alternatively, it may be desirable for the external weight to not extend past a sole wrap portion of the strike plate when the club head is an orientation similar to the orientation shown in FIG. **48**. Both of these external

weight configurations may be desirable so the external weight is not cantilevered and unsupported, which may cause ringing or attenuation when the club head impacts a golf ball. This would need to be balanced with other goals, such as achieving a certain swing weight or CG placement as discussed above. Any increase in the mass of external weight **243** will lower Zup and tend to increase launch or make it easier to get a ball airborne, and, as the mass moves more forward, may reduce spin.

[0591] However, in other instances, it may be desirable for the tip end of the shaft of the external weight **243** to extend all the way to and engage the face, which may help with durability by providing additional support to the face. The external weight in this instance would typically have a blunted end, rounded end, wedge shaped end, or an end that matches the contours of the face, and may include a cap made of secondary material that incorporates these shapes, such as a polymer cap, elastomeric cap, polycarbonate, hard rubber cap, or even a metal material (e.g. aluminum). The end cap may be designed to rotate or spin relative a shaft axis of the external weight **243**. Having the end of the external weight engage or contact the face may help with damping or durability by providing extra support. For example, an end cap, with hardness ranges from between 10 Shore OO and 80 Shore D, may help with both damping and durability, more preferably from 35 Shore A to 80 Shore D, and even more preferably 35 Shore A up to 50 Shore D may help with damping and durability. In addition to controlling damping and durability, the hardness may also be selectively tuned to help control COR. For example, controlling hardness may help to decrease the COR of the strike face if it's too high (e.g. one could reduce the COR by between 0.001 to 0.005 COR points (e.g. a club head measuring beyond a prescribed COR limit e.g. 0.008 COR points above a baseline COR calibration plate could be reduced to 0.003 COR points above the baseline calibration plate) using this methodology). The end cap may be capable of compressing between 0.05 mm to 0.55 mm, preferably 0.10 to 0.25 mm engagement to allow for good engagement with the rear surface of the face. A torque limiting wrench may help set the amount of interference.

[0592] Another potential use would be to influence shot tendency. For example, if the tip or end of the shaft of the external weight contacts on a toe side of the face (e.g. toward of a midplane passing through a geometric center of the strike face) this would tend to cause a more left tendency and may even impart a draw spin for a right handed club. Alternatively, positioning the tip to contact low on the face (e.g. below balance point or below a geometric center or less than 16.5 mm from a leading edge (e.g. **179**) such as 12-15 mm from a leading edge (see e.g. FIG. 2 reference numeral **179** for proper club head orientation and what is meant by below 16.5 mm)) may promote a higher launch and/or greater back spin. One may also position the contact point to be a combination of both low and toe (e.g. 12-15 mm from a leading edge and 2-10 mm toward of a geometric center). Additionally, the tip of the external weight may be shaped to match shape of the rear surface (e.g. the internal surface **259**) to increase contact area, and the tip end may be free to rotate about a shaft axis of the external weight to allow for better placement.

[0593] According to certain examples, a total mass of the external weight **243** is between, and inclusive of, 2 grams and 10 grams, such as between, and inclusive of, 3 grams

and 6 grams, and more particularly between, and inclusive of, 3 grams and 4 grams. In certain examples, a ratio of the total mass of the external weight **243** and the total mass of the golf club head **200** is between, and inclusive of, 0.014 and 0.017, such as between, and inclusive of, 0.015 and 0.016. According to some examples, the total mass of the golf club head **200** is between, and inclusive of, 210 g and 270 g, such as between, and inclusive of, 220 g and 255 g or between, and inclusive of, 225 g and 251 g.

[0594] The internal cavity **232** is enclosed by the body **202**, the strike plate **252**, the external weight **243**, and the filler material **251** in the slot **250**. The golf club head **200** additionally includes a filler material **233** within the internal cavity **232**. The filler material **233** fills the entirety of the internal cavity **232** in some examples. In certain examples of the golf club head **200**, the total volume of the golf club head **200** is between, and inclusive of, 58 cc and 75 cc, such as between, and inclusive of, 58 cc and 60 cc, or between, and inclusive of, 73 cc and 75 cc. In such examples, the total volume of the internal cavity **232**, as well as the total volume of the filler material **233** in the internal cavity **232**, is between, and inclusive of, 15 cc and 45 cc, such as between, and inclusive of, 18 cc and 37 cc, or between, and inclusive of, 19 cc and 21 cc.

[0595] In some implementations, the filler material **233** is made from a non-metal, such as a thermoplastic material, thermoset material, and the like. According to some examples, the filler material **233** is initially a viscous material that is injected or otherwise inserted into the golf club head **200** through the weight port **241** before the external weight **243** is inserted into the weight port **241**. In still other examples, the filler material **233** may be pre-formed and placed into the golf club head **200** and sealed in place before the strike plate **252** is attached to the body **202**. The filler material **233** can be similar to the filler material of the golf club head **100**, in that the filler material **233** has a density that is less than the density of the material forming the body **202**, the strike plate **252**, and the external weight **243**. In some examples, the filler material **233** has a density that is less than the density of the filler material **251** in the sole slot **250**. In some examples, the filler material **233** has a density between, and inclusive of, 0.03 g/cc and 0.30 g/cc, such as between, and inclusive of, 0.03 g/cc and 0.25 g/cc, between, and inclusive of, 0.03 g/cc and 0.19 g/cc, between, and inclusive of, 0.03 g/cc and 0.17 g/cc, or between, and inclusive of, 0.09 g/cc and 0.17 g/cc, or between, and inclusive of 0.144 g/cc and 0.15 g/cc. The total volume of the internal cavity **232** and the total volume of the filler material **233** is such that, in certain examples, the total mass of the filler material **233** is between, and inclusive of, 2 g and 5 g, such as between, and inclusive of, 2.5 g and 4 g, or between, and inclusive of, 2.5 g and 3.5 g, or between, and inclusive of, 2.6 g and 3 g. According to some examples, a ratio of the total mass of the filler material **233** to a total volume of the golf club head **200** is between, and inclusive of, 0.027 g/cc and 0.086 g/cc, such as between, and inclusive of, 0.034 g/cc and 0.067 g/cc.

[0596] According to additional examples, as shown in FIGS. 71-75, a golf club head **300A** includes at least some features similar to the features of the golf club head **200**, with like numbers referring to like features. For example, the golf club head **300A** is a hollow-body-type golf club head that includes a body **302**, which defines a toe portion **314**, a heel portion **312**, a top portion **316** (e.g., topline portion), a

sole portion 318 (e.g., bottom portion), and a rear portion 322 of the golf club head 300A. The body 302 also defines a face opening 377 of a front portion 320 of the golf club head 300A (see, e.g., FIG. 72). Additionally, the body 302 includes a hosel 308 that extends from the heel portion 312 of the body 302. The hosel 308 can be configured to receive an adjustable head-shaft connection system to help adjust the launch characteristics of the golf club head 300A.

[0597] The front portion 320 includes a strike face 306 designed to impact a golf ball during a normal golf swing. The strike face 306 includes grooves 311. The rear portion 322 includes a rear wall 362 opposite the strike face 306. The rear wall 362 is co-formed with the rest of the body 302 such that the body 302 form a one-piece, seamless, and unitary monolithic construction. In one example, the rear wall 362 has a thickness between, and inclusive of, 1 mm and 2 mm, such as between, and inclusive of, 1.07 mm and 1.4 mm, or between, and inclusive of, 1.07 mm and 1.37 mm. The body 302 further includes an internal cavity 332 that is defined as the space between the toe portion 314, the heel portion 312, the top portion 316, the sole portion 318, the front portion 320, and the rear portion 322.

[0598] At least a majority, if not all, of the strike face 306 of the golf club head 300A is formed separately from the body 302 and attached to the body 302, such as via a weld. More specifically, the strike face 306 is defined by a strike plate 352 that is separately formed from the body 302 and attached to the body 302 at the front portion 320. The strike plate 352 has a variable thickness T_{face} . In some examples, a maximum thickness of the strike plate 352 is between, and inclusive of, 2.0 mm and 3.0 mm, such as between, and inclusive of, 2.25 mm and 2.4 mm, or between, and inclusive of, 2.4 mm and 2.7 mm. In these examples, a minimum thickness of the strike plate 352 is between, and inclusive of, 1.2 mm and 1.9 mm, such as between, and inclusive of, 1.45 mm and 1.60 mm or between, and inclusive of, 1.6 mm and 1.9 mm.

[0599] The body 302 is configured to mate with a portion of an outer peripheral edge of the strike plate 352, which is welded to the body 302 via a peripheral weld 353. The face opening 377 is defined between the toe portion 314, the heel portion 312, the top portion 316, and the sole portion 318 of the golf club head 300A. Generally, the face opening 377 receives the strike plate 352 and helps to secure the strike plate 352 to the body 302. The face opening 377 extends entirely through the front portion 320 and is open to the internal cavity 332 of the golf club head 300A. When attached to the body 302, the strike plate 352 in effect covers the face opening 377, thus helping to enclose the internal cavity 332 and form the hollow body of the golf club head 300A.

[0600] In the illustrated examples, the peripheral weld 353 is peripherally discontinuous (extends about less than all of the outer periphery of the strike plate 352 such that at least one portion of the outer peripheral edge of the strike plate 352 is not welded to the body 302), however in other examples the peripheral weld may be continuous, and in other examples the strike plate 252 may not include a sole wrap portion 223 and may have a continuous peripheral weld. For example, as shown in FIGS. 71-74, the strike plate 352 includes a sole wrap portion 323 that is not welded to the body 302. More specifically, an outer peripheral edge, or perimeter, of the strike plate 352 defined along the sole wrap portion 323 of the strike plate 352 is not welded to the body

302. The sole wrap portion 323 effectively wraps around and under the strike face 306 to define a portion of the bottom or the sole portion 318 of the golf club head 300A. Accordingly, the sole wrap portion 323 is angled relative to the strike face 306. In some examples, the sole wrap portion 323 does not wrap around (e.g., is not vertically underneath, when the strike face 306 is perpendicular to a ground plane) an internal shelf of the sole portion 318.

[0601] Referring to FIG. 73, in some examples, the strike plate 352 also includes a jog feature 370 (e.g., notch) on a toward side of the strike plate 352. The jog feature 370 defines a transition from the peripheral edge of the sole wrap portion 323 to the peripheral edge of the toe portion of the strike plate 352. The jog feature 370 promotes a part of the strike plate 352 extending further toward than the sole wrap portion 323. Accordingly, the jog feature 370 defines an intersection between the sole wrap portion 323 and the part of the strike plate 352 that extends toward of the sole wrap portion 323. The jog feature 370 is defined by an inwardly-facing surface 345.

[0602] Not only is the outer peripheral edge of the sole wrap portion 323 of the strike plate 352 not welded to the body 302, but the outer peripheral edge 335 of the sole wrap portion 323 is spaced apart from the body 302 such that a gap is defined between the outer peripheral edge 335 of the strike plate 352 and the body 302. The gap defines a sole slot 350 of the golf club head 300A. Generally, the sole slot 350 is a groove or channel formed in a sole of the golf club head 300A. The sole slot 350 is elongated in a lengthwise direction, substantially parallel to the strike face 306. In the illustrated examples, the sole slot 350 is a through-slot, or a slot that is open on a sole portion side of the sole slot 350 and open on an internal cavity side or interior side of the sole slot 350. In some examples, the sole slot 350 is filled with a filler material 351, which can be the same type of material as the slot filler material of the golf club head 100.

[0603] Referring to FIG. 74, according to certain examples, the strike plate 352 includes a weld guide 390 protruding from an internal surface 359 of the strike plate 352. The weld guide 390 extends along the outer peripheral edge of the strike plate 352 at a location offset from the outer peripheral edge. The weld guide 390 provides a visual guide for locating an ideal weld bead and helps an inspector determine the quality of the peripheral weld 353 by comparing the peripheral weld 353 to the weld guide 390 using an X-Ray machine.

[0604] The body 302 further includes a weight port 341 at the sole portion 318 of the golf club head 300A. The weight port 341 is configured to receive and retain an external weight 343 of the golf club head 300A. In some examples, the external weight 343 is selectively insertable into and releasable from the weight port 341, such as when a player desired to change the mass characteristics of the golf club head 300A by replacing one external weight with another external weight having a different mass. For example, the weight port 341 can include internal threads and the external weight 343 can include external threads, which threadably engage the internal threads of the weight port 341 to retain the external weight 343 in the weight port 341. The external weight 343 can be released from the weight port 341 by engaging a release tool with a socket of the external weight 343 and rotating the release tool to rotate and release the external weight 343. In other examples, the external weight 343 is not selectively releasable from the weight port 341

and is permanently retained in the weight port **341** via an adhesive or via a welding technique. The external weight **343** is made of a material that may be different or the same as (e.g., more dense or less dense e.g. 1 g/cc to 19 g/cc) the material of the body **302** depending on various considerations already discussed above with regard to the external weight **243**. In one example, the body **302** is made of steel and the external weight **343** is made of tungsten. The discussion above with respect to external weight **243** discussing various purposes, ratios, and applications applies equally to the external weight **343**, and it is recommended to refer to the above discussion for further information and potential uses of the external weight **343**. This includes any and all ratios of mass of the weight and filler material compared to volumes of the club head.

[0605] According to certain examples, a total mass of the external weight **343** is between, and inclusive of, 2 grams and 10 grams, such as between, and inclusive of, 3 grams and 6 grams, and more particularly between, and inclusive of, 3 grams and 4 grams. In certain examples, a ratio of the total mass of the external weight **343** and the total mass of the golf club head **300A** is between, and inclusive of, 0.014 and 0.017, such as between, and inclusive of, 0.015 and 0.016. According to some examples, the total mass of the golf club head **300A** is between, and inclusive of, 210 g and 270 g, such as between, and inclusive of, 220 g and 255 g or between, and inclusive of, 225 g and 251 g.

[0606] The internal cavity **332** is enclosed by the body **302**, the strike plate **352**, the external weight **343**, and the filler material **351** in the slot **350**. The golf club head **300A** additionally includes a filler material **333** within the internal cavity **332**. The filler material **333** fills the entirety of the internal cavity **332** in some examples. In certain examples of the golf club head **300A**, the total volume of the golf club head **300A** is between, and inclusive of, 58 cc and 75 cc, such as between, and inclusive of, 58 cc and 60 cc, or between, and inclusive of, 73 cc and 75 cc. In such examples, the total volume of the internal cavity **332**, as well as the total volume of the filler material **333** in the internal cavity **332**, is between, and inclusive of, 15 cc and 45 cc, such as between, and inclusive of, 18 cc and 37 cc or between, and inclusive of, 33 cc and 34 cc.

[0607] In some implementations, the filler material **333** is made from a non-metal, such as a thermoplastic material, thermoset material, and the like. According to some examples, the filler material **333** is initially a viscous material that is injected or otherwise inserted into the golf club head **300A** through the weight port **341** before the external weight **343** is inserted into the weight port **341**. In still other examples, the filler material **333** may be pre-formed and placed into the golf club head **300A** and sealed in place before the strike plate **352** is attached to the body **302**. The filler material **333** can be similar to the filler material of the golf club head **100**, in that the filler material **333** has a density that is less than the density of the material forming the body **302**, the strike plate **352**, and the external weight **343**. In some examples, the filler material **333** has a density that is less than the density of the filler material **351** in the sole slot **350**. In some examples, the filler material **333** has a density between, and inclusive of, 0.03 g/cc and 0.30 g/cc, such as between, and inclusive of, 0.03 g/cc and 0.25 g/cc, between, and inclusive of, 0.03 g/cc and 0.19 g/cc, between, and inclusive of, 0.03 g/cc and 0.17 g/cc, or between, and inclusive of, 0.09 g/cc and 0.17 g/cc, or between, and

inclusive of, 0.10 g/cc and 0.12 g/cc. The total volume of the internal cavity **332** and the total volume of the filler material **333** is such that, in certain examples, the total mass of the filler material **333** is between, and inclusive of, 2 g and 5 g, such as between, and inclusive of, 2.5 g and 4 g, or between, and inclusive of, 2.5 g and 3.5 g, or between, and inclusive of, 3.9 g and 4.1 g. According to some examples, a ratio of the total mass of the filler material **333** to a total volume of the golf club head **300A** is between, and inclusive of, 0.027 g/cc and 0.086 g/cc, such as between, and inclusive of, 0.034 g/cc and 0.067 g/cc.

[0608] According to additional examples, as shown in FIGS. **76-79**, a golf club head **400** includes at least some features similar to the features of the golf club head **200** and the golf club head **300A**, with like numbers referring to like features. For example, the golf club head **400** is a hollow-body-type golf club head that includes a body **402**, which defines a toe portion **414**, a heel portion **412**, a top portion **416** (e.g., topline portion), a sole portion **418** (e.g., bottom portion), and a rear portion **422** of the golf club head **400**. The body **402** also includes a face opening **477** of a front portion **420** of the golf club head **400** (see, e.g., FIG. **80**). Additionally, the body **402** includes a hosel **408** that extends from the heel portion **412** of the body **402**. The hosel **408** can be configured to receive an adjustable head-shaft connection system to help adjust the launch characteristics of the golf club head **400**.

[0609] The front portion **420** includes a strike face **406** designed to impact a golf ball during a normal golf swing. The strike face **406** includes grooves **411**. The body **402** includes a rear wall **462** that is opposite the strike face **206** and defines a portion of the rear portion **422**. The rear wall **462** is co-formed with the rest of the body **402** such that the body **402** form a one-piece, seamless, and unitary monolithic construction. Referring to FIG. **101**, the rear wall **462** includes an exterior surface **461** and an interior surface **463**. In one example, the rear wall **462** has a thickness between, and inclusive of, 0.65 mm and 2 mm, such as between, and inclusive of, 0.75 mm and 1.4 mm, or between, and inclusive of, 1.1 mm and 1.4 mm. The body **402** helps define an internal cavity **432** of the golf club head **400** that is defined as the space between the toe portion **414**, the heel portion **412**, the top portion **416**, the sole portion **418**, the front portion **420**, and the rear portion **422**. In this manner, the internal cavity **432** is enclosed at least partially by the body **402**, a strike plate **452** of the golf club head **400**, and an internal weight **440** of the golf club head **400** (see, e.g., FIG. **81**).

[0610] At least a majority, if not all, of the strike face **406** of the golf club head **400** is formed separately from the body **402** and attached to the body **402**, such as via a weld. More specifically, the strike face **406** is defined by the strike plate **452** of the golf club head **400**, which is separately formed from the body **402** and attached to the body **402** at the front portion **420**. The strike plate **452** can have a variable thickness as described above in association with the golf club head **100** and the golf club head **200**.

[0611] The body **402** is configured to mate with a portion of an outer peripheral edge of the strike plate **452**, which is welded to the body **402** via a peripheral weld as described above. As best shown in FIGS. **80** and **81**, which shows the strike plate **452** removed, the face opening **477** is defined between the toe portion **414**, the heel portion **412**, the top portion **416**, and the sole portion **418** of the golf club head

400. Generally, the face opening **477** receives the strike plate **452** and helps to secure the strike plate **452** to the body **402**. The face opening **477** extends entirely through the front portion **420** and is open to the internal cavity **432** of the golf club head **200**. When attached to the body **402**, the strike plate **452** in effect covers the face opening **477**, thus helping to enclose the internal cavity **432** and form the hollow body of the golf club head **400**.

[0612] As shown in FIGS. **76**, **78**, and **79**, in some examples, the strike plate **452** includes a sole wrap portion **423** that is not welded to the body **402**. More specifically, an outer peripheral edge, or perimeter, of the strike plate **452** defined along the sole wrap portion **423** of the strike plate **452** is not welded to the body **402**. The sole wrap portion **423** effectively wraps around a face-to-sole transition region and under the strike face **406** to define a portion of the bottom or the sole portion **418** of the golf club head **400**. Accordingly, the sole wrap portion **423** is angled relative to the strike face **406**. In some examples, the strike plate **452** also includes a jog feature (e.g., notch) on a toward side of the strike plate **452**, similar to the jog feature **270** of the golf club head **200**. According to certain examples, the strike plate **452** can include a weld guide similar to the weld guide **290** of the strike plate **252**.

[0613] Similar to the golf club head **200**, a gap is defined between the outer peripheral edge of the strike plate **452**, at the sole portion **418**, and the body **402**. The gap defines a sole slot **450** of the golf club head **400**. Generally, the sole slot **450** is a groove or channel formed in a sole of the golf club head **400**. The sole slot **450** is elongated in a lengthwise direction, substantially parallel to the strike face **406**. In the illustrated examples, the sole slot **450** is a through-slot, or a slot that is open on a sole portion side of the sole slot **450** and open on an internal cavity side or interior side of the sole slot **450**. In some examples, the sole slot **450** is filled with a filler material **451** (see, e.g., FIG. **78**), which can be the same type of material as the filler material of the golf club head **100**.

[0614] According to some examples, the body **402** additionally includes an injection port **407** (see, e.g., FIGS. **76-78**) located on the toe portion **414** of the golf club head **400**. However, in other examples, the injection port **407** can be located anywhere on the body **402** of the golf club head **400**, including at the top portion **416**, the rear portion **422** (e.g., in the rear wall **462**), the sole portion **418**, the heel portion **412**, or the toe portion **414**. In the illustrated examples, the injection port **407** is located in the toe portion **414** defined by the body **402**, which includes a build-up of mass to accommodate the injection port **407**. A portion of the mass build-up forms a protrusion **413** of an internal-weight cavity **441** or internal-weight pocket (see, e.g., FIG. **86**). The injection port **407** has a length equal to a thickness of the toe portion **414**, which in some examples is at least 7.0 mm. In certain examples, the injection port **407** is oriented such that a central axis extending through the injection port **407** does not intersect the front portion **420**. However, in other examples, the injection port **407** is oriented such that its central axis intersects the front portion **420**. Additionally, according to certain examples, the central axis of the injection port **407** is oblique (e.g., non-parallel) relative to the grooves **411** of the strike face **406**.

[0615] The internal cavity **432** is enclosed by at least the body **402**, the strike plate **452**, and the filler material **451** in the slot **450**. The golf club head **400** additionally includes a

cavity filler material **433** (see, e.g., FIGS. **103** and **104**) within the internal cavity **432**. This cavity filler material **433** of the golf club head **400** can be similar to or the same as the filler material **233** of the golf club head **200**. Moreover, the cavity filler material **433** fills the entirety of the internal cavity **432** in some examples. According to certain examples, the density of the cavity filler material **433** used in golf club heads of a correlated set of golf club heads (see below) depends on the loft of the golf club head. For example, the density of the cavity filler material **433** within the internal cavity of a golf club head having a loft that is not less than 35-degrees (or not less than 40-degrees) can be higher than the density of the cavity filler material within the internal cavity of a golf club head having a loft that is less than 35-degrees (or less than 40-degrees). In one particular example, the density of the cavity filler material for golf club heads of a correlated set of golf club heads is a first density, for golf club heads having a loft that is less than 34-degrees, and a second density, for golf club heads having a loft that is more than 34-degrees, where the second density is greater than the first density. According to another example, the density of the cavity filler material for golf club heads of a correlated set of golf club heads is a first density, for golf club heads having a loft that is less than 29-degrees, and a second density, for golf club heads having a loft that is more than 29-degrees, where the second density is greater than the first density. In yet another example, the density of the cavity filler material for golf club heads of a correlated set of golf club heads is a first density, for golf club heads having a loft that is less than 38-degrees, and a second density, for golf club heads having a loft that is more than 38-degrees, where the second density is greater than the first density. In each of these examples, a ratio of the second density to the first density can be between, and inclusive of, 2.0 and 5.0, or between, and inclusive of, 3.0 and 4.5.

[0616] In certain examples of the golf club head **400**, the total volume of the golf club head **400** is between, and inclusive of, 38 cc and 75 cc, such as between, and inclusive of, 43 cc and 60 cc, or between, and inclusive of, 45 cc and 55 cc. In such examples, the total volume of the internal cavity **432**, as well as the total volume of the cavity filler material **433** in the internal cavity **432**, is between, and inclusive of, 9 cc and 45 cc, such as between, and inclusive of, 10 cc and 21 cc, or between, and inclusive of, 11 cc and 18 cc.

[0617] In some implementations, the cavity filler material **433** is similar to or the same as the filler material **233** of the golf club head **200**. Moreover, according to some examples, the cavity filler material **433** is initially a viscous material that is injected or otherwise inserted into the golf club head **400** through the injection port **407**. The total volume of the internal cavity **432** and the total volume of the cavity filler material **433** is such that, in certain examples, the total mass of the cavity filler material **433** is between, and inclusive of, 2 g and 5 g, such as between, and inclusive of, 2.5 g and 4 g, or between, and inclusive of, 2.5 g and 3.5 g, or between, and inclusive of, 2.6 g and 3 g. According to some examples, a ratio of the total mass of the cavity filler material **433** to a total volume of the golf club head **400** is between, and inclusive of, 0.027 g/cc and 0.086 g/cc, such as between, and inclusive of, 0.034 g/cc and 0.067 g/cc.

[0618] The golf club head **400** further includes a plug **405** in some examples. The injection port **407** is sealed with the plug **405** after the cavity filler material **433** is injected

through the injection port **407** into the internal cavity **432**. In one example, the plug **405** is the same as or similar to the plug of the golf club head **200** described above.

[0619] The golf club head **400** includes an internal weight that is made of a material (e.g., tungsten) having a density greater than the density of the material of the body **402**. In some examples, the internal weight is configured to accommodate the injection of the cavity filler material **433** through the internal weight. For example, referring to FIGS. **80**, **81**, **86**, and **87**, the golf club head **40** can include an internal weight **440** that has a filler-injection channel **434**. The filler-injection channel **434** passes entirely through the internal weight **440**. Referring to FIG. **85**, in one example, the filler-injection channel **434** passes entirely through the internal weight **440** from a first side **497** of the internal weight **440** to a second side **499** of the internal weight **440** that is opposite the first side **497**. The filler-injection channel **434** is not circumferentially closed in the illustrated examples. In other words, the filler-injection channel **434** can have an open side along the length of the filler-injection channel **434**. The open side of the filler-injection channel **434** can face and be at least partially enclosed by an interior surface of the strike plate **452**. Incorporating the filler-injection channel **434** into the internal weight **440** enables the internal weight **440** and the injection port **407** to be advantageously placed low and toward on the body **402**, thus promoting a relatively low vertical position (e.g., Z-up) of the center-of-gravity of the golf club head **400** and a relatively high moment of inertia, while still enabling the injection of cavity filler material **433** into the internal cavity **432**.

[0620] The filler-injection channel **434** of the internal weight **440** is aligned with the toe port **407**. For example, a central axis of the filler-injection channel **434** is coaxial with a central axis of the toe port **407**. Alignment between the filler-injection channel **434** and the toe port **407** enables the cavity filler material **433** to flow through the toe port **407** and directly into the filler-injection channel **434**. The cavity filler material **433** then flows through the filler-injection channel **434** and into the internal cavity **432**. In some examples, the toe port **407** and the filler-injection channel **434** are angled (such as at an oblique angle) relative to the plurality of grooves **411**. In a direction extending from the toe portion **414** to the heel portion **412** (i.e., a toe-to-heel direction), the toe port **407** and the filler-injection channel **434** are angled upwardly toward the top portion **416**. The angling of the toe port **407** and the filler-injection channel **434** in this manner enables the toe port **407** to be formed in a thicker part of the body **402**, while still facilitating the flow of the cavity filler material **433** into the internal cavity **432**.

[0621] The body **402** additionally includes an internal-weight cavity **441** configured to receive the internal weight **440** in seated engagement. Accordingly, in some examples, the size and shape of the internal-weight cavity **441** corresponds with the size and shape of at least a portion of the internal weight **440**. In the illustrated examples, the internal-weight cavity **441** and the internal weight **440** are elongated in a top-to-sole direction. Moreover, in the top-to-sole direction, the depth of the internal-weight cavity **441** and a thickness of the internal weight **440** increase. For example, referring to FIG. **87**, the depth of the internal-weight cavity **441** increases from a minimum depth D1 to a maximum depth D2 in the top-to-sole direction. Similarly, referring to FIG. **84**, in some examples, the thickness of the internal weight **440**, along a plane perpendicular to the strike face

406, increases from a minimum thickness T1 to a maximum thickness T2 in the top-to-sole direction. The diverging nature of the internal-weight cavity **441** and the internal weight **440**, along the plane perpendicular to the strike face **406**, promotes a positioning of more mass lower on the golf club head **400**. Referring to FIG. **85**, along a plane parallel to the strike face **406**, the internal weight **440** has a substantially parallelogram shape in the illustrated example, but could have a rectangular, triangular, or other shape in alternative examples.

[0622] Referring to FIG. **86**, in certain examples, the internal-weight cavity **441** includes a protrusion **413** or keying feature that surrounds the toe port **407** on an interior portion of the toe port **407**. Additionally, in such examples and referring to FIGS. **82** and **85**, the internal weight **440** includes a notch **495**. The notch **495** is sized and shaped to receive the protrusion **413** in seated engagement. Seated engagement between the notch **495** and the protrusion **413** helps to fix the internal weight **440** in the internal-weight cavity **441** and locate the internal weight **440** in a proper position relative to the internal-weight cavity **441**.

[0623] As shown in FIGS. **81**, **86**, and **87**, the internal-weight cavity **441** is defined by an outer perimeter of the body **402**, along the toe portion **414** and the sole portion **418** of the golf club head **400**, and by internal walls **487** of the body **402**. Accordingly, the outer perimeter of the body **402** defines toward and lower sides of the internal-weight cavity **441**, and the internal walls **487** define upper and heelward sides of the internal-weight cavity **441**. The internal walls **487** includes a horizontal wall, extending in a generally heel-to-toe direction, and an upright wall, extending in a generally top-to-sole direction. In some examples, one of the internal walls **487** (e.g., the upright wall) includes a second filler-injection channel **446** that is aligned with the filler-injection channel **434** of the internal weight **440**. Therefore, the cavity filler material **433** can flow from the filler-injection channel **434** into the second filler-injection channel **446**, and from the second filler-injection channel **446** into the internal cavity **432**. Like the filler-injection channel **434**, the second filler-injection channel **446** may not be circumferentially closed, such that the second filler-injection channel **446** has an open side along the length of the second filler-injection channel **446**. The open side of the second filler-injection channel **446** can face and be at least partially enclosed by an interior surface of the strike plate **452**.

[0624] Referring to FIGS. **80**, **81**, **86**, and **87**, the body **402** additionally includes a primary beam **467** (e.g., a primary bar or primary sound bar) that is coupled to the interior surface **463** of the rear wall **462**. The primary beam **467** extends from the internal walls **487**, which define the internal-weight cavity **441**, to a variable thickness region **445** of the rear wall **462**, which will be described in more detail below. Accordingly, the primary beam **467** is also directly coupled to at least one of the internal walls **487**, at a first or lower end of the primary beam **467**, and directly coupled to the variable thickness region **445**, at a second or upper end of the primary beam **467**. As shown, in certain examples, the first end of the primary beam **467** is directly coupled to both the internal walls **487**, at an intersection of the internal walls **487**. The first end of the primary beam **467** is below Z-up of the golf club head **400**, in some examples, or above Z-up of the golf club head **400**, in other examples.

[0625] The primary beam and/or the secondary beam (or other beam) of any one of the examples of golf club heads

disclosed herein has a mass per unit length of the corresponding beam. In some examples, the mass per unit length is between, and inclusive of, 0.09 g/mm and 0.40 g/mm, such as between, and inclusive of, 0.09 g/mm and 0.25 g/mm. According to yet some examples, the mass per unit length of the primary beam and/or the secondary beam is between, and inclusive of, 0.09 g/mm and about 0.35 g/mm, such as between, and inclusive of, 0.09 g/mm and 0.30 g/mm, such as between, and inclusive of, 0.09 g/mm and 0.25 g/mm, such as between, and inclusive of, 0.09 g/mm and 0.20 g/mm, such as between, and inclusive of, 0.09 g/mm and 0.17 g/mm, or such as between, and inclusive of, 0.1 g/mm and 0.2 g/mm. In some examples, the primary beam and/or the secondary beam has a mass per unit length less than 0.25 g/mm, such as less than 0.20 g/mm, such as less than 0.17 g/mm, such as less than 0.15 g/mm, such as less than 0.10 g/mm. In one example, the primary beam and/or the secondary beam has a mass per unit length of 0.16 g/mm. According to certain examples, the mass of the primary beam and/or the secondary beam is less than about 12 grams and, more preferably, less than about 8 grams. In certain examples, the primary beam and/or the secondary beam is sized, shaped, and configured in a manner similar to or the same as the bridge bars described in U.S. Pat. No. 11,338,183, which is incorporated herein by reference.

[0626] In some examples, the primary beam 467 is elongated and extends lengthwise, in an upward direction away from the toe portion 414, at an angle relative to the plurality of grooves 411 of the strike plate 452. Accordingly, in some examples, the first end of the primary beam 467 is more toward than the second end of the primary beam 467. In some examples, the primary beam 467 is angled such that the primary beam 467 defines an angle, relative to the topline of the golf club head 400, that is between, and inclusive of, 75-degrees and 105-degrees (such as 90-degrees). The primary beam 467 promotes stiffness and strength of the golf club head 400 by structurally connecting the variable thickness region 445 with the internal walls 487. In other words, the primary beam 467 improves the stiffness and strength of the golf club head 400 by transferring loads and vibrations from a thinner region of the golf club head 400 to a thicker region of the golf club head 400. In some examples, a ratio of a maximum value of the thickness T8 (see, e.g., FIG. 104) of the primary beam 467, above Z-up or above the interior-weight cavity 441, to a maximum value of the thickness T7 (see, e.g., FIGS. 103 and 104) of the strike plate 452 at the strike face 406 (excluding the weld guides) is between, and inclusive of, 0.75 and 3.5, between, and inclusive of, 0.85 and 3.25, between, and inclusive of, 0.9 and 2.95, or between, and inclusive of, 0.25 and 3.5. According to certain examples, the maximum value of the thickness T8 of the primary beam 467 is greater than the maximum value of the thickness T7 of the strike plate 452. The thickness T8 of the primary beam 467 is measured from the interior surface 463 of the back wall 462 proximate where the primary beam 467 meets/joins the back wall 462 to an outermost surface of the primary beam 467 distal the back wall 462, and measured primarily along the y-axis of the head origin coordinate system.

[0627] As shown, in some examples, the primary beam 467 extends upwardly and terminates at the variable thickness region 445. In other words, in certain examples, the primary beam 467 does not extend up to an interior surface of the top portion 416 of the golf club head 400, such that

a gap is defined between the interior surface of the top portion 416 and the upper end of the primary beam 467. However, as shown in FIG. 107, in some examples, the primary beam 467 of a golf club head 400F extends up to the interior surface of the top portion 416. In certain examples, the golf club head 400F includes a topline rib 471 and the primary beam 467 extends upwardly from an internal wall 487, defining the internal-weight cavity 441, to an interior surface of the topline rib 471.

[0628] Referring to FIG. 104, in some examples, the cavity filler material 433 at least partially surrounds the primary beam 467 on a first side (e.g., heelward side facing the heel of the golf club head) and a second side (e.g., toward side facing the toe of the golf club head) that is opposite the first side. According to certain examples, the cavity filler material 433 fully surrounds a perimeter of the primary beam 467 or fully surrounds all exposed surfaces of the primary beam 467.

[0629] As shown in FIGS. 80, 81, 86, and 87, the body 402 additionally includes an internal mass pad 465 located at an upper-toe region of the golf club head 400. The internal mass pad 465 includes a large protrusion extending from (e.g., formed in) the interior surface 463 of the rear wall 462. In effect, the internal mass pad 465 increases the thickness of the rear wall 462 such that the thickness of the rear wall 462 at the internal mass pad 465 (i.e., at the upper-toe region of the body 402) is greater than the thickness of the rear wall 462 within the variable thickness region 445. In some examples, the thickness of the rear wall 462 at the internal mass pad 465 (e.g., the maximum thickness of the rear wall 462 within the variable thickness region 445 plus the thickness of the internal mass pad 465) is between, and inclusive of, 1.1 mm and 1.6 mm, 1.3 mm and 1.9 mm, 1.5 mm and 2.1 mm. The internal mass pad may be used to strategically locate concentrated mass in the high toe region to increase inertia about a Z-axis and/or raise Zup in higher lofted irons e.g. having lofts greater than 30 degrees. It may be desirable to increase Zup with increasing loft and the internal mass pad and or the primary rib 467 may be used to strategically control CG placement and inertia.

[0630] Referring to FIGS. 88-91, and according to some examples of the present disclosure, a golf club head 400A is shown. The golf club head 400A includes features similar to the features of the golf club head 400, with like numbers referring to like features. Accordingly, unless otherwise noted below, the description of the features of the golf club head 400 apply to the analogous features of the golf club head 400A.

[0631] Like the golf club head 400, the golf club head 400A includes an internal weight that is made of a material having a density greater than the density of the material of the body 402 and is located closer to the toe of the golf club head 400A than the heel, and closer to the sole of the golf club head 400A than the topline. However, unlike the internal weight 440 of the golf club head 400, the internal weight 440A of the golf club head 400A does not have a filler-injection channel 434 that accommodates the injection of cavity filler material 433 into the internal cavity 432. The internal weight 440A has a different shape and is located in a different portion of the golf club head 400A than the internal weight 440. For example, the internal weight 440A is thinner and longer than the internal weight 440. Moreover, the internal weight 440 is elongated in the toe-to-heel direction, rather than the top-to-sole direction. Accordingly,

more of the mass of the internal weight 440A is located closer to the sole portion 418 of the golf club head 400A and closer to the heel portion 412 than the internal weight 440. Locating more of the mass of the internal weight 440A lower and heelward than the internal weight 440, promotes a reduction in the z-up value and a reduction in the moment of inertia of the golf club head 400A, compared to the golf club head 400. Furthermore, because the internal weight 440A is lower than the internal weight 440, the internal weight 440A is situated below the toe port 407, and thus the internal weight 440A, even without a filler-injection channel, does not interfere with the flow of cavity filler material 433 through the toe port 407.

[0632] The body 402 of the golf club head 400A additionally includes an internal-weight cavity 441A configured to receive the internal weight 440A in seated engagement. Accordingly, in some examples, the size and shape of the internal-weight cavity 441A corresponds with the size and shape of at least a portion of the internal weight 440A. In the illustrated examples, the internal-weight cavity 441A and the internal weight 440A are elongated in the heel-to-toe direction. Moreover, in the heel-to-toe direction, the depth of the internal-weight cavity 441A and the thickness T3 of the internal weight 440A (see, e.g., FIG. 92) can be constant. Referring to FIG. 93, along a plane parallel to the strike face 406, the internal weight 440A has a substantially parallelogram shape (with a constant height H) in the illustrated example, but could have a rectangular, triangular, or other shape in alternative examples. The parallelogram shape of the internal weight 440A is formed by adjacent sides of the internal weight 440A that are angled at an oblique angles.

[0633] Referring to FIGS. 88-91, the body 402 of the golf club head 400A additionally includes the primary beam 467 that is coupled to the interior surface 463 of the rear wall 462. Like the primary beam 467 of the golf club head 400, the primary beam 467 of the golf club head 400A extends from the internal walls 487, which define the internal-weight cavity 441A, to the variable thickness region 445 of the rear wall 462. However, the location of the primary beam 467 of the golf club head 400A is more soleward and heelward than that of the primary beam 467 of the golf club head 400. This is because the location of the internal-weight cavity 441A is more soleward and extends more heelward than the internal-weight cavity 441. Additionally, because of the more soleward location of the internal-weight cavity 441A, the distance between the variable thickness region 445 and the internal-weight cavity 441A is greater with the golf club head 400A, and thus the length of the primary beam 467 of the golf club head 400A is longer. Also, in certain examples, in addition to the length, the overall shape of the primary beam 467 of the golf club head 400A is different than that of the primary beam 467 of the golf club head 400. For example, a width of the primary beam 467 of the golf club head 400A can decrease in the top-to-sole direction (so the primary beam 467 is wider at its top than its bottom), whereas the primary beam 467 of the golf club head 400 has a width that increases in the top-to-sole direction (so the primary beam 467 is wider at its bottom than at its top).

[0634] Notwithstanding the different location and shape of the primary beam 467 of the golf club head 400A, the primary beam 467 is elongated and extends lengthwise, in an upward direction away from the toe portion 414, at an angle relative to the plurality of grooves 411 of the strike plate 452. Accordingly, like the primary beam 467 of the golf club

head 400, the primary beam 467 of the golf club head 400A promotes stiffness and strength of the golf club head 400A by structurally connecting the variable thickness region 445 with the internal walls 487.

[0635] According to certain examples, a total mass of each one of the internal weight 440 and the internal weight 440A is between, and inclusive of, 25 grams and 45 grams, such as between, and inclusive of, 30 grams and 40 grams. In other examples, the mass of the internal weight 440 may be at least 20 grams, 25 grams, 30 grams, or 35 grams and the internal weight 440 may have a mass no more than 100 grams, 90 grams, 80 grams, 70 grams, 60 grams, 50 grams, or 40 grams. Accordingly, the internal weight 440 may have a mass within any of the above ranges, e.g. between 25 grams and 60 grams as one of the many examples. In further embodiments, the mass of the internal weight 440 may be at least 20 grams and no more than 48 grams. In certain examples, a ratio of the total mass of each one of the internal weight 440 and the internal weight 440A and the total mass of the corresponding one of the golf club head 400 and the golf club head 400A is between, and inclusive of, 0.014 and 0.017, such as between, and inclusive of, 0.015 and 0.016. According to some examples, the total mass of each one of the golf club head 400 and the golf club head 400A is between, and inclusive of, 210 g and 270 g, such as between, and inclusive of, 220 g and 255 g or between, and inclusive of, 225 g and 251 g.

[0636] The rear wall 462 of each one of the golf club head 400 and the golf club head 400A has a variable thickness region 445, which is located above the center of gravity of the golf club head (e.g., above Zup) in some examples. The variable thickness region 445 is defined as a region of the rear wall 462 where the thickness of the rear wall 462 varies. In the illustrated examples, the thickness of the rear wall 462 varies by varying the contour of the interior surface 463 of the rear wall 462 while holding steady the contour of the exterior surface 461 of the rear wall 462. In this manner, the variable thickness nature of the rear wall 462 within the variable thickness region 445 is imperceptible when viewing the golf club head from an exterior of the golf club head. Referring to FIG. 94, the variable thickness region 445 occupies an upper portion of the rear wall 462 defined, in a top-to-sole direction, between a topline plane 489 and a sole-bar plane 485, and in a heel-to-toe direction, between, and the heel and the toe of the golf club head. The topline plane 489 is co-planar with the topline (e.g., an uppermost edge of the golf club head) and the sole-bar plane 485 is co-planar with an upper bound of a sole bar feature (e.g., having a relatively large thickness relative to the thickness of the strike plate 452) of the rear wall 462. Both the topline plane 489 and the sole-bar plane 485 are parallel with the y-axis of the golf club head origin coordinate system.

[0637] When the golf club head is in a proper address position on the ground plane 111, the sole-bar plane 485 is angled relative to the ground plane 111 (e.g., relative to the plurality of grooves 411) at a first angle ϕ_1 , and the topline plane 489 is angled relative to the sole-bar plane 485 at a second angle ϕ_2 . Accordingly, when the golf club head is in the proper address position on the ground plane 111, the topline plane 489 is angled relative to the ground plane 111 at an angle equal to the sum of the first angle ϕ_1 and the second angle ϕ_2 . In some examples, the first angle ϕ_1 is between, and inclusive of, 5-degrees and 30-degrees, between, and inclusive of, 10-degrees and 20-degrees, or

between, and inclusive of, 13-degrees and 17-degrees. In these and other examples, the second angle ϕ_2 is between, and inclusive of, 3-degrees and 20-degrees, between, and inclusive of, 5-degrees and 15-degrees, or between, and inclusive of, 7-degrees and 12-degrees. The sole-bar plane **485** and the topline plane **489** are upwardly angled in the heel-to-toe direction in some examples. Accordingly, the variable thickness region **445** is also upwardly angled in the heel-to-toe direction in those examples. In other words, the variable thickness region **445** extends from a lower location on the golf club head at the heel portion up to an upper location on the golf club head at the toe portion. Moreover, because the topline plane **489** is angled relative to the sole-bar plane **485**, the variable thickness region **445** diverges in the heel-to-toe direction, such that the area of the variable thickness region **445** is greater at the toe portion **414** than at the heel portion **412**.

[0638] The following description of the features of the variable thickness region **445** will proceed with reference to the golf club head **400**, but, unless otherwise noted, apply to the golf club head **400A** and other golf club heads of the present disclosure. Referring to FIGS. **80**, **81**, **98**, **100**, and **101**, the variable thickness region **445** includes a pattern of recesses **447** formed in the interior surface **463** of the rear wall **462**. The recesses **447** promote a reduction in mass within the variable thickness region **445**. By reducing the mass of the upper portion of the rear wall **462** via the variable thickness region **445**, more mass can be located lower on the golf club head **400**, which promotes a lower CG of the golf club head. In certain examples, the particular pattern of the recesses **447** promotes strength and rigidity despite the reduced mass. The recesses **447** are spaced apart from each other by corresponding webs **449** of the rear wall **462** so that the minimum distance d_2 between adjacent ones of the recesses **447** (e.g., width of a web **449**) is the same. Additionally, the size of each recess **447** of the pattern of recesses **447** can be defined by a distance d_1 between opposite walls of the recess **447**. As shown in FIG. **101**, the portions of the rear wall **462** within the variable thickness region **445** and defined by the recesses **447** have a thickness T_6 , and the portions of the rear wall within the variable thickness region **445** and between the recesses **447** have a thickness T_5 . The thickness T_6 of the rear wall **462** defined by the recesses **447** is less than the thickness T_5 of the rear wall **462** between the recesses **447**. Each one of the thickness T_5 and the thickness T_6 is less than 1.5 mm, in some examples, and greater than 0.3 mm, in some examples. In some examples, the thickness T_5 , which can be a maximum thickness of the variable thickness region **445**, is between, and inclusive of, 1.1 mm and 1.4 mm, between, and inclusive of, 0.3 mm and 1.25 mm, between, and inclusive of, 0.4 mm and 1.1 mm, or between, and inclusive of, 0.45 mm and 0.95 mm, or between, and inclusive of, 0.55 mm and 0.85 mm, and the thickness T_6 , which can be a minimum thickness of the variable thickness region **445**, is between, and inclusive of, 0.65 mm and 0.75 mm, between, and inclusive of, 0.6 mm and 1.2 mm, between, and inclusive of, 0.7 mm and 1.0 mm.

[0639] Although not required, the recesses **447** may form a pattern where at least one feature is repeated throughout the grouping of the recesses **447**. In one example, the repeating feature includes the constant minimum spacing between adjacent recesses **447**. In other words, the minimum spacing (e.g., minimum distance d_2) between adjacent

recesses **447** being the same throughout the variable thickness region **445** makes the grouping of the recesses **447** into a pattern.

[0640] According to another example, the repeating feature alternatively or additionally includes adjacent recesses having the same shape. Referring to FIG. **100**, in some examples, at least two, at least three, at least four, at least five, or at least six of the recesses **447** have the same shape. In the illustrated example, the repeating shape of the recesses **447** is a polygonal shape (e.g., a hexagonal shape, a square shape, a rectangular shape, a triangular shape, a pentagonal shape, an octagonal shape, and the like). In view of the repeating pattern of hexagonally-shaped recesses, in some examples, the recesses **447** of the variable thickness region **445** form a honeycomb pattern. According to certain examples, the repeating shape of the recesses **447** can be any of various shapes other than polygonal, such as circular, oval, or shapes having rounded corners.

[0641] In yet another example, the repeating feature of the pattern of recesses **447** is that each recess **447** of the pattern is confined within an imaginary boundary **479** having the same-shape and the same size, where the imaginary boundary **479** associated with each one of the recesses **447** does not overlap with the imaginary boundary **479** of any other one of the recesses. In other words, although potentially having a different size or shape, every one of the recesses **447** of the pattern of recesses **447** can fit within (i.e., not extend beyond of) a same-shaped and same-sized imaginary boundary **479** that does not overlap with any other imaginary boundary **479**. Accordingly, each one of the recesses **447** of the pattern of recesses **447** fits within a respective one of multiple identically-sized and identically-shaped non-overlapping imaginary boundaries **479**. An imaginary boundary **479** can have any of various shapes each having a maximum dimension. Because the recess **447** is contained within the imaginary boundary **479**, a maximum dimension d_{max} of the recess **447** is less than or equal to the maximum dimension of the imaginary boundary **479**.

[0642] In the illustrated example of FIG. **100**, the imaginary boundary **479** is a circle having a diameter. The diameter of the circle, which is the maximum dimension of the circle, is equal to or greater than the maximum dimension d_{max} (associated with a maximum value of the distance d_1) of the recess **447** confined within the circle. In some examples, such as shown, the recess **447** is inscribed within the circle such that the maximum dimension d_{max} of the recess **447** is equal to the diameter of the circle. However, in other examples, the recess **447** is confined within, but not inscribed by (e.g., touching), the circle. In various examples, the diameter of the circle is between, and inclusive of, 6 millimeters and 10 millimeters. In some examples, a maximum dimension of at least one of the recesses **447** is between, and inclusive of, 6 millimeters and 10 millimeters. Although various patterns of similarly shaped recesses **447** are discussed above and shown, as presented herein, a repeating pattern of similarly shaped recesses is unnecessary to accomplish the combination of weight savings and strategic structural stiffness accomplished by the recesses. In other words, the recesses could take on a variety of organic shapes or geometric shapes provided that there are multiple transitions (two or more) from a thickened region to a thinned region along the rear wall **462**.

[0643] In some examples, the variable thickness region along the rear wall **462** may be described using areal mass

in units of grams per mm^2 . Referring to FIG. 94, an areal mass of the rear wall 462 of the golf club head between the topline portion, the sole bar plane 485, the toe portion 414, and the heel portion 412 is between 0.0005 g/mm^2 and 0.00925 g/mm^2 , such as, for example, about 0.0037 g/mm^2 . Additionally or alternatively, referring to FIG. 94, an areal mass of the rear wall 462 of the golf club head between the topline portion, the toe portion 414, and the heel portion 412, and the portion of the rear wall above Zup is between 0.0005 g/mm^2 and 0.00925 g/mm^2 , such as, for example, between 0.0030 g/mm^2 and 0.0080 g/mm^2 , preferably between 0.0050 g/mm^2 and 0.0090 g/mm^2 , even more preferably between 0.0070 and 0.0085 . Generally, the areal mass of the variable thickness region of the rear wall 462 is the mass per unit area of the area defined by the region of the rear wall 462 above the sole bar plane 485 or above Zup and bounded by the topline portion 416, toe portion 414, and heel portion 412. In some implementations, the area of the variable thickness region of the rear wall 462 varies between 500 mm^2 to $1,800 \text{ mm}^2$, preferably the area of the variable thickness region of the rear wall 462 is between 650 mm^2 and $1,200 \text{ mm}^2$, even more preferably the area of the variable thickness region of the rear wall 462 varies between 900 mm^2 to $1,500 \text{ mm}^2$. In some example, the rear wall 462 may be aperture free within the variable thickness region i.e. the rear wall 462 may be solid within the variable thickness region without any openings.

[0644] An alternative solution may be to include openings or lightening ports to further reduce the mass above Zup. This could be done in combination with a variable thickness region with webbing for stiffness. The rear wall may include one to three openings and the openings would be preferably covered by a lightweight badge formed of a material having a density between 0.5 g/cc to 5 g/cc . This method may achieve similar weight savings, but would present greater aesthetic difficulties compared to a solid rear wall iron. A solid rear wall iron is desirable over one with a badge because the iron has the appearance of looking like a muscle back iron or better players club, whereas clubs with badges are typically seen as more game improvement irons for higher handicap golfers.

[0645] Each one of the recesses 447 has a depth relative to the interior surface 463 of the rear wall 462. The depth of the recesses 447 is equal to the difference between the thickness T5 of the rear wall 462 between the recesses 447 and the thickness T6 of the rear wall 462 at the recesses 447. In some examples, the depth of all the recesses 447 of the pattern of recesses 447 is the same, which can be another repeating feature that defines the grouping of recesses 447 as a pattern. According to some examples, a ratio of the thickness T6 of the rear wall 462 at the recesses 447 to a minimum thickness of the strike plate 452 at the strike face 406 is between, and inclusive of, 0.1 and 0.7, between, and inclusive of, 0.15 and 0.65, between, and inclusive of, 0.2 and 0.6, between, and inclusive of, 0.25 and 0.55, or between, and inclusive of, 0.3 and 0.6. In some examples, a ratio of the thickness T5 of the rear wall 462 to a maximum thickness of the strike plate 452 at the strike face 406 is between, and inclusive of, 0.1 and 0.7, between, and inclusive of, 0.15 and 0.65, between, and inclusive of, 0.2 and 0.6, or between, and inclusive of, 0.25 and 0.55. The above thicknesses of the strike plate 452 may or may not include a thickness of a weld guide 490.

[0646] According to some examples, a ratio of the depth of a recess 447 of the pattern of recesses 447 to a maximum

thickness of the strike plate 452 at the strike face 406 is between, and inclusive of, 0.06 and 0.18, or between, and inclusive of, 0.07 and 0.13. In such examples, the Zup of the golf club head can be no more than 20 mm, such as between, and inclusive of, 12 mm and 19 mm. The minimum thickness and the maximum thickness of the strike plate 452 are located on a portion of the front portion 420 that is formed separate from the body 402 and attached to the body 402. In some examples, the minimum thickness and the maximum thickness of the strike plate 452 are located within a region of the strike face 406 that includes the grooves 411. According to some examples, the minimum thickness and the maximum thickness of the strike plate 452 are located within the central region of the strike face 406, which can be defined by a $18 \text{ mm} \times 36 \text{ mm}$ rectangular area centered on the geometric center of the strike face 406.

[0647] According to some examples, a ratio of the depth of a recess 447 of the pattern of recesses 447 to a Zup of the golf club head 400 is between, and inclusive of, 0.011 and 0.025, or between, and inclusive of, 0.014 and 0.019. In certain examples, a ratio of the thickness T6 of the rear wall 462 at the recesses 447 to a minimum value of the thickness T7 of the strike plate 452 at an upper portion of the strike face 406, proximate the topline of the golf club head 400, is between, and inclusive of, 0.30 and 0.70, or between, and inclusive of, 0.45 and 0.68 (see, e.g., FIG. 103). In many of the examples, this is for a thin face iron e.g. and iron having a minimum face thickness no more 2.25 mm, 2.15 mm, 2.00 mm, 1.90 mm, or no more than 1.80 mm. Similarly a maximum face thickness of a thin face is no more than 3.35 mm, 3.15 mm, 3.00 mm, 2.90 mm, or no more than 2.80 mm. The volume of each one of the recesses 447 of the pattern of recesses 447 is defined by the multiplication of a total area of the recess by the depth of the recess. In some examples, the total area of each one of at least two recesses of the pattern of recesses 447 is at least 224 mm^2 , such as between, and inclusive of, 225 mm^2 and 450 mm^2 , or between, and inclusive of, 270 mm^2 and 400 mm^2 . In certain examples, the volume of each one of at least two recesses of the pattern of recesses 447 is at least 100 mm^3 or at least 170 mm^3 , such as between, and inclusive of, 100 mm^3 and 300 mm^3 (e.g., between, and inclusive of, 112 mm^3 and 293 mm^3). According to some examples, a thickness of the strike plate 452 is variable and varies between, and inclusive of, 1.1 mm and 3.5 mm, or between, and inclusive of, 1.4 mm and 3.3 mm.

[0648] According to some examples, at least some recesses 447 of the pattern of recesses 447 are arranged in at least one row that extends at an angle relative to the plurality of grooves 411 of the strike plate 452, and parallel to the topline plane 489, parallel to the sole-bar plane 485, or non-parallel to both the topline plane 489 and the sole-bar plane 485. In the illustrated example, the pattern of recesses 447 includes a first row 446A. The first row 446A is parallel to the topline plane 489 in the illustrated example. The recesses 447 of the first row 446A form a row because a plane, which passes through the variable thickness region 445 and is parallel to the y-axis of the golf club head origin coordinate system, passes through each one of the recesses 447 of the first row 446A. In certain examples, at least some recesses 447 of the pattern of recesses 447 are arranged in at least two rows, such as shown in FIG. 94. The two rows include the first row 446A and a second row 446B, where the second row 446B extends parallel to and below the first row

446A. In yet alternative examples, such as shown in FIG. 107, at least some recesses **447** of the pattern of recesses **447** are arranged in at least three rows, such as the first row **446A**, the second row **446B**, and a third row **446C**, which extends parallel to and below the second row **446B**. According to other examples, such as shown in FIGS. 108 and 109, at least some recesses **447** of the pattern of recesses **447** are arranged in at least four rows, such as the first row **446A**, the second row **446B**, the third row **446C**, and a fourth row **446D**, which extends parallel to and below the third row **446C**.

[0649] Referring to FIG. 101, the variable thickness region **445** can be configured such that, in a heel-to-toe direction along the variable thickness region **445**, the thickness of the rear wall **462** transitions from a thick portion to a thin portion and back to a thick portion at least four times, at least five times, at least six times, or at least seven times. More specifically, in the heel-to-toe direction, the thickness of the rear wall **462** transitions from the thickness of the rear wall **462** between the recesses **447** to the thickness of the rear wall **462** defined by the recesses **447** and from the thickness of the rear wall **462** defined by the recesses **447** to the thickness of the rear wall **462** between the recesses **447** at least four times, at least five times, at least six times, or at least seven times.

[0650] Stress from an impact with a golf ball is distributed from the strike face **406**, through the strike plate **452**, to the top portion **416** of the golf club head **400**. From the top portion **416**, the stress is further distributed to the rear wall **462**, including to the variable thickness region **445**. As presented above, the variable thickness region **445** is configured to remove mass from upper portion of the rear wall **462** so that mass can be redistributed lower on the golf club head **400** for lowering the CG of the golf club head **400**. However, with less mass in the upper portion of the rear wall **462**, there is less material to absorb the stress from an impact with a golf ball. Accordingly, in some examples, as shown in FIG. 80, to help reduce the amount of stress absorbed by the variable thickness region **445**, the body **402** includes a topline rib **471** located in the top portion **416** of the golf club head **400**. The topline rib **471** is positioned at an intersection of the topline of the golf club head **400** and the rear wall **462**. Moreover, the topline rib **471** includes a thickened region of the top portion **416**. The added thickness provided by the topline rib **471** promotes strength and rigidity along the top portion **416** and helps absorb stress from impacts with a golf ball, which reduces stress absorbed by the rear wall **462**.

[0651] Referring now to FIGS. 94-99, the present disclosure includes some examples of a correlated set of iron-type golf club heads. As defined herein, a correlated set of iron-type golf club heads means a set of at least two iron-type golf club heads that are sold together. Although only two golf club heads sold together can form a correlated set of golf club heads, conventional correlated sets of iron-type golf club heads include more than two golf club heads. For example, a correlated set can include nine club set (e.g., 3I-AW (3-iron to gap wedge)), an eight club set (e.g., 3I-PW (3-iron to pitching wedge) or 4I-AW (4-iron to gap wedge)), a seven club set (e.g., 5I-AW (5-iron to gap wedge) or 4I-PW (4-iron to pitching wedge)), or any one of a between, and inclusive of, three to six club set. As used herein, and unless otherwise noted, the features of the golf

club heads of a correlated set of golf club heads means the features are embodied in at least two clubs that are sold together as a set.

[0652] According to certain examples, a correlated set of iron-type golf club heads includes at least two, at least three, at least four, at least five, or at least six of the golf club head **400A** of FIG. 94, a golf club head **400B** of FIG. 95, a golf club head **400C** of FIG. 96, a golf club head **400D** of FIG. 97, the golf club head **400** of FIG. 98, and a golf club head **400E** of FIG. 99. Each one of the golf club heads **400B-400E** includes features similar to the features of the golf club head **400** and the golf club head **400A**, with like numbers referring to like features. Accordingly, unless otherwise noted below, the description of the features of the golf club head **400** and the golf club head **400A** apply to the analogous features of the golf club heads **400B-400E**. For example, like the golf club head **400A**, the golf club head **400B** and the golf club head **400C** has an internal-weight cavity **441A** and an internal weight **440A**, and the golf club head **400D** has an internal-weight cavity **441** and an internal weight **440**. However, unlike the golf club head **400** and the golf club head **400A**, the golf club head **400E** does not have an internal-weight cavity (or an internal weight) and does not have a beam (sound bar). In the illustrated examples of the correlated set of golf club heads, the golf club head **400A** is a 3-iron, the golf club head **400B** is a 4-iron, the golf club head **400C** is a 5-iron, the golf club head **400D** is a 6-iron, the golf club head **400** is a 7-iron, and the golf club head **400E** is an 8-iron. Although not shown, in certain examples, the correlated set of golf club heads also includes a 9-iron, a pitching wedge, and a gap wedge each having a configuration similar to that of the golf club head **400E**, but without a variable thickness region. Accordingly, in some examples, golf club heads, of a correlated set of golf club heads, that have a loft greater than 38-degrees do not have a beam (sound bar), a variable thickness region, nor an internal-weight cavity, a loft greater than 34-degrees and less than 38-degrees do not have a beam or an internal-weight cavity, but do have a variable thickness region, and a loft less than 34-degrees have a beam, a variable thickness region, and an internal-weight cavity (with weight).

[0653] Each golf club head of the correlated set of golf club heads described herein includes the variable thickness region **445** having the pattern of recesses **447** formed in the interior surface **463** of the rear wall **462**. However, the pattern of recesses **447** of any one golf club head of the correlated set of golf club heads is different than the pattern of recesses **447** of any other one golf club head of the correlated set of golf club heads. According to some examples, the number of recesses **447** that have the same shape is different amongst different golf club heads of the same correlated set of golf club heads. In one example, at least one golf club head has at least two recesses with the same shape, while at least one other golf club head of the same correlated set has no recesses with the same shape. For example, the golf club head **400A** has three recesses with the same hexagonal shape and the golf club head **400D** has only one recess with a hexagonal shape. In another example, the golf club head **400A** has only three recesses with the same hexagonal shape and the golf club head **400** has six recesses with the same hexagonal shape.

[0654] In some examples, the pattern of recesses **447** of one golf club head of a correlated set is different than another golf club head of the correlated set because,

although a number of recesses per unit length of the pattern of recesses 447 for both golf club heads is the same, a shape of at least one recess of the pattern of recesses is different. For example, the number of recesses per unit length of the pattern of recesses 447 of the golf club head 400A and the golf club head 400B is the same. However, some of the recesses 447 of the golf club 400A have shapes that are different than some of the recesses 447 of the golf club head 400B.

[0655] In addition to having a different pattern of recesses 447, in some examples, at least two of the golf club heads of a correlated set of golf club heads have internal weights that have different sizes/shapes and are located in different locations within the golf club heads. For example, the golf club head 400A and the golf club head 400 can be in the same correlated set, and the internal-weight cavity 441A and the internal weight 440A of the golf club head 400A is shaped and located differently than the internal-weight cavity 441 and the internal weight 440 of the golf club head 400.

[0656] In addition to having a different pattern of recesses 447, in some examples, at least two of the golf club heads of a correlated set of golf club heads have beams with a different size/shape and/or location, or alternatively a different number of beams. For example, the golf club head 400A and the golf club head 400 can be in the same correlated set, and the primary beam 467 of the golf club head 400A is longer, has a diverging shape, and is more heelward than the primary beam 467 of the golf club head 400. Even where the internal weights have the same shape and location, such as with the golf club head 400A and the golf club head 400B of the same correlated set, the primary beam 467 of the golf club head 400A is shaped differently at both ends of the primary beam 467 than the primary beam 467 of the golf club head 400B. Moreover, the primary beam 467 of the golf club head 400D, which has an internal weight 440 that is shaped the same as the internal weight 440 of the golf club head 400, is shorter and angled differently than the primary beam 467 of the golf club head 400.

[0657] In some examples, a golf club head of the present disclosure also includes a secondary beam, in addition to the primary beam. Referring to FIG. 97, in some examples, the golf club head 400D includes a secondary beam 469 that is located heelward of the primary beam 467. The secondary beam 469 includes a first end, coupled to the sole portion 418 heelward of the internal-weight cavity 441, the toe portion 414, and/or the toe port 407, and a second end, coupled to the variable thickness region 445 of the rear wall 462. The secondary beam 469 that is coupled to the interior surface 463 of the rear wall 462 and extends from the sole portion 418 (e.g., a sole bar) to the variable thickness region 445. In this manner, and like the primary beam 467, the secondary beam 469 structurally connects the variable thickness region 445 with the sole portion 418, which promotes stiffness and strength of the golf club head 400D by transferring loads and vibrations from a thinner region of the golf club head 400D to a thicker region of the golf club head 400D. According to one example, the lower end of the primary beam 467 is located above Z-up of the golf club head 400D, and the lower end of the secondary beam 469 is located below Z-up of the golf club head 400D.

[0658] The secondary beam 469 can have a width (measured in a heel-to-toe direction) that diverges and then converges in a sole-to-top direction. Accordingly, in some examples, one or more golf club head of the present disclo-

sure includes at least two beams (sound bars). In certain examples, one or more golf club heads of the present disclosure includes at least three beams (see, e.g., FIGS. 116 and 117). The golf club head 400D forms part of a correlated set of golf club heads, with at least one other of the golf club heads 400-400C and 400E, in some examples. Accordingly, in these examples, at least one golf club head of a correlated set of golf club heads has the secondary beam 469 and at least one other golf club head of the correlated set of golf club heads does not have the secondary beam 469. Or, alternatively, at least two of the golf club heads have a secondary beam with the secondary beams being configured differently from one another.

[0659] In some examples, such as shown in FIGS. 108-110, some golf club heads of the present disclosure include at least a third beam. Referring to FIG. 108, in one example, a golf club head 400G, which includes at least some features similar to the features of the golf club head 400, includes the primary beam 467, the secondary beam 469, and a third beam 473. The primary beam 467 and the secondary beam 469 are connected to and extend upwardly from an upper one of the internal walls 487. Moreover, each one of the primary beam 467 and the secondary beam 469 terminates at the variable thickness region 445 at a respective upper end of the primary beam 467 and the secondary beam 469. The third beam 473 is connected to and extends between the upper ends of the primary beam 467 and the secondary beam 469. The internal wall 487, the primary beam 467, the secondary beam 469, and the third beam 473 form a quadrilateral shape and define a pocket 475 in the rear wall 462, having a corresponding quadrilateral shape, therebetween. In some examples, the interior surface 463 of the rear wall 462 within the pocket 475 does not have a pattern of recesses 447. In other words, the pattern of recesses 447 formed in the interior surface 463 does not extend into the pocket 475. However, in other examples, the pocket 475 includes a portion of the pattern of recesses 447. In the example of FIG. 108, the primary beam 467 extends upwardly away from the heel portion 412 and the secondary beam 469 extends upwardly toward the heel portion 412. In other words, the primary beam 467 and the secondary beam 469 extend upwardly away from each other. In this manner, an area of the pocket 475 can increase in the sole-to-top direction. The third beam 473 extends at an angle, relative to the grooves 411, in some examples. For example, the third beam 473 can be parallel to a topline plane of the golf club head 400G.

[0660] Referring to FIG. 109, in one example, a golf club head 400H, which includes features similar to the features of the golf club head 400G, includes the primary beam 467, the secondary beam 469, and a third beam 473. Like the golf club head 400G, the internal wall 487, the primary beam 467, the secondary beam 469, and the third beam 473 of the golf club head 400H form a quadrilateral shape and define a similarly shaped pocket 475 in the rear wall 462. However, unlike the golf club head 400G, the primary beam 467 of the golf club head 400H extends upwardly toward the heel portion 412 and the secondary beam 469 extends upwardly away from the heel portion 412. In other words, the primary beam 467 and the secondary beam 469 extend upwardly toward each other. In this manner, an area of the pocket 475 can decrease in the sole-to-top direction. The third beam 473 of the golf club head 400H extends at an angle, relative to the grooves 411, in some examples. For example, the third

beam 473 of the golf club head 400H can be parallel to a sole-bar plane of the golf club head 400H.

[0661] Referring to FIG. 110, in one example, a golf club head 400I, which includes features similar to the features of the golf club head 400G and the golf club head 400H, includes the primary beam 467, the secondary beam 469, and a third beam 473. However, unlike the golf club head 400G and the golf club head 400H, the golf club head 400I includes a fourth beam 478 connected to and extending between the upper ends of the third beam 473 and the secondary beam 469. The internal wall 487, the primary beam 467, the secondary beam 469, the third beam 473 and the fourth beam 478 form a pentagonal shape and define a pocket 475 in the rear wall 462, having a corresponding pentagonal shape, therebetween. In some examples, the interior surface 463 of the rear wall 462 within and outside of the pocket 475 does not have a pattern of recesses 447.

[0662] Referring to FIG. 110, in one example, a golf club head 400I, which includes features similar to the features of the golf club head 400G and the golf club head 400H, includes the primary beam 467, the secondary beam 469, and a third beam 473. However, unlike the golf club head 400G and the golf club head 400H, the golf club head 400I includes a fourth beam 478 connected to and extending between the upper ends of the third beam 473 and the secondary beam 469. The internal wall 487, the primary beam 467, the secondary beam 469, the third beam 473 and the fourth beam 478 form a pentagonal shape and define a pocket 475 in the rear wall 462, having a corresponding pentagonal shape, therebetween. In some examples, the interior surface 463 of the rear wall 462 within and outside of the pocket 475 does not have a pattern of recesses 447.

[0663] Referring to FIGS. 111-114, in one example, a golf club head 400J, which includes features similar to the features of the golf club heads 400G-400I, the golf club head 400J includes a primary beam 467, a secondary beam 469, and a third beam 473. However, unlike the golf club heads 400G-400I, the third beam 473 is not connected to and does not extend between the upper ends of the primary beam 467 and the secondary beam 469. Rather, the third beam 473 is located heelward of the secondary beam 469. Accordingly, the primary beam 467, the secondary beam 469, and the third beam 473 are spaced apart relative to each other in a toe-to-heel direction. Furthermore, like the secondary beam 469, the third beam 473 is coupled to the sole portion 418 at a first end or lower end. A second end or upper end of the third beam 473 terminates at or within the variable thickness region 445 of the golf club head 400J. Accordingly, like the primary beam 467 and the secondary beam 469, the third beam 473 of the golf club head 400J structurally connects the variable thickness region 445 with the sole portion 418, which promotes stiffness and strength of the golf club head 400J by transferring loads and vibrations from a thinner region of the golf club head 400J to a thicker region of the golf club head 400J.

[0664] The golf club head 400J additionally includes a rear injection port 401 through which cavity filler material can be injected into the internal cavity 432 of the golf club head 400J. However, unlike the injection port 407 of the golf club head 400, the rear injection port 401 is formed in a rear wall 462 of the body 402 of the golf club head 400J. After the cavity filler material is injected into the internal cavity 432, the rear injection port 401 can be sealed with an external weight 443, which can be threadably engaged with

the rear injection port 401 to seal the rear injection port 401. The discussion above with respect to the external weight 243 discussing various purposes, ratios, and applications applies equally to the external weight 443, and it is recommended to refer to the above discussion for further information and potential uses of the external weight 443. This includes any and all ratios of mass of the weight and filler material compared to volumes of the club head.

[0665] According to some examples, before the cavity filler material is injected into the internal cavity 432 through the rear injection port 401, a vibration damper 457 may be inserted into the internal cavity 432 through the rear injection port 401. The vibration damper 457 may be made of a material similar to the material of the damper 280 described above. Additionally, the vibration damper 457 may be made of materials similar to those discussed above that are relevant to the external weight 243 and the materials discussed for the optional end cap on the end of the external weight 243. The applications of the vibration damper 457 may also be similar to those of the external weight 243 discussed above (e.g. the vibration damper 457 may be used for damping, improving durability, shot dispersion, selective stiffening of the face and potentially selective removal of higher density material in place of lower density damper material, and/or tuning or reducing the COR of the club head). According to the illustrated example, the vibration damper 457 has a cylindrical shape (e.g., puck shape). To help position and retain the vibration damper 457 within the internal cavity 432, the body 402 further includes a damper pocket 459 within the internal cavity 432. The damper pocket 459 is laterally offset, towards the toe portion 414, relative to the rear injection port 401. Accordingly, the vibration damper 457 can be inserted through the rear injection port 401 and slid into engagement with the damper pocket 459, which is shaped to nestably engage the vibration damper 457. When nestably engaged by the damper pocket 459, the vibration damper 457 is configured to engage an interior surface of the strike plate of the golf club head 400J, such that the vibration damper 457 is sandwiched between the strike plate and the rear wall 462 of the golf club head 400J. Engagement with the strike plate helps to dampen vibrations in the strike plate resulting from an impact with a golf ball. In some examples, the strike plate of the golf club head 400J is attached to the body 402 when the vibration damper 457 is inserted into the internal cavity 432 such that sliding the vibration damper 457 into engagement with the damper pocket 459 includes compressing the vibration damper 457 between the strike plate and the rear wall 462. The vibration damper 457 may be capable of compressing between 0.05 mm to 0.55 mm, preferably 0.10 to 0.25 mm when engaged with the rear surface of the front portion to allow for good engagement with the rear surface of the front portion and the rear wall 462 within the internal cavity 432. In some examples, as shown, the damper pocket 459 is formed in the secondary beam 469, which is located between the rear injection port 401 and the primary beam 467.

[0666] Typically, the strike plate is already joined to the body when the damper is installed and is shown removed in these views to aid in understanding and describing the embodiments.

[0667] Generally, after sliding the damper into place, then foam may optionally be injected to further secure the damper in place. However, the compression alone of the damper between the rear surface of the front portion of the

golf club head or the strike plate and an internal rear wall may be sufficient for holding the damper in place. The amount of interference fit of the damper between the rear surface of the front portion and the rear wall may be tuned to aid in installation and damping. Greater interference will make installation more challenging, but doing so will ensure the damper is more secure. Additionally, greater interference may provide greater damping however it may also reduce the COR of the strike face, especially for impact locations near the damper, if the interference is too great. In some instances, it may be intentional to reduce the COR using the damper. In some embodiments, the strike face may have a corresponding recessed surface to aid in installation which would allow for an additional level of securing the damper in place and may ensure correct placement. The installer would slide the damper over and experience some difficulty and then the damper would find the recess and more easily slide into place, become seated, and still remain in a state of compression. The vibration damper 457 may contact the rear surface of the front portion at one or more contact areas and may have cut-outs or relief portions where it does not contact the rear surface of the front portion, such that gaps are defined between the contact areas or regions.

[0668] In some embodiments, the damper may be made of a certain material that provides more than damping. If the material is hard enough, it may allow for a thinning of the face and the damper may take some of the load. A damper is typically in the range of 0.2 g/cc to 4 g/cc, preferably 0.5 g/cc to 2 g/cc and a face is typically 4.3 g/cc to 8 g/cc, and more typically 7 g/cc to 8 g/cc, so thinning of the face can save mass on the face, which can be located elsewhere in the club head.

[0669] Once the damper is in place the foam can optionally be injected, and then the external weight 443 may be installed to close the rear injection port 401.

[0670] According to some examples, at least two of the golf club heads of a correlated set of golf club heads have differently shaped and/or sized heel pockets. Referring to FIGS. 115 and 114, a golf club head 400K of a correlated set of golf club heads includes a heel pocket 453. The heel pocket 453 is located in the heel portion 412 of the golf club head 400K and is defined between the rear wall 462 and the front portion 420 of the golf club head 400K defined by the body 402. In other words, the heel pocket 453 is heelward of a separately-formed strike plate of the golf club head 400K. Put another way, at least a portion of the heel pocket 453 is heelward of the par line of the golf club head 400K. The heel pocket 453 is positioned behind the portion of the strike face defined by the body 402. The heel pocket 453 has front-to-rear depth that decreases in a sole-to-top direction. Therefore, the heel pocket 453 is deeper proximate the sole portion 418 than proximate the top portion 416. In some examples, the heel pocket 453 has a height such that the heel pocket 453 extends at least as high as the variable thickness region 445 of the golf club head 400K. The heel pocket 453 reduces mass in the heel portion 412, which can be relocated to other portions of the golf club head 400K to improve the performance or sound characteristics of the golf club head 400K.

[0671] Referring to FIGS. 117 and 118, in some examples, a golf club head 400L of a correlated set of golf club heads includes features similar to the golf club head 400K. For example, the golf club head 400L includes a heel pocket 453. Additionally, the golf club head 400L can include a

primary beam 467 that has one or more beam recesses 455. The primary beam 467 includes a toeward surface that faces the toe portion 414 of the golf club head 400L (see, e.g., FIG. 117) and a heelward surface that faces the heel portion 412 of the golf club head 400L (see, e.g., FIG. 118). In one example, a beam recess 455 is formed in the toeward surface of the primary beam 467 and/or a beam recess 455 is formed in the heelward surface of the primary beam 467. The beam recess 455 defines a thinned region of the primary beam 467. In some examples, an entirety of the beam recess 455 is offset from an outer perimeter of the primary beam 467 such that a thickness of the primary beam 467 along an entirety of its outer perimeter is greater than the thickness of the beam recess 455. The beam recess 455 reduces mass in the primary beam 467, which can be relocated to other portions of the golf club head 400L to improve the performance or sound characteristics of the golf club head 400L. Referring to FIG. 119, the primary beam 467 extends up to and is directly connected to an interior surface of the topline portion 416 of golf club head 400L. In some examples, the primary beam 467 extends up to and is directly connected to a topline rib 471 formed in the topline portion 416. The beam recess 455 is optional, such that, in some examples, the primary beam 467 of the golf club head 400L does not have any beam recesses.

[0672] In addition to having a different pattern of recesses 447, in some examples, at least two of the golf club heads of a correlated set of golf club heads have differently shaped and/or sized mass pads 465. For example, the mass pad 465 of the golf club head 400A can be larger, per unit mass, volume, or size of the golf club head, than the mass pad 465 of the golf club head 400.

[0673] Referring to FIGS. 94-98, in addition to having a different pattern of recesses 447, in some examples, at least two of the golf club heads of a correlated set of golf club heads have differently shaped and/or sized topline rib 471. For example, the shape and/or size of the topline rib 471 of one of the golf club heads 400-400D is different than any other one of the golf club heads 400-400D. In one example, the length of the topline rib 471 of the golf club head 400A of FIG. 94 is longer (per unit length of the golf club head) than the topline rib 471 of the golf club head 400B of FIG. 95. Moreover, the topline rib 471 of the golf club head 400A extends from a heel portion 412 of the golf club head 400A to a location at least halfway along the topline portion 416, whereas the topline rib 471 of the golf club head 400B is spaced apart from the heel portion 412.

[0674] The different configuration of the pattern of recesses, the internal weights, the primary beam, the topline rib, and/or the inclusion or omission of the secondary beam of the golf club heads of the correlated set of golf club heads is based on an optimization of the performance of each one of the golf club heads of the correlated set. In other words, the features of each one of the golf club heads of the set is optimized, independently of any other one of the golf club heads of the set, to achieve certain preset performance parameters specific to the golf club head. For example, the performance parameters of a long iron, such as a 3-iron or 4-iron, are different than a mid-iron, such as a 5-iron, 6-iron, or 7-iron, or a short iron, such as an 8-iron, a 9-iron, a pitching wedge, or a gap wedge. Because each golf club head in the set of golf club heads has different performance parameters (e.g., launch angle (loft), spin rate, workability

(inertia (Izz)), CGx, Zup, audiology, etc.), the optimized distribution of mass is different for each golf club head of the set of golf club heads.

[0675] Referring to FIG. 102, according to some examples, the golf club heads of the correlated sets of golf club heads disclosed herein can have the properties and corresponding values identified in the chart 800. General definitions of at least some of the properties can be found in U.S. Pat. No. 10,493,335 (see, e.g., FIGS. 2F, 8 and 10), which is incorporated herein by reference in its entirety. In addition to the golf club heads 400-400E of the illustrated example of correlated set of golf club heads, the chart 800 identifies, by skew (e.g., 9-iron and pitching wedge (PW)), examples of other golf club heads of the correlated set of golf club heads. With reference to the chart 800, the mass properties and structural configuration of the golf club heads of the correlated set of golf club heads are optimized to provide a lower Z-up in long irons and higher inertia in mid-irons. For example, the Z-up value for the golf club head 400 is intentionally higher than the golf club head 400D, the rearward location of the CG (e.g., D1) is intentionally greater than the golf club head 400D, and the inertia (e.g., Izz) is intentionally higher than the golf club head 400D. In some examples, for golf club heads with lofts greater than 29-degrees, Z-up is less than 20 mm, Izz is greater than 240 kg*mm², and D1 is greater than 9 mm. As shown in the chart 800, the value for Z-up increases with increasing loft of the golf club heads. In some examples, the value for D1 increases at a rate of at least 0.5 mm for every 3.0 to 5.5 degrees of loft. According to certain examples, the Z-up of at least two golf club heads with a loft ranging between, and inclusive of, 30-degrees and 46-degrees is greater than

[0676] Referring to FIG. 106, according to some examples, the features of the golf club heads of the correlated sets of golf club heads disclosed herein can have thickness values identified in the chart 900. The thickness values provided in the chart 900 are in millimeters (mm). Moreover, each thickness value for a given feature in the chart 900 is a baseline thickness value for that feature. Accordingly, the thickness value for any given feature in the chart 900 can fall within a range encompassing the baseline thickness and defined by the tolerance range (TOL) associated with the feature. For example, the baseline thickness value for a center portion F1 of the strike plate 452 of the golf club head 400A is 2.34 mm, but the thickness value of the center portion F1 can be any of various thickness values within 0.15 mm greater than or less than 2.34 mm (i.e., between, and inclusive of, 2.19 mm and 2.49 mm). In addition to the golf club heads 400-400E of the illustrated example of correlated set of golf club heads, the chart 900 identifies, by skew (e.g., 2-iron (2), 9-iron (9), pitching wedge (P), and gap wedge (A)), examples of other golf club heads of the correlated set of golf club heads. The features identified in the chart 900 includes the center portion F1, an upper-toe portion F2, an upper-heel portion F3, a lower-heel portion F4, and a lower-toe portion F5 of the strike plate 452. Additionally, the chart 900 identifies thickness values for a lower-toe portion F6, an upper-toe portion F7, an upper-heel portion F8, and a lower-heel portion F9 of the front portion 420 defined by the body 402 and adjacent the strike plate 452. The chart 900 additionally identifies thickness values for a thickened portion of the top portion 414 (i.e., T9) and a thinned portion of the top portion 416 (i.e., T10) defined

by the body 402 (see, e.g., FIG. 105). Furthermore, the chart 900 includes thickness values for the strike plate 452 at the sole wrap-around portion S1 of the strike plate 452, the thickness T5 of the rear wall 462, and the thickness T6 of the rear wall 462.

[0677] FIGS. 123 and 124 are examples that show a golf club head in the address position. As shown in FIGS. 123 and 124, positioning a golf club head in the address position lends itself to creating a coordinate system 1185 tied to the ground plane. The axes of the coordinate system 1185 can be tied to the ground plane, a first vertical plane perpendicular to the ground plane, and a second vertical plane perpendicular to both the ground plane and the first vertical plane. Generally, in most examples, the first vertical plane is parallel to the hosel axis 1193 (or the hosel axis 1193 lies in the first vertical plane). An x-axis of the coordinate system passes through a designated origin and either lies along the first vertical plane or is parallel to the first vertical plane. A y-axis of the coordinate system passes through the designated origin and either lies along the ground plane or is parallel to the ground plane. The z-axis of the coordinate system passes through the designated origin and either lies along the second vertical plane or is parallel to the second vertical plane. Additionally, the USGA methodology may be used to measure the various parameters described throughout this application including head height and club head center of gravity CG location. According to the illustrated example of FIGS. 123 and 124, the club head origin coordinate system 1185 is defined with an origin at the ground plane intersection point (GPIP), which is described in more detail below. The positive x-axis is heelward of the GPIP and the negative x-axis is toward of the GPIP. The positive y-axis is rearward of the GPIP and the negative y-axis is forward of the GPIP. The positive z-axis is upward of the GPIP and the negative z-axis is downward of the GPIP.

[0678] Referring to FIGS. 120-124, and according to one example, a golf club head 1100, which is an iron-type golf club head, includes a body 1102 and a hosel 1108 that is coupled to and extends from the body 1102. Some features of the golf club head 1100 are similar to the features of the iron-type golf club head shown and described in U.S. patent application Ser. No. 15/706,632, filed Sep. 15, 2017. The body 1102 has a toe portion 1114, a heel portion 1112, a top-line portion 1116 (e.g., top portion), a sole portion 1118 (e.g., bottom portion), a front portion 1104, and a rear portion 1105. The toe portion 1114 defines a side of the body 1102 that is opposite a side of the body 1102 defined by the heel portion 1112 in a direction parallel to an x-axis of the club head origin coordinate system 1185. The top-line portion 1116 defines a side of the body 1102 that is opposite a side of the body 1102 defined by the sole portion 1118 in a direction parallel to a z-axis of the club head origin coordinate system 1185. The front portion 1104 defines a side of the body 1102 that is opposite a side of the body 1102 defined by the rear portion 1105 in a direction parallel to a y-axis of the club head origin coordinate system 1185.

[0679] The front portion 1104 of the body 1102 includes a strike face 1106 designed to impact a golf ball during a normal golf swing. The strike face 1106 has a planar surface that is angled relative to a ground plane 111 when the golf club head 1100 is in the address position to define a loft of the golf club head 1100. In other words, the strike face 1106 of the golf club head 1100 does not include a curved surface. Accordingly, the strike face 1106 of the golf club head 1100

is defined as the portion of the front portion 1104 with an outwardly facing planar surface. Although the front portion 1104 may include a curved surface, such as an outer surface of a sole wrap portion, the strike face 1106 does not include or is not defined by such a curved surface. In contrast, the strike face of a metal-wood, driver, or hybrid golf club head does have a curved surface that curves around a substantially upright axis. The strike face 1106 has a face length that is equal to the distance between a par line of the golf club head 1100 and a towardmost point of the golf club head 1100. The par line is defined as a theoretical line defining the transition on the front portion 1104 between a flat surface to a curved surface generally proximate to the heel end of the golf club head 1100. Put another way, the par line defines where the flat surface of the front portion 1104 ends and the curved surface of the front portion 1104 begins.

[0680] The front portion 1104 further includes grooves 1107 (e.g., scorelines), formed in the strike face 1106, which help promote desirable flight characteristics (e.g., backspin) of a golf ball upon impacted by the strike face 1106. The grooves 1107 are vertically spaced apart from, and parallel to, each other and can extend across all or just a portion of the width of strike face 1106 at their respective vertical locations on the strike face 1106.

[0681] The front portion 1104 also includes a front interior surface 1121, which is opposite the strike face 1106 and separated from the strike face 1106 by a thickness FT of the front portion 1104 (see, e.g., FIGS. 121, 122, and 129). In some examples, the thickness FT is variable across the strike face 1106, as described below. In some examples, such as those show in FIGS. 120-129, the portion of the front portion 1104 that defines the strike face 1106 is co-formed with the top-line portion 1116, the sole portion 1118, the heel portion 1112, the toe portion 1114, and at least a portion of the rear portion 1105, such that these portions of the body 1102 and the strike face 1106 form a one-piece, monolithic, seamless, and unitary, construction. Accordingly, in these examples, the strike face 1106 is not defined by a separate strike plate attached to the rest of the body 1102, but instead the body 1102 and the strike face 1106 are formed from the same manufacturing process, such as being co-cast or co-machined together in certain examples. However, in other examples, the strike face 1106 is defined by a strike plate that is formed separately from and attached to (e.g., welded to, adhered to, etc.) the rest of the front portion 1104 of the body 1102. For example, the front portion 1104 can include a front opening and a strike plate can be attached to the front portion 1104 over the front opening. In some implementations, a strike plate can include an entirety of the strike face 1106, or just a portion of the strike face 1106. The strike face 1106 includes a leading edge, which is defined as the forwardmost portion or edge of the strike face 1106.

[0682] The hosel 1108 extends from the heel portion 1112 of the body 1102. The hosel 1108 is configured to receive and engage with a shaft and grip combination of a golf club (not shown). The shaft extends from the hosel 1108 and the grip is secured to the shaft at a location on the shaft opposite that of the golf club head 1100. Referring to FIGS. 123 and 124, the hosel 1108 defines a hosel axis 1197, which is a central axis of the hosel 1108. The hosel axis 1197 helps define a ground plane intersection point GPIP, which is the point at which the hosel axis 1197 intersects the ground plane 111 when the golf club head 1100 is in an address position on the ground plane 111.

[0683] The body 1102 can have a multiple-piece construction with each piece being made from the same or a different material, and/or formed by the same or a different process. Separately forming and attaching together different parts of the body 1102 and/or making different parts of the body 1102 from different materials, enables allows flexibility in the types of manufacturing processes and materials used, promotes the ability to make a golf club head that achieves a wide range of performance, aesthetic, and economic results.

[0684] The rear portion 1105 is coupled to and extends rearwardly from the sole portion 1118. The rear portion 1105 is also coupled to and extends rearwardly from lower parts of the heel portion 1112 and the toe portion 1114. The body 1102 includes a sole bar 1127 formed in part of the rear portion 1105 and part of the sole portion 1118 (see, e.g., FIGS. 128 and 129). The sole bar 1127 is located in a low, rearward portion of the golf club head 1100. The sole bar 1127 has a relatively large thickness in relation to the strike face 1106 of the front portion 1104, and other portions of the golf club head 1100, thereby accounting for a significant portion of the mass of the golf club head 1100, and thereby shifting the CG of the golf club head 1100 relatively lower and rearward.

[0685] Referring to FIG. 120, the body 1102 of the golf club head 1100 further includes a sole slot 1150 formed in the sole portion 1118 of the body 1102. Generally, the sole slot 1150 is a groove or channel formed in a sole of the golf club head 1100. The sole is defined as the exterior surface of the sole portion 1118. The sole slot 1150 is elongated in a lengthwise direction substantially parallel to the strike face 1106. In the illustrated examples, the sole slot 1150 is a through-slot, or a slot that is open on a sole portion side of the sole slot 1150 and open on an internal cavity side or interior side of the sole slot 1150. However, in other implementations, the sole slot 1150 is not a through-slot, but rather is closed on an internal cavity side or interior side of the sole slot 1150. The sole slot 1150 extends from the sole upwards towards the interior cavity 1152. Along a plane that is perpendicular to the strike face 1106, the sole slot 1150 extends from the sole at an angle relative to the strike face 1106. According to some examples, the angle is between, and inclusive of, 1° and 5°, such as 3°.

[0686] In some examples, the sole slot 1150 is filled with a filler material 1151, which can be the same material as or a material similar to the filler material described above. The filler material 1151 is made from a non-metal, such as a thermoplastic material, thermoset material, and the like, in some implementations. In other implementations, the sole slot 1150 is not filled with the filler material 1151, but rather maintains an open, vacant, space within the sole slot 1150. In some embodiments, the filler material 1151 may be press-fit or adhesively bonded into a slot, channel, or other flexible boundary structure. In other embodiments, a filler material may poured, injected, or otherwise inserted into a slot or channel and allowed to cure in place, forming a sufficiently hardened or resilient outer surface. In still other embodiments, a filler material may be placed into a slot or channel and sealed in place with a resilient cap or other structure formed of a metal, metal alloy, metallic, composite, hard plastic, resilient elastomeric, or other suitable material.

[0687] In some examples, as shown in FIGS. 147-149, the sole portion 1118 includes a slot mass pad 1250 within the interior cavity and located directly adjacent to the sole slot

1150 on a forward side (see, e.g., FIG. **147**) and rearward side (see, e.g., FIG. **149**) of the sole slot **1150**. The length LP of the slot mass pad **1250**, in a heel-toe direction, is less than the length SSL of the sole slot **1150**, in the heel-toe direction. In this manner, the sole slot **1150** is deeper (e.g., has a depth ST2) along the slot mass pad **1250** than heelward and toward of the slot mass pad **1250** (e.g., depth ST1). Similarly, as shown in FIG. **148**, the filler material **1151** has a greater thickness (e.g., second filler thickness FT2) along the slot mass pad **1250** than heelward and toward of the slot mass pad **125** (e.g., first filler thickness FT1).

[0688] Referring to FIGS. **120** and **121**, the rear portion **1105** of the body **1102** includes an opening **1178** and a badge **1160** (e.g., a rear panel or rear fascia) coupled to and enclosing the opening **1178**. The opening **1178** is open to an internal cavity **1152** of the golf club head **1100**. In other words, the internal cavity **1152** is accessible through the opening **1178** when uncovered. The rear portion **1105** includes a lip **1180** that continuously surrounds and defines the perimeter of the opening **1178**. The lip **1180** is recessed relative to adjacent surfaces of the rear portion **1105** and configured to receive the badge **1160** in seated engagement. The badge **1160** is affixed to the lip **1180** via any of various techniques, such as welding, brazing, gluing, adhering, fastening, and the like. The size and shape of the outer periphery of the badge **1160** complements the size and shape of the lip **1180**, such that when in seated engagement with the lip **1180**, the badge **1160** covers the entirety of the opening **1178**. In this manner, the badge **1160** encloses the opening **1178**, as well as the internal cavity **1152**, which is defined as the cavity defined between internal surfaces of the heel portion **1112**, the toe portion **1114**, the top-line portion **1116**, the sole portion **1118**, the front portion **1104** and the rear portion **1105**, including the badge **1160**. Accordingly, when the badge **1160** is in seated engagement with the lip **1180**, the body **1102** of the golf club head **1100** is hollow (i.e., the internal cavity **1152** is enclosed). For this reason, the iron-type golf club head **1100** is considered to have a hollow-body design. The material of the badge **1160** can be the same as or similar to the material of the badge described in one or more of the above examples.

[0689] As shown in FIGS. **125-128**, the toe portion **1114** of the golf club head **1100** includes an unsupported badge-engagement wall **1125**. The unsupported badge-engagement wall **1125** includes a forward-facing surface that is spaced apart from, and faces, the front interior surface **1121**, and a rearward-facing surface opposite the forward-facing surface and facing away from the front interior surface **1121**. A thickness of the unsupported badge-engagement wall **1125** is defined between the forward-facing surface and the rearward-facing surface. The rearward-facing surface is bonded to the badge **1160**. The badge **1160** includes a toe wrap portion **1161** that is supported on a toe recessed portion **1124** of the golf club head **1100**. The lip **1180** includes a badge-mounting surface **1193** to which the badge **1160** can be attached. Moreover, a topline rear overhang portion **1132** can engage the badge **1160** such that an outer perimeter portion of the badge **1160** is positioned between the topline overhang portion **1132** and the badge-mounting surface **1193**.

[0690] The rear portion **1105** also includes a rear wall **1162**, which extends from a lower periphery of the opening **1178** down to the sole in some examples. Accordingly, in certain examples, the rear wall **1162** is integrated into the

sole bar **1127** and defines a lower portion of the lip **1180**. The rear wall **1162** defines a rear interior surface **1177** within the interior cavity **1152** that faces the front interior surface **1121**. The rear wall **1162** is spaced apart from the front interior surface **1121**, such that a space is defined between the rear interior surface **1177** and the front interior surface **1121**.

[0691] Referring to FIGS. **121**, **122**, **125**, **128**, **129**, and **131-136C**, in some examples, the golf club head **1100** further includes a damper assembly **1201** within the internal cavity **1152**. The damper assembly **1201** includes an internal damper **1170** and a backstop **1200**. The internal damper **1170** provides acoustic dampening of the golf club head **1100**. For example, the internal damper **1170** helps to counteract unpleasant sounds and vibration frequencies produced by various features of the golf club head when the golf club head **1100** impacts a golf ball. Moreover, the internal damper **1170** is configured so that a coefficient of restitution COR of the golf club head **1100** remains above a desired COR while providing acoustic dampening of the golf club head **1100**. The internal damper **1170** is positioned within the internal cavity **1152** so that a portion of the internal damper **1170** is in contact with the front interior surface **1121** of the front portion **1104**, at the strike face **1106**, and a portion of the internal damper **1170** is in contact with the rear interior surface **1177** of the rear wall **1162**. In some examples, the sole portion **1118** includes a sole bar **1127** and the rear interior surface **1177** and the rear wall **1162** is formed in the sole bar **1127**. In some examples, the internal damper **1170** is made of a material that is at least one of less stiff (e.g., softer) than the material of the front portion **1120** and the material of the rear wall **1162**, has a lower modulus of elasticity than that of the material of the front portion **1120** and the material of the rear wall **1162**, or has a lower hardness than that of the material of the front portion **1120** and the material of the rear wall **1162**. The material of the internal damper **1170** can be the same as or similar to the material of the damper described in one or more of the above examples.

[0692] In certain examples, the internal damper **1170** includes front ribs **1174A** (e.g., front projections) on a front facing surface of the internal damper **1170** and rear ribs **1174B** (e.g., rear projections) on a rear facing surface of the internal damper **1170**. The front ribs **1174A** are spaced apart in a heel-to-toe direction and the rear ribs **1174B** are spaced apart in the heel-to-toe direction. Gaps **1175** (e.g., cutouts) are defined between adjacent ones of the front ribs **1174A** and adjacent ones of the rear ribs **1174B**. Only the front ribs **1174A** contact the front interior surface **1121** of the front portion **1104** and only the rear ribs **1174B** contact the rear interior surface **1177**. In this manner, acoustic dampening can be achieved without negatively affecting the COR properties of the golf club head **1100**. In some examples, the front ribs **1174A** are attached to the front interior surface **1121** of the front portion **1104**, and the rear ribs **1174B** are attached to the rear interior surface **1177** via an adhesive or glue.

[0693] Referring to FIG. **135**, the internal damper **1170** has a height IDH (e.g., a top-to-bottom dimension), generally parallel to the strike face **1106** and a top-to-sole direction, and a width (e.g., a front-to-rear dimension), generally perpendicular to the strike face **1106**. A length of the internal damper **1170** is generally parallel to the strike face **1106** and a heel-to-toe direction. In some examples, one or both of the height IDH and the width of the internal damper **1170** varies

along a length of the internal damper **1170**. For example, in a heel-to-toe direction, the width of the internal damper **1170** increases such that the internal damper **1170** widens in the heel-to-toe direction. Such a configuration provides increased acoustic dampening toward of a geometric center CF of the strike face **1106**, wherein effects of acoustic vibrations may be more pronounced, relative to heelward of the geometric center CF. As defined herein, the geometric center CF can be an ideal impact location on the strike face **1106**. Additionally, or alternatively, the geometric center CF can be defined by the USGA center face template method, as is known in the art. According to some examples, the geometric center CF is an ideal impact location on the strike face **1106**, which is defined as a point on the strike face **1106** that lies in a horizontal plane (parallel to the ground plane **111**) that passes through the CG, and lies in a vertical plane (perpendicular to the ground plane **111**) that passes through the CG and is perpendicular to the strike face **1106**, when the golf club head is in the address position on the ground plane **111**. Additionally, or alternatively, as shown, in the heel-to-toe direction, the height IDH of the internal damper **1170** increases for at least a portion of the length of the internal damper **1170** so that the internal damper **1170** gets taller for a first portion of the internal damper **1170** and shorter for a second portion of the internal damper **1170**. Where the internal damper **1170** is taller, the internal damper **1170** contacts more surface area than where the internal damper **1170** is shorter.

[0694] The backstop **1200** is at least partially embedded in the material of the internal damper **1170**. As used herein, a first part or first feature is embedded (i.e., fully embedded) in a second part when the first part or first feature is fully enclosed closely in the material of the second part. Accordingly, the backstop **1200**, being at least partially embedded in the material of the internal damper **1170**, means at least one portion, feature, or surface of the backstop **1200** is not fully enclosed closely in the material of the internal damper **1170**. The backstop **1200** is made of a material that is at least one of stiffer than the material of the internal damper **1170**, but less stiff than the material of the front portion **1104** and the material of the rear wall **1162**, harder than the internal damper **1170**, but softer than the material of the front portion **1104** and the material of the rear wall **1162**, or stronger (i.e., has a high modulus of elasticity) than the material of the internal damper **1170**, but weaker than the material of the front portion **1104** and the material of the rear wall **1162**. However, in certain examples, the modulus of elasticity of the material of the backstop **1200** can be greater than the modulus of elasticity of the material of the front portion **1104**. In some examples, the internal damper **1170** is made of a first type of polymeric material, the backstop **1200** is made of a second type of polymeric material, different than the first type of polymeric material, and the front portion **1104** and the rear wall **1162** are made of a metallic material. However, in other examples, the internal damper **1170** is made of a polymeric material, the backstop **1200** is made of a first type of metallic material (e.g., aluminum), and the front portion **1104** and the rear wall **1162** are made of a second type of metallic material (e.g., steel or titanium), different than the first type of metallic material. According to certain examples, the hardness of the material of the internal damper **1170** is less than 5 on the Shore D scale, and the hardness of the material of the backstop **1200** is greater than 40 on the Shore D scale.

[0695] In some examples, the modulus of elasticity of the material of the backstop **1200** is at least 0.75 megapascals (MPa) or 108.7 pounds per square inch (psi) and up to 1220 gigapascals (GPa), preferably the modulus of elasticity of the material of the backstop **1200** is at least 0.90 GPa and up to 300 GPa, preferably 0.90 GPa and up to 125 GPa, 0.90 GPa and up to 75 GPa. In these or other examples, a ratio of the modulus of elasticity of the material of the front portion **1104** (e.g., the strike face **1106**) to the modulus of elasticity of the material of the backstop **1200** is between, and inclusive of, 0.004 and 6.200, even more preferably between 0.004 and 1.428, between 0.004 and 0.620, between 0.004 and 0.380, 0.004 and 0.024. According to certain examples, a ratio of the modulus of elasticity of the material of the internal damper **1170** to the modulus of elasticity of the material of the backstop **1200** is preferably less than 1 and greater than 0, and more preferably between, and inclusive of, 0.000000625 and 0.070, more preferably between, and inclusive of, 0.000000625 and 0.035. Preferably the modulus of elasticity of the material of the internal damper is less than the modulus of elasticity of the material of the strike face, and preferably modulus of elasticity of the material of the internal damper is less than the modulus of elasticity of the material of the backstop. Preferably, the modulus of elasticity of the material of the backstop is less than the modulus of elasticity of the material of the strike face, however the modulus of elasticity of the material of the backstop may be greater than the modulus of elasticity of the material of the strike face. Preferably, a ratio of a maximum front to back dimension of the internal damper **1170** (depth of damper) (as measured perpendicular to the front surface of the strike face) to the maximum front to back dimension of the backstop **1200** (depth of backstop) (as measured perpendicular to the front surface of the strike face) is between, and inclusive of, 1.15 and 2.25. The depth of the backstop is measured from a forwardmost surface of the backstop **1200** to a rearward most surface of the backstop **1200**. Similarly, the depth of the internal damper **1170** is measured from a forwardmost surface of the internal damper **1170** to a rearward most surface of the internal damper **1170**. In the case of a backstop co-formed with the body of the golf club head, an internal damper preferably has a maximum dimension that extends 15% to 225% of a maximum depth of the co-formed backstop. In other words, at least a portion of the internal damper is more proximate the strike face than the backstop.

[0696] Referring to FIGS. **132** and **134**, the backstop **1200** includes a base plate **1204**, a first protrusion **1206A** protruding from the base plate **1204**, and a second protrusion **1206B** protruding from the base plate **1204**. In some examples, the base plate **1204**, the first protrusion **1206A**, and the second protrusion **1206B** are made of the same material, are co-formed, and share a one-piece monolithic and seamless construction.

[0697] The base plate **1204** has a height (e.g., in a top-to-sole direction) that is greater than a height of the first protrusion **1206A** and the second protrusion **1206B**. Accordingly, the base plate **1204** includes an upper section that extends upwardly toward the top portion **1116** away from the first protrusion **1206A** and the second protrusion **1206B**, and the base plate **1204** includes a lower section that extends downwardly toward the sole portion **1118** away from the first protrusion **1206A** and the second protrusion **1206B**. In some examples, at least a portion of the upper section and

the lower section of the base plate **1204** is embedded in the material of the internal damper **1170**. According to one example, the height of the base plate **1204** is such that a portion of the upper section of the base plate **1204** is not embedded in the material of the internal damper **1170** and extends out from the material of the internal damper **1170** upwardly toward the top portion **1116**. In some examples, an entirety of the lower section of the base plate **1204** is embedded in the material of the internal damper **1170**. The [0698] In some examples, the first protrusion **1206A** and the second protrusion **1206B** each has a width (e.g., in a toe-to-heel direction) that is the same as or smaller than the width of the base plate **1204**. The width of the upper section of the base plate **1204** can be the same as the width of the lower section of the base plate **1204**. However, as shown in FIG. 146, the width of the lower section can be different than the width of the upper section. In one example, as shown, the width of the lower section of the base plate **1204** increases or diverges in a downwardly direction toward the sole portion **1118**. Such a diverging lower section helps to stabilize and secure the backstop **1200** within the internal damper **1170**. Additionally, as shown in FIG. 140, the length of the lower section can be equal to or greater than the length of the upper section. The added length of the lower section can help with stabilization and retainment as mentioned above.

[0699] The first protrusion **1206A** protrudes from the base plate **1204** in a direction opposite that of the second protrusion **1206B**. Moreover, each one of the first protrusion **1206A** and the second protrusion **1206B** protrudes a certain distance away from the base plate **1204**, which defines a height of the protrusions. Accordingly, each one of the first protrusion **1206A** and the second protrusion **1206B** has a proximal end connected to base plate **1204**, and a distal end away from the base plate **1204** and separated from the base plate **1204** by the height of the corresponding protrusion.

[0700] The backstop **1200** includes a discrete deflection impeding surface **1202A** and a rear engagement surface **1202B**. The discrete deflection impeding surface **1202A** is defined by the distal end of the first protrusion **1206A**. The rear engagement surface **1202B** is defined by the distal end of the second protrusion **1206B**. The shape of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** corresponds with shape of the first protrusion **1206A** and the second protrusion **1206B**, respectively. In the illustrated example of FIGS. 132 and 134, each one of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** has a circular shape. However, in other examples, such as the backstop **1200C** shown FIG. 139, each one of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** has a non-circular shape, such as a half circle or partial circle. In some examples, each one of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** has the same shape and/or size. However, in other examples, the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** are differently sized and/or shaped.

[0701] In the illustrated examples, each one of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** is flat. The size (i.e., surface area) of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** corresponds with the size of the perimeter of the corresponding protrusion in some

examples. Additionally, in some examples, the size (i.e., surface area) of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** is also dependent on the presence or absence of manufacture-enabling features formed in the backstop **1200**. For example, in some examples, the backstop **1200** includes a first blind hole **1208A** formed in the discrete deflection impeding surface **1202A** and a second blind hole **1208B** formed in the rear engagement surface **1202B**. The blind holes are configured to receive pins that hold the backstop **1200** in place within a mold as the material of the internal damper **1170** is deposited into the mold and around the backstop **1200**. After forming the internal damper **1170**, the pins are removed from the first blind hole **1208A** and the second blind hole **1208B**. In some examples, the blind holes are located at the respective centers of the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B**, such that the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** form an annular shape (e.g., annulus) or continuous shape around the blind holes.

[0702] In some examples, the discrete nature of the discrete deflection impeding surface **1202A** corresponding with a relatively small volume of the backstop **1200** relative to the volume of the internal cavity of the golf club head. For example, the volume of the backstop **1200** is no more than 5%, 10%, 15% or 20% of the volume of the internal cavity. The volume of the backstop **1200** can be between, and inclusive of, 0.1% and 5%, 1% and 3%, or 1.2% and 2.5% of the volume of the internal cavity of the golf club head. In certain examples, the total volume of the damper assembly **1201**, which is a summation of the volumes of the internal damper **1170** and the backstop **1200**, is no more than 40% of the volume of the internal cavity. The total volume of the damper assembly **1201** can be between, and inclusive of, 10% and 35% or 15% and 28% of the volume of the internal cavity of the golf club head. In certain examples, the volume of the backstop **1200** is between, and inclusive of, 3% and 15% or 5% and 10% of the total volume of the damper assembly **1201**.

[0703] The backstop **1200** is at least partially embedded in a backstop portion **1176** of the internal damper **1170**. In some examples, the backstop portion **1176** is in an intermediate position along the internal damper **1170** between opposing ends of the internal damper **1170** (e.g., spaced away from the opposing ends). However, the location of the backstop portion **1176** and thus the backstop **1200** is dependent on a location **1109** on the front interior surface **1121** designated to engage (e.g., contact) the discrete deflection impeding surface **1202A** under a predetermined load or corresponding deflection. In certain examples, the backstop portion **1176** does not include vertical projections that extend an entire height of the backstop portion **1176**. Rather, in some examples, the backstop portion **1176** does not include vertical projections on a rear-facing surface of the backstop portion **1176** and includes front ribs **1230** (i.e., vertical projections) on a front-facing surface of the backstop portion **1176** that extend less than an entire height of the backstop portion **1176** (see, e.g., FIGS. 131 and 135). The “shorter” vertical projections or front ribs **1230** of the backstop portion **1176** are spaced apart a distance greater than the distance between the other vertical projections or front ribs **1174A** of the internal damper **1170**. The spacing between the vertical projections or front ribs **1230** of the backstop portion **1176** is such that the ribs **1230** straddle the

discrete deflection impeding surface **1202A** (e.g., an entirety of the discrete deflection impeding surface **1202A** is situated between the projections). The “shorter” vertical projections or front ribs **1174A** of the backstop portion **1176** help to align and position the discrete deflection impeding surface **1202A** relative to the location on the front interior surface **1121** designated to contact the discrete deflection impeding surface **1202A**.

[0704] Referring to FIGS. **129** and **136A**, the damper assembly **1201** is located between the rear interior surface **1177** of the rear wall **1162** and the front interior surface **1121** of the front portion **1104**. More specifically, in certain examples, the damper assembly **1201** is located within the interior cavity **1152** so that the internal damper **1170** of the damper assembly **1201** is wedged between the rear interior surface **1177** and the front interior surface **1121**, the rear engagement surface **1202B** engages the rear interior surface **1177**, and the discrete deflection impeding surface **1202A** is spaced apart from the front interior surface **1121** such that a gap **G2** is defined between the front interior surface **1121** and the discrete deflection impeding surface **1202A**. The gap **G2** is perpendicular to the strike face **1106** in some examples. In other words, in some examples, the discrete deflection impeding surface **1202A** is parallel to the strike face **1106**. In certain examples, the rear interior surface **1177** is not parallel to the strike face **1106**, such that a first angle $\phi 1$ is defined between the strike face **1106** and the rear interior surface **1177** and the gap **G2** is not perpendicular to the rear interior surface **1177**. Accordingly, in these examples, the discrete deflection impeding surface **1202A** defines a second angle $\phi 2$ with the rear engagement surface **1202B**. The second angle $\phi 2$ is equal to the first angle $\phi 1$. Moreover, each one of the first angle $\phi 1$ and the second angle $\phi 2$ is an oblique angle.

[0705] Referring to FIGS. **131**, **133**, **135**, and **136A-136C**, in some examples, the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** are not embedded in the material of the internal damper **1170**. In other words, no portion of the material of the internal damper **1170** is on or covers the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B**. Accordingly, the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** are exposed and directly accessible from outside the internal damper **1170**. In these examples, when the damper assembly **1201** is wedged between the rear interior surface **1177** and the front interior surface **1121**, engagement between the rear engagement surface **1202B** and the rear interior surface **1177** is an abutment (e.g., direct contact) between the rear engagement surface **1202B** and the rear interior surface **1177**, and the discrete deflection impeding surface **1202A** is positioned to directly contact the rear interior surface **1177** when the strike face **1106** is sufficiently deflected.

[0706] According to the examples associated with FIGS. **131**, **133**, **135**, and **136A-136C**, the size of the gap **G2**, in the direction perpendicular to the strike face **1106**, is such that the discrete deflection impeding surface **1202A** impedes deflection of the strike face **1106** only for impacts with the strike face **1106** above a threshold force. In FIG. **136A**, the strike face **1106** is shown experiencing zero impact force and therefore no deflection. Accordingly, the gap **G2** is a maximum gap. Referring to FIG. **136B**, the strike face **1106** is experiencing an impact force **V1** that is less than the threshold force. The impact force **V1** on the strike face **1106**

deflects the strike face **1106** towards the discrete deflection impeding surface **1202A** a certain amount that is less than the size of the gap **G2** before deflection. Accordingly, in FIG. **136B**, the size of the gap **G2** is less than the size of the gap **G2** in FIG. **136A**. Referring to FIG. **136C**, the strike face **1106** is experiencing an impact force **V2** that is more than the threshold force. The impact force **V2** on the strike face **1106** deflects the strike face **1106** towards and into contact with the discrete deflection impeding surface **1202A**, such that there is no gap between the front interior surface **1121** and the discrete deflection impeding surface **1202A** (i.e., the gap **G2** is zero). Contact between the front interior surface **1121** and the discrete deflection impeding surface **1202A** impedes (e.g., slows down) further deflection of the strike face **1106**. In other words, after the front interior surface **1121** contacts the discrete deflection impeding surface **1202A**, the strike face **1106** continues to deflect but the rate of deflection is decreased compared to deflection of the strike face **1106** before contacting the discrete deflection impeding surface **1202A** (see, e.g., FIG. **154**, which is discussed in more detail below).

[0707] According to some examples, to accomplish the above functionality, the size of the gap **G2** is between and inclusive of, 1.0 mm and 2.0 mm, between, and inclusive of, 1.1 mm and 1.9 mm, or between, and inclusive of, 1.2 mm and 1.6 mm. The size of the gap **G2** can also be defined in terms of its ratio with a minimum face thickness **FT** of the front portion **1104** along the strike face **1106**. Because the discrete deflection impeding surface **1202A** reduces the rate of deflection for impacts above a force threshold **1189**, the front portion **1104** can be thinner than would otherwise be recommended for conventional golf club heads. In some examples, the gap **G2** is between, and inclusive of, 75% and 125% of the minimum face thickness **FT** of the front portion **1104**. In other examples the gap **G2** is between, and inclusive of, 50% and 225% or between, and inclusive of, 45% and 255%, of the minimum face thickness **FT** of the front portion **1104**, in other examples the gap **G2** is between, and inclusive of, 60% and 175% of the minimum face thickness **FT** of the front portion **1104**. In some examples, a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 0.75 and 1.25, in other examples a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 0.45 and 2.55, in other examples a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 0.60 and 1.70. In some examples, a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 1.75 and 2.55, in other examples a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 0.45 and 0.95, in other examples a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 1.30 and 1.65. In some examples, a ratio of the gap **G2** and the minimum face thickness **FT** of the front portion **1104** is between, and inclusive of, 0.85 and 1.15.

[0708] The size of the gap **G2** can also be selected to define a desired maximum deflection of the strike face **1106** relative to an impact force. For example, the maximum deflection of the front interior surface **1121**, when the strike face **1106** is impacted by a 10 kN force, can be between, and inclusive of, 155% and 185%, or between, and inclusive of, 125% and 225% of the gap **G2**. In other examples, the front

interior surface **1121** deflects between, and inclusive of, 75% and 225% of the minimum face thickness FT when the strike face **1106** is impacted by a 10 kN force. In yet other examples, the front interior surface **1121** deflects between, and inclusive of, 75% of the minimum face thickness FT and 75% of a maximum face thickness when the strike face **1106** is impacted by a 10 kN force.

[0709] Alternatively, in some examples, as shown in FIGS. 137A-138C, the rear engagement surface **1202B** of a backstop **1200A** of a damper assembly **1201A** is not embedded in the material of an internal damper **1170A** of the damper assembly **1201A**, but the discrete deflection impeding surface **1202A** of the backstop **1200A** is embedded in the second material of the internal damper **1170A**. In other words, the material of the internal damper **1170A** covers the discrete deflection impeding surface **1202A**, but does not cover the rear engagement surface **1202B**. Accordingly, the discrete deflection impeding surface **1202A** is not exposed and is not accessible, except through a layer of the material of the internal damper **1170A**. The layer of the material covering the discrete deflection impeding surface **1202A** has a thickness T7, and a gap G3, perpendicular to the strike face **1106**, is defined between the material covering the discrete deflection impeding surface **1202A** and the front interior surface **1121**. The thickness T7 is between, and inclusive of, 0.5 mm and 1.5 mm. Accordingly, the gap G2 between the front interior surface **1121** and the discrete deflection impeding surface **1202A** is equal to the summation of the thickness T7 and the gap G3. In these examples, when the damper assembly **1201** is wedged between the rear interior surface **1177** and the front interior surface **1121**, the rear engagement surface **1202B** abuts (e.g., directly contacts) the rear interior surface **1177**, but the discrete deflection impeding surface **1202A** is positioned to indirectly contact the rear interior surface **1177**, via the material of the internal damper **1170** covering the discrete deflection impeding surface **1202A**, when the strike face **1106** is sufficiently deflected. In other words, in these examples, the rear engagement surface **1202B** is in direct contact with the rear interior surface **1177**, and the discrete deflection impeding surface **1202A** is configured to merely indirectly contact the front interior surface **1121**.

[0710] According to the examples associated with FIGS. 137A-138C, like the examples associated with FIGS. 131, 133, 135, and 136A-136C, the size of the gap G2, in the direction perpendicular to the strike face **1106**, is such that the discrete deflection impeding surface **1202A** impedes deflection of the strike face **1106** only for impacts with the strike face **1106** above a threshold force. However, the size of the gap G2 in the examples associated with FIGS. 137A-138C is at least partially dependent on the thickness T7 and the characteristics of the material of the internal damper **1170A** covering the discrete deflection impeding surface **1202A**. In certain examples, the greater the thickness T7 and/or the stiffer the internal damper **1170A**, the larger gap G2. In FIG. 138A, the strike face **1106** is shown experiencing zero impact force and therefore no deflection. Accordingly, the gap G2 is a maximum gap. Referring to FIG. 138B, the strike face **1106** is experiencing an impact force V1 that is less than the threshold force. The impact force V1 on the strike face **1106** deflects the strike face **1106** towards the discrete deflection impeding surface **1202A** a certain amount that is less than the size of the gap G3 before deflection. Accordingly, in FIG. 138B, the size of the gap G2

is less than the size of the gap G2 in FIG. 138A. Referring to FIG. 138C, the strike face **1106** is experiencing an impact force V2 that is more than the threshold force. The impact force V2 on the strike face **1106** deflects the strike face **1106** towards and into contact with the material of the internal damper **1170A**, such that there is no gap between the front interior surface **1121** and the material covering the discrete deflection impeding surface **1202A** (i.e., the gap G3 is zero). Contact between the front interior surface **1121** and the material of the internal damper **1170A** covering the discrete deflection impeding surface **1202A** impedes (e.g., slows down) further deflection of the strike face **1106**. In other words, after the front interior surface **1121** contacts the material of the internal damper **1170A** covering the discrete deflection impeding surface **1202A**, the material compresses and the strike face **1106** continues to deflect. However, because of the material of the internal damper **1170A**, and particularly because of the closeness of the discrete deflection impeding surface **1202A** relative to the front interior surface **1121**, the rate of deflection is decreased compared to deflection of the strike face **1106** before contacting the material of the internal damper **1170A** covering the discrete deflection impeding surface **1202A** (see, e.g., FIG. 154, which is discussed in more detail below).

[0711] In yet certain alternative examples, such as shown in FIG. 138D, both the rear engagement surface **1202B** and the discrete deflection impeding surface **1202A** of a backstop **1200B** of a damper assembly **1201B** are embedded in the second material of an internal damper **1170B** of the damper assembly **1201B**. In other words, the material of the internal damper **1170B** covers both the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B**. Accordingly, the discrete deflection impeding surface **1202A** and the rear engagement surface **1202B** are not exposed and are not accessible, except through a corresponding layer of the material of the internal damper **1170B**. The layer of the material covering the rear engagement surface **1202B** has a thickness T8. Accordingly, the gap G2 between the front interior surface **1121** and the discrete deflection impeding surface **1202A** is equal to the summation of the thickness T8 and the gap G3. In these examples, when the damper assembly **1201** is wedged between the rear interior surface **1177** and the front interior surface **1121**, engagement between the rear engagement surface **1202B** and the rear interior surface **1177** is facilitated by an abutment (e.g., direct contact) between the material of the internal damper **1170B**, covering the rear engagement surface **1202B**, and the rear interior surface **1177**. Additionally, the discrete deflection impeding surface **1202A** is positioned to indirectly contact the rear interior surface **1177**, via the material of the internal damper **1170B** covering the discrete deflection impeding surface **1202A**, when the strike face **1106** is sufficiently deflected. In other words, in these examples, the rear engagement surface **1202B** is in indirect contact with the rear interior surface **1177** via the material of the internal damper **1170B** covering the rear engagement surface **1202B**, and the discrete deflection impeding surface **1202A** is configured to merely indirectly contact the front interior surface **1121**.

[0712] In some examples, the gap G2 is greater than the thickness T8. According to certain examples, a summation of the gap G2 and the thickness T8 is between, and inclusive of, 0.9 mm and 2.0 mm. In some examples, a ratio of a

summation of the gap G2 and the thickness T8 to the minimum face thickness FT is between, and inclusive of, 0.49 and 1.48.

[0713] As shown in FIGS. 139-140, in some examples, a damper assembly 1201C includes a backstop 1200C with a base plate 1204 that extends downwardly away from the discrete deflection impeding surface 1202A a distance equal to or greater than the distance the portion of the base plate 1204 extends upwardly away from the discrete deflection impeding surface 1202A. In certain examples, the extra length of the lower portion of the base plate 1204 extending below the deflection impeding surface 1202A, which is embedded within an internal damper 1170C, can help to stabilize the backstop 1200C relative to the internal damper 1170C, with the lower portion of the base plate 1204 acting as an anchor.

[0714] Referring to FIGS. 142-143, according to some examples, a golf club head 1100D includes a damper assembly 1201D. The damper assembly 1201D includes a backstop 1200D partially embedded in an internal damper 1170D. The internal surface of the front portion of the golf club head 1100D includes a discrete protrusion 1210, as defined in more detail below. A rear wall 1162 of the golf club head 1100D includes a tab 1163 that protrudes upwardly from a lip of a rear opening formed in the rear wall 1162. The tab 1163 is sized and shaped to engage with the rear engagement surface 1202B of the backstop 1200D. For example, in the illustrated implementation, the tab 1163 has a semi-circular shape. The tab 1163 is aligned, in a direction perpendicular to the strike face of the golf club head 1100D, with the discrete protrusion 1210 in some examples. Referring to FIG. 149, in some examples, the rear wall 1162 includes a tab support portion 1244 that extends from the sole slot 1150 up to and forms part of tab 1163. The tab support portion 1244 has an increased thickness (e.g., mass pad) that helps to strengthen and stiffen the tab 1163.

[0715] The baseplate 1204 of the backstop 1200D extends upwardly away from the rear engagement surface 1202B. Additionally, the baseplate 1204 includes a diverging portion 1242 that has a width that increases in a direction downwardly away from the rear engagement surface 1202B. Accordingly, the diverging portion 1242 has a maximum width at a maximum distance away from the rear engagement surface 1202B and a minimum width at the rear engagement surface 1202B. The diverging portion 1242 has a fan-shape in some examples. The diverging nature of the diverging portion 1242 helps to stabilize and retain the backstop 1200D relative to the internal damper 1170D, with the diverging portion 1242 acting as an anchor with a relatively larger surface area than in the example of the backstop 1200C. According to some examples, the internal damper 1170D includes an aperture 1242 below the rear engagement surface 1202B. The diverging portion 1242 is accessible through the aperture 1242 in some examples. To accommodate various sizes of the internal damper 1170D, and reduce the overall size of the damper assembly 1201D, in some examples, the backstop portion of the internal damper 1170D includes a retaining ring 1240 that extends above the internal damper 1170D on both sides of the retaining ring 1240. In some examples, the retaining ring 1240 is taller than the tallest portion of the internal damper 1170D. The retaining ring 1240 is configured to help secure and retain the backstop 1200D in the internal damper 1170D.

[0716] Referring back to FIGS. 123 and 129, the discrete deflection impeding surface 1202A is located, relative to the front interior surface 1121, so that the discrete deflection impeding surface 1202A is in line (e.g., aligned) with the location 1109 on the front interior surface 1121 that is designated to engage (e.g., contact) the discrete deflection impeding surface 1202A under a predetermined load or corresponding deflection. In some examples, the discrete deflection impeding surface 1202A is in line with the location 1109 when a line 1212, passing through a center of the discrete deflection impeding surface 1202A and perpendicular to the strike face 1106, passes through the location 1109 on the front interior surface 1121 (see, e.g., FIGS. 129 and 136A). The line 1212 can define a central axis of the deflection impeding surface 1202A.

[0717] The location 1109 on the front interior surface 1121 is predetermined to promote a desired reduction in the deflection rate of the strike face 1106 for impact forces to the strike face 1106 that are above a desired threshold force. For golf club heads, the desired threshold force is associated with the force created by a standard golf ball when impacted by the golf club head on the center face CF when traveling at a desired club-head speed. Accordingly, the location 1109 on the front interior surface 1121 is predetermined to promote a desired reduction in the deflection rate of the strike face 1106 for club-head speeds that are above a desired club-head speed threshold.

[0718] According to one example, the location 1109 is the location on the front interior surface 1121 corresponding with a maximum deflection (MD) point of the front portion 1104 (e.g., the strike face 1106). The MD point is the point on the front interior surface 1121 that deflects (e.g., moves) the most toward the rear portion 1105 when the strike face 1106 is deflected in response to an impact on the center face CF. The MD point has a corresponding point on the strike face 1106, which is defined as the point at which a line, passing through the MD point and perpendicular to the strike face 1106, intersects the strike face 1106. In some examples, the MD point is the center face CF (e.g., geometric center or balancing point) of the strike face 1106. However, as shown in FIG. 123, the MD point is offset from the center face CF. For example, the MD point can be heelward and soleward of the center face CF, as shown in FIG. 123. However, in other examples, the MD point can be one or more of heelward, toward, soleward, or upward of the center face CF. According to certain examples, the location 1109 is no more than 10 mm toward and no more than 10 mm heelward of the center face CF, and no more than 10 mm above and no more than 10 mm below the center face CF.

[0719] In some examples, the front interior surface 1121 of the front portion 1104 includes a discrete feature specifically configured to engage the discrete deflection impeding surface 1202A. The discrete feature includes a backstop-engaging surface that defines a portion of the front interior surface 1121. The backstop-engaging surface of the discrete feature is the surface that faces the discrete deflection impeding surface 1202A. The backstop-engaging surface is sized and shaped to engage the discrete deflection impeding surface 1202A. In some examples, the size and/or shape of the discrete feature, and thus the size and/or shape of the backstop-engaging surface, matches the size and/or shape of the discrete deflection impeding surface 1202A. For example, the discrete feature and the backstop-engaging surface can be circular or non-circular to match a circular or

non-circular shape, respectively, of the discrete deflection impeding surface **1202A**. In some examples, the backstop-engaging surface and the discrete deflection impeding surface **1202A** are flat, but can be non-flat in certain examples. When the interior surface **1121** includes the discrete feature, the gap **G2** is measured from the backstop-engaging surface of the discrete feature.

[0720] Referring to FIGS. **121**, **126-129**, and **138A**, according to certain examples, the discrete feature of the front interior surface **1121** is a discrete protrusion **1210** that protrudes rearwardly, and towards the discrete deflection impeding surface **1202A**, from the immediately adjacent portion of the front interior surface **1121**. The discrete protrusion **1210** can protrude from the immediately adjacent portion of the front interior surface **1121** a distance that is between, and inclusive of, 5% and 30% of the minimum face thickness (FT). In some examples, the distance that the discrete protrusion **1210** protrudes from the immediately adjacent portion of the front interior surface **1121** is between, and inclusive of, 0.1 mm and 0.3 mm. This distance can vary across a width of the discrete protrusion **1210**. In this manner, the backstop-engaging surface of the discrete protrusion **1210**, which is a distal end surface of the discrete protrusion **1210**, can be parallel to the discrete deflection impeding surface **1202A** when the immediately adjacent portion of the front interior surface **1121** is not parallel to the discrete deflection impeding surface **1202A**. Referring to FIG. **160**, a chart **1260** shows the face thickness FT of the strike face **1106**, at various identified locations on the front interior surface **1121**, for three examples of iron-type golf club heads (e.g., iron-type golf club head **1100A**, iron-type golf club head **1100B**, and iron-type golf club head **1100C**) having a discrete protrusion **1210** at the MD point of the heads.

[0721] In some examples, as shown in FIG. **158**, the discrete deflection impeding surface **1202A** of a backstop **1200J** of a damper assembly **1201F** is defined by the recessed surface of a recess **1205** formed in the base plate **1204** of the backstop **1200J**. The recess **1205** is configured to nestably receive the discrete protrusion **1210** when the strike face **1106** is sufficiently deflected.

[0722] Referring to FIG. **157**, according to certain examples, the discrete feature of the front interior surface **1121** is a discrete recess **1203** formed in the front interior surface **1121**. The recessed surface of the discrete recess **1203** defines the backstop-engaging surface, which is configured to engage the discrete deflection impeding surface **1202A** of a damper assembly when the strike face **1106** is sufficiently deflected. The damper assembly can be a damper assembly **1201E** having a backstop **1200I** where the discrete deflection impeding surface **1202A** protrudes from the internal damper **1170**.

[0723] Similar to Referring to FIG. **157**, as shown in FIG. **159**, in certain examples, the discrete protrusion **1210** includes a recess **1211**. However, unlike the discrete recess **1203**, which is formed into the front interior surface **1121** such that a thickness of the front portion **1104**, at the discrete recess **1203** is thinner than immediately adjacent to the discrete recess **1203**, the recess **1211** is formed into a thickened portion of the front portion **1104** (shown as ridges in cross-section) so that the thickness of the front portion **1104**, at the recess **1211**, is not substantially thinner than immediately adjacent the thickened portion. The recessed surface of the recess **1211** defines the backstop-engaging

surface of the discrete protrusion **1210**. The recess **1211** is configured to nestably receive the discrete deflection impeding surface **1202A**, which can protrude from the internal damper **1170**, when the strike face **1106** is sufficiently deflected.

[0724] Although in previous examples, the discrete deflection impeding surface is formed in a backstop of a damper assembly, in other examples, such as shown in FIG. **150**, the discrete deflection impeding surface is co-formed with or forms a one-piece monolithic construction with the rear wall of the golf club head. Referring to FIG. **150**, a golf club head **1400** is shown. The golf club head **1400** includes some features that are similar to the golf club head **400**. For example, the golf club head **1400** is a hollow-body-type golf club head that includes a body **1402**, which defines a top portion **1416** (e.g., topline portion), a sole portion **1418** (e.g., bottom portion), and a rear portion **1422** of the golf club head **1400**. The golf club head **1400** further includes a front portion **1420**. The front portion **1420** includes a strike face **1406** designed to impact a golf ball during a normal golf swing. The strike face **1406** includes grooves **1411**. The body **402** further includes a rear wall **1462** that is opposite the strike face **1406**, defined by a strike plate **1452**, and defines a portion of the rear portion **1422**. The rear wall **1462** includes a variable thickness region **1445** that includes a pattern of recesses **1447**. The rear wall **1462** is co-formed with the rest of the body **1402**. The body **1402** helps define an internal cavity **1432** of the golf club head **1400** that is defined as the space between a toe portion, a heel portion, the top portion **1416**, the sole portion **1418**, the front portion **1420**, and the rear portion **1422**. In this manner, the internal cavity **1432** is enclosed at least partially by the body **1402**, the strike plate **1452**, and an internal weight **1440** of the golf club head **400**. The golf club head **1400** additionally includes a cavity filler material **1433** within the internal cavity **1432**.

[0725] The rear wall **1462** includes a backstop **1200E** that includes a discrete deflection impeding surface **1202A**. Accordingly, the backstop **1200E** and the discrete deflection impeding surface **1202A** of the backstop **1200E** are formed in the rear wall **1462** of the golf club head **1400** to form a one-piece monolithic seamless construction with the rear wall **1462**. A gap **G2**, as described above, is defined between the discrete deflection impeding surface **1202A** and a front interior surface **1121** of the front portion **1420**, which can be defined by a protrusion **1210**. Accordingly, the backstop **1200E** and the discrete deflection impeding surface **1202A** function the same way as the backstop and discrete deflection impeding surface of the above examples to promote a desired reduction in the deflection rate of the strike face **1406** for impact forces to the strike face **1406** that are above a desired threshold force.

[0726] Although in previous examples, the discrete deflection impeding surface is formed in a backstop of a damper assembly, where the damper assembly includes both an internal damper and a backstop embedded in the internal damper, in other examples, such as shown in FIG. **151**, the internal damper forms the backstop and the discrete deflection impeding surface is defined by the internal damper. Referring to FIG. **151**, the golf club head **1100E** includes an internal damper **1457**. The internal damper **1457** can be inserted into the internal cavity of the golf club head **1100E**, and onto an internal shelf or into an internal pocket, via an opening in the golf club head, such as an opening in a rear

wall of the golf club head. For example, the internal damper **1457** can be inserted through a rear opening and slide along an internal pocket until the internal damper **1457** is secured within the internal cavity via engagement with the internal pocket.

[0727] The internal damper **1457** forms a backstop **1200K** of the golf club head **1100E**. Moreover, the backstop **1200K** defines a discrete deflection impeding surface **1202A** of the golf club head **1100E**. Accordingly, the discrete deflection impeding surface **1202A** is defined by the material of the internal damper **1457**. In some examples, the golf club head **1100E** includes an internal shelf or channel on which the internal damper **1457** is supported. When supported on the internal shelf, the internal damper **1457** is positioned so that a gap **G2**, as described above, is defined between the discrete deflection impeding surface **1202A** and a front interior surface **1121** of the front portion **1104**. The backstop **1200K** and the discrete deflection impeding surface **1202A** function the same way as the backstop and discrete deflection impeding surface of the above examples to promote a desired reduction in the deflection rate of the strike face **1406** for impact forces to the strike face **1406** that are above a desired threshold force.

[0728] Referring to FIGS. **152** and **153**, in one example of a golf club head **1100F**, and similar to the golf club head **1400**, a rear wall **1162** of the rear portion **1105** and a backstop **1200G**, which defines a discrete deflection impeding surface **1202A**, form a one-piece monolithic seamless construction with the rear wall **1162**. However, unlike the golf club head **1400**, the rear wall **1162** includes damper retention features, such as tabs, protrusions, etc., configured to retain an internal damper **1170**. Accordingly, as shown, the backstop **1200G** can be co-formed with the body of the golf club head **1100F** and the internal damper **1170** can be separately formed and attached to and at least partially around the backstop **1200G**. The internal damper **1170** can include backstop retention features, such as channels, pockets, recesses, and the like, which engage the damper retention features to help retain the internal damper **1170** to the backstop **1200G**.

[0729] Referring to FIGS. **155** and **156**, in certain examples, the backstop is co-formed with a badge of the golf club head. In FIG. **155**, a golf club head **1100G** includes a badge **1160A** that covers a rear opening of the rear portion **1105** of the golf club head **1100G**. The badge **1160A** is similar to the badge **1160**, however, the badge **1160A** includes a backstop **1200H**, which defines a discrete deflection impeding surface **1202A**. The backstop **1200H** is made of the same material as the rest of the badge **1160A**, and can be co-formed with the badge **1160A** to form a one-piece monolithic construction with the badge **1160A**, in some examples. But in other examples, the backstop **1200H** is made of a different material than the rest of the badge **1160A**, and is separately formed and attached to the rest of the badge **1160A**.

[0730] The badge **1160A** is attached to the rear wall of the golf club head **1100G** in a spaced apart manner relative to the front interior surface **1121**, such that the gap **G2** is defined between the discrete deflection impeding surface **1202A** and the front interior surface **1121**. However, in an alternative example of a golf club head **1100H**, as shown in FIG. **156**, a badge **1160B**, similar to the badge **1160A**, is attached directly to the front interior surface **1121**, and thus moves in a rearwardly direction when the strike face **1106**

deflects. The backstop **1200H** of the badge **1160B** is also directly attached to the front interior surface **1121**, in some examples, such that the discrete deflection impeding surface **1202A** is attached directly to the front interior surface **1121** and no gap exists between the discrete deflection impeding surface **1202A** and the front interior surface **1121**. However, the badge **1160B** is designed such that a gap **G2** is defined between the rear engagement surface **1202B** of the backstop **1200H** and the rear interior surface **1177** when the badge **1160B** is attached to the front interior surface **1121**. As the strike face **1106** is deflected and the badge **1160B** moves rearwardly, the gap **G2** is reduced until the rear engagement surface **1202B** of the backstop **1200H** contacts the rear interior surface **1177** and the deflection rate of the strike face **1106** is reduced.

[0731] The discrete feature of the front interior surface **1121**, which includes the backstop-engaging surface, and the discrete deflection impeding surface **1202A** are discrete because they have a relatively small surface area compared to a total area of the front interior surface **1121**, a total area of the strike face **1106**, or a total height of the strike face **1106**. In some examples, the total area of the discrete deflection impeding surface **1202A**, which can be the same as the total area of the backstop-engaging surface of the discrete feature of the front interior surface **1121**, is between, and inclusive of, 1% and 25%, between, and inclusive of, 1% and 5%, or is between, and inclusive of, 6% and 10% of the total area of the front interior surface **1121**. According to certain examples, the total area of the discrete deflection impeding surface (**1202A**) is between, and inclusive of, 30 mm² and 1000 mm², between, and inclusive of, 30 mm² and 100 mm², or between, and inclusive of, 35 mm² and 65 mm².

[0732] Referring to the chart **1181** of FIG. **154**, a thickness of the strike face **1106**, a material of the strike face **1106**, and a size of the gap **G2** of the above-referenced examples of a golf club head is such that a first displacement-force rate **1185A** of the strike face **1106** is greater than a second displacement-force rate **1185B** of the strike face **1106**. The first displacement-force rate **1185A** is defined as the displacement of the strike face **1106**, per impact force to the strike face **1106** at a given location on the strike face **1106**, for impact forces below a force threshold **1189**. In contrast, the second displacement-force rate **1185B** is defined as the displacement of the strike face **1106**, per impact force to the strike face **1106** at the given location on the strike face **1106**, for impact forces above the force threshold **1189**. The transition from the first displacement-force rate **1185A** to the second displacement-force rate **1185B** initiates when the strike face **1106** deflects enough that the front interior surface **1121** engages the discrete deflection impeding surface **1202A**. As presented above, engagement between the front interior surface **1121** and the discrete deflection impeding surface **1202A** can include direct contact, in some examples, or indirect contact, such as via direct engagement between the internal damper **1170** and the discrete deflection impeding surface **1202A**.

[0733] According to some examples, the first displacement-force rate **1185A** of the strike face **1106** is at least 30% greater, at least 40% greater, at least 50% greater, at least 60% greater, or at least 70% greater than the second displacement-force rate **1185B**.

[0734] As shown, in certain examples, each one of the first displacement-force rate **1185A** and the second displacement-

ment-force rate **1185B** is a constant rate, which can be a rate that does not vary by more than 5-10%.

[0735] The force threshold **1189** is greater than or equal to 6,000 Newtons (N) in some examples, and greater than or equal to 7,000 Newtons (N) in other examples.

[0736] In certain examples, the strike face **1106** deflects at least 0.8 mm for impact forces less than the force threshold **1189**, and deflects at least 0.8 mm for impact forces greater than the force threshold **1189**. However, the amount of deflection for impact forces less than the force threshold **1189** is more than the amount of deflection for impact forces more than the force threshold **1189**.

[0737] In some examples, a thickness of the strike face **1106**, a material of the strike face **1106**, and a size of the gap **G2** is such that (i) the discrete deflection impeding surface **1202A** impedes deflection of the strike face **1106** for first impact forces with a golf ball so that a maximum coefficient of restitution (COR) value of the strike face **1106**, when receiving the first impact forces, remains below an upper COR threshold value; (ii) and the discrete deflection impeding surface **1202A** does not impede deflection of the strike face **1106** for second impact forces with the golf ball so that the COR value of the strike face **1106**, when receiving the second impact forces, is higher than the upper COR threshold value. The second impact forces are lower than the first impact forces.

[0738] Any charts and tables disclosed herein, including but not limited to those contained in FIGS. **102**, **106**, and **154** which give exact values, are to be interpreted as also disclosing an embodiment where each of the values is $\pm 10\%$ of the value indicated, and in further embodiments each of the values is $\pm 7.5\%$, $\pm 5\%$, or $\pm 2.5\%$, thereby disclosing distinct upper values for each, distinct lower values for each, as well as closed ranges having upper and lower limiting values.

[0739] As with all the relationships disclosed herein, these relationships are more than mere optimization, maximization, or minimization of a single characteristic or variable, and are often contrary to conventional design thinking yet have been found to achieve a unique balance of the trade-offs associated with competing criteria such as durability, weight distribution, CG placement, impact dynamics, COR and CT characteristics, and desired moments of inertia. The aforementioned balance requires trade-offs among the competing characteristics recognizing key points of diminishing returns. Therefore, this disclosure contains a unique combination of relationships that produce enhanced durability, sound, feel, and performance, and reduce the negative attributes associated with a large internal weight that is not constrained by the strike face. Further, the relative dimensions, including, but not limited to component length, width, depth, thickness, cross-sectional dimensions, their placement within the club head, and their relationships to one another and the other design variables disclosed herein, influence the aforementioned criteria. Additionally, many embodiments have identified upper and/or lower limits ranges. For embodiments outside these ranges or relationships, the performance may suffer and adversely impact the goals of the design. Further, the location and size of the primary beam **467**, and thus the associated mass within the club head, may be considered contrary conventional club head design thinking, particularly when combined with the

disclosed characteristics of the internal weight and the location of the maximum face thickness and/or minimum face thickness.

[0740] According to some examples, one or more of the golf club head **100**, the golf club head **300A**, and the golf club heads **400-400M** includes additional features or is made from additional processes described in one or more of U.S. Pat. No. 8,535,177, issued Sep. 17, 2013; U.S. Pat. No. 8,845,450, issued Sep. 20, 2014; U.S. Pat. No. 8,328,663, issued Dec. 11, 2012; U.S. patent application Ser. No. 14/565,057, filed Dec. 9, 2014; U.S. Pat. No. 9,975,018, issued May 22, 2018; U.S. Pat. No. 9,044,653, issued Jun. 2, 2015; U.S. Pat. No. 9,033,819, issued May 19, 2015; U.S. Pat. No. 6,811,496, issued Nov. 2, 2004; U.S. patent application Ser. No. 15/649,508, filed Jul. 13, 2017; U.S. patent application Ser. No. 15/859,274 filed Dec. 29, 2017; U.S. patent application Ser. No. 15/394,549, filed Dec. 29, 2016; U.S. patent application Ser. No. 15/706,632, filed Sep. 15, 2017; U.S. patent application Ser. No. 16/059,801, filed Aug. 9, 2018; U.S. patent application Ser. No. 16/161,337, filed Oct. 16, 2018; U.S. patent application Ser. No. 16/434,162, filed Jun. 6, 2019; U.S. patent application Ser. No. 15/681,678, filed Aug. 21, 2017; U.S. Pat. No. 8,088,025, issued Jan. 3, 2012; U.S. Pat. No. 10,155,143, issued Dec. 18, 2018; U.S. Pat. No. 9,731,176, issued Aug. 15, 2017, which are all incorporated herein by reference in their entirety. Some features of the golf club head **100**, the golf club head **200**, the golf club head **300A**, and the golf club heads **400-400M** are similar to the features of the iron-type golf club head shown and described in U.S. patent application Ser. No. 15/706,632, filed Sep. 15, 2017, which is incorporated herein in its entirety.

[0741] Features, properties, characteristics, materials, values, ranges, or groups described in conjunction with a particular aspect, embodiment or example of the disclosure are to be understood to be applicable to any other aspect, embodiment or example described herein unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract, and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The disclosure is not restricted to the details of any foregoing embodiments. The disclosure extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract, and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

[0742] Reference throughout this specification to “one example,” “an example,” or similar language means that a particular feature, structure, or characteristic described in connection with the example is included in at least one example of the present disclosure. Appearances of the phrases “in one example,” “in an example,” and similar language throughout this specification may, but do not necessarily, all refer to the same example. Similarly, the use of the term “implementation” means an implementation having a particular feature, structure, or characteristic described in connection with one or more examples of the present disclosure, however, absent an express correlation to indicate otherwise, an implementation may be associated with one or more examples.

[0743] The schematic flow chart diagrams included herein are generally set forth as logical flow chart diagrams. As such, the depicted order and labeled steps are indicative of one example of the presented method. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more steps, or portions thereof, of the illustrated method. Additionally, the format and symbols employed are provided to explain the logical steps of the method and are understood not to limit the scope of the method. Although various arrow types and line types may be employed in the flow chart diagrams, they are understood not to limit the scope of the corresponding method. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the method. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted method. Additionally, the order in which a particular method occurs may or may not strictly adhere to the order of the corresponding steps shown.

[0744] In the above description, certain terms may be used such as “up,” “down,” “upper,” “lower,” “horizontal,” “vertical,” “left,” “right,” “over,” “under” and the like. These terms are used, where applicable, to provide some clarity of description when dealing with relative relationships. But, these terms are not intended to imply absolute relationships, positions, and/or orientations. For example, with respect to an object, an “upper” surface can become a “lower” surface simply by turning the object over. Nevertheless, it is still the same object. Further, the terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to” unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive and/or mutually inclusive, unless expressly specified otherwise. Directions and other relative references (e.g., inner, outer, upper, lower, etc.) may be used to facilitate discussion of the drawings and principles herein, but are not intended to be limiting. For example, certain terms may be used such as “inside,” “outside,” “top,” “down,” “interior,” “exterior,” and the like. Such terms are used, where applicable, to provide some clarity of description when dealing with relative relationships, particularly with respect to the illustrated embodiments. Such terms are not, however, intended to imply absolute relationships, positions, and/or orientations. For example, with respect to an object, an “upper” part can become a “lower” part simply by turning the object over. Nevertheless, it is still the same part and the object remains the same. As used herein, “and/or” means “and” or “or,” as well as “and” and “or.”

[0745] The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise. Further, the term “plurality” can be defined as “at least two.” Accordingly, as used herein, the terms “a,” “an,” and “at least one” encompass one or more of the specified element. That is, if two of a particular element are present, one of these elements is also present and thus “an” element is present. The terms “a plurality of” and “plural” mean two or more of the specified element. As used herein, the term “and/or” used between the last two of a list of elements means any one or more of the listed elements. For example, the phrase “A, B, and/or C” means “A,” “B,” “C,” “A and B,” “A and C,” “B and C,” or “A, B, and C.” As used herein, the term “coupled” generally means physically coupled or linked and does not exclude the presence of intermediate elements between the

coupled items absent specific contrary language. The inventive features include all novel and non-obvious features disclosed herein both alone and in novel and non-obvious combinations with other elements. As used herein, the phrase “and/or” means “and,” “or” and both “and” and “or”. As used herein, the singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise. As used herein, the term “includes” means “comprises.” Any use of terminology such as “at least one of A and B” shall be interpreted to mean “at least one of A or B,” and is not meant to exclude having both A and B, unless noted otherwise.

[0746] The term “about” in some embodiments, is defined to mean within $\pm 5\%$ of a given value, however in additional embodiments any disclosure of “about” may be further narrowed and claimed to mean within $\pm 4\%$ of a given value, within $\pm 3\%$ of a given value, within $\pm 2\%$ of a given value, within $\pm 1\%$ of a given value, or the exact given value. Further, when at least two values of a variable are disclosed, such disclosure is specifically intended to include the range between the two values regardless of whether they are disclosed with respect to separate embodiments or examples, and specifically intended to include the range of at least the smaller of the two values and/or no more than the larger of the two values. Additionally, when at least three values of a variable are disclosed, such disclosure is specifically intended to include the range between any two of the values regardless of whether they are disclosed with respect to separate embodiments or examples, and specifically intended to include the range of at least the A value and/or no more than the B value, where A may be any of the disclosed values other than the largest disclosed value, and B may be any of the disclosed values other than the smallest disclosed value.

[0747] Additionally, instances in this specification where one element is “coupled” to another element can include direct and indirect coupling. Direct coupling can be defined as one element coupled to and in some contact with another element. Indirect coupling can be defined as coupling between two elements not in direct contact with each other, but having one or more additional elements between the coupled elements. Further, as used herein, securing one element to another element can include direct securing and indirect securing. Additionally, as used herein, “adjacent” does not necessarily denote contact. For example, one element can be adjacent another element without being in contact with that element.

[0748] As used herein, the phrase “at least one of”, when used with a list of items, means different combinations of one or more of the listed items may be used and only one of the items in the list may be needed. The item may be a particular object, thing, or category. In other words, “at least one of” means any combination of items or number of items may be used from the list, but not all of the items in the list may be required. For example, “at least one of item A, item B, and item C” may mean item A; item A and item B; item B; item A, item B, and item C; or item B and item C. In some cases, “at least one of item A, item B, and item C” may mean, for example, without limitation, two of item A, one of item B, and ten of item C; four of item B and seven of item C; or some other suitable combination. Unless otherwise indicated, the terms “first,” “second,” etc. are used herein merely as labels, and are not intended to impose ordinal, positional, or hierarchical requirements on the items to

which these terms refer. Moreover, reference to, e.g., a “second” item does not require or preclude the existence of, e.g., a “first” or lower-numbered item, and/or, e.g., a “third” or higher-numbered item.

[0749] Unless otherwise indicated, the terms “first,” “second,” etc. are used herein merely as labels, and are not intended to impose ordinal, positional, or hierarchical requirements on the items to which these terms refer. Moreover, reference to, e.g., a “second” item does not require or preclude the existence of, e.g., a “first” or lower-numbered item, and/or, e.g., a “third” or higher-numbered item.

[0750] As used herein, a system, apparatus, structure, article, element, component, or hardware “configured to” perform a specified function is indeed capable of performing the specified function without any alteration, rather than merely having potential to perform the specified function after further modification. In other words, the system, apparatus, structure, article, element, component, or hardware “configured to” perform a specified function is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the specified function. As used herein, “configured to” denotes existing characteristics of a system, apparatus, structure, article, element, component, or hardware which enable the system, apparatus, structure, article, element, component, or hardware to perform the specified function without further modification. For purposes of this disclosure, a system, apparatus, structure, article, element, component, or hardware described as being “configured to” perform a particular function may additionally or alternatively be described as being “adapted to” and/or as being “operative to” perform that function.

[0751] In view of the many possible embodiments to which the principles of the disclosure may be applied, it should be recognized that the illustrated embodiments are only preferred examples and should not be taken as limiting the scope of the disclosure. Various modifications may be made thereto without departing from the broader spirit and scope of the disclosure as set forth. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense. Accordingly, the scope of the disclosure is at least as broad as the full scope of the following exemplary claims and their equivalents.

[0752] The present subject matter may be embodied in other specific forms without departing from its spirit or essential characteristics. The described examples are to be considered in all respects only as illustrative and not restrictive. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

1. An iron-type golf club head, comprising:
 - a heel portion;
 - a toe portion;
 - a sole portion;
 - a top portion;
 - a front portion comprising a strike face and a front interior surface opposite the strike face;
 - a rear portion; and
 - a discrete deflection impeding surface formed in or attached to the rear portion and spaced apart from the front interior surface such that a gap, perpendicular to the strike face, is defined between the front interior surface and the discrete deflection impeding surface,

wherein the gap is between, and inclusive of, 75% and 125% of a minimum face thickness of the front portion.

2-7. (canceled)

8. The iron-type golf club head according to claim 1, wherein a total area of the discrete deflection impeding surface is between, and inclusive of, 1% and 25% of a total area of the front interior surface.

9. The iron-type golf club head according to claim 8, wherein the total area of the discrete deflection impeding surface is between, and inclusive of, 1% and 5% of the total area of the front interior surface.

10. The iron-type golf club head according to claim 8, wherein the total area of the discrete deflection impeding surface is between, and inclusive of, 6% and 10% of the total area of the front interior surface.

11-13. (canceled)

14. The iron-type golf club head according to claim 8, wherein:

the front interior surface of the front portion further comprises a discrete protrusion protruding rearwardly from an immediately adjacent portion of the front interior surface;

the discrete protrusion protrudes from the immediately adjacent portion a distance that is between, and inclusive of, 5% and 30% of the minimum face thickness.

15-20. (canceled)

21. The iron-type golf club head according to claim 8, wherein a ratio of a size of the gap to a maximum height of the strike face is at most 0.05 and at least 0.02.

22-26. (canceled)

27. The iron-type golf club head according to claim 21, wherein:

the rear portion further comprises a rear opening and a badge at least partially covering the rear opening.

28-30. (canceled)

31. The iron-type golf club head according to claim 21, wherein the rear portion comprises a rear wall and the discrete deflection impeding surface forms a one-piece monolithic seamless construction with the rear wall.

32-44. (canceled)

45. An iron-type golf club head, comprising:

a heel portion;

a toe portion;

a sole portion;

a top portion;

a front portion comprising a strike face, made of a first material having a first modulus of elasticity, and a front interior surface opposite the strike face;

a rear portion comprising a rear wall that defines a rear interior surface; and

a damper assembly comprising:

an internal damper made of a second material having a second modulus of elasticity that is less than the first modulus of elasticity; and

a backstop at least partially embedded in the second material of the internal damper made of a third material having a third modulus of elasticity less than the first modulus of elasticity, wherein the backstop comprises a discrete deflection impeding surface and a rear engagement surface;

wherein:

the damper assembly is located between the rear interior surface of the rear wall and the front interior surface of the front portion such that the internal

damper is wedged between the rear interior surface and the front interior surface, the rear engagement surface engages the rear interior surface, and the discrete deflection impeding surface is spaced apart from the front interior surface such that a gap, perpendicular to the strike face, is defined between the front interior surface and the discrete deflection impeding surface; and

the gap is between, and inclusive of, 75% and 125% of a minimum face thickness of the front portion.

46-48. (canceled)

49. The iron-type golf club head according to claim **45**, wherein a ratio of a size of the gap to a maximum height of the strike face is at most 0.05 and at least 0.02.

50. The iron-type golf club head according to claim **45**, wherein a ratio of the first modulus of elasticity to the third modulus of elasticity is between, and inclusive of, 0.000000625 and 0.070.

51. The iron-type golf club head according to claim **50**, wherein a ratio of the second modulus of elasticity to the third modulus of elasticity is between, and inclusive of, 0.000000625 and 0.070.

52. The iron-type golf club head according to claim **45**, wherein:

the backstop further comprises a base plate, a first protrusion protruding from the base plate in a first direction, and a second protrusion protruding from the base plate in a second direction, opposite the first direction; and

the discrete deflection impeding surface is defined by a distal end of the first protrusion; and

the rear engagement surface is defined by a distal end of the second protrusion.

53. The iron-type golf club head according to claim **52**, wherein:

the base plate comprises an upper section extending upwardly toward the top portion away from the first protrusion and the second protrusion;

the base plate comprises a lower section, extending downwardly toward the sole portion away from the first protrusion and the second protrusion; and

at least a portion of the upper section and the lower section of the base plate is embedded in the second material of the internal damper.

54-56. (canceled)

57. The iron-type golf club head according to claim **45**, wherein:

the discrete deflection impeding surface and the rear engagement surface are not embedded in the second material; and

the rear engagement surface abuts the rear interior surface.

58. The iron-type golf club head according to claim **45**, wherein:

the discrete deflection impeding surface and the rear engagement surface are both embedded in the second material; and

the rear engagement surface is spaced apart from the rear interior surface such that a thickness, perpendicular to the rear interior surface, of the second material is between the rear interior surface and the rear engagement surface.

59. The iron-type golf club head according to claim **58**, wherein the gap is greater than the thickness of the second material.

60. (canceled)

61. The iron-type golf club head according to claim **59**, wherein a ratio of a summation of the gap and the thickness of the second material to the minimum face thickness is between, and inclusive of, 0.49 and 1.48.

62-67. (canceled)

68. The iron-type golf club head according to claim **45**, wherein:

a hardness of the third material is greater than hardness of the second material; and

the hardness of the third material is less than the hardness of the first material.

69. The iron-type golf club head according to claim **68**, wherein:

the hardness of the second material is less than 5 Shore D; and

the hardness of the third material is greater than 40 Shore D.

70-72. (canceled)

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